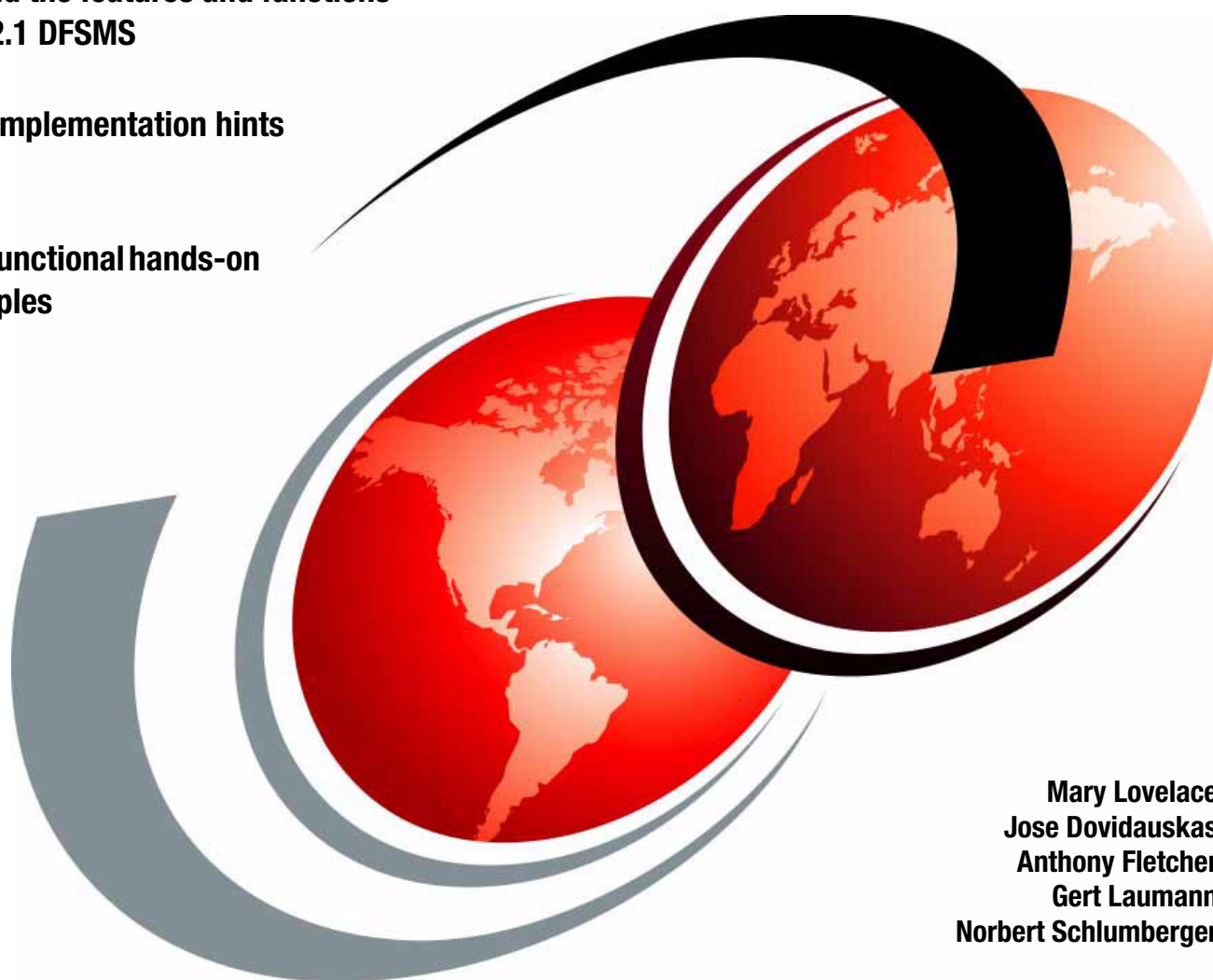


z/OS V2.1 DFSMS Technical Update

Understand the features and functions
of z/OS V2.1 DFSMS

Contains implementation hints
and tips

Provides functional hands-on
code samples



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Redbooks



International Technical Support Organization

z/OS V2.1 DFSMS Technical Update

June 2014

Note: Before using this information and the product it supports, read the information in “Notices” on page xi.

First Edition (June 2014)

This edition applies to Version 2, Release 1 of z/OS (product number 5650-zos).

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Preface

Each release of IBM z/OS® DFSMS builds upon the previous version to provide enhanced storage management, data access, device support, program management, and distributed data access for the z/OS platform in a system-managed storage environment.

This IBM® Redbooks® publication provides a summary of the functions and enhancements integrated into z/OS V2.1 DFSMS. It provides you with the information that you need to understand and evaluate the content of this DFSMS release, along with practical implementation hints and tips.

This book is written for storage professionals and system programmers who have experience with the components of DFSMS. It provides sufficient information so that you can start prioritizing the implementation of new functions and evaluating their applicability in your DFSMS environment.

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Gert Laumann is an IT Specialist in Integrated Technology Delivery, Server Systems Operations / Storage Management in Denmark. He has 28 years of experience in z/OS, working with storage management since 1989. He is the team leader for the Danish mainframe storage team. His experience is mostly with IBM products (DFSMSdftp, DFSMSdss, DFSMShsm, and DFSMSrmm), but he has also worked with OEM software and hardware products. His focus has mainly been automation and standardization across mainframe customer platforms. He currently helps customers share IBM DASD and tape hardware.

Norbert Schlumberger is an IT Architect with IBM Germany. He has 35 years of experience in storage software and storage management for IBM and customer systems, including 24 years of experience with DFSMSrmm. He has developed, delivered, and taught DFSMSrmm education to customers around the world. He has many tools available, such as a Tape Copy Tool to support the DFSMSrmm business. His areas of expertise include performing conversions from vendor tape management products to DFSMSrmm and new DFSMSrmm implementations. Norbert provides marketing support for DFSMSrmm, including IBM 3494 and IBM 3495 ATs, VTSs and TS7700 Virtual Engines. He has worked at IBM for 39 years.

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DFSMS V2.1 release summary

This chapter provides a summary of the features and functions in z/OS V2.1 DFSMS. The enhancements are listed based on the component of DFSMS. The implementation and use of the enhancements are described in the remaining chapters of this publication.

This chapter covers the following topics:

- ▶ DFSMS catalog enhancements
- ▶ VSAM and VSAM RLS enhancements
- ▶ PDSE enhancements
- ▶ SMS enhancements
- ▶ DFSMSdfp enhancements
- ▶ DFSMSdss enhancements
- ▶ DFSMShsm enhancements
- ▶ DFSMSrmm enhancements
- ▶ Advanced Copy Services enhancements
- ▶ DADSM/CVAF enhancements
- ▶ Object Access Method enhancements

1.1 DFSMS catalog enhancements

The catalog component of DFSMS V2.1 provides the following enhancements:

- ▶ VSAM record-level sharing (RLS) directory only caching

This enhancement adds the new DIRONLY parameter to DATACLAS RLSCFCACHE which specifies that RLS not cache the data or index part of the VSAM data set in the coupling facility cache structure.

- ▶ Generation data set (GDS) support for PDSE data sets

This enhancement removes the restriction against defining an SMS-managed partitioned data set extended (PDSE) as a generation data set (GDS). Both allocating a PDSE and defining a generation data group with generation data sets, including PDSEs, is unchanged. Details are provided in Chapter 2, “Catalog enhancements” on page 11.

The LISTCAT ENTRY output is enhanced to indicate when a generation data set is a PDSE by adding the DSNTYPE field with a value of LIBRARY.

- ▶ New Catalog Search Interface (CSI) field names

You can now access the following fields using the Catalog Search Interface:

- ASSOC
- ASSOCSYB
- BUFND
- BUFN
- HILVLRBA
- NDXLVLS
- SEQSTRBA
- STRNO
- TRACKS

- ▶ JES3 allocation assist tape TS7700

For scratch and specific allocations, this enhancement allows you to use JES3 to direct the allocations to candidate clusters for scratch mounts or to particular distributed library clusters for specific mounts in the TS7700 Virtualization Engine.

- ▶ Validate and remove an incorrect DEB address from the DEB table

This function introduces the new PURGE,PURGE=FORCE option for the DEBCHK macro that tells catalog to validate and remove an incorrect DEB address from the DEB table. This is used when a DEB is FREEMAINED, but for some reason the DEB table was not updated to remove that DEB address from the table.

- ▶ IDCAMS support for large block interface (LBI)

This enhancement allows IDCAMS REPRO and PRINT commands to perform on data sets with a blocksize larger than 32K, up to the maximum that the LBI interface supports, if the LBI feature is enabled. The blocksize is still limited to 32K when the LBI feature is not enabled.

- ▶ Catalog contention detection enhancements

The new MODIFY CATALOG,CONTENTION command can be used to specify a new wait time or action (or both) for one of the reason classes or catalog resources for which contention detection is available (ALLOCLCK, SYSIGGV2, SYSZTIOT, and SYSZVVDs).

- ▶ Generation data group enhancements

You can now specify the order in which the generation data set list is to be returned for data set allocation when the generation data group (GDG) name is supplied on the DD

statement. GDG entries can now be returned in either FIFO (oldest GDS defined to the newest GDS) or LIFO (newest GDS defined to the oldest GDS) order for concatenation.

- ▶ Catalog alias enhancements
 - IDCAMS now resolves the symbolic related name for an alias to make sure requests are oriented to the correct catalog. Previously, orientation was to the master catalog, which could cause unexpected results. The restriction on the IDCAMS DEFINE ALIAS command that the resolved value for entryname must be a catalog entry that is located in the same catalog that contains the value for aliasname has been removed.
 - IDCAMS DEFINE ALIAS command will record the alias creation date. This date can be helpful when cleaning up obsolete high level qualifiers. If an alias has no associated data sets, the alias creation date can be used to determine whether this is a new alias for which no data sets have been created yet, or this is an obsolete alias that should be deleted.
 - IDCAMS will now check when deleting a catalog entry that has an associated alias to verify that the alias is related to the entry being deleted, before deleting the alias record. For example, non-VSAM record A has alias association C, but alias C has association D in its X record. In this case, the alias C should not be deleted when data set A is deleted. This check is done for all non-VSAM, GDS, and UCON records.

1.2 VSAM and VSAM RLS enhancements

z/OS V2.1 introduces the following enhancements for VSAM and VSAM RLS:

- ▶ The restriction that data sets accessed by VSAM RLS could not use dynamic volume count is removed. Dynamic volume count enables the dynamic addition of volumes to a DASD data set without increasing the number of candidate volumes stored in the catalog.
- ▶ The SHOWCB macro has added two new sub-parameters, BUFNOL and BUFUSE. These enhancements are used to display fields of an Access Method Control Block (ACB).
- ▶ A new data set attribute identifies whether a data set is eligible for VSAM replication.
- ▶ Expanded and new keywords for specifying record access bias and ACB RMODE31 values for a data class.

1.3 PDSE enhancements

z/OS V2R1 adds the following PDSE enhancements:

- ▶ Increased PDSE member size limits

The maximum PDSE member size increases from 15,728,639 records to 2,146,435,071 records. The larger size limit applies to PDSEs being accessed with various sets of characteristics, including DSORG and MACRF values, whether they are open for input or output, and whether BLOCKTOKENSIZE=LARGE is specified.
- ▶ New PDSE version

z/OS V2.1 introduces a new version of PDSE data sets that can provide for improved performance, reduced path lengths, and improved index searches. New data sets can be allocated as belonging to the new version (version 2) by specifying a new positional parameter in the DSNTYPE keyword of the DD statement or TSO/E ALLOCATE command, or by specifying a new PARMLIB option (PDSE_VERSION) in IGDSMSnn members. Unless version 2 is specified, new allocations continue to create the current

version 1 PDSE data sets. Externally, version 1 and version 2 PDSEs look the same, and both versions can be open for input/output with no changes for the users.

1.4 SMS enhancements

In z/OS V2R1SMS introduces these enhancements:

- ▶ Exploiting clusters and extent pools in SMS volume selection
SMS improves volume selection to exploit clusters and extent pools. SMS now prefers volumes that are in the same cluster when:
 - Allocating or extending a multi-volume data set if the accessibility attribute is CONTINUOUS or CONTINUOUS PREFERRED.
 - Allocating the target data set for the data set fast replication function. When allocating a striped data set, SMS now attempts to allocate the stripes across separate extent pools.
- ▶ Providing accurate space statistics
You can now use a VARY SMS command to update space statistics in the ACDS for a pool storage group or a DASD volume.
- ▶ Providing an option to suppress SMS messages
SMS provides a new keyword in the IGDSMSxx PARMLIB member that allows an installation to suppress specific SMS messages.
- ▶ Providing an option to guarantee that a PDS is created
SMS provides a new keyword in the IGDSMSxx PARMLIB member and SETSMS command that allow you to direct SMS to honor any value for DSNTYPE that specifies that a PDSE is to be allocated, regardless of whether directory blocks have been requested.

1.5 DFSMSdfp enhancements

z/OS V2.1 adds the following DFSMSdfp enhancements:

- ▶ The IEBCOPY utility is enhanced with user exit capabilities for specifying control statements and for specific member selection.
- ▶ The IEBCOPY group copy function is expanded to include PDS to PDS member group copies. A new statement, COPYGROUP, provides the same functions as the existing COPYGRP statement, and expands it to support group copies when both the input and the output data set are PDS format.
- ▶ The IEBCOPY SELECT statement has been enhanced to allow wild card characters in the specification of member names, when used with the COPYGROUP statement. Member name filter pattern masking, using the asterisk (*) and percent (%) characters, allows you to specify a wide range of similar member names. If you use member name filter pattern matching on a SELECT statement with COPYGROUP, you can also code a corresponding EXCLUDE statement with member name filter pattern matching.
- ▶ The IEBCOPY utility is enhanced to provide an ABEND code and the associated reason code in a structure returned in register 0, for certain ABENDs. Starting in z/OS V2.1, IEBCOPY returns this information if an ABEND occurs in the FAMS subcomponent.
- ▶ IEBCOPY introduces a new COPYGROUP statement that provides the same function as the COPYGRP statement but also performs that same function for a PDS to PDS copy.

With COPYGRP, when the input and output data sets are both PDSEs, the operation is treated as a simple COPY operation, not a group copy.

1.6 DFSMSDss enhancements

z/OS V2.1 introduces the following enhancements for DFSMSDss:

- ▶ A RESET keyword has been added to the RESTORE FULL and RESTORE TRACKS commands. It specifies whether the data-set-changed indicator is reset for the data sets on the volume being restored. You can protect the use of RESET with the RESTORE command. In addition, you can now protect the use of RESET with the DUMP command.
- ▶ A DEBUG(SMSMSG) option has been added to the CONVERTV command, the COPY command for logical and physical data sets, and the RESTORE command for logical and physical data sets. It instructs DFSMSDss to include ACS WRITE statements in the job output.
- ▶ The FCCGVERIFY keyword on the CGCREATED command now accepts multiple volume serials.
- ▶ The REPLACEUNCONDITIONAL keyword on the RESTORE command now works for physical data sets.
- ▶ The RENAMEUNCONDITIONAL keyword on the RESTORE command, which previously worked only on non-VSAM physical data sets, will now work on VSAM physical data sets, as well.

1.7 DFSMShsm enhancements

In z/OS V2R1, DFSMShsm is enhanced with the following new functions:

- ▶ Storage tiers

In previous releases, DFSMShsm treated all data in Level 0 (L0) as being in one single tier in the overall storage hierarchy, with no policies to enable automated data movement within that L0 tier. In this release, DFSMShsm is enhanced to move data from one class of devices to another within the L0 hierarchy.

- ▶ Increased tape limit

To allow DFSMShsm to migrate and back up larger data sets, the DFSMShsm limit of the number of tapes that a DFSMShsm migration or backup data set can span has been increased from 40 to 254. RECYCLE will now also process connected sets of up to 254 volumes.

- ▶ Migration subtasking

A MIGRATIONSUBTASKS(YES | NO) parameter has been added to the SETSYS command. It allows DFSMShsm to run multiple subtasks concurrently under each migration task for primary space management, on-demand migration, and interval migration on level 0 volumes that migrate data sets to ML1 or ML2 volumes.

The ADDITIONALSUBTASKS(nn) subparameter allows you to dynamically change the number of additional subtasks that the system can use, running under each migration task. These additional subtasks add to the number of subtasks that the system already uses when the MIGRATIONSUBTASKS parameter is specified. .

By processing data sets in migration subtasks for a level 0 volume migration task, the aggregate throughput of all the migration tasks is improved.

- ▶ Fast replication enhancements
 - Recovering a data set to a new name during fast replication data set recovery. A NEWNAME(newdsname) parameter has been added to the FRRECOV command. It allows DFSMSHsm to use a new, fully-qualified data set name for the recovered backup version or dump copy.
 - Recovering a data set to any volume during fast replication data set recovery
DFSMSHsm fast replication data set recovery will no longer be restricted to recovering data sets back to the original volumes. If DFSMSHsm is not able to recover a data set to the original volumes, it will instead select the most eligible volumes with the most free space within the storage group. There are no changes to the DFSMSHsm commands.
 - IBM FlashCopy® consistency groups
DFSMSHsm fast replication backup will now support FlashCopy consistency groups. A new FlashCopy consistency group option will be added to the SMS copy pool definition. If set to 'Yes' for the copy pool, it indicates that the copy pool backup version must be data-consistent. If the FlashCopy consistency group function fails, the FRBACKUP command will be terminated and the new or in-process copy pool backup version will be invalidated. The FlashCopy consistency group option can be used in combination with other FlashCopy options.
- ▶ Serviceability and usability enhancements:
 - Automatically generate a new recycle command when tape take-away occurs during recycle
A RECYCLETAKEAWAYRETRY(YES | NO) parameter has been added to the SETSYS command. It allows DFSMSHsm to automatically generate a new RECYCLE command for a tape when the original recycle must terminate due to the takeaway process, or when the tape is in use by the another DFSMSHsm task. Two additional subparameters, MAXRETRYATTEMPTS(nn) and DELAY(ssss), allow you to set the maximum number of recycle retry attempts and to set the delay interval in seconds between recycle attempts, respectively.

In addition, a SELECT(RECYCLETAKEAWAY) subparameter has been added to the LIST TAPETABLEOFCONTENTS command. It displays the volumes that were not completely recycled because they were taken away by recall or another DFSMSHsm task
 - Using recycle instead of tapecopy for failed alternate tape
A (TAPECOPY | RECYCLE) subparameter has been added to both the BACKUP and the MIGRATION parameters of the SETSYS DUPLEX command. It specifies whether a tapecopy or a recycle will be immediately attempted, after an error occurs on the duplex alternate tape and the alternate tape is demounted and discarded so that DFSMSHsm can continue to write to the original tape.
 - Resetting the data-set-changed indicator for data sets restored during full volume recover processing
A RECOVERRESET keyword has been added to the DEFINE DUMPCLASS command. It specifies whether the data-set-changed indicator in the VTOC entry is reset for all data sets that are restored during full volume recover processing.
 - Additional output from QUERY ACTIVE(TCBADDRESS)
The output returned from the QUERY ACTIVE(TCBADDRESS) command will now include the tape volser, device address, and task name.
 - SMSVSAM server errors

When an SMSVSAM server error occurs, DFSMSHsm will now detect the error and quiesce all CDS I/O activity. Once the SMSVSAM server initializes, DFSMSHsm will automatically close and reopen the CDSs, and resume all requests waiting for CDS I/O operations.

1.8 DFSMSrmm enhancements

The functional enhancements available with z/OS V2.1 DFSMSrmm provide you with these benefits:

- DFSMSrmm operational enhancements

- Retain data sets based on the number of days since they were last referenced.

Data sets managed by the EXPDT retention method can be retained or expired based on the number of days since the data set was last referenced. A new LastReferenceDays attribute is added to the data set record as a binary number.

The value is taken from the new LASTREF suboperand of the DFSMSrmm parmlib OPTION RM(EXPDT) operand.

If the volume set is retained by SET or VOLUME the LastReferenceDays data set attribute will be kept equal for all files of a multi-volume data set. The latest LastReferenceDays update to a single file in a multivolume data set is propagated to all files that belong to the same multi-volume data set.

However, for volume set retained by FIRSTFILE the LastReferenceDays data set attribute will be set but not equalized across the multi-volume data set because the expiration date depends only of the first file of the first volume, and its LastReferenceDays.

- EXPDT retention method now allows retention to be based on a single volume or volume set, or on a controlling first file.

You now have the option of retaining volumes with the EXPDT retention method based on a single volume or volume set, or on a controlling first file.

- DFSMS management class attributes.

You can now set the expiration date in DFSMSrmm for a tape data set with a DFSMS management class. When you enable use of management class attributes by DFSMSrmm, the management class expiration attributes (except the management class Expiration attribute Retention limit) are retrieved by DFSMSrmm during OPEN for output and used to set the expiration date for the tape data set, and also to set the LASTREF extra days in the tape data set record on retention method EXPDT managed volumes.

- Web page for DFSMSrmm conversion support.

Information on migrating to DFSMSrmm from other products, former found in the EDGCMM01 documentation member of SYS1.SAMPLIB, has been updated and moved the z/OS web site designated for “as-is” downloads as a zipped sequential XMIT file of a PDS. The z/OS downloads Web site can be found at:

<http://www.ibm.com/systems/z/os/zos/downloads/#asis>

1.9 Advanced Copy Services enhancements

z/OS V2.1 adds the Advanced Copy Services enhancement where XADDPAlRed primary volumes can remain offline when you issue the XSTART command for restart or the XADDPAlR command for suspended pairs. Utility volumes and secondary volumes must be online when you issue the XADDPAlR command. Refer to Chapter 9, “System Data Mover enhancements” on page 173 for additional information.

1.10 DADSM/CVAF enhancements

The LSPACE macro now allows the caller to have obtained the SYSVTOC ENQ resource on the volume prior to calling LSPACE for that volume. A new ENQHELD keyword on the LSPACE macro specifies whether or not the LSPACE caller's address space has already obtained the SYSVTOC resource.

1.11 Object Access Method enhancements

z/OS DFSMS V2.1 provides the following enhancements to the object access method (OAM):

- ▶ OAM now supports tape block sizes larger than 32760. A new TAPESDB keyword on the SETOAM statement in the CBROAMxx PARMLIB member can be set to enable larger block sizes. When the first object is written to an OAM tape volume, the maximum block size for the volume is established. If support for larger tape block sizes is enabled, that maximum block size is set to the optimal system-determined block size for the device. Otherwise, a maximum block size of 32760 is used. The maximum block size for all objects written to a tape volume is the maximum block size for the volume established when the first object was written regardless of the current SETOAM TAPESDB setting.
- ▶ OAM now provides a new ALLOCRETRYMINUTES keyword on the SETOAM statement in the CBROAMxx PARMLIB member. This can be set to control how long OAM makes retry attempts or to bypass retry processing entirely and issue message CBR6400D immediately. This can be used with Automated Tape Allocation Manager processing.
- ▶ The minimum object size required to utilize Store Sequence processing (STOREBEG, STOREPRT, STOREEND) has been reduced from the previous limit of 256MB+1 to 50MB+1 for all objects except those being written to an optical volume.
- ▶ Previously, when OSMC moved an object to a different management class, existing backup copies of the object were kept. If the object moves to a management class that requires fewer (or no) OAM backup copies, the extra backup copies remained, but were no longer needed and wasted storage space. OAM now provides a new BACKUPDELETE keyword on the SETOSMC statement in the CBROAMxx PARMLIB member which can be set to indicate that OSMC should delete all unneeded OAM backup copies when processing an object.
- ▶ OAM now provides a new SETTLIB statement for tape library settings. The optional SETTLIB statement and its associated keywords in the CBROAMxx PARMLIB member can be used to override the default behavior for some of the main cartridge entry messages in a system managed tape library environment.

The SETTLIB statement and keywords are processed when the OAM address space is started and cannot be updated by operator command. The SETTLIB statement can be used to specify:

- How OAM will display volume entry ignore messages during cartridge entry processing (DETAIL, SUMMARY, or SUPPRESS).
- Where OAM will display volume entry ignore messages during cartridge entry processing (on both the console and system log or only on the system log).
- Where OAM will display successful volume entry messages (CBR3610I) during cartridge entry processing (on both the console and system log or only on the system log).



Catalog enhancements

This chapter provides an overview of the new features and functions in catalog included in z/OS V2.1. The updates mainly address availability and performance limitations.

The following topics are covered in this chapter:

- ▶ Record Level Sharing support for catalog
- ▶ Preparing to migrate to an RLS enabled ICF Catalog
- ▶ Catalog maintenance considerations in RLS environment
- ▶ Catalog contention detection enhancements
- ▶ Catalog alias enhancements
- ▶ Catalog parmlib member enhancements
- ▶ Catalog DFSMS GDG enhancements
- ▶ Catalog RNLs Health Check
- ▶ MODIFY CATALOG,LIST enhancement
- ▶ Improved catalog recovery features

2.1 Record Level Sharing support for catalog

Record Level Sharing (RLS) access for VSAM is widely used today for transactional (DFSMSStvs) and non-transactional applications. RLS provides multisystem sharing at a record level across a sysplex using the coupling facility (CF). VSAM RLS will use a CF based Lock Manager and CF Cache Manager in the implementation of Record Level Sharing. The function requires data sets using RLS control to be DFSMS managed.

RLS is one of multiple access modes to VSAM and will be set at OPEN time. This can be done by using the JCL parameter RLS or by specifying MACRF=RLS in the ACB. Other types of access are Non Shared Resources (NSR), Local Shared Resources (LSR), and Global Shared Resources (GSR). These access types can be referred to as non-RLS access to distinguish RLS from other modes of access.

2.1.1 RLS for catalog overview

DFSMS V2.1 introduces RLS usage for ICF catalogs and for volume catalogs (VOLCATS). The purpose of the enhancement is to meet requirements for better performance and availability on catalogs. Using RLS will provide locking at a record level improving performance by not having to serialize on the SYSIGGV2 *bcsname*. SMSVSAM will hold SYSIGGV2 *bcsname* SHARE while a catalog is opened for RLS access. This ensures catalog data integrity from programs relying on SYSIGGV2 to serialize the catalog.

Moving to 64 bit buffering using RLS will reduce the I/O activity. In addition, availability will be improved, if you do not have to do disruptive maintenance on your ICF catalogs due to split or catalog reorganization on a regular basis.

The improvements discussed here are expected to reduce the need for customers having to split ICF catalogs to meet availability requirements and to solve performance bottlenecks.

2.1.2 Requirements for RLS

RLS exploitation for ICF catalogs requires the base RLS setup. This is an IBM Parallel Sysplex® and also the SMSVSAM address space to be active. As always required with VSAM RLS data sets, you need a storage class assigned with a *Cacheset* and *Lockset* identifying cache and lock structures in the coupling facility.

The SMSVSAM started task has to be up and running. The SMSVSAM address space automatically starts at IPL if the RLSINIT (YES) keyword is specified in the IGDSMSxx member of SYS1.PARMLIB. If SMSVSAM is not started, you can start the address space after IPL by issuing the **V SMS,SMSVSAM,ACTIVE** command from the MVS console.

You will need to define the SHCDS data sets which are key for RLS recovery as they hold vital information for this. For example, a SMSVSAM restart depends on the SHCDSs being available. The contents of the SHCDS is:

- ▶ The name of the coupling facility lock structure in use.
- ▶ A list of subsystems and their status.
- ▶ A list of open data sets using the coupling facility.
- ▶ A list of data sets with unbound locks.
- ▶ A list of data sets in permit non-RLS state.

SHCDSs are basically VSAM linear data sets. Define and activate two of these for normal use and also one for spare purpose.

The data sets in scope for RLS exploitation must have their attributes changed as related to recoverable and non-recoverable options. Alter to LOG(NONE) if non-recoverable or to LOG(UNDO) or LOG(ALL) if the data set is recoverable. Also add the storage class created for this RLS application to have the appropriate cache set and coupling facility Cache assigned.

Refer to *z/OS DFSMSdfp Storage Administration*, SC23-6860 for information on how to set up the initial RLS environment.

2.1.3 Support for ICF catalogs using RLS

New and existing parameters are needed to define new or alter existing ICF catalogs to support RLS access mode. The LOG parameter has to be defined, however in this implementation, only non-recoverable catalogs will be supported (LOG(NONE)). You also need parameters to set the type of initial access (non-RLS or RLS) at the first access of the catalog.

Catalogs may be defined or altered in one of four states; Eligible, RLS quiesced, RLS enabled or in RLS mode. These states will be discussed in detail in “ICF catalog possible RLS states” on page 13.

New operator commands and define parameters have also been provided to support transition of ICF Catalogs between RLS and non-RLS access. These commands will work sysplex wide even when submitted from one LPAR only and have the catalogs do a serialized close across the sysplex. The status will end up being RLS enabled or RLS quiesced, ready for being accessed in RLS mode. Catalog requests issued during transition will not be failed. However, they will be suspended while the QUIESCE or ENABLE process happens. Regard these commands as migration or fallback tools. These commands are not intended for daily usage.

Once an ICF catalog has been migrated to RLS mode, it should remain in RLS mode. The only exception is if you encounter an issue that forces you to revert to non-RLS mode.

Important: Moving to RLS accessed ICF catalogs from non-RLS accessed will only be supported for USERCATALOGS. It is not supported for MASTERCATALOGS.

2.1.4 ICF catalog possible RLS states

In an environment using RLS for the ICF catalogs, these can be in one of four states:

1. Eligible

The catalog has been defined or altered to have the LOG attribute. This is mandatory to be able to access the catalog in RLS mode (as for all other VSAM data sets). OPEN in this state will still happen as non-RLS.

2. RLS quiesced

RLS quiesced indicator is set to YES in the catalogs VVR entry. OPEN is only possible for non-RLS access.

3. RLS enabled

RLS enable indicator is set to YES in the catalogs VVR entry. As RLS enabled the catalog can only be opened in RLS mode.

4. RLS mode

A catalog in RLS mode can be defined as a catalog currently open in RLS mode or was last closed in RLS mode. The RLS-IN-USE indicator will also be set in the catalogs VVR. The catalog can be taken out of RLS mode by command. For pre-DFSMS V2.1 systems, this will either require SMSVSAM up and running or by using the IDCAMS **SHCDS CFRESETDS(user_cat_name)** command to reset the RLS-IN-USE indicator. In this case it will require all instances of SMSVSAM to be shut down.

2.1.5 Comparison with other enhanced catalog features

RLS for catalog can coexist with other advanced catalog features. In this section we compare enhanced catalog features with the RLS for catalog feature.

Enhanced Catalog Sharing

Enhanced Catalog Sharing places the Catalogs self describing VVR in the CF (SYSIGGCAC_ECS) to reduce I/O. Catalog Data Space Cache (CDSC) uses Virtual Lookaside Facility (VLF) and is intended for shared catalogs. Virtual storage is possible within a range from 256 to 2GB. Cached records are stored in a data space.

In Storage Cache

In Storage Cache has a maximum of 3041 BCS records. Storage is obtained within the catalog address space. It is intended for non-shared catalogs.

The enhanced catalog features can be replaced by RLS. For the master catalog it still makes sense to use enhanced catalog sharing since RLS mode is not supported for master catalogs at this point.

Catalogs using RLS will not need to access the VVR in cache. For catalogs in RLS mode the catalog records are placed in RLS local buffer pools or CF cache structures. The STORAGECLASS cacheset defines which cache structure to use. The DATACLASS controls buffer pool (64 bit) and caching options through the RLSOVETHEBAR and RLSCFCACHE settings.

Locking happens at a record level to serialize access to individual catalog records instead of enqueueing on the SYSIGGV2 bcsname resource (entire catalog).

For other tuning parameters like STRNO, BUFND, BUFNI on DEFINE USERCATALOG command, these settings will be ignored in an RLS environment. RLS will obtain buffers dynamically as they may be needed (utilizing System Managed Buffering).

2.1.6 New parameters on DEFINE USERCATALOG command

To support transition to RLS new parameters were added to the **DEFINE USERCATALOG** command. The parameters define the availability level for readiness to use RLS mode on the ICF catalogs.

Table 2-1 on page 15 is an overview of the parameters added in DFSMS V2.1. Note that for SUSPEND and RESUME the catalog will either be in SUSPEND state or RESUME state.

This is also the case for RLSQUIESCE and RLSENABLE, the two parameters directly related to RLS catalogs. The catalog status is either QUIESCED or ENABLED.

Table 2-1 Summary of new DEFINE USERCATALOG parameters in DFSMS V2.1

New DEFINE user catalog parameters	Description	Default
SUSPEND	Requests will be suspended until a RESUME has been issued. Mutually exclusive with LOCK parameter	RESUME
RESUME	Request for the catalog will be executed immediately. Command will release a SUSPEND state.	RESUME
RLSQUIESCE	The catalog will be accessed in non-RLS mode following the define.	RLSQUIESCE
RLSENABLE	The catalog will be accessed in RLS mode after the allocation.	RLSQUIESCE
RECONNECT	Request the new catalog to use existing ALIAS information.	N/A
LOG(NONE)	Catalog is eligible for access in RLS mode.	Value will appear as NULL

If the catalog is in LOCKED or SUSPENDED mode, an authorized user with READ access to the RACF STGADMIN profile can still access and repair a locked catalog. At the same time other operations against the catalog will be failed or queued.

If the IGG.CATLOCK profile is not defined, or the RACF FACILITY class is not active, you cannot LOCK, UNLOCK, SUSPEND or RESUME a catalog. An example of defining a USERCATALOG using the new define parameters is shown in Example 2-1 on page 15.

Example 2-1 DEFINE USERCATALOG using new DEFINE parameter

```

DEFINE USERCATALOG -
  (NAME(UCAT.RLSTST.BNC) ICFCATALOG -
  VOLUME(MHL1A0) TRK(50 50) -
  STORCLAS(SCRLS) -
  DATACLAS(WELCHRLS) -
  LOG(NONE) RECONNECT RLSENABLE -
  FREESPACE(20 20) -
  NOIMBED NOREPLICATE) -
  DATA (CISZ(32768))
  
```

Note, that LOG(NONE) is used to prepare for RLS, and RLSENABLE will enable OPEN in RLS mode.

2.2 Preparing to migrate to an RLS enabled ICF Catalog

When you have the base RLS environment in place, you can start migrating your first ICF catalog to use RLS mode. To summarize the basic requirements for RLS:

- ▶ SMSVSAM must be active.
- ▶ An RLS cache structure must have been defined in the Coupling Facility.

- An RLS Cache Set must have been defined.
- An RLS Lock Set Name must have been defined.
- The catalogs in scope must be DFSMS managed.

It is suggested that all systems in the sysplex have migrated to DFSMS V2.1 prior to starting the migration. This is because shared catalogs cannot be accessed in RLS mode if they are opened on a pre DFSMS V2.1 system.

2.2.1 Sizing the RLS catalog environment for RLS buffers and CF cache

To assure sufficient RLS local buffering and coupling facility cache size for your RLS managed ICF catalogs, look at the buffer sizes in your current environment. Add up the buffer sizes based on the number of catalogs and the type of caching. The values you should look into are these:

- Current Catalog Data Space Cache - using VLF.
Capture current MAXVIRT value (in 4K blocks). Sum up the total for the entire sysplex.
- In Storage Cache (ISC) - default.

Calculate 3041 records per catalog. BCS records vary in size. An average size chosen could be 2000 bytes (total is then 608,200 bytes per catalog). Regard this as a minimum and add additional as needed to meet performance requirements. Again, calculate the total for the sysplex.

Buffer calculation scenario

Let us calculate a sample scenario. You have a sysplex with four LPARs sharing 50 ICF catalogs (10 ISC managed and 40 in CDSC).

Your COFVLFxx member specifies 16384 for the MAXVIRT value (times 4K). This is a total of 67,108,864 bytes for the VLF managed catalogs. For four systems this is 268,435,456 bytes.

The 10 ICF catalogs using ISC cache currently uses (10*3041*2000) bytes = 60,820,000. For four LPARs this adds up to 24,3280,000 bytes.

The total requirement in buffering all existing catalogs will then total as shown in Table 2-2 on page 16.

Table 2-2 Buffering requirement calculation

Component	Current usage in bytes	Current usage in MB
COFVLFxx	268,435,456	262,144
ISC	243,280,000	2,375,781
Total in Bytes/MB	511,715,456	4.997,221

The current buffer need totals over 500 MB. A coupling facility cache size of 500 MB could then be allocated as a start and monitored as conversion of the ICF Catalogs to RLS access mode progresses.

RLsFixedPoolsize and RLsAboveTheBarMaxPoolSize parameters

To meet the requirement in the scenario above as to RLS local buffers, set RLsFixedPoolsize in IGDSMSxx to 500 MB. This value specifies the maximum number of buffers under and

above the bar. To limit use of storage above the bar use the *RlsAboveTheBarMaxPoolSize* parameter. Set this parameter to limit use for the RLS buffer (default is 0).

Coupling Facility Structure Sizer tool

The Coupling Facility Structure Sizer Tool (CFSizer) is a web-based application that will return structure sizes based on the latest CFLEVEL for the IBM products that exploit the coupling facility, CFSIZER can be used to generate the exact settings and the JCL for creating the Cache Structure. The CFSIZER can be found at:

<http://www-947.ibm.com/systems/support/z/cfsizer>

1. To size a structure, open the CFSizer web page. Click the **VSAM RLS** the product links in the left side navigation bar as shown in Figure 2-1.

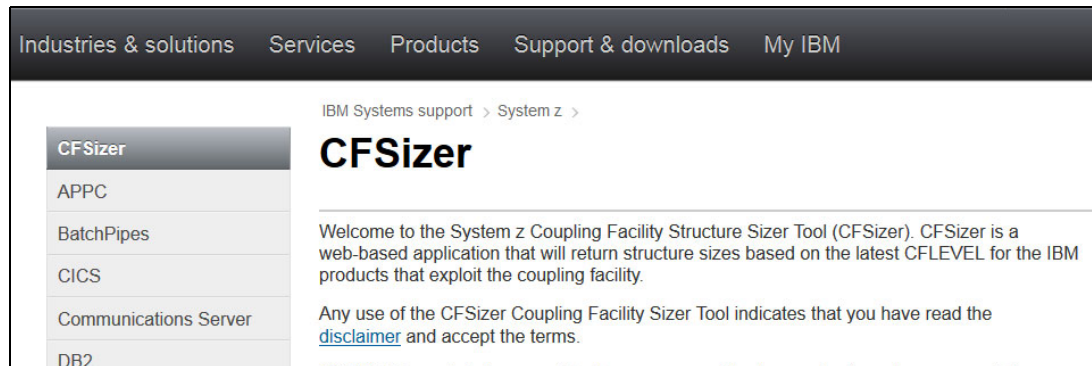


Figure 2-1 CFSizer tool main page

2. In the VSAM RLS page, select one or more structures to size by clicking the check box for the cache structure you want to size. At least one structure must be checked to submit a request for a structure size. Enter the required information and click **Submit** to continue.We selected the **VSAM RLS CACHE structure** as shown in Figure 2-2.

☐ VSAM RLS LOCK structure

[VSAM RLS LOCK structure help](#)

Number of systems	Max number of systems	Max number of connectors
8	16	32

☒ VSAM RLS CACHE structure

[VSAM RLS CACHE structure help](#)

Sum of buffer pool in MBs

Submit

Figure 2-2 Using CFSizer for building RLS cache structure

3. The CFSizer will present you with sample CFRM statements based on the input you provided. You can cut and paste these statements into your CFRM policy and modify them as necessary. Figure 2-3 shows the results of our input.

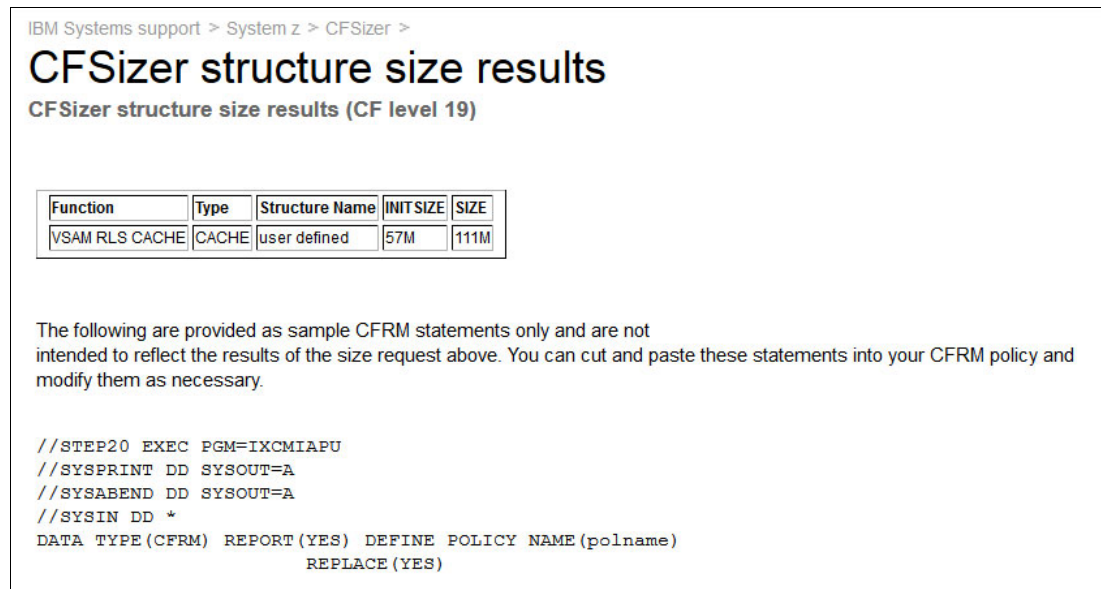


Figure 2-3 CFSizer results on VSAM RLS Cache structure for catalog

It is suggested that you create a dedicated structure for ICF catalogs. This is to assure that if catalog buffers are flushed from cache, it is based on the catalog activity, and not based on other activity in the coupling facility.

2.2.2 Migrating to RLS enabled ICF Catalog

After having upgraded all systems in the sysplex to DFSMS V2.1, you may start exploiting RLS mode on ICF catalogs. This is done by either defining a completely new ICF catalog for test purpose or by altering an existing ICF Catalog to have the LOG(NONE) attribute. The LOG(NONE) attribute will put the catalog in RLS quiesced mode. By default non-RLS catalogs will the value NULL in the LOG parameter.

An example of a DEFINE is shown in Example 2-2 on page 18. Note the LOG(NONE) option. Keywords like STRNO and BUFND and BUFNI has been removed as they will not be used in a RLS environment.

Example 2-2 DEFINE USERCATALOG example of ICF catalog for RLS support

```
DEFINE USERCATALOG -
  (NAME(UCAT.RLSTST) ICFCATALOG -
  VOLUME(MHL1A0) TRK(5 1) -
  STORCLAS(SCRLS) -
  DATACLAS(WELCHRLS) -
  LOG(NONE) -
  FREESPACE(20 20) -
  NOIMBED NOREPLICATE) -
  DATA (CISZ(4096))
```

A **LISTCAT** following immediately after the **DEFINE** will show that the user catalog has the required LOG(NONE) attribute and also is VSAM QUIESCED already. No RLS QUIESCE

command is needed at this point. Example 2-3 on page 19 is sample output from the **LISTCAT** command.

Example 2-3 LISTCAT output from newly defined ICF user catalog defined with LOG(NONE)

```
RLSDATA
  LOG -----NONE    RECOVERY REQUIRED --(NO)
  VSAM QUIESCED -----(YES)    RLS IN USE -----(NO)
```

To make the ICF catalog we defined fully RLS enabled, issue the **F CATALOG,RLSENABLE** command. Example 2-4 on page 19 shows the SYSLOG after issuing this command.

Example 2-4 How to enable an ICF Catalog for RLS access mode

```
F CATALOG,RLSENABLE(UCAT.RLSTST)
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC352I MODIFY CATALOG UCAT.RLSTST TO STATE RLSENABLE SUCCESSFUL
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
```

The **MODIFY CATALOG** command in Example 2-4 on page 19 is directed towards a specific ICF catalog. If you want a system wide scope to change all RLS enabled catalogs in one command across the sysplex, issue the **F CATALOG,RLSENABLE,SYSTEM** command. The same command syntax is valid for RLSQUIESCE.

A **LISTCAT** of the user catalog will now show the VSAM QUIESCED field changed from YES to NO. This means the catalog is no longer quiesced, but enabled for RLS. RLS enablement is also indicated by RLS IN USE field, which has changed to YES as shown in Example 2-5 on page 19.

Example 2-5 Extract from LISTCAT of user catalog in RLS enabled state

```
RLSDATA
  LOG -----NONE    RECOVERY REQUIRED --(NO)
  VSAM QUIESCED -----(NO)    RLS IN USE -----(YES)
```

Another way of displaying the RLS readiness of an ICF catalog is to use the **F CATALOG,OPEN** command. This command lists all open catalogs and displays settings for the open catalogs in this environment. Example 2-6 on page 19 shows sample output.

Example 2-6 Display of RLS enabled ICF Catalog using MODIFY CATALOG command

```
-F CATALOG,OPEN
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC348I ALLOCATED CATALOGS
*CAS*****
* FLAGS -VOLSER-USER-CATALOG NAME
* YSU-R- SBXHS8 0001 UCAT.RLSTST
*****
* Y/N-ALLOCATED TO CAS, S-SMS, V-VLF, I-ISC, C-CLOSED, D-DELETED,
* R-SHARED, A-ATL, E-ECS SHARED, K-LOCKED, U-RLS, W-SUSPENDED
*CAS*****
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
```

Note the U indicator for a RLS enabled catalog. If you have a lower level system in your sysplex, and a shared RLS candidate catalog is open in this system, you will get an error

message when trying to enable this catalog. The error message is shown in Example 2-7 on page 20.

Example 2-7 Error message, trying to RLS enable a catalog shared with lower level system

```
F CATALOG,RLSENABLE(UCAT.RLSTST)
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC353I SERVICE 'CLRC' RETURNED RC-236-030, QUIESOFF, UCAT.RLSTST
IEC353I MODIFY CATALOG UCAT.RLSTST TO STATE RLSENABLE UNSUCCESSFUL
IEC353I CATALOG ADDRESS SPACE MODIFY UNSUCCESSFUL
```

You will get a similar error message, when trying to do the RLS enable using IDCAMS **ALTER** command. The error message is shown in Example 2-8 on page 20.

Example 2-8 Error message in IDCAMS ALTER RLSENABLE on shared catalog

```
IDCAMS  SYSTEM SERVICES                                TIME: 17:37:27

      ALTER      -
              UCAT.RLSTST      -
              RLSENABLE
IDC3014I CATALOG ERROR
IDC3009I ** VSAM CATALOG RETURN CODE IS 236 - REASON CODE IS IGG0CLRC-30
IDC0532I **ENTRY UCAT.RLSTST NOT ALTERED
IDC0001I FUNCTION COMPLETED, HIGHEST CONDITION CODE WAS 8

IDC0002I IDCAMS PROCESSING COMPLETE. MAXIMUM CONDITION CODE WAS 8
```

2.2.3 Fallback from using RLS mode on ICF Catalogs

SMSVSAM recovery is essential for ICF catalogs using RLS and of course for other exploiters. SMSVSAM is the server for all RLS processing, and RLS processing will fail if SMSVSAM for some reason is not available.

z/OS V2R1.0 DFSMSdfp Storage Administration, SC23-6860 has a detailed description on recovery of SMSVSAM and the RLS environment.

For ICF catalogs using RLS, the process to disable the use of RLS should SMSVSAM and the RLS environment encounter problems is straightforward. Use the

F CATALOG,RLSQUIESCE(ucat name) command or the **F CATALOG,RLSQUIESCE,SYSTEM** command and catalog management will revert to normal catalog and VSAM I/O.

Before enabling RLS use on ICF catalogs, it is also suggested that you have a fallback plan in place. New catalog commands support fallback, but down level systems will need SMSVSAM active to switch catalogs out of RLS mode.

2.3 Catalog maintenance considerations in RLS environment

New keywords have been introduced in support of RLS enablement for catalogs. In this section we look at catalog maintenance in an RLS environment.

2.3.1 Backing up RLS managed catalogs

Backing up your RLS managed user catalogs with IDCAMS can be done as in prior releases. There are new IDCAMS keywords to manage RLS source or target catalogs; RLSSOURCE(NOIYESIQUIESCE) and RLSTARGET(NOIYESIQUIESCE). Using these parameters will influence processing as follows:

- ▶ NO
Indicates the source and target data sets will be opened using non-shared resources. (NSR).
- ▶ YES
Indicates that the source and target data sets will be opened using Record Level Sharing (RLS) and the data set will have consistent read integrity.
- ▶ QUIESCE
Indicates that the source and target data sets will be opened using Record Level Sharing (RLS) and the data set will be quiesced before processing any entries.

IDCAMS backup

Using IDCAMS for backing up a catalog, you can choose to access the catalog using RLS or not. When QUIESCE is specified RLS access will be QUIESCED during backup and will be resumed once IDCAMS finishes backup. Using this parameter will assure integrity, but also suspend availability during backup. Example 2-9 on page 21 shows a sample IDCAMS job with RLSSOURCE(QUIESCE).

Example 2-9 Sample IDCAMS export (backup) of a user catalog using RLSSOURCE parameter

```
//CAT004 EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//DD1      DD DISP=(,CATLG),
//          DSN=MHLRES1.BACKUP1.RLSTST1,
//          SPACE=(TRK,(2,1),RLSE)
//SYSOUT   DD SYSOUT=*
//SYSIN    DD *
EXPORT UCAT.RLSTST OUTFILE(DD1) TEMPORARY -
          RLSSOURCE(QUIESCE)
```

DFSMSdss backup

When using DFSMSdss for backing up RLS managed catalogs there is no specific keyword for QUIESCE. DFSMSdss will do the QUIESCE implicitly during backup and enable RLS once the backup is finished. Example 2-10 on page 21 is an example of DFSMSdss backing up a catalog.

Example 2-10 Example of catalog backup using DSS as the backup tool

```
//MHLRES1D JOB .....
//STEPT006 EXEC PGM=ADDRSSU,REGION=4096K
//SYSPRINT DD SYSOUT=*
//TAPE DD DISP=(NEW,CATLG),
//      DSN=MHLRES1.DSS.BACK2.MHLRES1A,SPACE=(TRK,(2,1)),
//      UNIT=3390,DCB=(BLKSIZE=27920)
//SYSIN DD *
DUMP DS(INC(UCAT.RLSTST      )) -
OUTDDNAME (TAPE)
```

/*

DFSMSHsm backup

DFSMSHsm is also able to backup catalogs. You need to ensure that DFSMSHsm has access to IGG.CATLOCK in order to be able to work with a locked catalog. Otherwise, DFSMSHsm is a privileged user and is able to backup catalogs. DFSMSHsm will backup aliases and restore those also when restoring the catalog.

Remember to LOCK or SUSPEND the catalog before doing the restore to control the integrity of that catalog.

2.3.2 Using REPRO MERGECAT on an RLS managed catalog

Running the REPRO MERGECAT function in a shared environment from a non-RLS enabled ICF catalog to an RLS enabled ICF catalog requires the QUIESCE done on the target catalog, before you are allowed to do a REPRO MERGECAT. Attempting to do the REPRO MERGECAT without having the target catalog quiesced for RLS access will end up giving the error in Example 2-11 on page 22.

Example 2-11 REPRO MERGECAT message when running from non-RLS to RLS managed catalog.

```
REPRO -
  INDATASET(UCAT.VSBOX01      ) -
  OUTDATASET(UCAT.RLSTST.MHLRES1A ) -
  LEVEL(MHLRES1A) -
  MERGECAT
IDC3300I  ERROR OPENING UCAT.RLSTST.MHLRES1A
IDC3351I  ** VSAM OPEN RETURN CODE IS 168
IDC3003I  FUNCTION TERMINATED. CONDITION CODE IS 12
```

2.3.3 Reporting on catalogs

To track your catalog efficiency there are several reporting options. This is the case for both non-RLS managed and RLS managed catalogs. In the following section, we will look into the reporting options.

I/O statistics of catalog address space report

Reporting on both non-RLS and RLS managed catalogs is done through **MODIFY CATALOG** commands or through standard z/OS tools.

Using the **F CATALOG,REPORT,PERFORMANCE** command will get you the I/O statistics of the catalog address space since the last IPL or since the last reset of these statistics through the **MODIFY CATALOG,REPORT,PERFORMANCE(RESET)** command. An example of the output from this report is shown in Example 2-12 on page 22. Note the 'Statistics since' time stamp, that tells you the starting time for this summary report.

Example 2-12 Example of MODIFY CATALOG,REPORT,PERFORMANCE output

```
F CATALOG,REPORT,PERFORMANCE
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC359I CATALOG PERFORMANCE REPORT
*CAS*****
*  Statistics since  9:51:36.31 on  09/26/2013      *
*  -----CATALOG EVENT-----  --COUNT--  ---AVERAGE---  *
```


* Entries to Catalog	2,134K	5.491 MSEC	*
* BCS ENQ Shr Sys	3,624K	0.837 MSEC	*
* BCS ENQ Exc1 Sys	1,317K	1.933 MSEC	*
* BCS DEQ	4,559K	0.037 MSEC	*
* VVDS RESERVE CI	1,190K	0.052 MSEC	*
* VVDS DEQ CI	1,190K	0.056 MSEC	*
* VVDS RESERVE Shr	5,522K	0.097 MSEC	*
* VVDS RESERVE Exc1	24,667	0.380 MSEC	*
* VVDS DEQ	5,543K	0.058 MSEC	*
* SPHERE ENQ Exc1 Sys	452,247	0.084 MSEC	*
* SPHERE DEQ	452,251	0.074 MSEC	*
* CAXWA ENQ Shr	71,017	0.011 MSEC	*
* CAXWA ENQ Exc1	71	0.002 MSEC	*
* CAXWA DEQ	71,093	0.002 MSEC	*
* VDSPM ENQ	4,803K	0.027 MSEC	*
* VDSPM DEQ	4,803K	0.001 MSEC	*
* RPL ENQ	92,968	0.136 MSEC	*
* RPL DEQ	92,968	0.004 MSEC	*
* BCS Get	1,807K	0.312 MSEC	*
* BCS Put	430,989	1.040 MSEC	*
* BCS Erase	719	0.314 MSEC	*
* VVDS I/O	6,789K	0.257 MSEC	*
* VLF Delete Minor	23	0.006 MSEC	*
* VLF Define Major	98	0.001 MSEC	*
* VLF Identify	618	0.001 MSEC	*
* RMM Tape Exit	823	0.000 MSEC	*
* OEM Tape Exit	823	0.000 MSEC	*
* BCS Allocate	8,617	5.499 MSEC	*
* BCS Deallocate	2,432	1.019 MSEC	*
* SMF Write	580,422	0.019 MSEC	*
* CAS Cache Lock	248	0.081 MSEC	*
* CAS Alloc Lock	14	41.258 MSEC	*
* CAS MLA Lock	38	8.472 MSEC	*
* VVDS Format	8	6.910 MSEC	*
* IXLCONN	3	97.797 MSEC	*
* IXLCACHE Read	5	16.676 MSEC	*
* Resolve Symbolic	10	0.029 MSEC	*
* MVS Allocate	2,576	12.752 MSEC	*
* MVS Deallocate	2,478	0.943 MSEC	*
* Capture UCB	7,985	0.003 MSEC	*
* Uncapture UCB	16	0.007 MSEC	*
* SMS Active Config	88	0.171 MSEC	*
* SMS Mgmt Class ACS	4	0.733 MSEC	*
* RACROUTE Auth	977,727	0.046 MSEC	*
* RACROUTE Define	17,062	0.039 MSEC	*
* DADSM Scratch	645	6.006 MSEC	*
* DADSM Rename	4	6.305 MSEC	*
* DADSM Allocate	17,044	1.538 MSEC	*
* DADSM Allocate SMS	36	4.201 MSEC	*
* Get LatchShr Æ	7,667K	0.010 MSEC	*
* IXLREBLD	1	4.866 MSEC	*
* IXLEERSP	1	0.011 MSEC	*
* ENQ SYSZPCCB	2,329K	0.001 MSEC	*
* DEQ SYSZPCCB	2,329K	0.000 MSEC	*
* Release Latch Æ	7,681K	0.000 MSEC	*

```

* Capture to Actual          63,555      0.002 MSEC *
* Get LatchExc Æ            14,663      5.100 MSEC *
* ENQ SYSIGGV1                3          0.004 MSEC *
* DEQ SYSIGGV1                3          0.001 MSEC *
* ENDREQ                      1,519      0.001 MSEC *
* Purge Latch                 77         0.071 MSEC *
* BCS Close                   4,132      0.732 MSEC *
* SYSVSAM S ENQ Excl         8,625     13.932 MSEC *
* SYSVSAM S DEQ              8,623      0.049 MSEC *
* SYSVSAM D ENQ Shr          8,615      0.196 MSEC *
* SYSVSAM D DEQ              8,615      0.059 MSEC *
* CAS*****
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED

```

The REPORT PERFORMANCE command should be issued regularly to identify potential performance bottlenecks, as your catalogs are a key component in I/O activity.

Cache data space efficiency

The **F CATALOG,REPORT,CACHE** command will report on the cache data space efficiency. The hit rate is reported and should be at least be 20% to reach a reasonable value. Statistics have to be measured over time to be valid. Measuring for an hour may not represent your normal catalog access.

You can reset the statistics by issuing **F CATALOG,CLOSE** command. When the catalog is reopened after this, a new cache structure is built. Through this all counters have been reset except for the purge values. An example of the cache report is shown in Example 2-13 on page 24. This will only have ISC statistics, but you will also see statistics from VLF managed catalogs if there are any in your environment.

Example 2-13 Example of output from F CATALOG,REPORT,CACHE command

```

F CATALOG,REPORT,CACHE
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC359I CATALOG CACHE REPORT 545
* CAS*****
* HIT% -RECORDS- -SEARCHES --FOUND-- -DELETES- -SHR UPD- --PURGE-- *
*                                                                    *
* UCAT.RLSTST.BNC                                (ISC)                *
*   0      4,757   800,025          11          0          0      1,648 *
* UCAT.ZOSR1A                                    (ISC)                *
*   0          1          1          0          0          0          0 *
* UCAT.VTFMTAPE                                (ISC)                *
*   0          1          1          0          0          0          0 *
* UCAT.VSTEST1                                  (ISC)                *
*  50%          1          2          1          0          0          0 *
* MCAT.SANDBOX.Z1C.SBOX00                      (ISC)                *
*  59%          84   10,026   5,990          0          0          17 *
* CAS*****
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED

```

I/O statistics of specific user catalog

The **MODIFY CATALOG,REPORT,CATSTATS(catname)** will give you I/O statistics including BUFND, BUFNI and STRNO information on a specific user catalog as specified in the catname parameter.

Example 2-14 Sample report using MODIFY CATALOG,REPORT,CATSTATS parameter

```
CATALOG,REPORT,CATSTATS(UCAT.RLSTST)
EC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
EC359I CATALOG I/O STATS REPORT
CAS*****
      ADDS  UPDATES      GETS  GETUPD  DELETES  BUFNI  BUFND STRNO  *
                                           *
UCAT.RLSTST                                           *
      0      0      88      4      0      1      2    180  *
CAS*****
EC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
```

The new CATSTATX parameter on the **MODIFY CATALOG,REPORT,CATSTATX(catname)** command will report the same information as MODIFY CATALOG,CATSTATS, but it includes CA-reclaim and CA-reuse information as well. This command can report on one or more catalogs with the use of '*' or wild cards in the catname specification.

Example 2-15 on page 25 is an example of catalog reporting using the CATSTATX parameter.

Example 2-15 Report on user catalog using the new CATSTATX parameter

```
CATALOG,REPORT,CATSTATX(UCAT.RLSTST)
EC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
EC359I EXTENDED CATALOG STATS
CAS*****
  CATALOG NAME      = UCAT.RLSTST                      *
  INSERTS (ADDS)    =      0                            *
  UPDATES           =      0                            *
  RETRIEVES         =     88                            *
  RETRIEVES FOR UP  =      4                            *
  ERASES (DELETES)  =      0                            *
  CA-RECLAIMS       =     N/A                          *
  CA-REUSES         =     N/A                          *
  BUFNI SETTING    =      1                            *
  BUFND SETTING    =      2                            *
  STRNO SETTING    =     180                            *
  AVG ELAPSED TIME  =     2.139 MSEC                    *
  AVG CPU TIME      =    144.232 USEC                    *
CAS*****
EC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
```

The information on CA-reclaim, AVG ELAPSED TIME and AVG CPU TIME has been added in z/OS V2.1.

Reporting on RLS managed catalogs

RMFMON III will have RLS activity reports. Choose SYSPLEX option, there are several reports on RLS. See RMFMON III Sysplex panel with these reports displayed in Figure 2-4 on page 26.

```

RMF Sysplex Report Selection Menu
Selection ==>

Enter selection number or command for desired report.

Sysplex Reports
  1 SYSSUM   Sysplex performance summary          (SUM)
  2 SYSRTD   Response time distribution            (RTD)
  3 SYSWKM   Work Manager delays                  (WKM)
  4 SYSEQ    Sysplex-wide Enqueue delays          (ES)
  5 CFOVER   Coupling Facility overview           (CO)
  6 CFSYS    Coupling Facility systems            (CS)
  7 CFACT    Coupling Facility activity           (CA)
  8 CACHSUM  Cache summary                        (CAS)
  9 CACHDET  Cache detail                        (CAD)
 10 RLSSC    VSAM RLS activity by storage class   (RLS)
 11 RLSDS    VSAM RLS activity by data set        (RLD)
 12 RLRLRU   VSAM LRU overview                   (RL)

Data Index
  D DSINDEX  Data index                          (DI)

```

Figure 2-4 RMF3 reports on RLF activity

Possible selections related to RLF are

- ▶ VSAM RLS activity by storage class
- ▶ VSAM RLS activity by data set
- ▶ VSAM LRU overview

Coupling facility lock structure report

The **D SMS,CFLS** command displays information about the coupling facility lock structure. This information includes the lock rate, lock contention rate, false contention rate, average number of requests waiting for locks, the lock structure size, and primary structure information. If the lock structure is in duplex mode, secondary structure information will also be displayed. Output from this command will look similar to Figure 2-16 on page 26.

Example 2-16 Output from D SMS,CFLS command

```

D SMS,CFLS
IEE932I 853
IGW320I 07:42:24 Display SMS,CFLS(IGWLOCK00 )
PRIMARY STRUCTURE:IGWLOCK00 VERSION:CBE13E285CE22FD3 SIZE:14336K
RECORD TABLE ENTRIES:27365 USED:6
LOCK STRUCTURE MODE: SIMPLEX STATUS: ENABLE
System   Interval   LockRate   ContRate   FContRate   WaitQLen
SC64     1 Minute    0.0        0.000      0.000       0.00
SC64     1 Hour      0.0        0.000      0.000       0.00
SC64     8 Hour      0.0        0.000      0.000       0.00
SC64     1 Day       0.0        0.000      0.000       0.00
(04)     1 Minute    0.0        0.000      0.000       0.00
(04)     1 Hour      0.0        1.923      0.000       0.00
(04)     8 Hour      0.0        5.048      0.000       0.00
(04)     1 Day       0.0        4.968      0.000       0.00

```

***** LEGEND *****

LockRate = number of lock requests per second
CONTRATE = % of lock requests globally managed
FCONTRATE = % of lock requests falsely globally managed
WaitQLen = Average number of requests waiting for locks

SMF reports

SMF is another source of information. You will need a reporting tool that can process the SMF data. Choose the option you have in your shop.

SMF type 42 subtypes 15 through 19 can be used for summary reporting on RLS use. These subtypes are available:

- ▶ Subtype 15. VSAM RLS Storage Class Response Time Summary
- ▶ Subtype 16. VSAM RLS Data Set Response Time Summary
- ▶ Subtype 17. VSAM RLS Coupling Facility Lock Structure Usage
- ▶ Subtype 18. VSAM RLS CF cache partition usage
- ▶ Subtype 19. VSAM RLS local buffer manager LRU statistics summary

Use command **V SMS,MONDS(spherename),ON** to collect subtype 16 statistics. Additionally data set collection for SMF type 42 must be turned on, when using IBM RMF™ III.

For catalogs and VSAM in general, all the catalog specific SMF records can be used. These are:

- ▶ Type 60. VVR Updated
- ▶ Type 61. ICF Define
- ▶ Type 62. VSAM OPEN
- ▶ Type 64. VSAM CLOSE (new for catalogs in z/OS 1.13)
- ▶ Type 65. ICF Delete
- ▶ Type 66. ICF Alter

2.3.4 Migration considerations

It is required that SMSVSAM is enabled on all systems in the sysplex exploiting RLS mode for catalogs. This is valid for DFSMS V2.1 systems as for lower level systems participating in the sysplex.

It is also required that all systems in the sysplex have migrated to DFSMS V2.1 before implementing RLS mode for shared ICF catalogs. Even if you put on toleration support on the downlevel systems, a candidate catalog to move to RLS will only be usable in a down level system in RLS Quiesced mode. It will not be usable when the catalog enters RLS Enabled mode.

2.3.5 Benchmarks for ICF catalogs using RLS mode

Benchmarks comparing VLF managed and RLS managed catalogs shows significant improvements in performance. One particular benchmark on massive DELETES showed that elapse time was reduced 90%, while CPU usage was reduced over 75%. On CREATES numbers were 56% and 81% in reduction. For GENERIC READS however the saving was 32% in elapse time, but CPU usage was higher; over 15%.

For DIRECT READS the corresponding figures are a 24% reduction in elapse time and a CPU consumption that is 14% higher.

We did a small study creating and deleting 300,000 data sets in a non-RLS managed environment compared to an RLS managed environment.

An RLS quiesced ICF catalog was created, and the 300.000 data set creates and deletes were done in eight batch jobs sequentially. Elapse time and CPU time was captured on those. Next the ICF catalog was RLS enabled and the same 300.000 data sets were created and deleted in the same way, again capturing elapse time and CPU time.

In the graphical presentation in Figure 2-5 on page 28, the improvement on elapse time using RLS managed catalogs is obvious in our case. In average elapse time on data set creates were reduced by 57.2%.

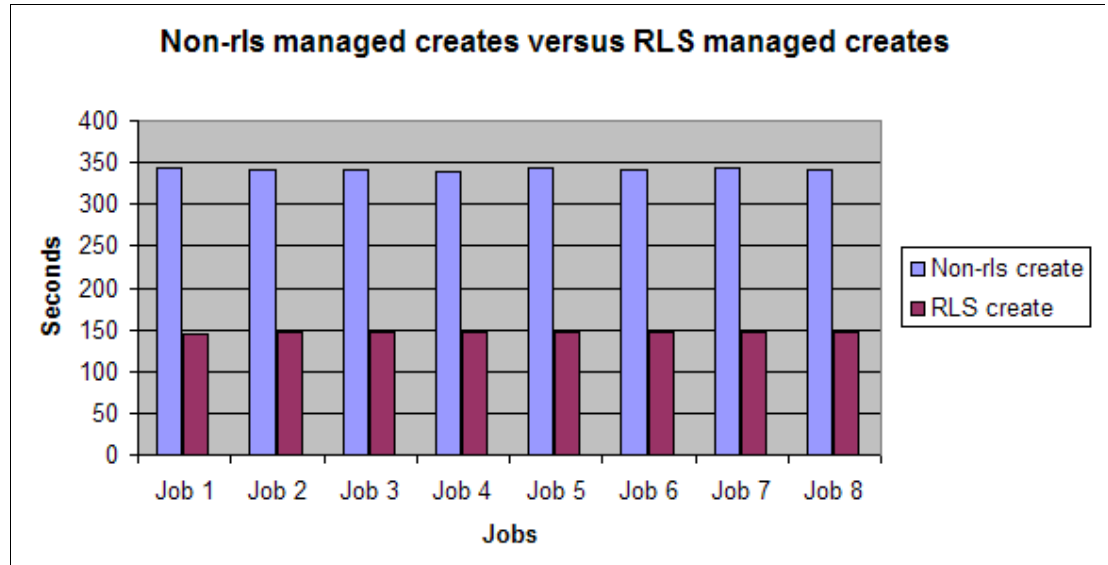


Figure 2-5 Graphical view of elapse time creating 300K data sets (Non-rls and RLS managed)

Deleting the same 300.000 data sets, the comparison between non-RLS managed and RLS managed deletes showed a 31.8% reduction in elapse time moving from a non-RLS managed ICF catalog to a RLS managed ICF catalog. This is shown in Figure 2-6 on page 28.

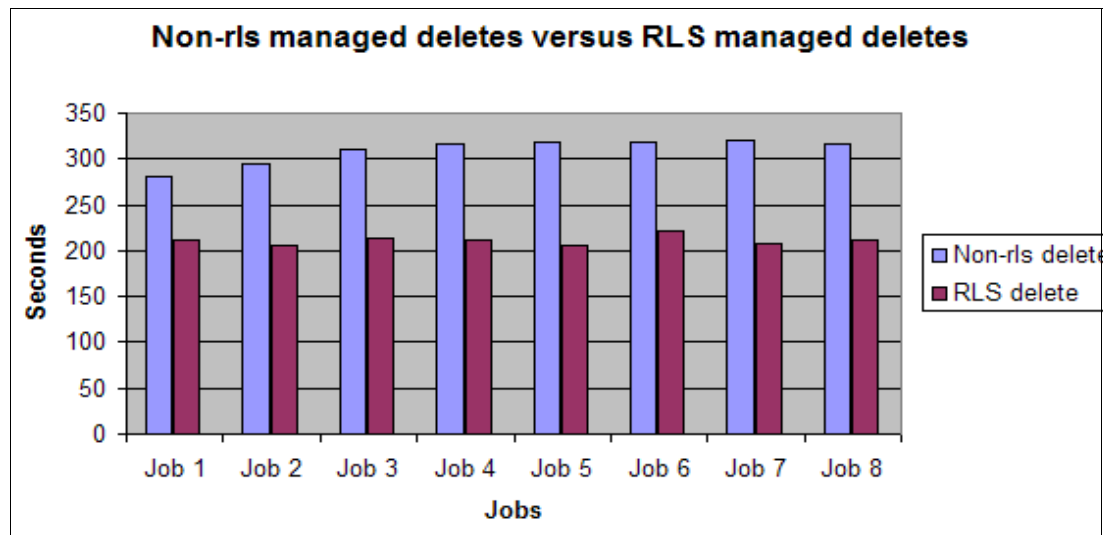


Figure 2-6 Graphical view of elapse time deleting 300K data sets (Non-rls and RLS managed)

Comparing the CPU consumption on the 300.000 data set creates and data set deletes in a non-RLS managed environment with creates and deletes in an RLS managed environment only showed a little improvement moving from a non-RLS managed ICF catalog to an RLS managed ICF catalog in our scenario.

2.3.6 Tolerations APARs related to ICF catalogs and RLS

The following APARs are required on down level systems when accessing catalogs which may have been opened for RLS on z/OS V2.1:

- ▶ OA36403
- ▶ OA36409
- ▶ OA36916
- ▶ OA36492
- ▶ OA36422
- ▶ OA36414

2.3.7 Documentation

Planning for implementation of RLS for catalogs, you should consult the following documentation:

- ▶ *z/OS DFSMS Managing Catalogs*, SC23-6853
- ▶ *z/OS DFSMS Access Method Services Commands*, SC23-6846
- ▶ *z/OS DFSMSdfp Storage Administration*, SC23-6860

2.4 Catalog contention detection enhancements

The initial version of catalog contention detection (introduced in DFSMS V1.12) provided the framework for checking contention of resources between ICF catalogs (initially only on the task input/output table SYSZTIOT). The function only gave a means to notify the operator which tasks waited outside the specified wait-time (default 10 minutes) and provided the information required to address the potential catalog contention.

A SYMREC record would be written to LOGREC and message IEC393I displaying information about the waiting task or tasks would appear. This procedure would next happen repeatedly at preset intervals.

Also a new CATALOG MODIFY command was introduced: **F CATALOG,CONTENTION**. This command can both display current settings and change these dynamically.

The enhancement to Contention Detection will give better diagnostics whenever a contention problem occurs. Immediate action will improve the probability of a successful recovery.

2.4.1 Enhancement overview

DFSMS V2.1 introduces monitoring of additional resources and also the possibility of triggering an action per resource monitored. Wait-time and actions by resource has to be set in the IGGCATxx member of SYS1.PARMLIB catalog member.

In addition to notification, the system can also re-drive. When re-drive is active the first time a service task with an active resource passes the contention threshold, the service task is abended and the request is resubmitted to catalog for processing.

The new resources now being monitored are SYSZVVDS, SYSIGGV2, SYSZTIOT, and ALLOCLCK.

2.4.2 Contention detection problems

The resources monitored are essential resources used in catalog management. The function of each individual resource is explained in more detail in this section.

- ▶ SYSZVVDS. Used to serialize access to associated VVDS records.
- ▶ SYSIGGV2. Used to serialize access to associated catalog resources.
- ▶ SYSZTIOT. Used to control access to task input/output table resources.
- ▶ ALLOCLCK. A catalog Address Space internal lock that will assure serialization of access to most catalog allocation events

If the catalog parmlib is not activated, the default for all resources is a wait-time of 10 minutes and notification only, no action. The default wait-time of 10 minutes can be overwritten by IGGCATxx member of SYS1.PARMLIB settings in the format *resource(minutes,action)*.

An example of the settings is shown in Example 2-17.

Example 2-17 Catalog parmlib settings for monitored resources

```
SYSZTIOT(11,N)
SYSZVVDS(12,R)
SYSIGGV2(13,R)
ALLOCLCK(0,N)
```

Valid actions are:

- ▶ N for notify. This is the default action and cannot be disabled.
- ▶ R for re-drive.

In this example ALLOCLCK notification is disabled (set to 0). For the remaining resources, monitoring and an action is enabled. The remaining parameters in IGGCATxx can be changed through command (only some resources), through IPL or through a catalog restart. Validation will happen through a catalog restart and if one parameter is in error, it will be skipped and maintain the previous value.

Resource monitor

The catalog detection function will monitor resources looking into wait-time, and alert those exceeding wait-time. Below is the action sequence that will start based on the extended wait-time.

1. A symptom record will be written to LOGREC, and message IEC393I will be issued providing information about the task.
2. If the same task is still pending after another five minutes, message IEC393I is issued again.
3. Issuing message IEC393I next will continue at 15 minute intervals until the wait has ended. At end the status will be updated in the next system check (within 30 seconds).

Re-drive action

In addition to the notification above, a re-drive action can be triggered. If this action is set, the task having exceeded wait-time will be cancelled and also resubmitted for execution. Remember re-drive does not fail the job, just resubmits the catalog requests. The abend issued is 91A-13 and will be accompanied by a dump with reason code 246. A LOGREC record with the same reason code will be generated.

At abend time the re-drive action will free all previously allocated resources and through this remove the contention without operator intervention. The proactive and automatic handling of this catalog problem will reduce the risk of having multiple resources queue up, some of which could have a critical impact on the entire system.

The resubmitted request may successfully complete. If it should hang again the re-drive action will not happen. Only the notification part (LOGREC write and message IEC393I) will occur.

Re-driving a catalog request is not always possible as the initial request may have completed part of the request. Re-driving the request again may fail due to this. A manual clean up would in this case be required.

2.4.3 Example of catalog contention

Seeing the message IEC393I may not necessarily mean you have a critical problem. A way to assure you have captured the documentation is to take a dump using the **F CATALOG, TAKEDUMP** command. Example 2-18 shows a sample command and the resulting output.

Example 2-18 Example of catalog,takedump command

```
F CATALOG,TAKEDUMP
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEA045I AN SVC DUMP HAS STARTED AT TIME=13.32.04 DATE=09/30/2013 681
FOR ASIDS(003B,0001)
QUIESCE = YES
IEA794I SVC DUMP HAS CAPTURED: 682
DUMPID=005 REQUESTED BY JOB (CATALOG )
DUMP TITLE=CAS DYNAMIC DUMP-IGGOCLGA RC246 RSN100
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
IEF196I IGD101I SMS ALLOCATED TO DDNAME (SYS00014)
IEF196I          DSN (DUMP.D130930.H17.SC64.CATALOG.S00005          )
IEF196I          STORCLAS (SCDUMP) MGMTCLAS (MCDB22) DATACLAS (
)
IEF196I          VOL SER NOS= SBOXZ1
IEF196I IGD104I DUMP.D130930.H17.SC64.CATALOG.S00005          RETAINED,
IEF196I DDNAME=SYS00014
```

The new IEC393I message provides more information on the problem that occurred. The format of the message is shown in Example 2-19.

Example 2-19 Example of contents in message IEC393I

```
IEC393I CATALOG CONTENTION 778
WARNING: CATALOG TASK CONTENTION WAIT-TIME WAS EXCEEDED.
RESOURCE(SYSZTIOT)JOBNAME(HZSPROC )ASID(0014)TCB(007CA968)
```

Using the IEC393I message, you can extract the job name and use the command **F CATALOG,LISTJ(jobname)** to get more information from this display. Also issue the command **F CATALOG,LIST** to have information on all current catalog tasks. It is always a good idea to have output from these commands in the SYSLOG for diagnostics and problem solving or later error reporting, for example in a PMR.

Intervention at this point could be to issue the command **F CATALOG,END(taskid),REDRIVE** to try to re-drive the active catalog task, if this was not already automated. Should this not be possible the command **F CATALOG,ABEND(taskid)** can abend the task and remove the critical contention.

Additionally, to identify tasks outside of CAS causing contention, the command **D GRS,RES=(SYSZTIOT,*)** is useful for the diagnostics.

2.4.4 Changing catalog contention detection settings

The contention detection settings in the IGGCATxx catalog parmlib member can be dynamically changed if you want to make changes to defaults or values already set. This can be done with a catalog modify command as shown in Example 2-20.

Example 2-20 Example of changing contention detection setting

```
MODIFY CATALOG,CONTENTION(SYSZTIOT,,R)
```

The setting above will enable a re-drive action on the SYSZTIOT resource if a wait-time violation should occur. Wait-time setting will in this case remain unchanged. Remember that this setting will not be permanent, but will be reset to the IGGCATxx value during IPL or catalog restart, if a value is set in IGGCATxx. To have an dynamic setting become permanent, you have to change the setting in IGGCATxx parmlib member.

Tip: Valid wait-time settings are zero for disabled and otherwise between 5 and 999 minutes.

Current settings can always be displayed with the **F CATALOG,CONTENTION** command. Example 2-21 is an example of output display from this command.

Example 2-21 Display of catalog contention settings

```
F CATALOG,CONTENTION
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC359I CATALOG CONTENTION REPORT
*CAS*****
* RESOURCE          THRESHOLD(MIN)    ACTION(S)                *
*CAS*****
* SYSZTIOT           11                 N                          *
* SYSZVVDs           12                 NR                         *
* SYSIGGV2           13                 NR                         *
* ALLOCLCK           INACTIVE           N                          *
*CAS*****
* ACTION KEY                                                *
* N = NOTIFY OPERATOR  R = REDRIVE REQUEST                *
*CAS*****
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
```

Migration considerations

There are no special considerations for migration. Changes can be implemented dynamically and in the IGGCATxx parmlib member.

Documentation

IGGCATxx parmlib member setup is documented in *MVS Initialization and Tuning Reference z/OS V2.1*, SA23-1380. Also consult *z/OS Managing Catalog*, SC23-6853 for more information.

2.5 Catalog alias enhancements

Catalog clean up is sensitive as access to data depends on the full integrity in catalogs. Need for specific data varies over the years and data set entries may decline and even disappear. This can leave aliases behind without data sets connected at them. You can identify un-necessary aliases, but you also need to verify that they are obsolete. Doing this verification can be difficult though, as data set aliases do not have the creation date stored historically.

With DFSMS V2.1 a new alias creation will store the alias record with a creation date in the master catalog record (type X). This will ease later identification as to when the alias was created and if it is potentially a completely new one that was never used.

Example 2-22 is a LISTCAT of an alias created in an pre-DFSMS V2.1 release.

Example 2-22 LISTCAT of a data set alias created before DFSMS V2.1

```
ALIAS ----- MHLRES1
IN-CAT --- MCAT.SANDBOX.Z1C.SBOX00
HISTORY
RELEASE-----2      CREATION-----0000.000
ASSOCIATIONS
USERCAT--UCAT.VSBOX01
```

The CREATION date field s show zeros only and in theory this alias can have been created recently or many years back.

For comparison, a LISTCAT was issued on an alias created on a DFSMS V2.1 system where the alias creation date now is being stored. See Example 2-23 on page 33.

Example 2-23 LISTCAT of data set alias created under DFSMS V2.1

```
ALIAS ----- MHLRES1
IN-CAT --- MCAT.SANDBOX.Z1C.SBOX00
HISTORY
RELEASE-----2      CREATION-----2013.259
ASSOCIATIONS
USERCAT--UCAT.VSBOX01
```

Only data sets created with DFSMS V2.1 and after will have the creation date available.

2.5.1 Additional verification on delete ALIAS

Prior to DFSMS V2.1 a delete of an alias was honored, as long as the alias name could be found. In DFSMS V2.1 additional verification will be done on the association cell of the base record, assuring that the backwards pointer is correct.

If the resolved name is not found in the master catalog, if the resolved name is a symbolic name, the symbolic will be resolved and the code will re-orient to the correct user-catalog depending on the high-level qualifier (HLQ).

2.5.2 Catalog Search Interface enhancements

Catalog Search Interface (CSI) was created to offer a faster interface to ICF catalogs compared to CAMLST and IDCAMS search. CSI can offer generic search through filtering and multiple entries in the program in one search. CSI can be invoked from high level programs through a callable interface (GUPI) and can return search results for one or more entries at a time.

Updates in DFSMS V2.1

In DFSMS V2.1 new fields will be externalized to improve the usability of CSI. The new fields are listed here.

- ▶ STRNO. Number of concurrent requests.
- ▶ BUFND. Number of buffers requested for Data component.
- ▶ BUFNI. Number of buffers requested for Index component.
- ▶ INDXLVLS. Number of index levels seq-set-rba.
- ▶ HILVLRBA. RBA of High Level Index Record.
- ▶ ASSOC. A repeating list of catalog records associated with this entry.
- ▶ ASSOCSYB. Indicates if the entry is symbolic-relate.
- ▶ TRACKS. Total tracks per volume.

Basically the CSI feature is one load module IGGCSI00 that can be called, linked to or linked into a user module. Note that IGGCSI00 cannot be used in these environments:

- ▶ In a dynamic allocation exit
- ▶ During an OPEN/CLOSE/EOV exit
- ▶ In cross-memory mode
- ▶ In SRB mode
- ▶ In disabled mode

2.5.3 Sample programs and REXX for CSI reporting

Doing a CSI search will require fields describing options that control CSI processing, filtering criteria, types of catalog entries to be returned, generic (or specific) filter key for entry names to be processed or catalog field names to be returned.

To get started with CSI, you will find three assembler programs and a REXX in SYS1.SAMPLIB. The assembler programs only have to be assembled and linked to be ready for use.

You may have specific reporting needs and may consequently need to change the source code accordingly. The programs and description of each are described here.

- ▶ IGGCSILC. Input is the catalog name. Output is a listing of all entry names in the catalog printed along with a summary of the number of each entry type found and the total of all entries found.
- ▶ IGGCSIVG. Input is the catalog name. Output is a summary for each catalog name showing the total unused space by VSAM type. A total for all catalog names supplied is printed after the last catalog name is processed.
- ▶ IGGCSIVS. Input is a six character volser number, followed by the catalog name. The output is a listing of all data sets that reside in the given catalog and are on the given volume.
- ▶ IGGCSIRX. IGGCSIRX is a REXX exec that will prompt you for a search key and return you the data set name, its type, and volume serial numbers.

2.5.4 Running reports with CSI programs

Below are examples of how to run the provided CSI programs in batch jobs. The corresponding output reports are displayed.

IGGCSILC sample program

The first example in Example 2-24 on page 35 shows how a report using the IGGCSILC program is created.

Example 2-24 Sample job running CSI program IGGCSILC

```
//CSI0001 EXEC PGM=IGGCSILC
//*
/*THE INPUT TO IGGCSILC IS AN 80-BYTE SYSIN DD RECORD. THE CATALOG NAME
/*OF THE CATALOG TO BE LISTED SHOULD BE LEFT JUSTIFIED IN COLUMN 1 OF
/*THE RECORD.
/*
/*A LISTING OF ALL ENTRY NAMES IN THE CATALOG IS PRINTED ALONG WITH A
/*SUMMARY OF THE NUMBER EACH ENTRY TYPE FOUND AND THE TOTAL OF ALL
/*ENTRIES FOUND.
//SYSOUT DD SYSOUT=*
//SYSIN DD *
UCAT.DBOBTARG
```

As shown in Example 2-25 on page 35, one or more catalog names are input for the report. The output will show a list of all data set names in the catalogs listed plus a summary at the end of the report.

Example 2-25 Sample output from IGGCSILC program

```
LISTING FROM CATALOG -- UCAT.DBOBTARG
CLUSTER ----- 0000000000000000000000000000000000000000000000000000000
  DATA ----- UCAT.DBOBTARG
  INDEX ----- UCAT.DBOBTARG.CATINDEX
NONVSAM ----- DBOBT.ADSNLOAD
NONVSAM ----- DBOBT.ADSNMACS
NONVSAM ----- DBOBT.DBOBWLMJ.JAVAENV
..... more data set entries ..... and next the summary >>>
THE NUMBER OF ENTRIES PROCESSED WAS:
    AIX -----0
```

ALIAS	-----0
CLUSTER	-----2
DATA	-----2
GDG	-----0
INDEX	-----1
NONVSAM	-----121
PAGESPACE	-----0
PATH	-----0
USERCATALOG	-----0
TOTAL	-----126

IGGCSIVG sample program

The IGGCSIVG program creates a summary of unused space per VSAM data set type by ICF catalog. The catalog names are entered as input. Example 2-26 on page 36 is a sample batch job for this program.

Example 2-26 Sample job for running catalog report using CSI program IGGCSIVG

```
//CSI0001 EXEC PGM=IGGCSIVG
//*INPUT
//*THE INPUT TO IGGCSIVG IS AN 80-BYTE SYSIN DD RECORD. THE CATALOG NAME
//*OF THE CATALOG TO BE PROCESSED MUST BE LEFT JUSTIFIED AND START IN
//*COLUMN 1 OF THE INPUT RECORD. MULTIPLE INPUT RECORDS CAN BE SUPPLIED.
//*OUTPUT
//*UNUSED SPACE BY VSAM TYPE. A TOTAL FOR ALL CATALOG NAMES
//*SUPPLIED IS PRINTED AFTER THE LAST CATALOG NAME IS PROCESSED.
//SYSOUT DD SYSOUT=*
//*SYSOUT DD DISP=SHR,DSN=MHLRES1.CSI.PRINT.VBA133
//SYSIN DD *
UCAT.DBOADATA
```

The output from the IGGCSIVG program will have a summary per catalog on unused VSAM space as shown in Example 2-27 on page 36. At the end of the report there will be a summary of all the catalogs input to the program.

Example 2-27 Sample output from IGGCSIVG program

CATALOG NAME: UCAT.DBOADATA

	BYTES UNUSED	BYTES ALLOCATED	# OF DS	MIN
LDS	36,187,750,400	48,436,510,720	3,186	0
KSDS	376,832	425,984	5	0
ESDS	0	0	0	0
RRDS	40,796,160	40,992,768	1	40,796,160
TOTAL	36,228,923,392	48,395,091,968	3,192	0

CATALOG NAME: UCAT.DBOAIMAG

	BYTES UNUSED	BYTES ALLOCATED	# OF DS	MIN
0100 0004 FS				
LDS	143,515,062,272	89,557,762,048	257	0
KSDS	0	0	0	0
ESDS	0	0	0	0
RRDS	0	0	0	0

TOTAL	143,515,062,272	89,557,762,048	257	0
-------	-----------------	----------------	-----	---

ALL CATALOGS:

	BYTES UNUSED	BYTES ALLOCATED	% OF DS	MIN
TOTAL	179,743,985,664	41,162,670,080	3,449	0

IGGCSIVS sample program

The third provided CSI report program is IGGCSIVS. The function of this program is to list all data sets in a given user catalog that has entries on a specific volume.

Input will be the volser and the catalog name in scope as shown in Example 2-28 on page 37.

Example 2-28 Sample job for running catalog report using CSI program IGGCSIVS

```
//CSI0001 EXEC PGM=IGGCSIVS
//*INPUT
//*THE INPUT TO IGGCSIVS IS AN 80-BYTE SYSIN DD RECORD. THE SIX-
//*CHARACTER VOLUME SERIAL NUMBER SHOULD BE LEFT JUSTIFIED AND STARTS IN
//*IN COLUMN 1 OF THE INPUT RECORD. THE CATALOG NAME OF THE CATALOG TO
//*BE SEARCHED STARTS IN COLUMN 7. MULTIPLE INPUT RECORDS CAN BE
//*SUPPLIED.
//*OUTPUT
//*THE OUTPUT IS A LISTING OF ALL DATA SETS THAT RESIDE IN THE GIVEN
//*CATALOG AND ARE ON THE GIVEN VOLUME.
//*
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SBOXA8UCAT.VSBOX01
```

Example 2-29 on page 37 contains sample output from the IGGCSIVG program listing all data sets on volser SBOXA8 belonging to ICF catalog UCAT.VSBOX01 as requested in the SYSIN DD statement in Example 2-28 on page 37.

Example 2-29 Sample output from IGGCSIVS program

```
DATA SETS FOUND IN CATALOG UCAT.VSBOX01 HAVING VOLSER SBOXA8

ADHUSER.DB9A.ATEXT.D090306.I01240.DATA          CLUSTER ADHUSER.DB9A.ATEXT.D090306
ADHUSER.DB9A.D.D090305.H19.M42.Z2
ADHUSER.DB9A.SD.D090224.H19.M03.Z5
ADHUSER.DB9A.SP.D090210.H14.M54.Z1
ADHUSER.DB9A.T.D090220.H17.M11.Z15
ADIS.SC63.ISPF42.ISPPROF
```

IGGCSIRX CSI REXX

The IGGCSIRX REXX is found in SYS1.SAMPLIB. It is used to display fully qualified data set names, an alias, a GDG entry or any catalog entry type. For the data set name, the catalog name, and the entry type the volser is returned. Example 2-30 on page 38 is an example. The REXX is called, you are requested to enter a key (data set name and alias). The response is returned to your display.

Example 2-30 Running IGGCSIRX CSI REXX in foreground

ENTER FILTER KEY

MHLRES1.CSI.PRINT.VBA133

CATALOG UCAT.VSBOX01

NONVSAM MHLRES1.CSI.PRINT.VBA133

MLD52B

If you type in an alias, a GDG name or other types of catalog entries than a data set name, the catalog name and type is returned for this key.

CATSRCHI - improved search facility

An improved version of IGGSRCIX REXX CATSRCHI is now offered 'as is'. This REXX will have full data set names or data set filters as input and will generate a list based on the selection criteria.

The CATSRCHI REXX is available for download from the IBM FTP site:

ftp://ftp.software.ibm.com/ - userid: anonymous and password: your email address

The full path in our environment is:

ftp://ftp.software.ibm.com/servers/storage/support/software/dfsms/II14316

Going to this web-site, you will find two members there:

- ▶ II14316.clist.clist. File holding the CATSRCHI rexx code.
- ▶ II14316.tso.cntl. File holding the JCL running the REXX.

An example of the search input could be is shown in Example 2-31 on page 38. The example shows a generic search on a high level qualifier level. The fieldname parameter specifies the fields on which information is requested.

Example 2-31 Running a generic search using CATSRCHI in a batchjob

```
//MHLRES1A JOB (999,POK),MSGLEVEL=1,NOTIFY=MHLRES1
/*JOBPARM SYSAFF=SC64
/*
//STEP1      EXEC PGM=IKJEFT1B
//SYSPROC    DD DISP=SHR,DSN=MHLRES1.TSO.CLIST
/*POINT TO YOUR CLIST DATA SET.
//SYSPRINT   DD SYSOUT=*
//SYSTSPRT   DD SYSOUT=*
//SYSTSIN    DD *
%CATSRCHI MHLRES4.** -
FIELDNAMES(XHARBADS XHURBA XACIFLAG VOLSER -
           COMPIND NOEXTNT AMDKEY PHYBLKSZ EXTENT)
/*
```

The output from the CATSRCHI REXX will display the externalized field names. For non-VSAM files only the data set name and the VOLSER will be displayed. A VSAM file will appear with cluster, data component and index component as shown in Example 2-32 on page 38.

Example 2-32 Sample output from CATSRCHI rexx displaying externalized fields

```
CLUSTER  MHLRES4.NSMS.CARENO.TEST01
XHARBADS = no data
XHURBA   = no data
XACIFLAG = no data
```


COMPIND = no data
NOEXTNT = no data
AMDKEY = no data
PHYBLKSZ = no data
EXTENT = no data

DATA MHLRES4.NSMS.CARENO.TEST01.DATA
XHARBADS = 55296
Volume MHLS1A XHURBA = 55296
XACIFLAG - Data set is not EA (extended addressing)
On volser(s) SBOXEF
COMPIND - No flags set
Data set has 1 extents on 1 volumes
Volume SBOXEF NOEXTNT = 1
Relative key position = 41
Key length = 10
PHYBLKSZ = x'00004800'

Data set has 1 extents on SBOXEF

Extent Seq number	= 1	Extent tracks	= 1
Extent low cchhr	= 000D000B	Extent Start RBA	= 0
Extent high cchhr	= 000D000B	Extent End RBA	= 55295

Number of extents from all VVR entries is 1

INDEX MHLRES4.NSMS.CARENO.TEST01.INDEX
XHARBADS = 25088
Volume SBOXEF XHURBA = 512
XACIFLAG - Data set is not EA (extended addressing)
On volser(s) SBOXEF
COMPIND - No flags set
Data set has 1 extents on 1 volumes
Volume SBOXEF NOEXTNT = 1
Relative key position = 41
Key length = 10
PHYBLKSZ = x'00000200'

Data set has 1 extents on SBOXEF

Extent Seq number	= 1	Extent tracks	= 1
Extent low cchhr	= 000D000E	Extent Start RBA	= 0
Extent high cchhr	= 000D000E	Extent End RBA	= 25087

Number of extents from all VVR entries is 1

Documentation on CSI

Additional information on CSI can be found in *z/OS V2R1.0 DFSMS Managing Catalogs*, SC23-6853.

2.6 Catalog parmlib member enhancements

The catalog parmlib member IGGCATxx was introduced in z/OS V1.13. The main purpose of this new feature was to remove the bottlenecks in the current 80 character line specification in

SYS1.NUCLEUS or SYS1.IPLPARM. It is also to make catalog changes more dynamic and independent of having to do an IPL.

The parameters listed here were part of the initial implementation:

- ▶ VVDSSPACE
- ▶ TASKMAX
- ▶ NOTIFYEXTENT
- ▶ DELFORCEWNG
- ▶ DSNCHECK
- ▶ SYMREC
- ▶ UPDTFAIL
- ▶ VVRCHECK
- ▶ DELFORCEWNG
- ▶ EXTENDEDALIAS

In DFSMS V2.1 additional parameters have been added to the IGGCATxx catalog parmlib.

2.6.1 How to prepare for catalog parmlib

If not already done, you have to populate the IGGCATxx member in SYS1.PARMLIB with the options you want to move to the PARMLIB function. It is suggested to define all possible parameters once you decide to implement the catalog parmlib function to visualize all options available. But initially perhaps only use default values. You have the possibility to quickly change these values easily having the defaults available and visible in the IGGCATxx catalog parmlib.

Example 2-33 on page 40 is a sample where all options in the catalog parmlib are displayed with the corresponding default value.

Example 2-33 Example of IGGCATxx member showing options with default value

```
ALIASLEVEL(1)
AUTOADD(OFF)
CATMAX(1024)
DELFORCEWNG(OFF)
DELRECOVWNG(YES)
DSNCHECK(YES)
DUMP(OFF)
EXTENDEDALIAS(NO)
GDGFIFOENABLE(NO)
NOTIFYEXTENT(80)
SYMREC(YES)
SYS%(OFF)
TAPEHLQ(SYS1)
TASKMAX(180)
TASKMIN(60)
TASKTABLESIZE(200)
UPDTFAIL(YES)
VVDSSPACE(40,40)
```

Most of these options can be changed dynamically by **MODIFY CATALOG** commands. The system applies all IGGCATxx values at IPL time as well as at a restart of the Catalog Address Space.

Activating a new IGGCATxx catalog parmlib configuration change will also validate the individual settings. Example 2-34 on page 41 is a sample Modify Catalog Restart command, where the IGGCAT00 member contains an error.

Example 2-34 Example of error message after restart of Catalog Address Space.

```
IEC355I IDACAT14, CATALOG ADDRESS SPACE IS RESTARTING
IEE252I MEMBER IGGCAT00 FOUND IN SYS1.PARMLIB
IEC386W INVALID KEYWORD DETECTED IN IGGCAT00 AT LINE:DELFORCEWNG(OFF)
```

All valid settings will be updated in IGGCATxx after a Catalog restart. The settings in error will remain unchanged.

2.6.2 Activating IGGCATxx catalog parmlib

To have the system start using the IGGCATxx member in SYS1.PARMLIB, you have to specify the IGGCATxx member suffix in IEASYSxx member by specifying CATALOG=xx. If CATALOG=01 is specified the system will look for a IGGCAT01 member in SYS1.PARMLIB.

If there is no CATALOG specification in the IEASYSxx member, the system will, by default, look for an IGGCAT00 member. If this is not found the old setup using SYS1.NUCLEUS or SYS1.IPLPARM will be used.

Activation will happen either by a Catalog restart command or through IPL.

2.6.3 IGGCATxx new options in DFSMS V2.1

In this section we look at the new options and their function made available in DFSMSV2.1.

- ▶ **TASKMIN(n)**. Catalog service task lower limit (number of tasks, that can run concurrently). Default is 60. Minimum is 24 and maximum is 180.
- ▶ **TAPEHLQ(name)**. High level qualifier of a Tape Volume catalog. Default is SYS1. Valid character set includes alphabetic, numerics and the national characters @, #, \$.
- ▶ **SYS%(ONIOFF)**. Specifies whether SYS% to SYS1 conversion is enabled. Default is OFF.
- ▶ **AUTOADD(ONIOFF)**. Specifies whether Enhanced Catalog Sharing (ECS) Autoadd is to be turned on or off. Default is OFF. When ON, catalogs will be added to ECS on their next reference if they are eligible for ECS.
- ▶ **DUMP(ONIOFF)**. Specifies whether dynamic dumping is to be turned on or off. Default is OFF.
- ▶ **ALIASLEVEL(n)**. Specifies the MLA search level. ALIASLEVEL has a default value of 1, with a minimum of 1 and a maximum of 4.
- ▶ **DUMPON(rc,rsn,mod,cnt)**. Specifies the return (*rc*) and reason codes (*rsn*) to take a dump on if a match is found in a valid catalog module (*mod*). The optional *cnt* value specifies how many times to skip before taking the dump. The default for *cnt* is 1.
- ▶ **TASKTABLESIZE(n)**. Specifies the maximum possible tasks running at a particular time. This includes both catalog and non-catalog tasks. TASKTABLESIZE is an IPL only parameter. TASKTABLESIZE is ignored on a CAS restart (no error message is issued).

TASKTABLESIZE has a default value of 200, with a minimum of 200 and a maximum of 400.

- CATMAX(n). The maximum number of catalogs that can be open at any given time on the system. The default is 1024. Valid values between 1 and 9999. When maximum number of open catalogs is reached - close of least recently used catalogs will start.
- VVRCHECK(YES/NO). Specifies if you want to enable enhanced VVR checking on VVDS I/O. The default is NO.

2.6.4 How to display settings on Catalog parmlib

Display of current settings in catalog address space (CAS) can be done through the **F CATALOG,REPORT** command. The display output will show the IGGCATxx settings except for resource values as shown in Example 2-35 on page 42.

Example 2-35 IGGCATxx settings displayed by F CATALOG,REPORT command

```

F CATALOG,REPORT
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC359I CATALOG REPORT OUTPUT
*CAS*****
* CATALOG COMPONENT LEVEL   = HDZ2210                      *
* CATALOG ADDRESS SPACE ASN = 0098                          *
* SERVICE TASK UPPER LIMIT  = 180                           *
* SERVICE TASK LOWER LIMIT  = 60                             *
* HIGHEST Æ SERVICE TASKS    = 3                             *
* Æ ATTACHED SERVICE TASKS   = 3                             *
* MAXIMUM Æ OPEN CATALOGS    = 1,024                         *
* ALIAS TABLE AVAILABLE     = YES                            *
* ALIAS LEVELS SPECIFIED     = 1                              *
* SYS% TO SYS1 CONVERSION    = OFF                           *
* CAS MOTHER TASK            = 00790680                      *
* CAS MODIFY TASK            = 007FC558                      *
* CAS ANALYSIS TASK          = 007FC0F8                      *
* CAS ALLOCATION TASK         = 007FC328                      *
* CAS ASYNC TASK             = 0077AE88                      *
* CAS SYSPLEX COMMAND TASK   = 0077AA28                      *
* CAS SYSPLEX QUIESCE TASK   = 0077AC58                      *
* VOLCAT HI-LEVEL QUALIFIER  = SYS1                           *
* NOTIFY EXTENT              = 80%                            *
* DEFAULT VVDS SPACE         = ( 10, 10) TRKS                *
* ENABLED FEATURES           = DSNCHECK DELFORCEWNG SYMREC   *
* ENABLED FEATURES           = UPDTFAIL                      *
* DISABLED FEATURES          = VVRCHECK AUTOTUNING BCSHECK   *
* DISABLED FEATURES          = DELRECOVWNG EXTENDEDALIAS     *
* DISABLED FEATURES          = ECS AUTOADD DUMPON GDGFIFO     *
* INTERCEPTS               = (NONE)                         *
*CAS*****
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED

```

To see which catalog parmlib you are using on this system, use the **D IPLINFO,CATALOG display** command. This will show which IGGCATxx parmlib member is active. The output from the command is shown in Example 2-36 on page 43.

Example 2-36 D IPLINFO,CATALOG display output

```
D IPLINFO,CATALOG
IEE255I SYSTEM PARAMETER 'CATALOG': 00
```

The IGGCATxx catalog parmlib suffix on this system is '00'. In this case the catalog parmlib member name is IGGCAT00.

Documentation

See *MVS Initialization and Tuning Reference z/OS V2.1*, SA23-1380 and *z/OS Managing Catalogs*, SC23-6853 for details on the catalog parmlib feature.

2.7 Catalog DFSMS GDG enhancements

Current GDG processing is from newest to oldest (LIFO). Given the processing order GDG data sets are presented from oldest (lowest generation data set number) to newest in sequence. Based on requirements DFSMS V2.1 has added a new attribute so that GDG data sets can be processed in reverse order. The new attribute is GDGFIFOENABLE.

Processing GDG data sets in FIFO order will often reduce the need for doing SORT, as data often are time stamped and need to be processed accordingly.

2.7.1 Enable FIFO processing of GDG processing

To activate the FIFO processing of GDG data sets, you will first have to enable this in catalog parmlib with the parameter GDGFIFOENABLE(YES). The default for this parameter is NO, so if you try to create a new GDG using a new keyword FIFO without changing the default you will see the error message in Example 2-37 on page 43.

Example 2-37 Error message returned if you try to define GDG without activating GDGFIFOENABLE.

```
DEFINE GENERATIONDATAGROUP -
      (NAME(MHLRES1.TEST.FIFO.GDG) -
      EMPTY FIFO -
      SCRATCH -
      LIMIT(10) )
IDC0010I GDG FIFO ALLOCATION ORDER FEATURE NOT ENABLED.
IDC3003I FUNCTION TERMINATED. CONDITION CODE IS 12
```

GDGFIFOENABLE(YES) has to be set in catalog parmlib to be able to create a GDG with the FIFO option. You can activate this setting by restarting CAS or doing an IPL after having updated the IGGCATxx member. There is no dynamic MODIFY command available for this setting.

The FIFO option is available on a DEFINE GDG statement or on the IDCAMS alter statement.

FIFO processing scenario

To illustrate the change in processing order we create a scenario.

1. Two GDG entries were created, one with default option (LIFO), the other one with the FIFO option provided. Two data sets were allocated in each GDG.

2. Next we did an IDCAMS print to process the data sets and the allocation of the DD statements shows the two different processing orders. The data sets were processed from newest to oldest with the LIFO option as seen in Example 2-38 on page 44.

Example 2-38 LIFO processing order of GDG data set with default option

IGD104I MHLRES1.TEST.LIFO.GDG.G0002V00	RETAINED, DDNAME=DD1
IGD104I MHLRES1.TEST.LIFO.GDG.G0001V00	RETAINED, DDNAME=

3. For the FIFO option the GDG data sets are processed in reverse order as seen in Example 2-39 on page 44.

Example 2-39 Processing order of GDG data set with FIFO option enabled.

IGD104I MHLRES1.TEST.FIFO.GDG.G0001V00	RETAINED, DDNAME=DD1
IGD104I MHLRES1.TEST.FIFO.GDG.G0002V00	RETAINED, DDNAME=

2.7.2 Setting processing through JCL

The JCL parameter GDGORDER can be used to influence the concatenation order. This keyword will overwrite any LIFO or FIFO setting on the GDG. Example 2-40 on page 44 shows an example of the GDGORDER=FIFO JCL parameter.

Example 2-40 Overwriting LIFO/FIFO setting on GDG using JCL parameter.

//SMFDATA DD DSN=MHLRES1.TEST.LIFO.GDG,DISP=SHR,GDGORDER=FIFO

Valid settings for GDGORDER parameter are USECATLG, FIFO and LIFO.

Documentation

See more detail in *z/OS V2R1.0 MVS JCL Reference*, SA23-1385.

2.8 Catalog RNLs Health Check

We suggest converting the resources SYSIGGV2/SYSZVVDS/SYSVTOC in the GRS RNL to avoid serious deadlocks involving these. The exception can be for resources placed on disk outside the sysplex.

With DFSMS V2.1 a new health check (Catalog_Rnls) will now be monitoring these resources and will create an alert, if suggested conversions are not met. The customer will benefit from this, if incorrect settings are detected and can be changed before facing a hang-up situation.

2.8.1 SDSF Health Checker screen

The new health check can be found in the Health Checker screen in SDSF.

1. To reach this, go to the SDSF main menu and type **CK** to enter the health checker menu. The Catalog_RNLS is to be found as shown in Figure 2-7 on page 45.

SDSF HEALTH CHECKER DISPLAY SC64			LINE 1-19 (199)	
COMMAND INPUT ==>			SCROLL ==> CSR	
NP	NAME	CheckOwner	State	Statu
	ALLOC_ALLC_OFFLN_POLICY	IBMALLOC	ACTIVE(ENABLED)	SUCCE
	ALLOC_SPEC_WAIT_POLICY	IBMALLOC	ACTIVE(ENABLED)	SUCCE
	ALLOC_TIOT_SIZE	IBMALLOC	ACTIVE(ENABLED)	SUCCE
	ASM_LOCAL_SLOT_USAGE	IBMASM	ACTIVE(ENABLED)	SUCCE
	ASM_NUMBER_LOCAL_DATASETS	IBMASM	ACTIVE(ENABLED)	SUCCE
	ASM_PAGE_ADD	IBMASM	ACTIVE(ENABLED)	SUCCE
	ASM_PLPA_COMMON_SIZE	IBMASM	ACTIVE(ENABLED)	EXCEP
	ASM_PLPA_COMMON_USAGE	IBMASM	ACTIVE(ENABLED)	SUCCE
	CATALOG_IMBED_REPLICATE	IBMCATALOG	ACTIVE(ENABLED)	EXCEP
	CATALOG_RNLS	IBMCATALOG	ACTIVE(ENABLED)	SUCCE
	CNZ_AMRF_EVENTUAL_ACTION_MSGS	IBMCNZ	ACTIVE(ENABLED)	SUCCE

Figure 2-7 SDSF Health Checker screen with the new Catalog_RNLS health check

2. Run the Health Checker real time by selection the health check in scope. In this scenario select the **Catalogs_RNLS**. The output from the health check is shown in Example 2-41 on page 45. If successful, the status will be updated accordingly.

Example 2-41 Result of running Catalog_RNLS health check

```

CHECK(IBM Catalog,Catalog_RNLS)
SYSPLEX:   SANDBOX   SYSTEM: SC64
START TIME: 09/13/2013 23:35:08.042348
CHECK DATE: 20120827  CHECK SEVERITY: LOW

```

```

IGGHC110I CHECK(IBM Catalog,Catalog_RNLS) ran successfully and found no
exceptions.

```

```

END TIME: 09/13/2013 23:35:08.044066  STATUS: SUCCESSFUL

```

3. During the health check, if any of the resources are not converted a notification will be created, that a potential deadlock situation can occur and on which of the resources. The user is suggested to follow instructions in the health check as to how to resolve and prevent the potential deadlock situation.

Documentation

The book *z/OS V2R1.0 MVS Planning: Global Resource Serialization, SA23-1389* will explain how to convert resources in more detail. As to Catalog RNL reserves, refer to chapter 2, page 44.

2.9 MODIFY CATALOG,LIST enhancement

The latch number has been added to **F CATALOG,LIST** output. Message IEC347I has been changed to display latches, when any contention occurred.

Example 2-42 on page 45 is the pre- DFSMS V2.1 display.

Example 2-42 MODIFY CATALOG LIST display in previous releases

```

IEC347I LIST CATALOG TASK(S)
*CAS*****

```

```
* FLAGS - TASK ADDRESS - JOBNAME / STEPNAME - ELAPSED TIME - ID *
* ----- NOACTIVE / NONE 00.00.00 *
*****
* 0-OLDEST, W-WAIT, A-ABEND, E-ENQ, R-RECALL, L-RLS *
*CAS*****
```

Latch information has been added in DFSMS V2.1 as shown in Example 2-43 on page 46.

Example 2-43 MODIFY CATALOG LIST display having latch information added

```
IEC347I LIST CATALOG TASK(S)
*CAS*****
* FLAGS - TASK ADDR - JOBNAME / STEPNAME - ELAPSED TIME - ID *
* ----- 00891D78 DUMPSRV / DUMPSRV 00.01.55 02 *
* WAITING FOR Get LatchShr # 00001 FROM 09F06730 FOR 00.01.23 *
*****
* 0-OLDEST, W-WAIT, A-ABEND, E-ENQ, R-RECALL, L-RLS *
*CAS*****
```

2.10 Improved catalog recovery features

New parameters have been provided for in DFSMS V2.1 on the DELETE and DEFINE USERCATALOG statement to keep and maintain alias information on deletion and later creation of an ICF catalog. Using the new DELETE USERCATALOG NODISCONNECT and DEFINE USERCATALOG RECONNECT parameters, you will no longer have to maintain the ALIAS pointers yourself.

Reconnect will also support rebuilding alias information even if the ICF catalog is defined on another volume or device type.

Another significant enhancement for ICF Catalogs is the new **MODIFY RECOVER** command that has been added in DFSMS V2.1 to support catalog forward recovery procedures. This new command will work for both RLS managed and non-RLS managed catalogs:

2.10.1 F CATALOG,RECOVER,LOCK(ucat name)

This command will lock and close the catalog sysplex wide. The catalog will appear as LOCKED in a catalog display. Trying to access the catalog, you will get the return code 186, with message "UNABLE TO ACCESS LOCKED CATALOG". A LOCK will generally be used, when the catalog is being reorganized or recovered. Note that LOCK will fail any user request as long as the LOCK is active.

2.10.2 F CATALOG,RECOVER,UNLOCK(ucat name)

This command will release a LOCK set by a previous **RECOVER LOCK** command or by an **IDCAMS ALTER LOCK** command.

2.10.3 F CATALOG,RECOVER,SUSPEND(ucat name)

This command will do a serialized close of the catalog across the sysplex. The suspend will prevent unauthorized request to the catalog. In a catalog display, the catalog will appear as in suspended state.

Compared to the LOCK parameter, SUSPEND will queue up incoming requests in the users address space and will not fail the job.

2.10.4 F CATALOG,RECOVER,RESUME(ucat name)

The **RESUME** command will resume a suspended catalog, that was earlier brought in this state by a **RECOVER SUSPEND** command.

Note that the user catalog name can be specified generically or as a fully-qualified catalog name.

Example 2-44 on page 47 shows the messages when using the **RECOVER** command to prevent access and again opening for access. The first example shows a lock of the catalog.

Example 2-44 LOCK an ICF Catalog using a RECOVER LOCK command.

```
F CATALOG,RECOVER,LOCK(UCAT.RLSTST)
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC352I MODIFY CATALOG UCAT.RLSTST TO STATE LOCK SUCCESSFUL
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
```

To bring the ICF catalog out of a locked status use the **RECOVER UNLOCK** command as shown in Example 2-45 on page 47.

Example 2-45 UNLOCK an ICF Catalog from a LOCKED state by command.

```
F CATALOG,RECOVER,UNLOCK(UCAT.RLSTST)
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC352I MODIFY CATALOG UCAT.RLSTST TO STATE UNLOCK SUCCESSFUL
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
```

To set the ICF catalog in a suspended state before doing maintenance or recovery, use the **RECOVER SUSPEND** command as shown in Example 2-46 on page 47.

Example 2-46 Example of suspending an ICF Catalog using RECOVER SUSPEND command

```
F CATALOG,RECOVER,SUSPEND(UCAT.RLSTST)
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC352I MODIFY CATALOG UCAT.RLSTST TO STATE SUSPEND SUCCESSFUL
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
```

When the maintenance or recovery has been done to the suspended ICF catalog, resume the ICF catalog from the suspended state, issuing the **RECOVER RESUME** command. All catalog requests outside of the maintenance work will be queued up while this is ongoing. Example 2-47 on page 47 is the output from the **RESUME** command.

Example 2-47 Resume of ICF Catalog being in suspended state

```
F CATALOG,RECOVER,RESUME(UCAT.RLSTST)
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
```

```
IEC352I MODIFY CATALOG UCAT.RLSTST TO STATE RESUME SUCCESSFUL
IEC352I CATALOG ADDRESS SPACE MODIFY COMMAND COMPLETED
```

Remember to have the appropriate access to the IGG.CATLOCK resource to be able to issue the commands in this section.

Also note, that the IDCAMS ALTER LOCKISUSPEND command will now work Sysplex wide assuring full integrity.

2.10.5 Catalog define and delete parameters

New parameters NODISCONNECT|RECONNECT to be used along with the DELETE USERCATALOG command are available with DFSMS V2.1. NODISCONNECT will retain alias information at the deletion of a user catalog. If you use this parameter, the new parameter RECONNECT can be used on the DEFINE USERCATALOG command to have the newly defined catalog connected to the existing alias information so that you do not have to rebuild this information manually.

NODISCONNECT| RECONNECT subparameter scenario

For visualization of how the new NODISCONNECT | RECONNECT subparameters work, a DELETE of a user catalog is done using the keyword NODISCONNECT. The user catalog is defined again with the RECONNECT parameter. Example 2-48 on page 48 is an example of deleting a user catalog using NODISCONNECT parameter.

Example 2-48 Delete of user catalog using the NODISCONNECT parameter

```
//IDCAMS EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
        DELETE UCAT.RLSTST USERCATALOG RECOVERY NODISCONNECT
/*
```

Next the user catalog was defined again with the RECONNECT keyword to connect to the existing alias pointers as shown in Example 2-49 on page 48.

Example 2-49 Example of defining a user catalog using RECONNECT parameter

```
//IDCAMS EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
        DEFINE USERCATALOG -
            (NAME(UCAT.RLSTST ) ICFCATALOG -
            VOLUME(MHL1A0) TRK(5 1) -
            STORCLAS(SCRLS) -
            DATACLAS(WELCHRLS) -
            LOG(NONE) RECONNECT -
            FREESPACE(20 20) -
            NOIMBED NOREPLICATE) -
            DATA (CISZ(4096))
/*
```

The only ALIAS in this user catalog was RLSTST with one data set created. When we looked up the alias using ISPF 3.4 after the define of the usercatalog, the alias was visible. At this point there are no with no data sets. A restore from a backup had to be done. In our scenario it is a DFSMSdss backup.

The backup job was started but failed immediately with the error in Example 2-50 on page 49.

Example 2-50 DSS restore of a user catalog fails due to missing access to IGG.CATLOCK

```
ICH408I USER(MHLRES1 ) GROUP(SYS1      ) NAME(MHLRES1/MARY LOVELAC)
      IGG.CATLOCK CL(FACILITY)
      INSUFFICIENT ACCESS AUTHORITY
      ACCESS INTENT(READ  ) ACCESS ALLOWED(NONE  )
```

The example above shows that DFSMSDss does an implicit LOCKISUSPEND of a user catalog before importing data from backup.

To correct the error READ access to the IGG.CATLOCK facility class is provided. The restore was restarted and ended successfully this time as shown in Example 2-51 on page 49.

Example 2-51 Sysout from DSS restore of user catalog pre-allocated with keyword RECONNECT

```
ADR442I (001)-FRLB0(01), DATA SET UCAT.RLSTST PREALLOCATED, IN CATALOG MCAT.....
ADR360I (001)-TDVSM(02), PROCESSING SUSPENDED USER CATALOG UCAT.RLSTST
ADR963I (001)-TDLOG(02), CLUSTER UCAT.RLSTST WAS DUMPED USING RECORD LEVEL SHARI
                        GMT TIMESTAMP IS: 2013.275 00:04:50.8
ADR489I (001)-TDLOG(02), CLUSTER UCAT.RLSTST WAS RESTORED
                        CATALOG          MCAT.SANDBOX.Z1C.SBOX00
                        COMPONENT        UCAT.RLSTST
                        COMPONENT        UCAT.RLSTST.CATINDEX
ADR372W (001)-TDNVS(03), ALIAS RLSTST NOT RESTORED FOR USER CATALOG UCAT.RLSTST,
ADR480W (001)-TDLOG(01), THE FOLLOWING DATA SETS WERE NOT PROCESSED FROM THE LOG
                        RLSTST
ADR454I (001)-TDLOG(01), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED
```

Note that before processing the restore DFSMSDss does a suspend of the catalog, The catalog is protected against concurrent updates. DFSMSDss also has the awareness that the dump was taken using RLS access (implicit QUIESCE before doing the dump, and RESUME after the DUMP had finished).

DFSMSDss finds the user catalog preallocated and consequently does not restore the alias, as this information is already in place.

Note that the SUSPEND issued by DFSMSDss is resumed again after successful restore.

2.10.6 Catalog forward recovery using new features

Forward recovery of a catalog is done differently using the newest DFSMS V2.1 enhancements. Of course you will still need backups as your recovery checkpoint and SMF data to perform forward recovery from this checkpoint to a current point in time. In this section we describe the steps for a catalog forward recovery at the DFSMS V2.1 level.

1. Using DFSMSDss as the backup tool, the RLS managed catalogs will be quiesced. DFSMSDss will enqueue on SYSIGGV2 for the non-RLS managed catalogs as done in DFSMS prior to V2.1.
2. Forward recovery to a new volume should be preceded by issuing the **F CATALOG,RECOVER,SUSPEND(ucat name)** command. This will queue up catalog requests against this catalog without failing them. Most importantly, the SUSPEND will close the catalog across the Sysplex ensuring integrity.

3. You are now ready to delete the catalog from its current location (knowing you have a valid backup). Delete should happen with the new NODISCONNECT parameter to maintain alias pointers. Only DSCBs and VVRs are deleted.
4. Next redefine the catalog using the new RECONNECT parameter (counterpart to NODISCONNECT). Use parameters VOLSER(vvvvvv), LOG(NONE), SUSPEND and RLSQUIESCE or RLSENABLE along with this.

Using the VOLSER, this will be updated across the Sysplex and RECONNECT will connect existing alias information, while the SUSPEND parameter assures integrity.
5. Next restore can happen using DFSMSdss. DFSMSdss will sense the suspended status and will restore the empty catalog without using additional serialization except from the SUSPEND.
6. When the restore has been successfully completed, you will of course need to do forward recovery from the time the catalog backup was taken up until the current point in time. This can happen by means of the ICFRU tool, that will use selected SMF records for the forward recovery.
7. The only outstanding action is now to release the SUSPEND of the catalog. Do this with the **MODIFY CATALOG,RECOVER,RESUME(ucat name)** command.

2.10.7 Catalog recovery using BCSRECOVER

This section will explain how to use DFSMSdss as the backup and restore tool for the ICF catalog using the new BCSRECOVER parameter for DFSMSdss. BCSRECOVER should be used along with the SUSPEND parameter to assure best possible integrity and availability.

As mentioned earlier, DFSMSdss will QUIESCE a catalog in use by RLS when backing up the catalog. For non-RLS managed catalogs, DFSMSdss will enqueue on the SYSIGGV2 resource.

Restoring an ICF catalog using DFSMSdss can be done using the BCSRECOVER parameter with the subparameter SUSPEND. The BCSRECOVER(SUSPEND) parameters will do a sysplex wide suspend that will suspend incoming requests while restore is ongoing. Once this has finished the requests can be resumed.

Using the BCSRECOVER(LOCK) parameter instead, DFSMSdss will invoke a sysplex wide lock of the catalog. Compared to SUSPEND, LOCK will fail all incoming unauthorized request while restore is ongoing. A sample job is shown in Example 2-52 on page 50.

Example 2-52 DSS restore job using BCSRECOVER SUSPEND parameters

```
//STEPT006 EXEC PGM=ADDRSSU,REGION=4096K
//SYSPRINT DD SYSOUT=*
//DASD DD DISP=SHR,UNIT=3390,VOL=SER=SBOX1A
//TAPE DD DISP=SHR,
// DSN=MHLRES1.DSS.BACK1.MHLRES1A,SPACE=(TRK,(2,1)),
// UNIT=3390,DCB=(BLKSIZE=27920)
//SYSIN DD *
  RESTORE DATASET(
    INCLUDE(
      UCAT.RLSTST ))
  INDDNAME(TAPE)
  OUTDDNAME(DASD)
  BCSRECOVER(SUSPEND)
```

Example 2-53 on page 51 shows the sysout from a BCSRECOVER. Alias and user catalog data is recovered.

Example 2-53 Sysout from DSS restore of user catalog and BCSRECOVER SUSPEND parameters

```
ADR360I (001)-TDVSM(01), PROCESSING LOCKED USER CATALOG UCAT.RLSTST
.....
ADR489I (001)-TDLOG(02), CLUSTER UCAT.RLSTST WAS RESTORED
                        CATALOG      MCAT.SANDBOX.Z1C.SBOX00
                        COMPONENT     UCAT.RLSTST
                        COMPONENT     UCAT.RLSTST.CATINDEX
ADR393I (001)-TDNVS(01), ALIAS RLSTST RESTORED FOR USER CATALOG UCAT.RLSTST
ADR454I (001)-TDLOG(01), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED
                        UCAT.RLSTST
ADR006I (001)-STEND(02), 2013.274 23:59:37 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.274 23:59:37 TASK COMPLETED WITH RETURN CODE 0000
```

After having restored the ICF catalog a check with the command **F CATALOG,OPEN** shows, that the catalog is no longer suspended. See Example 2-54 on page 51.

Example 2-54 Check of catalog after DSS restore using SUSPEND parameter

```
F CATALOG,OPEN
IEC351I CATALOG ADDRESS SPACE MODIFY COMMAND ACTIVE
IEC348I ALLOCATED CATALOGS 655
*CAS*****
*
*  FLAGS -VOLSER-USER-CATALOG NAME
*
*  YSU-R- SBXHS8 0001 UCAT.RLSTST
..... more catalogs listed .....
*****
*
*  Y/N-ALLOCATED TO CAS, S-SMS, V-VLF, I-ISC, C-CLOSED, D-DELET
*
*  R-SHARED, A-ATL, E-ECS SHARED, K-LOCKED, U-RLS, W-SUSPENDED
*
*CAS*****
```

Using SUSPEND ILOCK subparameters only applies to existing (pre-allocated) catalogs. If the catalog does not exist, DFSMSDss will define the catalog as locked to ensure the catalog is not accessible prior to DFSMSDss completing restore processing.

Programs using BCSRECOVER require READ access to RACF FACILITY CLASS resource IGG.CATLOCK



IDCAMS enhancements

This chapter provides an overview of the features and function in IDCAMS included in DFSMS V2.1.

IDCAMS is the program name for access method services (AMS). It is used to create and maintain catalogs and data sets.

These are the enhancements that we discuss in this chapter:

- ▶ Large Block Size for LBI Support
- ▶ IDCAMS support for RLS
- ▶ DELETE PDS or PDSE member with mask
- ▶ ALTER NULLIFY management class
- ▶ DIAGNOSE of GDGs

3.1 Large Block Size for LBI Support

The Large BlockSize Interface (LBI) was introduced around 10 years ago. It allows programs and utilities to use block sizes larger than 32760 bytes when recording data to tape. It uses less tape storage and provides fast data transfer.

Before DFSMS V2.1, IDCAMS did not support LBI. Any attempt to run IDCAMS PRINT or REPRO to print or copy a data set that was created with block size larger than 32760 bytes gets an error message IDC3300I, followed by IDC3321I with a return code 12. Also, an abend 013 with return code E1 is externalized in the job SYSOUT.

In DFSMS V2.1, IDCAMS PRINT and REPRO supports data sets with a block size up to the access method limit, which currently is 256K.

The block size is still limited to 32,760 bytes for data sets that do not support LBI. SYSIN and SYSPRINT do not support LBI.

3.1.1 Use of function

Example 3-1 shows a sample REPRO job that we used in a z/OS V2.1 system to REPRO the sequential data set MHLRES3.VSAM.UPDT1 on disk to the new data set MLHRES3.LBI.SC64.TEST on tape. Notice that we used a BLKSIZE of 240000.

We ran the same job in a z/OS V1.13 system. The job also run successfully, but the block size was adjusted to 32700 bytes, as IDCAMS on z/OS V1.13 does not support a LBI. The number of transferred blocks was much bigger than we run this job in the z/OSV 2.1 system.

Example 3-1 REPRO using LBI

```
//MHLRES3A JOB (999,POK),'MHLRES3',CLASS=A,MSGCLASS=T,
// NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM SYSAFF=*
//REPRO EXEC PGM=IDCAMS,REGION=0M
//SYSPRINT DD SYSOUT=*
//DD1 DD DSN=MHLRES3.VSAM.UPDT1,DISP=SHR
//DD2 DD DSN=MHLRES3.LBI.SC64.TEST,DISP=(,CATLG),
// DCB=(RECFM=FB,LRECL=300,BLKSIZE=240000),
// UNIT=VT3590,LABEL=(,SL)
//SYSIN DD *
        REPRO INFILE(DD1) -
              OUTFILE(DD2)
/*
```

3.1.2 Coexistence

If you create data sets on z/OS V2.1 system, and intend to read them on z/OS V1.13 or older systems, we suggest you not specify a block size larger than 32760 bytes for output data sets.

We tried to print the data set MHLRES3.LBI.SC64.TEST that we created in Example 3-1 in a z/OS V1.13 system, using the job shown in Example 3-2 on page 55.

Example 3-2 PRINT on z/OS 1.13 of a LBI data set

```
//MHLRES3P JOB (999,P0K), 'MHLRES3', CLASS=A, MSGCLASS=T,  
// NOTIFY=&SYSUID, TIME=1440, REGION=6M  
/*JOBPARM SYSAFF=*  
//PRINT EXEC PGM=IDCAMS, REGION=0M  
//SYSPRINT DD SYSOUT=*  
//SYSIN DD *  
        PRINT INDATASET(MHLRES3.LBI.SC64.TEST) -  
            COUNT(1)  
/*
```

The job failed with message IEC141I abend 013 rc E1. Return code E1 means that a magnetic tape data set with a block size greater than 32,760 is being opened for input or output extend processing, but the large block interface is not being used. Messages IDC3300I and IDC3321I are also shown in job log as shown in Example 3-3.

Example 3-3 Error messages associated with LBI not being used

```
IDC3300I ERROR OPENING MHLRES3.LBI.SC64.TEST  
IDC3321I ** OPEN/CLOSE/EOV ABEND EXIT TAKEN
```

3.2 IDCAMS support for RLS

In DFSMS V2.1 IDCAMS is enhanced to support commands PRINT, REPRO, EXPORT, and IMPORT to open VSAM data sets that are already open in RLS mode by other applications, for example IBM CICS®. In this section we describe the function in DFSMS V2.1 and provide samples.

3.2.1 Overview

Prior to DFSMS V2.1, IDCAMS can only open VSAM data sets using Non Shared Resources (NSR). Because of this, it had a limited support to access data sets already opened by other applications in RLS mode. You could use only PRINT, REPRO, and EXPORT commands to data sets that were defined with cross region SHAREOPTION 2. For example, you could make a copy of a VSAM data set opened in RLS mode to CICS, if the data set was created with SHAREOPTION(2 3). However, there is not any guarantee that the copy has integrity.

Now in DFSMS V2.1 IDCAMS can use the keywords RLSSOURCE and RLSTARGET to be able to open VSAM data sets using RLS. Here is a brief description of the syntax and the associated keywords:

► **RLSSOURCE**

For PRINT, REPRO, and EXPORT commands it specifies how the input data set is to be opened. The following are RLSSOURCE options:

- | | |
|----------------|---|
| NO | Tells IDCAMS to open the data set using NSR. This is the default. |
| YES | IDCAMS opens the data set in RLS mode, with consistent read integrity; |
| QUIESCE | IDCAMS quiesces a VSAM data set. It switches from RLS mode to non-RLS mode before processing any entry in the data set. |

► **RLSTARGET**

For REPRO and EXPORT commands it specifies how the target data set is to be opened, either NO, YES, or QUIESCE. The description for these options are the same as for RLSSOURCE.

3.2.2 Use

In this section we provide some examples on how to use the RLS enhancements that are provided in IDCAMS in DFSMS V2.1.

In Example 3-4 we show you a sample PRINT job. Here we print a VSAM KSDS data set that is already open in RLS mode for another application. We request IDCAMS to open the VSAM data set in RLS mode by using the RLSSOURCE(YES) parameter on the PRINT command.

Example 3-4 Printing a VSAM data set in RLS mode

```
//MHLRES3P JOB (999,P0K),'MHLRES3',CLASS=A,MSGCLASS=T,
// NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM SYSAFF=*
//PRINT EXEC PGM=IDCAMS,REGION=0M
//DD1 DD DSN=MHLRES3.VSAM.KSDS,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
        PRINT INFILE(DD1) -
              RLSSOURCE(YES) -
              COUNT(1)
/*
```

Example 3-5. shows the JCL to copy the MHLRES3.VSAM.KSDS data set contents to MHLRES3.VSAM.NEW.KSDS. We requested that IDCAMS opens MHLRES3.VSAM.KSDS in RLS mode by specifying RLSSOURCE(YES). We also requested our target data set, MHLRES3.VSAM.NEW.KSDS, to be open in RLS mode by specifying RLSTARGET(YES).

Example 3-5 REPRO of two VSAM KSDS data sets in RLS mode

```
//MHLRES3R JOB (999,P0K),'MHLRES3',CLASS=A,MSGCLASS=T,
// NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM SYSAFF=*
//REPRO EXEC PGM=IDCAMS,REGION=0M
//DD1 DD DSN=MHLRES3.VSAM.KSDS,DISP=SHR
//DD2 DD DSN=MHLRES3.VSAM.NEW.KSDS,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
        REPRO INFILE(DD1) -
              OUTFILE(DD2) -
              RLSSOURCE(YES) -
              RLSTARGET(YES)
/*
```

3.2.3 Coexistence

We do not find any coexistence issues for using these IDCAMS RLS enhancements in DFSMS V2.1 while having VSAM data sets opened in RLS mode by z/OS V1.13 or V1.12 systems.

However, trying to use the new keywords RLSSOURCE and RLSTARGET in a z/OS V1.12 or z/OS V1.13 results in a *IDC3211I KEYWORD IS IMPROPER* message, and the command is not executed.

3.3 DELETE PDS or PDSE member with mask

IDCAMS in z/OS 1.12 provided the support to delete all members of a partitioned data set (PDS or PDSE) in a single operation.

IDCAMS in DFSMS V2.1 enhances the DELETE command to be more flexible in performing the delete of members of a partitioned data set. It now allows a mask to be specified in a DELETE command for PDS and PDSE members.

3.3.1 Use

There are some rules for specifying a mask in deleting PDS or PDSE members in a IDCAMS DELETE command:

- ▶ A mask for a member name can contain an asterisk (*) or a percent sign (%).
 - An asterisk means match 0 or more characters in the member name.
 - A % means match one character.
- ▶ A single asterisk (*) tells IDCAMS to delete all members in a PDS/PDSE data set.
Double asterisks (**) also tells IDCAMS to delete all members in a PDS/PDSE data set. In addition, IDCAMS lists all deleted member names.

Here we provide some examples on how to use a mask for deleting selected members of a partitioned data set. In these examples, we assume we have to have a partitioned data set with the following members, before each DELETE command:

- ▶ ABC
- ▶ ABCDEFGC
- ▶ AC
- ▶ A1C
- ▶ DEF

DELETE example 1

In Example 3-6 we are using an IDCAMS DELETE command to delete only MEMBERS ABC and A1C from partitioned data set MHLRES3.PDSE.TEST.

Example 3-6 Using IDCAMS to delete members of a PDSE

```
//MHLRES33 JOB (999,POK),'MHLRES3',CLASS=A,MSGCLASS=T,  
// NOTIFY=&SYSUID,TIME=1440,REGION=6M  
//REPRO EXEC PGM=IDCAMS,REGION=0M  
//SYSPRINT DD SYSOUT=*  
//SYSIN DD *  
DELETE MHLRES3.PDSE.TEST(A%C)  
/*
```

The resulting job log output in Example 3-7 shows that ABC and A1C are deleted after running this job.

Example 3-7 Job log results of DELETE

```
DELETE MHLRES3.PDSE.TEST(A%C)
IDC0549I MEMBER ABC DELETED
IDC0549I MEMBER A1C DELETED
```

Members ABCDEFGC, AC and DEF are not deleted after running the job in Example 3-6.

DELETE example 2

You can use the following IDCAMS DELETE command to select all members with name starting with A and ending with C to be deleted:

```
DELETE MHLRES3.PDSE.TEST(A*C)
```

The resulting job log output would show the messages in Example 3-8 showing the deleted member names.

Example 3-8 Job log of DELETE command with mask

```
DELETE MHLRES3.PDSE.TEST(A*C)
IDC0549I MEMBER ABC DELETED
IDC0549I MEMBER ABCDEFGC DELETED
IDC0549I MEMBER AC DELETED
IDC0549I MEMBER A1C DELETED
```

3.3.2 Coexistence

You are able to delete members of partitioned data sets using the IDCAMS DELETE with a mask provided in DFSMS V2.1, even if they are shared between z/OS systems.

However, you are not able to run these DELETE with a mask commands in a z/OS system prior to V2.1. For example, if you try to run the same job of Example 3-6 on page 57 in a z/OS V1.13 system you get the message in Example 3-9 on page 58 and the job fails.

Example 3-9 Error message attempting to run DELET with mask on a system prior to DFSMS V2.1

```
IDC3203I ITEM 'MHLRES3.PDSE.TEST(A%' DOES NOT ADHERE TO RESTRICTIONS
```

3.4 ALTER NULLIFY management class

IDCAMS in DFSMS V2.1 allows you to use the ALTER command to remove the management class information of a SMS managed data set from the catalog in which it is cataloged.

3.4.1 Overview

For SMS managed data sets, the management class information resides in a NVR (for non-VSAM data sets) or VVR (for VSAM data sets) in each VVDS on the volumes where the extents for the data set are allocated. You can specify NULLIFY(MANAGEMENTCLASS) in the ALTER command in order to nullify the management class of a SMS managed data set. Its abbreviation is NULLIFY(MGMTCLAS).

3.4.2 Use

In Example 3-10 we show you the SMSDATA information that we take from a LISTCAT ALL of data set MHLRES3.CLIST. It shows that this data set has SMS management class MCDB22 assigned to it.

Example 3-10 LISTCAT prior to ALTER

SMSDATA	
STORAGECLASS ---STANDARD	MANAGEMENTCLASS---MCDB22
DATACLASS -----(NULL)	LBACKUP ---0000.000.0000

We decided to remove this management class information, by running an IDCAMS ALTER NULLIFY(MGMTCLAS) batch job, as shown in Example 3-11.

Example 3-11 ALTER NULLIFY(MGMTCLAS) example

```
//MHLRES3A JOB (999,POK), 'MHLRES3', CLASS=A, MSGCLASS=T,
// NOTIFY=&SYSUID, TIME=1440, REGION=6M
//ALTER EXEC PGM=IDCAMS, REGION=512K
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
ALTER MHLRES3.CLIST NULLIFY(MGMTCLAS)
/*
```

We took another LISTCAT ALL of this data set in order to confirm that the management class information was removed from the catalog, as seen in Example 3-12.

Example 3-12 LISTCAT after the ALTER NULLIFY(MGMTCLAS)

SMSDATA	
STORAGECLASS ---STANDARD	MANAGEMENTCLASS---(NULL)
DATACLASS -----(NULL)	LBACKUP ---0000.000.0000

3.4.3 Coexistence

You are to use the ALTER NULLIFY(MANAGEMENT CLASS) command on any system with DFSMS V2.1, even if they are shared with z/OS V1.13 or previous releases of z/OS.

However, if you try to use this command in a system with z/OS V1.13 or previous releases of z/OS you get the message IDC3211I KEYWORD 'MGMTCLAS' IS IMPROPER and the ALTER is not completed.

3.5 DIAGNOSE of GDGs

The IDCAMS DIAGNOSE command scans a basic catalog structure (BCS) or a VSAM volume data set (VVDS) components of an ICF catalog, to validate data structures and detect structure errors. You can use DIAGNOSE to check the structure and the content of a BCS or a VVDS, to cross check BCS and VVDS information, to detect missing records, and several types of mismatches and structural problems in a catalog.

3.5.1 Overview

DIAGNOSE has been enhanced in DFSMS V2.1 to detect a mismatch between the actual number of GDG extension cells and a count it keeps inside the record that describes a base GDG.

Extension records are created when the maximum record size of the BCS cannot contain a new component entry. This can occur as a result of the creation of a new generation data set. The first piece of information in a extension record is the extension cell. Catalog has to keep a counter inside the record that describes the base GDG; this counter holds the number of extension cells that exist for a GDG.

There may be error situations where the counter and the real number of extension cells do not match. This can lead to some processing errors that are undetectable until batch processing fails.

DIAGNOSE in DFSMS V2.1 helps users to quickly identify the cause of GDG processing errors.

3.5.2 Use

Example 3-13 on page 60 shows a sample job for a DIAGNOSE command. In this example, we check the BCS component of catalog MHLRES.TEST.UCAT for structural errors, like the mismatch between the number of extension cells, and the counter of extension cells.

Example 3-13 DIAGNOSE BCS

```
//MHLRES3D JOB 'JOE',NOTIFY=MHLRES3,
//          MSGCLASS=T
//*-----*
//DIAG      EXEC PGM=IDCAMS
//DD1       DD DSN=MHLRES3.TEST.UCAT,DISP=SHR
//SYSPRINT  DD SYSOUT=*
//SYSIN     DD *
           DIAGNOSE ICFCATALOG -
           INFILE(DD1)
```

If DIAGNOSE detects a mismatch in the number of GDG extension cells, it issues the following message in the job log:

```
IDC31379I GDG BASE EXT CELL COUNT DOES NOT MATCH THE EXT CELLS COUNT
```

If the count of number of extension cells is 0, but there are extension cells, DIAGNOSE presents the following message in the job log:

```
IDC31379I GDG BASE EXT CELL COUNT DOES NOT MATCH THE EXT CELLS COUNTV
```

Refer to manual *z/OS DFSMS Managing Catalogs Version 2 Release 1*, SC23-6853 for more information on using the DIAGNOSE command.

3.5.3 Coexistence

There are no coexistence issues between IDCAMS DIAGNOSE command on DFSMS V2.1 and z/OS 1.13 or earlier releases.

If you try to run the same DIAGNOSE command as shown in Example 3-13 in a system with z/OS 1.13 or older releases of z/OS, it does not detect this type of mismatch between the number of GDG extension cells and the counter of GDG extension cells.



Virtual Storage Access Method enhancements

This chapter provides an overview of the features and functions that apply to Virtual Storage Access Method (VSAM), introduced in DFSMS V2.1.

We also provide an overview of the IBM System z® High-Performance FICON® (zHPF) support for EXCP, and the support for sequential extended format (SAM EF) Version 2 data sets.

In this chapter we describe the enhancements that are available with DFSMS V2.1.

Several of the enhancements apply to VSAM RLS:

- ▶ VSAM RLS directory only caching
- ▶ VSAM RLS Buffer Management Facility 64 bit enhancement
- ▶ VSAM RLS Dynamic Volume Count.

There are enhancements that apply to VSAM access not in RLS mode:

- ▶ SHOWCB enhancements
- ▶ System managed buffering enhancements.

4.1 VSAM RLS enhancements

In this section we look at the enhancements in DFSMS V2.1 that apply to VSAM RLS.

4.1.1 VSAM RLS directory only caching

VSAM RLS uses one or more cache structures allocated in Coupling Facility images in the sysplex as a buffer shared between all systems in a sysplex. This buffer has VSAM data that has been accessed by one or more partitions in the sysplex.

RLS CF Cache value prior to DFSMS V2.1

You can control the amount of data to be cached by each VSAM RLS data set by assigning them different SMS data classes with different values for the RLS CF Cache Value. These are the valid values for DFSMS releases prior to V2.1:

ALL	VSAM data and index components will be cached in the CF. This is the default value.
NONE	Only the index will be cached in the CF.
UPDATESONLY	Only write requests will be cached in the CF.

Some VSAM RLS customers need a way to continue to use the CF cache structures as a sysplex data sharing mechanism but without writing any actual data to the cache structure.

RLS CF Cache value in DFSMS V2.1

In DFSMS V2.1, there is a new DIRONLY value for the RLS CF Cache specification in SMS data class, that tells RLS to not cache any data or index CI in RLS cache structures.

DIRONLY indicates that RLS will not cache the data or index parts of the VSAM data set in the coupling facility cache structures. In this case, RLS uses the cache structures to keep track of data that resides in DASD and in local buffer, but data or index CIs are not stored in the cache structure itself.

Installations that benefit from directory only caching:

- ▶ Installations with limited Coupling Facility storage but still need to share VSAM data sets across a parallel sysplex. They are able to define small cache structures and exploit them only to maintain data consistency.
- ▶ Installations with single system sysplex configurations also benefit from this feature. These installations do not need to have their data sets stored in the cache structure because their local cache buffers contains valid data at all times, since there are no other systems that update the VSAM data sets in RLS mode.
- ▶ Depending on their applications design and workload, some RLS users, including single system sysplex users, can also experience performance improvements when using directory only cache. This is because RLS skips writing the data to XCF cache structure every time the data is updated.
- ▶ Installations that want to prevent applications that utilize VSAM RLS to fill the cache structures with data that does not need to be shared among z/OS images in the sysplex.

Use and invocation

You need to define a new SMS data class, or Alter an existing one, to specify RLS CF Cache Value of D as shown in Figure 4-1 on page 65. You must also modify your ACS routines to assign new VSAM data sets to this data class accordingly.

```

Panel  Utilities  Scroll  Help
-----
                                DATA CLASS DEFINE                                Page 6 of 6

Command ==>

SCDS Name . . . : SYS1.SMS.V2R1.SCDS
Data Class Name : DCRLSNC

To DEFINE Data Class, Specify:
  Shareoptions Xregion . . .      (1 to 4 or blank)
                        Xsystem . . . (3, 4 or blank)
  Reuse . . . . . N                (Y or N)
  Initial Load . . . . . R        (S, R or blank)
  BWO . . . . .                   (TC, TI, NO or blank)
  Log . . . . .                   (N, U, A or blank)
  Logstream Id . . . . .
  FRlog . . . . .                 (A, N, R, U or blank)
  RLS CF Cache Value . . . . D    (A, N, U or D)
  RLS Above the 2-GB Bar . . N     (Y or N)
  Extent Constraint Removal . . N  (Y or N)
  CA Reclaim . . . . . Y          (Y or N)
  Log Replicate . . . . . N       (Y or N)
  Use ENTER to perform Verification; Use UP Command to View previous Panel;
  Use HELP Command for Help; Use END Command to Save and Exit; CANCEL to Exit

```

Figure 4-1 DIRONLY Define and Alter data class

Migration and coexistence consideration

Systems prior to z/OS V2.1 are not able to open VSAM RLS data sets that are created in z/OS V2.1 with an RLS CF Cache Value of DIRONLY.

We tried to run job MHRES3U in a z/OS V1.13 partition in our sysplex. This job runs a program that does updates to a VSAM RLS data set named MHLRES3.DIRONLY.KSDS. This data set was defined in a z/OS V2.1 partition, with a Data Class that has DIRONLY defined as RLS CF Cache Value. We received the error message in Example 4-1 when our program tried to open the data set.

Example 4-1 Error message on a pre-DFSMS V2.1 system

```

IEC161I 132-0814,MHLRES3U,UPDDIR,VSAM,,,MHLRES3.DIRONLY.KSDS
  where ccc = 814 - RLS Directory Only Cache

```

4.1.2 VSAM RLS Buffer Management Facility 64 bit enhancements

VSAM RLS customers have been able to take advantage of 64 bit storage for RLS buffering in a system in the sysplex. You can specify the amount of storage above the 2 GB bar that a system is going to use as local buffer by specifying RlsAboveTheBarMaxPoolSize in the IGDSMSxx parmlib member as shown in Example 4-2.

Example 4-2 Sample IGDSMSxx parmlib RlsAboveTheBarMaxPoolSize

```

[RlsAboveTheBarMaxPoolSize({ sysname, maxrls; ...})|(ALL,maxrls)]

```

You can specify a different above the bar amount for each system in the sysplex, or specify the same value for all the systems in the sysplex.

You also have to tell VSAM RLS that a data set is eligible for buffers allocated in the 64-bit storage by assigning them a SMS data class construct with RLS Above the 2 GB bar specified as shown in Example 4-3.

Example 4-3 RLS above the 2GB bar parameter

RLS Above the 2-GB Bar . . . : YES

As many users have increased the use of 64 bits buffers, the number of control blocks needed to support these additional buffers have increased also. The problem is that these control blocks resided in the limited storage space of SMSVSAM data space, competing for space with VSAM data allocated in the 31-bit buffers.

In DFSMS V2.1, some RLS related control blocks were moved from the VSAM data space into 64-bit storage. This brings RLS users two benefits:

- ▶ Relieve some space for the local 31-bit buffers.
- ▶ May improve performance when processing a large amount of VSAM RLS data, because SMSVSAM can reduce the premature flushing of the 31-bit buffers due to aging algorithms.

4.1.3 VSAM RLS Dynamic Volume Count

Dynamic Volume Count (DVC) provides the capability to dynamically add volumes to an SMS managed data set, for both VSAM and non-VSAM formats, when a data set extends.

During define processing, Dynamic Volume Count allows for a larger number of volumes to be considered without increasing the number of candidate volumes stored in the catalog. During existing data set allocation, it provides a way to increase the number of TIOT/JFCB entries that are created, so that more volumes can be dynamically allocated as required. This occurs as part of EOVS processing to extend to a new volume, during the lifetime of the allocation.

During extend to a new volume, SMS will check whether the data set has a candidate volume entry in the catalog. If there is no candidate volume entry for the data set, and the number of volumes for the data set is less than the Dynamic Volume Count value, SMS will add a candidate volume entry using the ALTER ADDVOLUME interface to the catalog for the selected volume. Thus the user application does not need to close the data set and perform ALTER ADDVOLUME to increase the volume count.

DFSMS V2.1 allows you now to specify to specify and effectively use DVC for VSAM data sets in RLS mode. This prevents space related abends when VSAM data sets need to grow while they are in RLS mode.

Usage

In order for your VSAM RLS data sets to take advantage of DVC, you must assign them a SMS data class with the following attributes:

- ▶ Space Constraint Relief = Y
- ▶ Dynamic Volume Count = x, where x represents a value from 1 to 59.

Figure 4-2 on page 67 shows the new attributes in the ISMF Data Class Alter panel.

DATA CLASS ALTER		Page 2 of 6
Command ==>		
SCDS Name . . . : SYS1.SMS.V2R1.SCDS Data Class Name : DCRLSNC		
To ALTER Data Class, Specify:		
Data Set Name Type	(EXT, HFS, LIB, PDS, Large or blank)	
If Ext	(P, R or blank)	
Extended Addressability . . N	(Y or N)	
Record Access Bias	(S, U, DO, DW, SO, SW or blank)	
RMODE31	(ALL, BUFF, CB, NONE or blank)	
Space Constraint Relief . . . Y	(Y or N)	
Reduce Space Up To (%) . . 0	(0 to 99 or blank)	
Dynamic Volume Count . . . 20	(1 to 59 or blank)	
System Managed Buffering . .	(1K to 2048M or blank)	

Figure 4-2 Dynamic Volume Count specification

Migration and coexistence

Down level systems in the sysplex are not be able to extend the data set without this enhancement. However, once the data set has been extended by Dynamic Volume Count on systems with DFSMS V2.1 or higher, the down level system will be able to recognize and utilize the new volume.

4.2 VSAM non-RLS enhancements

In this section we look at the enhancements in DFSMS V2.1 that apply to VSAM non-RLS access.

4.2.1 VSAM SHOWCB buffer option

You can use the SHOWCB macro to cause VSAM to move the contents of various fields in an access method control block (ACB), an exit list, or a request parameter list to a work area inside your program. For example, you may use this area to identify an error and print a message, or to keep statistic information about the VSAM data set, like the number of buffers being used for the data and index I/Os.

The SHOWCB macro is enhanced in DFSMS V2.1 to include two new fields that you can use to verify buffer utilization for a VSAM data set:

- | | |
|--------|---|
| BUFNOL | Number of I/O buffers allocated for the data component or index component during BLDVRP or system managed buffering (SMB) for LSR processing. |
| BUFUSE | Allows you to query the number of buffers in the LSR or NSR buffer pool that are currently being used. |

This information can be helpful for deciding about changing the size of a LSR buffer pool.

Both fields returns a value of 0, if you open your VSAM data set in RLS mode.

Use

Example 4-4 is an example of how you can code a SHOWCB macro on your application program in order to get the values of BUFNOL and BUFUSE.

Example 4-4 SHOWCB example

SHOW	SHOWCB	ACB=CONTROL,	x
		AREA=DISPLAY,	x
		FIELDS=(BUFNOL,	x
		BUFUSE),	x
		LENGTH=8	
DISPLAY	DS	OF	
BUFNOL	DS	F	
BUFUSE	DS	F	
RKP	DS	F	

The SHOWCB parameters that we used in this example are:

ACB	Specifies the address of the ACB (CONTROL in our example) for the data set that we need to see the buffer values.
AREA	Specifies the area where SHOWCB should store the FIELDS that we are asking for. We can externalize the information in AREA by using it in a WTO instruction, or recording it in a separate data set, for further analysis.
FIELDS	Specifies what type of information we want to extract. In our case, the BUFNOL and BUFUSE values.
LENGTH	Specifies 8 bytes, as both BUFNOL and BUFUSE are 4 bytes in size.

For more information about the SHOWCB macro, refer to the manual *z/OS DFSMS Macro Instructions for Data Sets Version 2 Release 1*, SC23-6852.

4.2.2 VSAM SMB specification in SMS data class

VSAM supports the use of system managed buffering (SMB) for VSAM data sets for batch processing. Prior to DFSMS V2.1, you can provide SMB access bias (ACCBIAS) and RMODE31 specifications in JCL, but not at a system level. If you need to change any of these options for a group of VSAM data sets, potentially you need to scan hundreds of JCL DD AMP in order to make changes to them.

Now in DFSMS V2.1 you have the ability to specify ACCBIAS and RMODE31 values in SMS data classes, that you can assign to your VSAM data sets as you need.

Use

You can let VSAM manage buffering by using the following specifications in a SMS data class, (refer to Figure 4-3 on page 70):

- Data Set Name Type EXT

The VSAM data set must be defined with the extend format attribute in order to be eligible for SMB.

- Record Access Bias

Specifies whether to let VSAM determine how many and which type of buffers to use when accessing VSAM extended format data sets by batch processing. This is known as system-managed buffering, and is available to VSAM data sets in any record organization which are allocated in the extended format. The values that you can use for Record Access Bias are the same that you can specify in JCL AMP statements:

- S (System)

Specifies VSAM to use SMB, determining the buffer algorithms based on the ACB MACRF macro and storage class specification.

- U (User)

Tells VSAM to obtain buffers the same way as if without SMB. This is the default value.

- DO

Uses SMB with direct access optimization.

- DW

SMB weighted for direct processing.

- SO

SMB with sequential optimization.

- SW

SMB weighted for sequential processing.

- RMODE31

Specifies whether or not for VSAM to allocate the buffers and control blocks in 31-bit addressable storage. You can use this field independently of SMB. With SMB the default location is in 31-bit addressable storage (above the 16-megabyte line). Without SMB, the default is in 24-bit addressable storage (below the line).

The values you may specify for RMODE31 in data class are:

- ALL

All buffers and control blocks reside above the line.

- BUFFER

Only buffers reside above the line.

- CB

Only control blocks reside above the line.

- NONE

Buffers and control blocks reside below the line.

Using a data class definition, you can modify these Record Access Bias and RMODE 31 values without having to edit individual JCL statements, and the modification is no longer limited to one single job step. See Figure 4-3 on page 70.

	DATA CLASS ALTER	Page 2 of 6
Command ==>		
SCDS Name . . . : SYS1.SMS.V2R1.SCDS		
Data Class Name : DCRLSNC		
To ALTER Data Class, Specify:		
Data Set Name Type	EXT	(EXT, HFS, LIB, PDS, Large or blank)
If Ext	R	(P, R or blank)
Extended Addressability . .	N	(Y or N)
Record Access Bias	S	(S, U, DO, DW, SO, SW or blank)
RMODE31	ALL	(ALL, BUFF, CB, NONE or blank)
Space Constraint Relief . . .	Y	(Y or N)
Reduce Space Up To (%) . .	0	(0 to 99 or blank)
Dynamic Volume Count . . .	20	(1 to 59 or blank)
System Managed Buffering . .		(1K to 2048M or blank)

Figure 4-3 Record Access Bias and RMODE31 specifications in Data Class

Migration and coexistence

The AMP parameter specifications in JCL override Record Access Bias and RMODE31 specifications in SMS data class. You can code data classes and ACS routines according to your needs, and modify or leave the JCL specifications as they are now if you happy with them.

4.2.3 Specifying eligibility for VSAM replication

DFSMS V2.1 provides a method to identify VSAM data sets eligible for replication. You can use a new data set attribute, LOGREPLICATE, to identify each VSAM data set that is eligible for replication. You can specify it in IDCAMS DEFINE or ALTER command, or in a SMS data class construct that you assign to the data set.

The VSAM replication in itself is performed by IBM InfoSphere® Classic Replication Server for VSAM, that is the VSAM feature for the IBM InfoSphere Classic Replication Server for z/OS product suite. Refer to Figure 4-4 on page 71. In that figure, we have a site where the Production workload is running and a Standby site. The standby does not run any production workload normally, but it has standby z/OS instances and subsystems already up and waiting for work. These sites can be thousands of miles apart from each other.

CICS on the production site performs reads, writes, deletes, and updates records in VSAM data sets residing on the same site. Those operations that modify the contents of the VSAM data sets are recorded in one or more logstreams in the production site.

The capture engine of IBM InfoSphere Classic Replication Server for VSAM captures those updates made to the logstreams, and send them, through a TCP/IP connection, to the standby site.

At the standby site, the apply engine receives the updates sent from the production site, and applies them to VSAM data sets residing at the standby site.

In case we have a disaster on our production site, the production workload can be routed to the standby site.

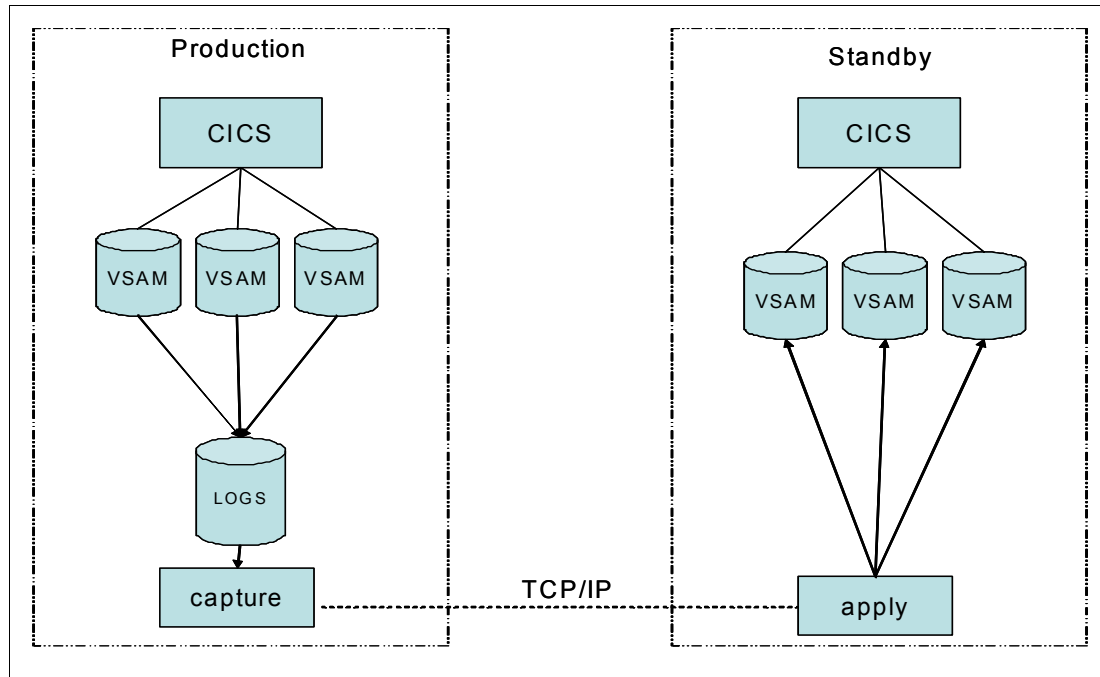


Figure 4-4 VSAM replication as performed by IBM InfoSphere Classic Replication Server for VSAM

This model environment is called Active-Standby. The implementation of this Active-Standby model allows production workload to run in only one site.

Use

You can specify the LOGREPLICATE attribute by specifying LOGREPLICATE in an IDCAMS DEFINE or ALTER command. The default of IDCAMS DEFINE command is NOLOGREPLICATE. Example 4-5 shows a sample job for a DEFINE of a VSAM KSDS data set with LOGREPLICATE attribute. This attribute tells VSAM that this data set is eligible for VSAM replication, and that updates made to this data set will be reflected in the replication logs pointed out by the LOGSTREAMID parameter.

Example 4-5 DEFINE of a VSAM KSDS data set with LOGREPLICATE attribute

```

//MHLRES3D JOB 'JOE',NOTIFY=MHLRES3,
//          MSGCLASS=T
//DEF      EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN    DD *
  DEFINE CLUSTER(NAME(MHLRES3.VSAMREP.KSDS) -
    KEYS(8,0) RECORDSIZE(300 300) SPEED -
    FREESPACE(15 15) -
    SHAREOPTIONS(3 3) -
    DATACLASS(DCEXTSEQ) -
    STORAGECLASS(SCRLS) -
    RECORDS(1000000 100000) -
    LOGREPLICATE -
    LOGSTREAMID(LOGA) -
    VOLUMES(VOL1 VOL2)) -
    DATA(CONTROLINTERVALSIZE(4096))
  
```

/*

As an alternative, you can also assign a SMS data class with Log Replicate Y. Figure 4-5 on page 72 shows an example of a data class construct that we called DCVSAMRE, with both Log Replicate Y and Logstream Id specified.

DATA CLASS DEFINE	Page 6 of 6
Command ==>	
SCDS Name . . . : SYS1.SMS.MHLRES3.SCDs	
Data Class Name : DCVSAMRE	
To DEFINE Data Class, Specify:	
Shareoptions Xregion . . . 3	(1 to 4 or blank)
Xsystem . . . 3	(3, 4 or blank)
Reuse N	(Y or N)
Initial Load R	(S, R or blank)
BWO	(TC, TI, NO or blank)
Log	(N, U, A or blank)
Logstream Id LOGA	
FRlog	(A, N, R, U or blank)
RLS CF Cache Value A	(A, N, U or D)
RLS Above the 2-GB Bar . . N	(Y or N)
Extent Constraint Removal N	(Y or N)
CA Reclaim Y	(Y or N)
Log Replicate Y	(Y or N)
Use ENTER to perform Verification; Use UP Command to View previous Panel;	
Use HELP Command for Help; Use END Command to Save and Exit; CANCEL to Exit	

Figure 4-5 Log Replicate Y

You can modify the LOGREPLICATE of an existing VSAM data set by using IDCAMS ALTER. Example 4-6 shows a sample job to assign the attribute LOGREPLICATE to the existing MHLRES3.VSAMR1.KSDS data set.

Example 4-6 ALTER LOGREPLICATE attribute

```
//MHLRES3A JOB (999,P0K),'MHLRES3',CLASS=A,MSGCLASS=T,
// NOTIFY=&SYSUID,TIME=1440,REGION=6M
//ALTER EXEC PGM=IDCAMS,REGION=512K
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
ALTER MHLRES3.VSAMR1.KSDS -
LOGREPLICATE -
LOGSTREAMID(LOGA) /*
```

If you assign a LOGREPLICATE attribute to a VSAM data set, you must also assign the a LOGSTREAMID to it, where updates to the VSAM data set are going to be logged.

Figure 4-6 shows the RLSDATA portion of a LISTCAT ENTRIES ALL IDCAMS command where you see the LOGREPLICATE attribute, and the LOGSTREAMID also.

```

RLSDATA
LOG -----(NULL)    RECOVERY REQUIRED --(NO)    FRLOG -----(NULL)
VSAM QUIESCED -----(NO)    RLS IN USE -----(NO)    LOGREPLICATE----- (YES)
LOGSTREAMID-----LOGA
RECOVERY TIMESTAMP LOCAL-----X'0000000000000000'
RECOVERY TIMESTAMP GMT-----X'0000000000000000

```

Figure 4-6 LISTCAT ENTRIES showing LOGREPLICATE

Coexistence

Support for the VSAM replication is also routed to z/OS V1.13, through a list of APARs. You must apply the PTFs for the following APARs, in order to be ready for VSAM replication support:

- ▶ OA38198 (Catalog)
- ▶ OA38200 (DFSMSHsm)
- ▶ OA38201 (IDCAMS)
- ▶ OA38202 (RLS)
- ▶ OA38203 (SMS)
- ▶ OA38204 (ISMF)
- ▶ OA38205 (NaviQuest)
- ▶ OA38209 (VSAM)
- ▶ OA38210 (DFSMSdss)
- ▶ OA38211 (RLS)
- ▶ OA38550 (FAMS).

4.3 DFSMS support for zHPF

IBM System z High Performance FICON (zHPF) is an enhancement to the Fibre Channel connection (FICON) protocol. It reduces the number of information units exchanged between a channel and the controller during an I/O operation. Sending small blocks of data over FICON involves additional handshaking between the channel engine and the FICON adapter in the control unit. zHPF has reduced the processor usage of this process. zHPF allows the collapsing of command chains and data-chained channel command word (CCW) strings into a single command called the transport control word (TCW). This configuration provides a substantial performance improvement in data transfer, especially in online environments.

z/OS V1.11 provided the initial support for zHPF was for data sets accessed using the media manager component of DFSMS, for data sets accesses using the media manager component of DFSMS, including VSAM data sets.

z/OS V1.13 added support for QSAM, BSAM, and BPAM and allowed EXCPVR callers to use zHPF channel programs.

DFSMS V2.1 extends zHPF support to EXCP callers. We expect to achieve significant I/O performance improvements for programs using EXCP. This support is also provided for z/OS V1.12 and z/OS V1.13 through the APAR OA38185.

4.3.1 zHPF requirements

These are the prerequisites for implementing zHPF in a z/OS system:

- ▶ Systems zEC12, z114, z196, IBM z10™, or newer processors.

- ▶ FICON Express8S cards on the host provide the most benefit, but older cards are also supported. The old FICON Express adapters are not supported.
- ▶ In order to have the EXCP support for zHPF your system must at least at z/OS V1.12 with PTF for APAR OA38185 applied. In DFSMS V2.1 support is on the base code.
- ▶ DS8700, DS8800, DS8870, or newer disk subsystems. These disk subsystems must have the zHPF Licensed Feature Key activated.

zHPF is transparent to applications. However, z/OS configuration changes are required. Hardware Configuration Definition (HCD) must have Channel path ID (CHPID) type FC defined for all the CHPIDs that are defined to the disk control units, which also supports zHPF.

For z/OS, you must set ZHPF=YES in IECIOSxx in SYS1.PARMLIB or issue the SETIOS ZHPF=YES command. ZHPF=NO is the default setting. We suggest you use the ZHPF=YES setting after the required configuration changes and prerequisites are implemented.

4.4 Sequential Extended Format Version 2

In previous z/OS releases, prior to z/OS V2.1, DFSMSdss is not able to use ESS or IBM DS8000® FlashCopy function in order to copy or move a single striped extend format sequential data set. This happens because each block has a suffix a block number, that delimits the boundaries of the data set in a disk volume. FlashCopy can not adjust those boundaries.

If you try to copy a sequential extended format (SAM EF) data set using DFSMSdss COPY function with the CONCURRENT(REQUIRED), the COPY succeeds, but without invoking FlashCopy function. You will see a message like Example 4-7 in the COPY job log:

Example 4-7 Error message FlashCopy is not invoked

```
ADR918I (001)-AMOVE(08), FAST REPLICATION COULD NOT BE USED FOR DATA SET
MHLRES3.SAM.TEST, RETURN CODE 12
```

Similarly, DFSMSdss is not able to call FlashCopy if the source data set is a single volume single striped SAM EF data set, but the target requires a multivolume allocation.

DFSMS V2.1 introduces a new format for extended format data sets: Format 2, also referred as Version 2 format in some publications. This new format does not have the volume boundary awareness on sequential extended format data sets.

4.4.1 Use

In this section we describe the steps you need to take in order to take advantage of the new Format 2 enhancement.

There is a new parameter PS_EXT_FORMAT(1|2) in IGDSMSxx PARMLIB member that tells the format in which the system should create extended format data sets:

- ▶ PS_EXT_FORMAT(1) is the default value, that tells the system to create the extended format in the old format.
- ▶ PS_EXT_FORMAT(2) tells the system to create any new extended format data set in the new format. The associated catalog entry indicates whether the data set is version 1 or version 2. The data set version will be displayed via IDCAMS LISTCAT and DCOLLECT.

Figure 4-7 on page 75 is a sample portion of a LISTCAT IDCAMS command issued against a version 2 extended format single stripe data set:

```

-----0
      VOLSER-----*      DEVTYPE-----X'00000000'
FSEQN-----
-----0
      VOLSER-----*      DEVTYPE-----X'00000000'
FSEQN-----
-----0
      VOLSER-----*      DEVTYPE-----X'00000000'
FSEQN-----
-----0
      ASSOCIATIONS----- (NULL)
      ATTRIBUTES
      VERSION-NUMBER-----2
      STRIPE-COUNT-----1
      EXTENDED

***

```

Figure 4-7 LISCAT sample

You can also issue the SETSMS PS_EXT_VERSION(2) command to active Version 2 extended data sets creation from the z/OS console.

The **D SMS,OPTIONS** command will show you what version your system is creating new extended format data sets.

Even if your system is set up to only create old version 1 extended format data sets by default, you can create Format 2 extended format data sets by explicitly coding DSNTYPE=(EXTREQ,2). An example is shown in Example 4-8. In this example, we create a new sequential data set, requesting it to be in Version 2 extend format, by specifying DSNTYPE=(EXTREQ,2) in the DD card.

For dynamic allocations, you can request the data set to be created in Version 2 format by specifying DALDSNV in the DYNALLOC macro. For more information about the DYNALLOC macro, refer to the publication *z/OS MVS Programming: Authorized Assembler Services Guide Version 2 Release 1*, SA23-1371.

Example 4-8 Requesting Version 2 format

```

//MHLRES3I JOB 'JOE',NOTIFY=MHLRES3,
//          COND=( (8,EQ),(12,EQ),(16,EQ),(888,EQ)),
//          MSGCLASS=T,REGION=32M
//*-----
//IEF      EXEC PGM=IEFBR14
//I        DD DSN=MHLRES3.SAM.TEST,DISP=(,CATLG),
//          SPACE=(TRK,(30,3),RLSE),UNIT=3390,
//          DCB=(RECFM=FB,LRECL=300,BLKSIZE=27900),
//          DATACLAS=DCEXTSEQ,STORCLAS=SCSDR0,
//          DSNTYPE=(EXTREQ,2)

```

We only specified DCEXTSEQ data class in order to get its Volume Count, instead of specifying several candidate volumes in JCL.

When Sustained Data Rate (MB/sec) in a Storage Class construct is 0 or blank, the system creates new extended format data sets with only a single stripe. Refer to Figure 4-8 on page 76.

The Accessibility C (CONTINUOUS) field in the Storage Class Alter panel tells the system to allocate and extend a data set in only one disk subsystem, that is capable of doing point-in-time copies. FlashCopy is the function that allows point-in-time copies in a DS8000 disk subsystem.

STORAGE CLASS ALTER		Page 1 of 2
Command ==>		
SCDS Name : SYS1.SMS.MHLRES3.SCDS		
Storage Class Name : SCSDR0		
To ALTER Storage Class, Specify:		
Description ==> STORCLASS FOR ALLOCATING A SAM DATA SET SINGLE STRIPE FOR		
==> FC TESTING		
Performance Objectives		
Direct Millisecond Response		(1 to 999 or blank)
Direct Bias		(R, W or blank)
Sequential Millisecond Response		(1 to 999 or blank)
Sequential Bias		(R, W or blank)
Initial Access Response Seconds		(0 to 9999 or blank)
Sustained Data Rate (MB/sec)	0	(0 to 999 or blank)
OAM Sublevel		(1, 2 or blank)
Availability	N	(C, P ,S or N)
Accessibility	C	(C, P ,S or N)
Backup		(Y, N or Blank)
Versioning		(Y, N or Blank)
Use ENTER to Perform Verification; Use DOWN Command to View next Page;		
Use HELP Command for Help; Use END Command to Save and Exit; CANCEL to Exit.		

Figure 4-8 Storage Class with Accessibility Continuous

After loading MHLRES.SAM.TEST data set with test data, we submitted the job shown in Example 4-9 to test FlashCopy usage.

Example 4-9 DFSMSdss COPY EXAMPLE

```
//MHLRES3C JOB 'JOE',NOTIFY=MHLRES3,
//          MSGCLASS=T,REGION=32M,CLASS=A
//*-----*
//STEP1    EXEC  PGM=ADRDSSU
//SYSPRINT DD   SYSOUT=*
//SYSIN    DD    *
COPY -
DATASET( INCLUDE( MHLRES3.SAM.TEST)) -
RENAMEUNCONDITIONAL((MHLRES3.SAM.TEST,MHLRES3.SAM.FLASH)) -
CONCURRENT(REQUIRED) -
DEBUG(FRMSG(DETAILED) ,SMSMSG)
/*
```

CONCURRENT(REQUIRED) tells DFSMSdss to use virtual concurrent copy, by invoking hardware functions like FlashCopy to perform the copy of the data set.

The DEBUG(FRMSG(DETAILED),SMSMSG) is just a way to verify if your copy is working as you planned. You can remove them if you are not trying to identify errors during COPY processing:

- ▶ FRMSG(DETAILED) give us details about the utilization of FlashCopy or SnapShot. You can use it to determine why some data sets are not being copied through fast replication techniques (FlashCopy or SnapShot).
- ▶ SMSMSG instruct DFSMSdss to display ACS WRITE statements to the job output.

Figure 4-9 shows a sample of messages that you see with DEBUG(FRMSG(DETAILED),SMSMSG) specified. Message ADR806I tell us that our data set MHLRES3.SAM.TEST was copied using a Fast Replication function (FlashCopy in our case).

```

ADR803I (001)-ACS (01), SMS DIAGNOSTIC MESSAGES:
  IGD01008I STORCLAS NOT NULL. SET TO SCSDRO
ADR803I (001)-VDSS (01), SMS DIAGNOSTIC MESSAGES:
  IGD01010I SG ACS GETS CONTROL &ACSENVIR=ALLOC
  IGD01010I &STORCLAS = SCSDRO
  IGD17332I DATA SET MHLRES3.SAM.FLASH WAS
  ALLOCATED ON VOLUME(S) WHICH ARE ELIGIBLE FOR FAST REPLICATION.
  PREFERRED FAST REPLICATION WAS SPECIFIED BY THE CALLER.
ADR809I (001)-VDSS (01), ADDITIONAL DIAGNOSTIC DATA FOR PRECEDING MESSAGE:
  SC=SCSDRO MC=MCDB22 DC=DCEXTSEQ
  REQPRI=0000087950KB REQSEC=0000000003TRK REQVOLS=10
ADR711I (001)-NEWDS(01), DATA SET MHLRES3.SAM.TEST HAS BEEN ALLOCATED WITH NEWNAME
MHLRES3.SAM.FLASH USING STORCLAS SCSDRO,
  DATACLAS DCEXTSEQ, AND MGMTCLAS MCDB22
ADR806I (001)-TOMI (01), DATA SET MHLRES3.SAM.TEST COPIED USING A FAST REPLICATION
FUNCTION

```

Figure 4-9 Fast Replication Function (FlashCopy) was used to copy

4.4.2 Migration and coexistence

We suggest you not set Version 2 extended format data sets in your IGDSMSxx parmlib member, or by using a SETSMS command, until all your sharing and backup systems are at z/OS V2.1 level, or compatibility APARS have been applied to older systems. The following are the compatibility APARS that you should apply on z/OS V1.12 and V1.13:

- ▶ OA39551
- ▶ OA39869
- ▶ OA39871

After applying these APARS to z/OS V1.12 and z/OS V1.13 systems, you are able to read and do any other operation other than create a Version 2 extended format data set.

Issue reported on APAR OA43701

At the time this book is being written, there is an issue with multi-volume single striped SAM Extended Format Version 2 data sets that are allocated using Guaranteed Space storage class attribute. If such a data set is opened for EXTEND or opened for OUTPUT or OUTIN when allocated DISP=MOD, the volume label on the second to last volume may be overwritten by application data.

If you have storage classes with the Guaranteed Space attribute, you should avoid implementing SAM Extended Format Version 2 before this APAR has PTFs ready for z/OS V2.1, V1.13, and also V1.12. You can do this by:

- ▶ Specify DSNTYPE=(EXT,1) or DSNTYPE=(EXT) on the JCL
- ▶ Specify EXT_VERSION(1) in IGDSMSxx parmlib member and do not specify DSNTYPE=(EXT,2) on the JCL.

If you have a volume with its label damaged by an occurrence of this issue, you can recover the volume by using the following procedure:

1. Vary the volume offline to all systems.
2. Run the ICKDSF job step in Figure 4-10 on page 78 to rebuild the volume label.

Example 4-10 ICKDSF job step to rebuild the volume label

```

//***** */
//* This step rebuilds CYL 0 HEAD 0 to point to VTOC */
//* Change the VOLID to the required volser, UNIT to device */
//* number, and VTOCCPTR to the CYLINDER and HEAD where the */
//* VTOC is */
//***** */
//RFMT EXEC PGM=ICKDSF
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
        REFORMAT UNIT(dddd) NVFY VTOCPTR(Cyl,Head) PURGE VOLID(vvvvvv)
/*

```

A successful vary online is a good indication that the volume label has been fixed.



PDSE enhancements

This chapter provides an overview of the features and function in PDSE included in z/OS V2.1. The following enhancements are described in this chapter:

- ▶ PDSE Version 2 scalability and usability
- ▶ PDSE member generation support
- ▶ Guaranteed PDSE data set format allocation
- ▶ PDSE larger member size
- ▶ Generation Data Group support

In addition the IEBCOPY utility has been enhanced to provide improved support for PDS and PDSE data sets. This is documented in “IEBCOPY COPYGROUP” on page 106.

5.1 PDSE Version 2 scalability and usability

z/OS V2.1 through DFSMS V2.1 provides version 2 of PDSE data sets that can provide for improved performance, reduced path lengths, and improved index searches.

New data sets can be allocated in version 2 format by specifying the new version on the DSNTYPE keyword on the JCL DD statement or on the TSO/E ALLOCATE statement, or through an option in the ISPF 3.2 panel.

There is also a new option PDSE_VERSION on the SYS1.PARMLIB member IGDSMSxx that makes all PDSE data sets created according to the PDSE_VERSION setting by default. If PDSE_VERSION is set, specific data sets can be created in the other format if necessary.

PDSE version 1 and version 2 formats are usable without changing access routines but a version 2 PDSE can only be created using DFSMS V2.1 or later.

5.1.1 Compatibility

PDSE Version 2 data sets can be used on DFSMS V1.12 and DFSMS V1.13 provided the compatibility fixes are installed.

PDSE Version 2 data sets cannot be created on systems prior to DFSMS V2.1.

5.2 PDSE member generation support

PDSE members that are deleted or replaced can not be retrieved. The function to retain and allow retrieval of old members is known as the PDSE member generation support.

The member generations support requires APAR OA42358. The text of APAR OA42358 as of the time of writing of this IBM Redbooks publication can be found in Appendix , “APAR OA42846 Single striped, multi-volume sam tailored compressed data set” on page 276.

5.3 Guaranteed PDSE data set format allocation

Prior to DFSMS V2.1, specifying DSNTYPE=LIBRARY, or DSNTYPE=(LIBRARY,x) does not guarantee that a PDSE is going to be created. You would have to specify DSORG=PO, or specify directory blocks in the space in allocation in order for a PDSE to be created. Otherwise, a sequential data set is created instead.

If a data set is allocated using a referback to an existing data set that is not in PDSE format, attributes could be picked up that cause a PDS (or PS) data set to be allocated even though DSNTYPE=LIBRARY was specified.

Now in DFSMS V2.1, SMS provides a new parameter in IGDSMSxx parmlib member that directs SMS to create a PDSE when you specify DSNTYPE=LIBRARY or (LIBRARY,x) in JCL, that is HONOR_DSNTYPE_PDSE. The syntax of the parameter is:

HONOR_DSNTYPE_PDSE(NO)	NO is the default, and processing remains as in releases prior to DFSMSV2.1.
HONOR_DSNTYPE_PDSE(YES)	Applies only when DSNTYPE=LIBRARY or DSNTYPE=(LIBRARY,x) is specified in the creation of

a new data set. z/OS creates a PDSE regardless of DSORG and directory blocks specification.

Alternatively, you can issue the **SETSMS HONOR_DSNTYPE_PDSE(YES)** z/OS console command in order to activate this enhancement.

You can confirm that HONOR_DSNTYPE_PDSE(YES) has been activated by looking at the output of **D SMS,OPTIONS** command. Figure 5-1 shows a portion of the output this command in a message IGD002I, where you can find the HONOR_DSNTYPE_PDSE information.

```
IGD002I 12:57:17 DISPLAY SMS
Rls_MaxCfFeatureLevel = Z
RlsAboveThebarMaxPoolSize = 0 RlsFixedPoolSize = 0
PDSE_MONITOR = (YES,0,0) PDSE1_MONITOR = (YES,0,0)
PDSE_DIRECTORY_STORAGE = 2000M PDSE1_DIRECTORY_STORAGE = 2000M
PDSE_BUFFER_BEYOND_CLOSE = NO PDSE1_BUFFER_BEYOND_CLOSE = NO
GDS_RECLAIM = YES DSSTIMEOUT = 0
BLOCKTOKENSIZE = NOREQUIRE FAST_VOLSEL = ON
USEEAV = YES BREAKPOINTVALUE = 21
OAMPROC = SUPPRESS_DRMSGs = NO
OAMTASK = PDSE_SYSEVENT_DONTSWAP = NO
DB2SSID = SAM_USE_HPF = YES
CA_RECLAIM = DATACLAS
PS_EXT_VERSION = 1
SUPPRESS_SMSMSG = IGD17054I(NO ) IGD17227I(NO ) IGD17395I(NO )
HONOR_DSNTYPE_PDSE = YES PDSE_VERSION = 1
```

Figure 5-1 D SMS,OPTIONS showing HONOR_DSNTYPE_PDSE = YES

5.3.1 Use

With this enhancement, you are able to allocate a PDSE when you specify DSNTYPE=LIBRARY or DSNTYPE=(LIBRARY,x) in JCL, whether or not DSORG or directory blocks is specified in space allocation. This simplifies the way you code your JCL.

Example 5-1 shows sample JCL with DSNTYPE=LIBRARY without DSORG specified. We did not specify directory blocks in our SPACE parameter.

Example 5-1 Sample allocation of a PDSE through IEFBR14

```
//MHLRES3I JOB 'JOE',NOTIFY=MHLRES3,
// MSGCLASS=T,REGION=32M
//IEF EXEC PGM=IEFBR14
//I DD DSN=MHLRES3.PDSE.TEST2,DISP=(,CATLG),
// SPACE=(TRK,(30,3),RLSE),UNIT=3390,VOL=SER=MLD201,
// DCB=(RECFM=FB,LRECL=300,BLKSIZE=27900),
// DATACLAS=DCEXTSEQ,STORCLAS=SCSDRO,
// DSNTYPE=LIBRARY
```

The PDSE created is shown in Figure 5-2.

```

Data Set Information
Command ==>

Data Set Name . . . . : MHLRES3.PDSE.TEST2

General Data
Management class . . : MCDB22
Storage class . . . : SCSDR0
Volume serial . . . : MLD231
Device type . . . . : 3390
Data class . . . . : DCEXTSEQ
Organization . . . : PO
Record format . . . : FB
Record length . . . : 300
Block size . . . . : 27900
1st extent tracks . : 30
Secondary tracks . : 3
Data set name type : LIBRARY
Data set version . : 1

Current Allocation
Allocated tracks . : 30
Allocated extents . : 1

Current Utilization
Used pages . . . . : 5
% Utilized . . . . : 1

Dates
Creation date . . . : 2013/10/01
Referenced date . . : 2013/10/01
Expiration date . . : ***None***
More:      +

```

Figure 5-2 HONOR_DSNTYPE_PDSE data set creation

5.3.2 Coexistence

Verify that you have APAR OA42239 (see Appendix A-4, “OA42239” on page 246) applied to your z/OS V2.1 systems before taking advantage of this enhancement. OA42239 fixes several DFSMS V2.1 problems, but in particular avoids allocating a PDSE if HONOR_DSNTYPE_PDSE is specified but the allocation request did not include DSNTYPE

Systems with z/OS 1.13 or older releases of z/OS do not recognize the HONOR_DSNTYPE_PDSE parameter, and do not show it in the output of a **D SMS,OPTIONS** command.

Running the job in Example 5-1 on page 81 in a system with z/OS V1.13 or older creates a sequential data set, instead of a PDSE.

If you try to issue the **SETSMS HONOR_DSNTYPE_PDSE(YES)** command on a z/OS V1.13 system, you get the message in Figure 5-2 and the command does not take effect.

Example 5-2 Error message on a pre-DFSMS V2.1 system

```

IGD029I ERROR FOR SETSMS COMMAND
ERROR IS INVALID KEYWORD: HONOR_DSNTYPE_PDSE

```

If you specify the HONOR_DSNTYPE_PDSE parameter in a IGDSMSxx parmlib member, and try to activate it through the SET SMS=xx command from a z/OS V1.13 or earlier system, you get messages IGD030I and IGD074D as shown in Figure 5-3.

```

T SMS=T6
IEE252I MEMBER IGDSMST6 FOUND IN SYS1.PARMLIB
IGD030I SMS PARAMETER RECORD IN MEMBER IGDSMST6 HAS A SYNTAX ERROR 047
ERROR IS INVALID KEYWORD: HONOR_DSNTYPE_PDSE
052 IGD074D REPLY WITH SMS VALUE, 'KEYWORD(VALUE)', OR REPLY 'D' TO
DEFAULT, 'C' TO CANCEL, OR 'S' TO SAVE ALL SMS VALUES

```

Figure 5-3 Trying to activate HONOR_DSNTYPE_PDSE in a z/OS V1.13 system

5.4 PDSE larger member size

DFSMS V2.1 increases the PDSE member size to approximately 2,146,435,071 records. This is an increase from the previous limit of 15,728,639 records. This is lower than the limit for PDS data set member size.

The DFSMS V2.1 member size limit is higher than the limit for PDS members.

5.4.1 Compatibility

The larger PDSE member size limit is available on releases prior to DFSMS V2.1 provided that the appropriate compatibility fixes are installed.

5.5 Generation Data Group support

The system catalog support has been enhanced to allow creation of a PDSE format data set within a Generation Data Group (GDG). The PDSE may be Version 1 or Version 2.

In this section we will show several scenarios for creating a PDSE within a GDG.

5.5.1 Scenario 1: Define a Generation Data Group

In this scenario we define a GDG. This is applicable for DFSMS V2.1 as well as prior releases. The GDG may be used for any type of data set that supports allocation in a GDG.

In Figure 5-3 on page 83 shows an example of a job to create a GDG. Even though this scenario is to show examples of the allocation of PDSE data sets in a GDG, the presence of the qualifier PDSE in the name is not significant. This GDG could be used to allocate any data set format if the data set name starts with MHLRES2.GDG.PDSE.

Example 5-3 JCL to allocate a GDG for tests

```

//MHLRES2G JOB      99990000,MHLRES2,CLASS=T,NOTIFY=&SYSUID.
// EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
DEFINE GDG(NAME(MHLRES2.GDG.PDSE) LIMIT(5)      )

```

5.5.2 Scenario 2: Define a PDSE version 1 in a GDG

In this scenario we allocate a PDSE version 1 in a GDG set in a DFSMS V2.1 system.

In Figure 5-4 on page 84 we show the job to allocate a PDSE as a member of GDG MHLRES2.GDG.PDSE. The data set is allocated as a PDSE because of the DSNTYPE statement specifying LIBRARY. Specifically a PDSE version 1 data set is wanted. The default setting on the running system is to allocate version 1 PDSE data set.

Example 5-4 JCL to allocate a PDSE data set as a member of a GDG

```
//MHLRES2G JOB (999,POK),'MHLRES2',CLASS=A,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
//S1      EXEC PGM=IEFBR14
//BIG1    DD DSN=MHLRES2.GDG.PDSE(+1),DISP=(NEW,CATLG),
//          SPACE=(TRK,(02,200)),UNIT=3390,
//          DSORG=PO,LRECL=80,BLKSIZE=27920,DSNTYPE=(LIBRARY,1),
//          RECFM=FB
```

In Example 5-5 on page 84 we show the messages resulting from the allocation of the PDSE as a member of GDG MHLRES2.GDG.PDS. There is no indication in the output that the data set is in version 1 format. The messages do show that the data set was allocated with data set name MHLRES2.GDG.PDSE.G0002V00. The G0002V00 suffix confirms that a member of the GDG was allocated.

Example 5-5 Log output showing allocation messages

```
1 //MHLRES2G JOB (999,POK),'MHLRES2',CLASS=A,          JOB13837
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M          00002000
//          IEFC653I SUBSTITUTION JCL -
(999,POK),'MHLRES2',CLASS=A,NOTIFY=MHLRES2,TIME=1440,REGION=6M
2 /*JOBPARM L=999,SYSAFF=*          00003000
3 //S1      EXEC PGM=IEFBR14          00004000
4 //BIG1    DD DSN=MHLRES2.GDG.PDSE(+1),DISP=(NEW,CATLG), 00005100
//          SPACE=(TRK,(02,200)),UNIT=3390,          00005213
//          DSORG=PO,LRECL=80,BLKSIZE=27920,DSNTYPE=(LIBRARY,1), 00005313
//          RECFM=FB          00005400
STMT NO. MESSAGE
-
4 IGD01010I SG ACS GETS CONTROL &ACSENVIR=ALLOC
4 IGD01010I &STORCLAS = STANDARD
ICH70001I MHLRES2 LAST ACCESS AT 13:43:05 ON FRIDAY, SEPTEMBER 20, 2013
IEF236I ALLOC. FOR MHLRES2G S1
IGD101I SMS ALLOCATED TO DDNAME (BIG1 )
      DSN (MHLRES2.GDG.PDSE.G0002V00 )
      STORCLAS (STANDARD) MGMTCLAS (MCDB22) DATACLAS ( )
      VOL SER NOS= MLD42B
IEF142I MHLRES2G S1 - STEP WAS EXECUTED - COND CODE 0000
IGD107I MHLRES2.GDG.PDSE.G0002V00          ROLLED IN, DDNAME=BIG1
```

The format of the data set can be checked using ISPF option 3.2.

In Example 5-6 on page 85 we show the output from ISPF 3.2. The data set version is shown in field Data set version. The fact that the data set is a PDSE is shown by the Data set name type being LIBRARY.

Example 5-6 ISPF listing of data set MHLRES2.GDG.PDSE.G0002V00

Data Set Information
Command ==>

Data Set Name . . . : MHLRES2.GDG.PDSE.G0002V00

General Data

Management class . . : MCDB22
Storage class . . . : STANDARD
Volume serial . . . : MLD32E
Device type : 3390
Data class : **None**
Organization . . . : PO
Record format . . . : FB
Record length . . . : 80
Block size : 27920
1st extent tracks . : 2
Secondary tracks . : 200
Data set name type : LIBRARY
Data set version . : 1

Current Allocation

Allocated tracks . : 2
Allocated extents . : 1
Maximum dir. blocks : NOLIMIT

Current Utilization

Used pages : 5
% Utilized : 20
Number of members . : 0

Dates

Creation date . . . : 2013/09/20
Referenced date . . : ***None***
Expiration date . . : ***None***

5.5.3 Scenario 3: Define a PDSE version 2 in a GDG

In this scenario we allocate a PDSE version 2 in a GDG set in DFSMS V2.1.

In Example 5-7 on page 85 we show the JCL to allocate a PDSE using version 2 format. The system default for PDSE allocation is version 1. The DSNTYPE=(LIBRARY,2) is required to specify version 2 format.

Example 5-7 JCL to allocate PDSE in version 2 format

```
//MHLRES2G JOB (999,POK),'MHLRES2',CLASS=A,  
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M  
/*JOBPARM L=999,SYSAFF=*  
//S1      EXEC PGM=IEFBR14  
//BIG1    DD DSN=MHLRES2.GDG.PDSE(+1),DISP=(NEW,CATLG),  
//          SPACE=(TRK,(02,200)),UNIT=3390,  
//          DSORG=PO,LRECL=80,BLKSIZE=27920,DSNTYPE=(LIBRARY,2),  
//          RECFM=FB
```

The job messages are essentially the same as in Example 5-5 on page 84.

In Example 5-8 on page 86 we show that as a result of specifying DSNTYPE=(LIBRARY,2) the Data set version is set to 2.

Example 5-8 ISPF 3.2 listing of information about data set MHLRES2.GDG.PDSE.G0003V00

Data Set Information

Command ==>

Data Set Name . . . : MHLRES2.GDG.PDSE.G0003V00

General Data

Management class . . : MCDB22
Storage class . . . : STANDARD
Volume serial . . . : MLD20C
Device type : 3390
Data class : **None**
Organization . . . : PO
Record format . . . : FB
Record length . . . : 80
Block size : 27920
1st extent tracks . : 2
Secondary tracks . : 200
Data set name type : LIBRARY
Data set version . : 2

Current Allocation

Allocated tracks . : 2
Allocated extents . : 1
Maximum dir. blocks : NOLIMIT

Current Utilization

Used pages : 5
% Utilized : 20
Number of members . : 0

Dates

Creation date . . . : 2013/09/20
Referenced date . . : ***None***
Expiration date . . : ***None***

5.5.4 Scenario 4: Access the PDSEs using DFSMS V1.13

In this scenario we verify that PDSE data sets allocated in DFSMS V2.1, in version 2 format and as a member of a GDG, can be accessed in DFSMS V1.13. This scenario requires that the necessary compatibility fixes are installed.

We accessed the following data sets using ISPF 3.2:

- ▶ MHLRES2.GDG.PDSE.G0002V00
- ▶ MHLRES2.GDG.PDSE.G0003V00

G0002V00 is version 1 as shown in Example 5-6 on page 85 and G0003V00 is version 2 as shown in Example 5-8 on page 86.

ISPF on releases prior to DFSMS V2.1, even with compatibility fixes installed, do not show the Data set version.

In Example 5-9 on page 86 we show that the Data set version information is not present. If it had been shown it would be after the Data set name type display.

Example 5-9 ISPF 3.2 listing of data set MHLRES2.GDG.PDSE.G000v00 using DFSMS V1.13

Data Set Information

Command ==>

Data Set Name . . . : MHLRES2.GDG.PDSE.G0003V00

General Data

Management class . . : MCDB22
Storage class . . . : STANDARD
Volume serial . . . : MLD20C

Current Allocation

Allocated tracks . : 2
Allocated extents . : 1
Maximum dir. blocks : NOLIMIT

Device type :	3390	
Data class :	**None**	
Organization . . . :	PO	Current Utilization
Record format . . . :	FB	Used pages : 5
Record length . . . :	80	% Utilized : 20
Block size :	27920	Number of members . : 0
1st extent tracks . :	2	
Secondary tracks . :	200	
Data set name type :	LIBRARY	Dates
		Creation date . . . : 2013/09/20
		Referenced date . . : 2013/09/20
		Expiration date . . : ***None***

5.5.5 Scenario 5: Allocate a PDSE version 1 in a GDG on DFSMS V1.13

In this scenario we attempt to allocate a PDSE version 1 in a GDG set on a DFSMS V1.13 system. In this scenario the allocation should fail.

We ran the same JCL as shown in Example 5-7 on page 85 on DFSMS V1.13 and as expected the job failed. In Example 5-10 on page 87 we show the messages resulting from running a job to allocate a PDSE to a GDG using DFSMS V1.13.

Example 5-10 Job messages showing catalog error when allocating PDSE to GDG on DFSMS V1.13

```
J E S 2   J O B   L O G   --   S Y S T E M   S C 6 3   --   N O D E

14.37.33 JOB13844 ---- FRIDAY,    20 SEP 2013 ----
14.37.33 JOB13844 IRR010I  USERID MHLRES2  IS ASSIGNED TO THIS JOB.
14.37.33 JOB13844 ICH70001I MHLRES2  LAST ACCESS AT 14:19:40 ON FRIDAY, SEPTEMB
14.37.33 JOB13844 $HASP373 MHLRES2G STARTED - INIT 1    - CLASS A - SYS SC63
14.37.33 JOB13844 IEF403I MHLRES2G - STARTED - TIME=14.37.33 - ASID=001E - SC63
14.37.33 JOB13844 -                                     --TIMINGS (MINS.)--
14.37.33 JOB13844 -JOBNAME STEPNAME PROCSTEP    RC    EXCP    CPU    SRB    CLOCK
14.37.33 JOB13844 -MHLRES2G          S1          FLUSH      0     .00     .00     .00
14.37.33 JOB13844 IEF453I MHLRES2G - JOB FAILED - JCL ERROR - TIME=14.37.33
14.37.33 JOB13844 -MHLRES2G ENDED.  NAME-MHLRES2          TOTAL CPU TIME=
14.37.33 JOB13844 -NOT_EXECUTED_STEP_TABLE BEGIN
14.37.33 JOB13844 -JOBNAME STEPNAME PROCSTEP STEPNO
14.37.33 JOB13844 -MHLRES2G          S1              1
14.37.33 JOB13844 -NOT_EXECUTED_STEP_TABLE END
14.37.33 JOB13844 $HASP395 MHLRES2G ENDED
----- JES2 JOB STATISTICS -----
      20 SEP 2013 JOB EXECUTION DATE
            8 CARDS READ
            55 SYSOUT PRINT RECORDS
            0 SYSOUT PUNCH RECORDS
            3 SYSOUT SPOOL KBYTES
      0.00 MINUTES EXECUTION TIME
1 //MHLRES2G JOB (999,POK),'MHLRES2',CLASS=A,
//              NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=SC63
IEFC653I SUBSTITUTION JCL - (999,POK),'MHLRES2',CLASS=A,NOTIFY=MHLRES2
2 //S1          EXEC PGM=IEFBR14
3 //BIG1        DD DSN=MHLRES2.GDG.PDSE(+1),DISP=(NEW,CATLG),
//              SPACE=(TRK,(02,200)),UNIT=3390,
```

```

//          DSORG=PO,LRECL=80,BLKSIZE=27920,DSNTYPE=(LIBRARY,2),
//          RECFM=FB
STMT NO. MESSAGE
-
      3 IGD01010I SG ACS GETS CONTROL &ACSENVIR=ALLOC
      3 IGD01010I &STORCLAS = STANDARD
ICH70001I MHLRES2  LAST ACCESS AT 14:19:40 ON FRIDAY, SEPTEMBER 20, 2013
IEF344I MHLRES2G S1 BIG1 - ALLOCATION FAILED DUE TO DATA FACILITY SYSTEM ERROR
IGD17102I CATALOG ERROR IN DEFINING NONVSAM DATA SET
MHLRES2.GDG.PDSE.G0004V00
RETURN CODE IS 48 REASON CODE IS 152 IGG0CLEJ
IGD306I UNEXPECTED ERROR DURING IGG0CLEJ PROCESSING
RETURN CODE 48 REASON CODE 152
THE MODULE THAT DETECTED THE ERROR IS IGDVTSCU
SMS MODULE TRACE BACK - VTSCU VTSCVT VTSCR SSIRT
SYMPTOM RECORD CREATED, PROBLEM ID IS IGD00013
IEF272I MHLRES2G S1 - STEP WAS NOT EXECUTED.
IEF373I STEP/S1      /START 2013263.1437
IEF032I STEP/S1      /STOP  2013263.1437
      CPU:      0 HR  00 MIN  00.00 SEC      SRB:      0 HR  00 MIN  00.00 SEC
      VIRT:      OK  SYS:      OK  EXT:      OK  SYS:      OK
IEF375I JOB/MHLRES2G/START 2013263.1437
IEF033I JOB/MHLRES2G/STOP  2013263.1437
      CPU:      0 HR  00 MIN  00.00 SEC      SRB:      0 HR  00 MIN  00.00 SEC

```

In Example 5-11 on page 88 we show the JCL used to allocate a PDSE using DFSMS V1.13 and specifying DSNTYPE=(LIBRARY,2). PDSE version 2 data sets cannot be *created* on releases prior to DFSMS V2.1, but they can be used on prior releases.

Example 5-11 JCL to create a PDSE version 2 on DFSMS V2.1

```

//MHLRES2G JOB (999,POK),'MHLRES2',CLASS=A,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=SC63
//S1      EXEC PGM=IEFBR14
//BIG1    DD DSN=MHLRES2.V2.PDSE,DISP=(NEW,CATLG),
//          SPACE=(TRK,(02,200)),UNIT=3390,
//          DSORG=PO,LRECL=80,BLKSIZE=27920,DSNTYPE=(LIBRARY,2),
//          RECFM=FB

```

In Example 5-12 on page 88 we show the messages from the job allocating a PDSE using DSNTYPE=(LIBRARY,2).

Releases prior to DFSMS V2.1 do not allocate PDSE version 2 format data sets, but the DSNTYPE=(LIBRARY,2) has been accepted although the format will be version 1.

Example 5-12 JOB messages resulting from allocating a PDSE on DFSMS V1.3 with LIBRARY,2

```

1 //MHLRES2G JOB (999,POK),'MHLRES2',CLASS=A,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
//          /*JOBPARM L=999,SYSAFF=SC63
IEFC653I SUBSTITUTION JCL - (999,POK),'MHLRES2',CLASS=A,NOTIFY=MHLRES2
2 //S1      EXEC PGM=IEFBR14
3 //BIG1    DD DSN=MHLRES2.V2.PDSE,DISP=(NEW,CATLG),
//          SPACE=(TRK,(02,200)),UNIT=3390,

```

```

//          DSORG=PO,LRECL=80,BLKSIZE=27920,DSNTYPE=(LIBRARY,2),
//          RECFM=FB
STMT NO. MESSAGE

      3 IGD01010I SG ACS GETS CONTROL &ACSENVIR=ALLOC
      3 IGD01010I &STORCLAS = STANDARD
CH70001I MHLRES2  LAST ACCESS AT 14:37:33 ON FRIDAY, SEPTEMBER 20, 2013
EF236I  ALLOC. FOR MHLRES2G S1
GD101I  SMS ALLOCATED TO DDNAME (BIG1      )
        DSN (MHLRES2.V2.PDSE              )
        STORCLAS (STANDARD) MGMTCLAS (MCDB22) DATACLAS (      )
        VOL SER NOS= MLDB35
        //          RECFM=FB
STMT NO. MESSAGE
-
      3 IGD01010I SG ACS GETS CONTROL &ACSENVIR=ALLOC
      3 IGD01010I &STORCLAS = STANDARD
ICH70001I MHLRES2  LAST ACCESS AT 14:37:33 ON FRIDAY, SEPTEMBER 20, 2013
IEF236I  ALLOC. FOR MHLRES2G S1
IGD101I  SMS ALLOCATED TO DDNAME (BIG1      )
        DSN (MHLRES2.V2.PDSE              )
        STORCLAS (STANDARD) MGMTCLAS (MCDB22) DATACLAS (      )
        VOL SER NOS= MLDB35
IEF142I  MHLRES2G S1 - STEP WAS EXECUTED - COND CODE 0000
IGD104I  MHLRES2.V2.PDSE                      RETAINED, DDNAME=BIG1
IEF373I  STEP/S1          /START 2013263.1440
IEF032I  STEP/S1          /STOP  2013263.1440
        CPU:      0 HR  00 MIN  00.00 SEC   SRB:      0 HR  00 MIN  00.00 SEC
        VIRT:     4K  SYS:   288K  EXT:      OK  SYS:   10636K
IEF375I  JOB/MHLRES2G/START 2013263.1440
IEF033I  JOB/MHLRES2G/STOP  2013263.1440

```

In Example 5-13 on page 89 we show that even though DSNTYPE=(LIBRARY,2) was specified when allocating the data set on DFSMS V1.13, the data set was allocated as version 1 because only version 1 data sets can be allocated prior to DFSMS V2.1.

Example 5-13 ISPF 3.2 listing showing version 2 data set allocated on DFSMS V1.13 is version 1

Data Set Information

Command ==>

Data Set Name . . . : MHLRES2.V2.PDSE

General Data

Management class . . : MCDB22
 Storage class . . . : STANDARD
 Volume serial . . . : MLDB35
 Device type : 3390
 Data class : **None**
 Organization : PO
 Record format . . . : FB
 Record length . . . : 80
 Block size : 27920
 1st extent tracks . : 2
 Secondary tracks . . : 200

Current Allocation

Allocated tracks . : 2
 Allocated extents . : 1
 Maximum dir. blocks : NOLIMIT

Current Utilization

Used pages : 5
 % Utilized : 20
 Number of members . : 0

Data set name type : LIBRARY	Dates
Data set version . : 1	Creation date . . . : 2013/09/20
	Referenced date . . : ***None***
	Expiration date . . : ***None***

In order to demonstrate that PDSE version 2 data sets created on DFSMS V2.1 can be accessed and updated on DFSMS V1.13, we put data in the data set using DFSMS V2.1, then read that data and added more data using DFSMS V1.13.

Then returned to DFSMS V2.1 and verified that both the original data and the updates using DFSMS V1.13 were in the data set.

In Example 5-14 on page 90 we show the data entered into a PDSE version 2 data set, then updated on DFSMS V1.13.

Example 5-14 PDSE version 2 data set as populated using DFSMS V2.1 then updated using DFSMS V1.13

```

EDIT      MHLRES2.GDG.PDSE.G0003V00(DEMODATA) - 01.01
Command ===>
***** ***** Top of Data *****
==MSG> -Warning- The UNDO command is not available until you change
==MSG>          your edit profile using the command RECOVERY ON.
000100 This data was entered using DFSMS V2.1
000110 and it is a version 2 data set
000200 This data was entered using DFSMS V2.1
000210 and it is a version 2 data set
000300 This data was entered using DFSMS V2.1
000310 and it is a version 2 data set
000400 This data was entered using DFSMS V2.1
000410 and it is a version 2 data set
000500 This data was entered using DFSMS V2.1
000510 and it is a version 2 data set
000600 This data was entered using DFSMS V2.1
000610 and it is a version 2 data set
000700 This data was entered using DFSMS V2.1
000710 and it is a version 2 data set
000800 This data was entered using DFSMS V2.1
000810 and it is a version 2 data set
000900 This data was entered using DFSMS V2.1
000910 and it is a version 2 data set
001000 This data was entered using DFSMS V2.1
001100 and it is a version 2 data set
001200 *****
001300 The above data was read successfully using DFSMS V1.13
001400 *****
001500 This data was entered using DFSMS V1.13
001600 and it is accepted even even thou it is a version 2 data set
001700 This data was entered using DFSMS V1.13
001800 and it is accepted even even thou it is a version 2 data set
001900 This data was entered using DFSMS V1.13
002000 and it is accepted even even thou it is a version 2 data set
002100 This data was entered using DFSMS V1.13

```

002200 and it is accepted even even thou it is a version 2 data set
002300 This data was entered using DFSMS V1.13
002400 and it is accepted even even thou it is a version 2 data set
002500 This data was entered using DFSMS V1.13
002600 and it is accepted even even thou it is a version 2 data set
002700 This data was entered using DFSMS V1.13
002800 and it is accepted even even thou it is a version 2 data set
002900 This data was entered using DFSMS V1.13
003000 and it is accepted even even thou it is a version 2 data set
003100 This data was entered using DFSMS V1.13
003200 and it is accepted even even thou it is a version 2 data set

Compatibility

The default setting for PDSE version can be set in DFSMS V2.1 systems to be Version 1 (default) or Version 2. This setting can then be overridden for a particular PDSE data set allocation to whatever is required for that PDSE.

The PDSE_VERSION(1|2) statement is in member IGDSMSxx of SYS1.PARMLIB.

Note: The PDSE_VERSION statement may not be specified in member IGSSMSxx of a system prior to DFSMS V2.1. There is no compatibility for this statement.

PDSE Version 2 data sets can not be allocated using releases prior to DFSMS V2.1.

If a data set is allocated using DFSMS V2.1 as a member of a GDG then it can be accessed and used from DFSMS V1.13

A PDSE data set may not be defined as a member of a GDG when using DFSMS releases prior to DFSMS V2.1.

Updates are required in DFSMS V1.12 and DFSMS V1.13 to use PDSE version 2 data sets. Examples of fixes required on pre-DFSMS V2.1 systems can be found in Appendix B, “APARs to be reviewed for DFSMS V1.13 or DFSMS V1.12” on page 309.

For IBM products a current listing of compatibility fixes can be obtained by loading the IBM enhanced HOLDDATA files, then using SMP/E to search, on the pre-DFSMS V2.1 system for the z/OS V2.1 compatibility fixes.



SMS enhancements

This chapter provides an overview of the features and function in SMS included in z/OS V2.1. They include:

- ▶ New IGDSMSxx parameters
- ▶ Provide accurate volume space statistics
- ▶ Alter ACDS and COMMDS to SHAREOPTIONS(3 3)
- ▶ SMS ACS read-only variable for EAVs
- ▶ DFSMS storage tiers
- ▶ Option to suppress specific SMS messages.

6.1 New IGDSMSxx parameters

DFSMS V2.1 introduces several new parameters that you specify in the IGDSMSxx parmlib member.

HONOR_DSNTYPE_PDSE	Specifies that a partitioned data set is to be created regardless of the values for other data set attributes, such as data set organization or directory blocks.
PDSE_VERSION	Specifies a default version number (1 or 2) for data sets that are allocated with DSNTYPE value of LIBRARY.
SUPPRESS_SMSMSG	Specifies an option to suppress messages IGD17054I, IDG17227I and IGD17395I.
PS_EXT_FORMAT	Indicates the format (1 or 2) in which the system creates extend format data sets.

6.1.1 IGD030I error message

If you try to activate any of the new parameters, through a **SET SMS=xx** z/OS console command, from a z/OS V1.13 or earlier system, you receive message IGD030I. You also receive message IGD030I when you IPL a z/OS V1.13 or earlier system pointing to an IGDSMSxx parmlib member that contains any of these new parameters in it.

For more information regarding the PDSE parameters, refer to Chapter 5, “PDSE enhancements” on page 79.

For information regarding the new sequential extended data set format 2, see 4.4, “Sequential Extended Format Version 2” on page 74.

6.2 Provide accurate volume space statistics

SMS in DFSMS V2.1 provides more accurate volume space statistics. Prior to DFSMS V2.1, SMS was enhanced to return accurate space statistics in the following situations:

- ▶ When Common Access VTOC Facility (CVAF) informs SMS that a volume space usage has changed.
- ▶ When a volume is varied online for the first time.
- ▶ When users issues IGDCNS calls to retrieve volume information that do not contain updated space statistics.
- ▶ When SMS issues LSPACE and updates volume information with most current space statistics.

However, there were some special cases where the space statistics remained outdated. This can happen, for example, when a user issues **ICKDSF REFORMAT REFVTOC** in a system outside the SMSplex.

6.2.1 Use

When a volume is resized, the disk subsystem sends an Unsolicited Stated Change Interrupt to all the systems. This interrupt causes device support code to be invoked in each system. Each system sends a Read Device Characteristics CCW in order to get the new size for that

device. Each system updates the size information that resides in the Device Class Extension (DCE), that is an extension for the UCB for that device.

Now in DFSMS V2.1, when SMS receives a request for space information, it compares the information in the DCE with the information inside the SMS address space. It automatically refreshes the space information inside the active configuration data set (ACDS) of SMS, returning accurate space statistics to the caller that requested this information.

VARY SMS command to update space statistics

In z/OS 2.1 there is a new **VARY SMS** command to force SMS to update space statistics for a volume in a storage group. The syntax of the command is shown in Example 6-1.

Example 6-1 VARY SMS syntax

```
V SMS,{STORGRP(storgrp) | SG(storgrp) |  
      VOLUME(volser) | VOL(volser)},  
      {SPACE | S}
```

where:

storgrp	Refers to a pool storage group
volser	A volser in a pool storage group.

For example, if you want your SMS active configuration to immediately get accurate space statistics for a storage group called BIGSTUFF, you issue the `V SMS,SG(BIGSTUFF),SPACE` command from a z/OS console.

6.2.2 Coexistence

If the caller that requested space information comes from a z/OS V1.13 or earlier system, it will not refresh the space statistics in the ACDS, if a volume has been resized from a system outside the SMSplex.

If the caller is a z/OS V2.1 system, then all the systems in the SMSplex benefit from this enhancement, as the ACDS is shared among all systems in the SMSplex.

6.3 Alter ACDS and COMMDS to SHAREOPTIONS(3 3)

SMS in DFSMS V2.1 now checks whether or not the ACDS or COMMDS were defined with `SHAREOPTIONS(3 3)` and tries to alter it, if it was defined with different sharing options. This enhancement helps users avoid potential problems due to incorrect sharing options.

Using `SHAREOPTIONS(3,3)` when allocating an ACDS or COMMDS allows full authority to read from and write to the ACDS or COMMDS from any system. The ACDS and COMMDS must be accessed from all systems in the complex simultaneously.

If SMS detects a lower value for `SHAREOPTIONS` when the CDS is activated, SMS attempts to alter the value to `SHAREOPTIONS(3,3)`, and issues a message to inform the user of the result of the change:

- ▶ If the change is successful, SMS issue new message IGD098I
- ▶ If the change is not successful, it issues message IGD099I

In both cases, the activation process continues as before.

6.3.1 Use

Example 6-2 shows a sample job for allocating a new ACDS. Note that we specified SHAREOPTIONS(2 3). This is not the SHAREOPTIONS that are adequate for the ACDS. We defined it this way to test the automatic alter to SHAREOPTIONS(3 3) that is going to be done by SMS.

Example 6-2 Sample ACDS definition with wrong SHAREOPTIONS(2 3)

```
//MHLRES3A JOB (999,P0K), 'MHLRES3', CLASS=A, MSGCLASS=T,
// NOTIFY=&SYSUID, TIME=1440, REGION=6M
//DEF1 EXEC PGM=IDCAMS, REGION=512K
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
DEFINE CLUSTER(
    NAME(SYS1.SMS.V2R1.ACDS) +
    LINEAR +
    VOLUME(SBOX12) +
    CYL(10 1) +
    SHAREOPTIONS(2,3) +
    DATA(
    NAME(SYS1.SMS.V2R1.ACDS.DATA) REUSE)
/*
```

After defining the new ACDS, we replaced the current ACDS with the newly defined ACDS by issuing the following commands:

1. SETSMS SAVEACDS(SYS1.SMS.V2R1.ACDS)

This command saves the current active SMS configuration into the newly defined ACDS.

2. SETSMS ACDS(SYS1.SMS.V2R1.ACDS)

This command activates the new ACDS.

During the activation process, SMS detected that our new ACDS was defined with SHAREOPTIONS(2,3), and automatically altered it to SHAREOPTIONS(3,3), as shown in Figure 6-1.

```
SETSMS ACDS(SYS1.SMS.V2R1.ACDS)
IGD098I ACDS SYS1.SMS.V2R1.ACDS 698
PREVIOUSLY DEFINED WITH SHAREOPTIONS(2,3) HAS BEEN AUTOMATICALLY
CORRECTED TO SHAREOPTIONS(3,3)
IEF196I IEF237I D26C ALLOCATED TO SYS00024
IEF196I IGD104I SYS1.SMS.ACDS.NEW RETAINED,
IEF196I DDNAME=SYS00003
IEF196I IEF237I A529 ALLOCATED TO SYS00025
IEF196I IEF285I WTSCPLX2.DFSMS.DATASET.SEP KEPT
IEF196I IEF285I VOL SER NOS= SBOXBF.
IEF196I IEF237I A529 ALLOCATED TO SYS00026
IEF196I IEF285I WTSCPLX2.DFSMS.DATASET.SEP KEPT
IEF196I IEF285I VOL SER NOS= SBOXBF.
IGD009I ACDS SWITCHED TO SYS1.SMS.V2R1.ACDS
```

Figure 6-1 SMS alters to SHAREOPTIONS(3,3) during activation process

6.3.2 Coexistence

There is not any coexistence issue with this DFSMS V2.1 enhancement when you share ACDS and COMMDs between systems at z/OS V2.1 and z/OS V1.13 or earlier releases of z/OS.

6.4 SMS ACS read-only variable for EAVs

DFSMS V2.1 provides a new ACS read-only variable *&EATTR* for EAV support. This variable contains the EATTR value as specified in JCL, IDCAMS DEFINE or ALLOCATE, dynamic allocations, and data class constructs. EATTR means the extend attributes for EAV.

ACS routines become more intelligent and easier to code. You can select the proper SMS constructs to your data sets, so the data sets can be allocated or not to an EAV storage group.

6.4.1 EAV overview

An extended address volume (EAV) is a volume with more than 65520 cylinders. An EAV increases the amount of addressable DASD storage per volume beyond 65520 cylinders by changing how tracks on IBM ECKD™ volumes are addressed.

The benefit of this support is that the amount of z/OS addressable disk storage is significantly increased. This helps customers who are approaching the 4-digit device number limit by providing constraint relief for applications using large amounts of data.

EAV support was first introduced in z/OS V1.10, and was limited to VSAM data sets. Also, the initial support for EAV allowed a maximum of 262,668 cylinders or 223 GB in volume size.

z/OS V1.11 extended this support to non-VSAM data sets, and also introduced a new attribute, EATTR, to be used at allocation time, for controlling the allocation of VSAM and non-VSAM data sets in regard to when extended address space (EAS) can be used. EAS is the area on EAV whose cylinder addresses are equal to or greater than 65,536, as shown in Figure 6-2 on page 98.

At the time the EATTR attribute was created, there was not an ACS read-only variable to be checked by the ACS routines.

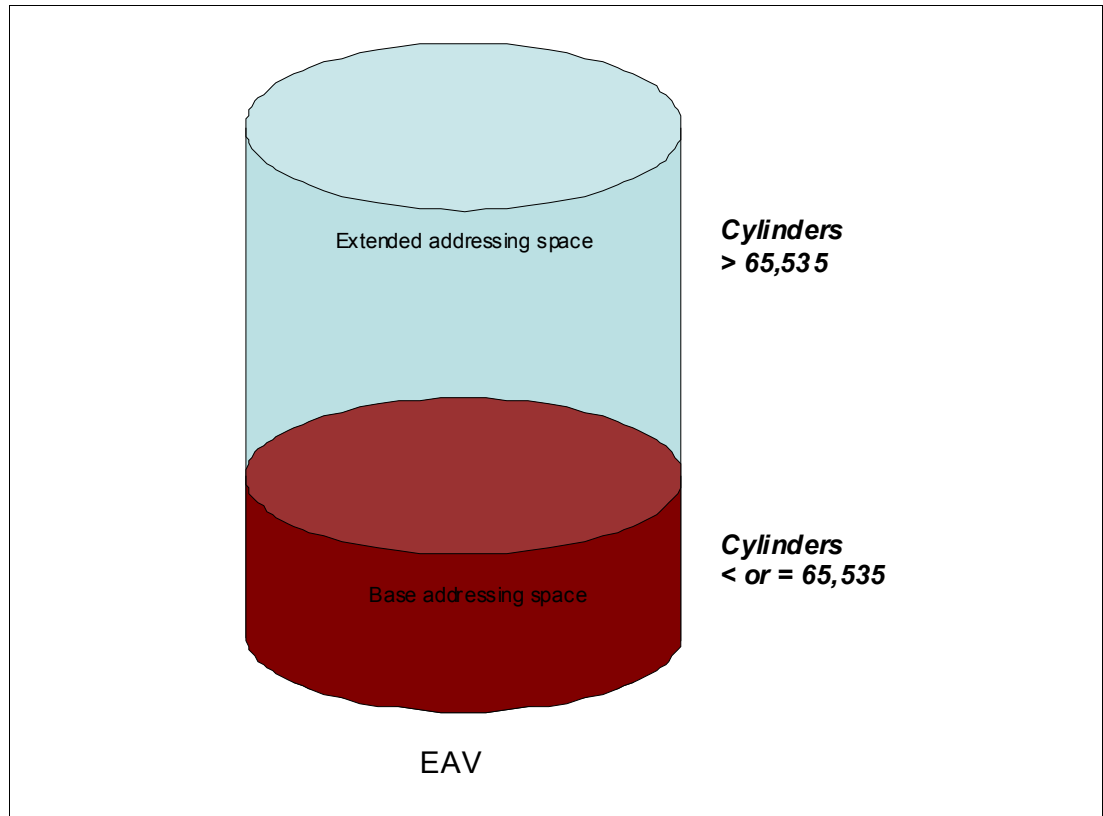


Figure 6-2 EAV layout

6.4.2 EATTR overview

You can specify this attribute EATTR in IDCAMS DEFINE and ALLOCATE, JCL, dynamic allocation, and data classes constructs. You can use IDCAMS ALTER to modify this attribute in existing data sets. There is no JCL override of the current EATTR value for existing data sets. To change the EATTR value after a new allocation requires you to run the IDCAMS ALTER command.

The &EATTR that are passed to the ACS routines contains the EATTR value that you can specify in IDCAMS, JCL, dynamic allocation and data classes are:

NO	No extended attributes. The data set can not have extended attributes (format 8 and 9 DSCBs) and cannot reside in EAS. This is the default behavior for non-VSAM data sets.
OPT	Extended attributes are optional. The data set can have extended attributes (format 8 and 9 DSCBs) and can optionally reside in EAS. This is the default behavior for VSAM data sets.
blank	No specification. It assumes the default behavior depending on the type of the data set; VSAM or non-VSAM.

6.4.3 Use

Example 6-3 on page 99 shows some lines of ACS code taken from a storage group ACS routine. We create a filtlist called ZOS21 that has the system names of the systems that are already at z/OS V2.1 in our SMSplex: SC63, SC64, and SC70. We assign a BIGSTUFF

storage group for data sets that were created with EATTR=OPT in JCL, dynamic allocation, IDCAMS, and TSO also. This storage group BIGSTUFF has only two 1 TB EAV volumes.

SC65 is still at z/OS 1.13 level, so we avoid executing the ACS code that tests the &EATTR read-only variable on that system.

Example 6-3 example of storage group ACS routine

```

FILTLIST MHLRES INCLUDE(MHLRES%.**)
FILTLIST ZOS21 INCLUDE('SC63','SC64','SC70')
.
.
.
WHEN (&DSN = &MHLRES AND &STORCLAS NE SCLIB%) DO
  SELECT(&SYSNAME)
    WHEN('SC65') DO
      SET &STORGRP = 'SGMHL01'
      EXIT
    END
  WHEN(&ZOS21) DO
    IF &EATTR = 'OPT' THEN
      SET &STORGRP = 'BIGSTUFF'
    ELSE
      SET&STORGRP = 'SGMHL01'
      EXIT
    END
  END
END
END

```

Example 6-4 shows a sample IEFBR14 job that we used to test this ACS routine.

Example 6-4 Sample IEFBR14 job with EATTR=OPT

```

//MHLRES3T JOB 'JOE',NOTIFY=MHLRES3,
//          MSGCLASS=T,REGION=32M
//IEF      EXEC PGM=IEFBR14
//I        DD DSN=MHLRES3.EATTR.TEST8,DISP=(,CATLG),
//          SPACE=(CYL,(300,300)),UNIT=3390,EATTR=OPT,
//          DCB=(RECFM=FB,LRECL=300,BLKSIZE=27900),
//          DATACLAS=DCEXTSEQ,STORCLAS=SCSDR0

```

The data set MHLRES3.EATTR.TEST8 was created successfully on one of our EAV volumes in storage group BIGSTUFF.

6.4.4 Coexistence

The read-only variable &EATTR is only available on z/OS V2.1 systems. SMS at z/OS V1.13 or lower releases does not recognize the &EATTR variable.

If you have systems at z/OS V1.13 or prior releases in your SMSplex, and plan to starting using the &EATTR read-only variable, you must code your ACS routines to branch around any reference to &EATTR when they run in a system with z/OS V1.13 or earlier releases of z/OS.

Example 6-3 on page 99 shows a sample piece of a storage group ACS code, where first test if the new allocation is going to occur in system SC65. This system is at z/OS V1.13. We avoid it testing the &EATTR when the new allocation happens in our system SC65.

6.5 DFSMS storage tiers

A storage tier is a class of devices that has a defined set of performance, availability, accessibility and capacity characteristics.

The three fundamental storage tiers are comprised of disk, optical and tape devices. Disk being defined as the highest tier and tape the lowest. Until recently, there have only been minor variations in the characteristics of the devices available within each of these fundamental tiers. But, with recent advances in storage technology, these fundamental tiers can be further divided into more discrete tiers. New technologies introduced significant variations in the disk storage tier. For example, we can have the following technologies available for allocation of our data sets in disk:

- ▶ Solid state drives (SSD)
- ▶ Hard disk drives (HDD)
- ▶ Serial Advanced Technology Attachment (SATA) drives

Each one of these technologies provides different performance levels to data access. Also, each one has a different cost to store your data. Before DFSMS V2.1, there was not a way of automatically move less active data between these technologies, from the fastest and more expensive, to the slowest and cheaper.

Figure 6-3 on page 101 shows the classic storage hierarchy that is around 30 years old. It is a three-tier hierarchy:

- ▶ Level 0 (L0) or Primary disks

This is the primary level that contains data that is directly accessible by users and applications. Data at this level is owned by users and applications and optionally managed by DFSMSHsm.

- ▶ Migration Level 1 (ML1)

When DFSMSHsm managed L0 data remains inactive for a specified (policy-based) duration of time, DFSMSHsm moves the data down in the hierarchy. Data at this level is in a proprietary format and cannot be directly accessed by users or applications. ML1 data is owned by DFSMSHsm. Storage devices at this level may only be disk. DFSMSHsm does not distinguish between any variations in disk technology used at this level.

- ▶ Migration Level 2 (ML2)

As the unreferenced age of data increases, the data is moved to the lowest level in the hierarchy, ML2. ML2 is generally offline storage and hence the least expensive. Similar to ML1, all ML2 devices are treated equally.

The *classic* DFSMS storage hierarchy (30+ years old)

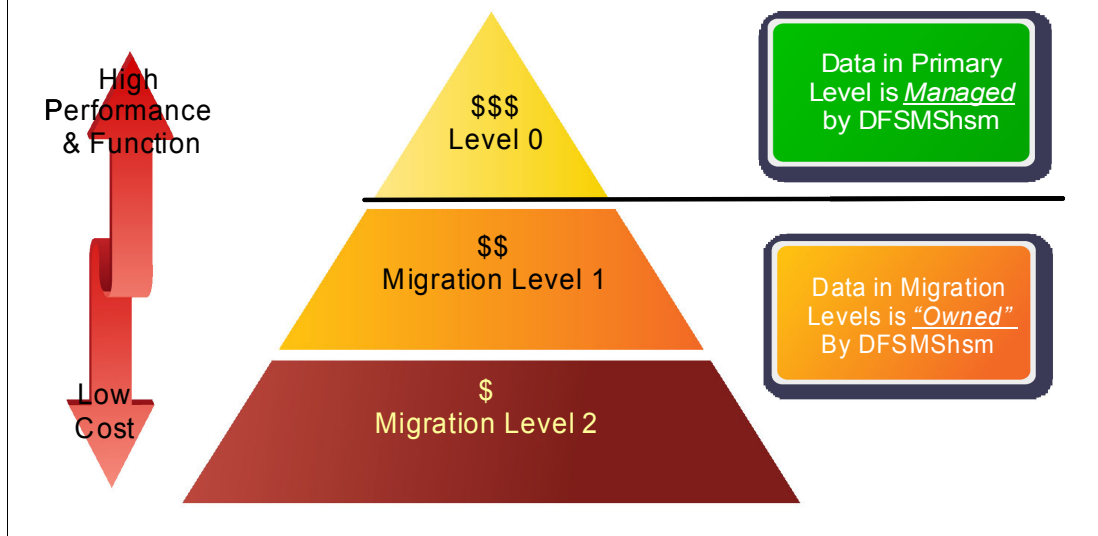


Figure 6-3 The classic DFSMS storage hierarchy

Today, with the possibility of several technologies present on the Level 0 tier, it comes the opportunity to break Level 0 into multiple tiers.

6.5.1 Storage tier hierarchies

The existing three-layer hierarchy is being separated into two distinct hierarchies; the primary hierarchy and the migration hierarchy, as seen in Figure 6-4 on page 102. The primary hierarchy is comprised of all the storage tiers that contain application and user data in its native format. The data in this hierarchy is directly accessible by applications and users and is comprised of tiers. The data in these tiers is DFSMSHsm managed.

Figure 6-4 on page 102 shows an example of a primary storage hierarchy subdivided into three tiers:

- ▶ SSD
- ▶ HDD
- ▶ SATA disks.

Data on primary storage hierarchy can be moved across the tiers in by DFSMSHsm, according to management class attributes assigned to SMS managed data sets. These attributes, also called class transition attributes, are not new to DFSMS. They have been used for class transition for OAM non structured data. DFSMS V2.1 allows these attributes to be used for class transitions of structured data, that is, your application data sets.

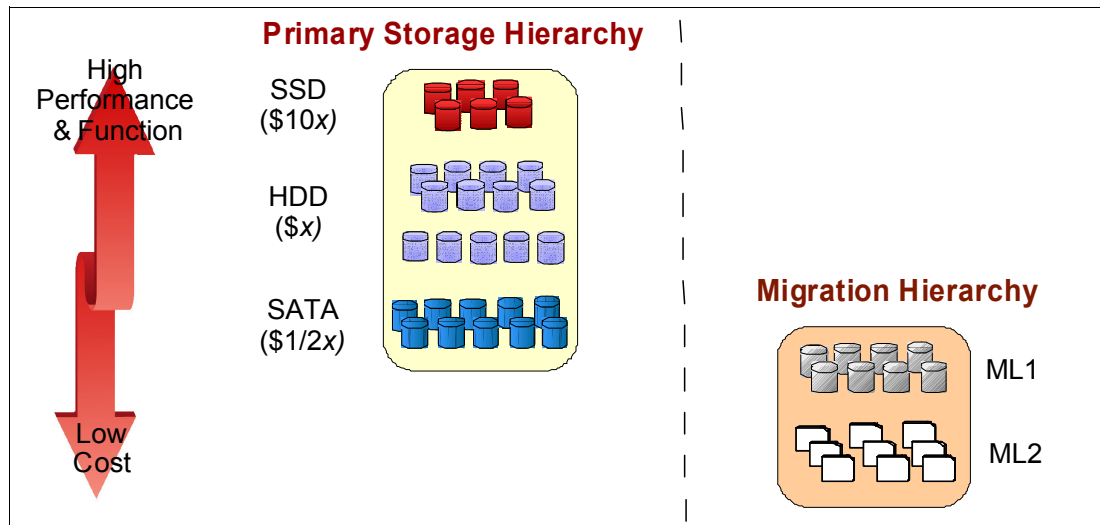


Figure 6-4 Two hierarchies of migration storage

For more information on how DFSMSShsm exploits these storage tiers, refer to 10.2, “DFSMSShsm storage tiers” on page 179.

6.6 Option to suppress specific SMS messages

DFSMS V2.1 provides a new *SUPPRESS_SMSMSG* option in IGDSMSxx parmlib member to suppress some specific informational messages. There are messages that may occur often, and some installations do not want to view these messages when they appear at the joblog. The messages that can be suppressed by using this option are:

- ▶ IGD17054I DATA SET NOT FOUND FOR DELETE/RENAME ON VOLUME volser DATA SET IS dsn
- ▶ IGD17227I JOBNAME jobname PROGRAM NAME progname STEPNAME stepname DDNAME ddname DATA SET dsn WAS ALLOCATED TO A SUBSEQUENT MULTI-TIERED STORAGE GROUP. ALLOCATED STORAGE GROUP WAS sg1. CANDIDATE STORAGE GROUPS ARE: sg2, sg3...
- ▶ IGD17395I DATA SET dsn WAS NOT ALLOCATED IN THE SAME STORAGE FACILITY IMAGE BECAUSE (text) MODID(modid) RC(rc) RSN(rsn)].

These messages can not be suppressed by the Message Processing Facility (MPF) of z/OS, nor can they be suppressed by installation exits.

6.6.1 Use

You can specify this new parameter *SUPPRESS_SMSMSG* in your IGDSMSxx parmlib member as shown in Example 6-5.

Example 6-5 Sample SUPPRESS_SMSMSG parameter

```
SUPPRESS_SMSMSG(YES,IGD17054I,IGD17227I,IGD17395I)
```

To activate this modification you can issue the **SET SMS=xx** command from the z/OS console.

The default behavior of SMS is not to suppress any of these messages.

You can also suppress those messages by using the SETSMS command. For example, you can tell SMS to suppress all occurrences of messages IGD17054I and IGD17227I by issuing the following command from the z/OS console:

SETSMS SUPPRESS_SMSMSG(YES,IGD17054I,IGD17227I)

You can use the **D SMS,OPTIONS** command to confirm that suppression for these SMS messages is active or not active. Figure 6-5 shows a piece of the output of the message IGD002I as a result of issuing this command. It shows that we are suppressing messages IGD17054I, and IGD17227I, and not suppressing IGD17395I in our SMSplex.

```
D SMS,OPTIONS
IGD002I 08:56:51 DISPLAY SMS 312
ACDS      = SYS1.SMS.ACDS.NEW
COMMDS    = SYS1.SMS.COMMDS
ACDS LEVEL = z/OS V2.1
.
.
.
CA_RECLAIM = DATACLAS
PS_EXT_VERSION = 1
SUPPRESS_SMSMSG = IGD17054I(YES) IGD17227I(YES) IGD17395I(NO )
```

Figure 6-5 Displaying SUPPRESS_SMSMSG

6.6.2 Coexistence

Activating the suppression of these SMS messages in z/OS V2.1 system has no effect on systems at DFSMS V1.13 or prior versions sharing the same ACDS.

If you issue the SETSMS SUPPRESS_SMSMSG from a system at DFSMS V1.13 or earlier, you receive message IGD029I, as it does not recognize the SUPPRESS_SMSMSG parameter.

If you try to activate an IGDSMSxx member with the SUPPRESS_SMSMSG parameter you receive message IGD030I.



DFSMSdfp enhancements

In this chapter we discuss the DFSMSdfp features and functions introduced in z/OS V2.1 DFSMS. The topics included in this chapter are:

- ▶ IEBCOPY COPYGROUP
- ▶ IEBCOPY user exit capabilities
- ▶ IEBCOPY return code feedback
- ▶ Open Close and End of volume (OCE) partial release
- ▶ OCE RAS enhancements
- ▶ IEAAPpx comments
- ▶ DCBE invalidation message IEC190I
- ▶ XTLOT HealthCheck

7.1 IEBCOPY COPYGROUP

Prior to DFSMS V2.1 IEBCOPY had a COPYGRP statement, which is still available, that did not process groups when using a PDS. That has been complemented by the new COPYGROUP command which has additional functions that makes it applicable to all types of PDS and PDSE formats and all operations between them.

The significant feature of COPYGROUP (and in the case of PDSE for COPYGRP) is that it can be used to ensure that a member and all its aliases are copied together, and that if an alias is selected the main member and any other aliases are copied together.

COPYGROUP may also be used with SELECT minor statements which specify a member or alias using wild cards. If SELECT is used with wild cards then EXCLUDE may be used to refine the selection list. The EXCLUDE minor statement operates on the set of members generated by the SELECT minor statement.

When no member selection is done then all aliases are copied whether using COPY, COPYGRP, or COPYGROUP.

COPYGROUP may be used to perform the following:

- ▶ COPY PDS to PDS, PDS to PDSE, PDSE to PDS, and PDSE to PDSE
- ▶ UNLOAD PDS to PS, and PDSE to PS
- ▶ LOAD PS to PDSE, and PS to PDS

A group consists of a member and all its aliases. If a member is selected, the member and all its aliases are included. The member may be the main member or any of the aliases, but they will all be copied, or none will be copied, if there is some condition preventing all from being copied.

Note: This is different from the COPYGRP function where *only* the selected member will be copied when the output data set is a PDS. If the output is a PDSE then the member and all its aliases will be copied.

The SELECT minor statement may be used with the COPYGROUP command to specify wild cards to facilitate member selection. Note that wild cards are not permitted when using the COPYGRP statement. When SELECT has been used with wild cards, then EXCLUDE may be used.

Note: In order to use the minor statements SELECT (and EXCLUDE where applicable) the INDD and OUTDD statements must be specified on the main statement such as COPY, COPYGRP, or COPYGROUP. The default DDNAMEs of SYSUT1 and SYSUT2 are not usable for this function.

7.1.1 Scenarios

We generated and ran a number of jobs to illustrate the basic capabilities of the COPYGROUP statement.

In the following examples, the data sets used are:

- ▶ SYS1.LINKLIB for INDD
- ▶ MHLRES2.SYS1.LINKLIB.IEBCOPY1 for OUTDD

Data set MHLRES2.SYS1.LINKLIB.IEBCOPY1 was allocated new before running the first job.

In Table 7-1 on page 107 we list a number of scenarios relating to the IEBCOPY COPYGROUP statement. We will describe these scenarios in this section.

Table 7-1 Scenarios provided to show COPYGROUP functions

Scenario number	Purpose
"Scenario 1" on page 108	Copy from PDS to empty PDS. Use COPYGROUP with INDD and OUTDD. SELECT a member that has an alias. Observe that the alias is copied as well as the main member.
"Scenario 2" on page 109	Copy from PDS to output PDS that is not empty. Use COPYGROUP with INDD and OUTDD. SELECT a member that has already been copied that has an alias. Observe that neither the member nor the alias is copied.
"Scenario 3" on page 109	PDS to PDS output not empty. COPYGROUP INDD,R OUTDD. SELECT a member that has an alias. Observe that both the member and the alias are copied.
"Scenario 4" on page 110	PDS to PDS output not empty. COPY INDD OUTDD. SELECT a member that has an alias. Observe that only the member is copied.
"Scenario 5" on page 111	PDS to PDS output not empty. COPYGROUP INDD OUTDD. SELECT the alias of a member. Observe that the alias is not copied because the member is present.
"Scenario 6" on page 112	PDS to PDS output not empty. COPYGROUP INDD, OUTDD. SELECT the alias of a member. Observe that neither the alias nor the main member is replaced because of a COPYGROUP conflict.
"Scenario 7" on page 114	PDS to PDS output not empty. COPY INDD,R OUTDD. SELECT the member and related alias of the member. Observe that neither the alias nor the main member is replaced because of a COPYGROUP conflict.
"Scenario 8" on page 115	Using the COPYGROUP statement demonstrate the use of SELECT minor statement with wildcard character to specify the primary members to be copied. Expect to see alias names included when their primary member is specified for inclusion.

Scenario number	Purpose
"Scenario 9" on page 118	Using the COPYGROUP statement demonstrate the use of SELECT and EXCLUDE minor statements with wildcard character to specify the member(s) to be copied. Expect to see alias names excluded when their primary member is specified for exclusion.
"Scenario 10" on page 121	COPYGRP limitation when working with PDS to PDS copying

Scenario 1

In this scenario copy from PDS to empty PDS. Use COPYGROUP with INDD and OUTDD. SELECT a member that has an alias. Observe that the alias is copied as well as the main member.

In Example 7-1 on page 108 you find the JCL that was run to show the use of the COPYGROUP statement with the SELECT minor statement to select a member.

Example 7-1 JCL to copy a member of a PDS to empty PDS

```
//MHLRES2C JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
/* example of use of COPYGROUP to contrast with COPYGRP
//S1      EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//INDD    DD DISP=SHR,DSN=SYS1.LINKLIB
//OUTDD   DD DISP=SHR,DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
//SYSIN   DD *
COPYGROUP INDD=INDD,OUTDD=OUTDD
SELECT MEMBER=(ARCABA)
```

In Example 7-2 on page 108 you find the IEBCOPY messages showing that the alias is copied together with the main member.

Example 7-2 IEBCOPY messages showing member ARCABA copied as well as alias ABA

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE      1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS    02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2C S1    19:50:39 MON 16 SEP 2013 PARM=' '
GRPCOPY COPYGROUP INDD=((INDD,R)),OUTDD=OUTDD                        00010006
        SELECT MEMBER=(ARCABA)                                       00020005
IEB1013I COPYING FROM PDS INDD=INDD VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I          TO PDS OUTDD=OUTDD VOL=MLD12F
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IGW01553I ALIAS ABA OF COPIED PRIMARY ARCABA HAS BEEN COPIED
IGW01552I MEMBER ARCABA HAS BEEN COPIED AND REPLACED
IGW01550I 2 OF 2 SPECIFIED MEMBERS WERE COPIED
IEB147I END OF JOB - 0 WAS HIGHEST SEVERITY CODE
```

Scenario 2

In this scenario copy from PDS to an output PDS that is not empty. Use COPYGROUP with INDD and OUTDD. SELECT a member that has already been copied that has an alias. Observe that neither the member nor the alias is copied.

In Example 7-3 on page 109 you find the JCL that was run to show the use of the COPYGROUP statement with the SELECT minor statement to select a member that already exists in the output PDS.

Example 7-3 JCL to copy a member of a PDS to a PDS that already has the member

```
//MHLRES2C JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
/* example of use of COPYGROUP to contrast with COPYGRP
//S1      EXEC PGM=IEBCOPY
//SYSPRINT DD  SYSOUT=*
//INDD    DD  DISP=SHR,DSN=SYS1.LINKLIB
//OUTDD   DD  DISP=SHR,DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
//SYSIN   DD  *
COPYGROUP INDD=INDD,OUTDD=OUTDD
SELECT MEMBER=(ARCABA)
```

In Example 7-4 on page 109 you find the IEBCOPY messages showing that the alias is not copied because the main member already exists.

Example 7-4 IEBCOPY messages showing neither the member nor the alias is copied

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS    02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2C S1    13:34:14 TUE 17 SEP 2013 PARM=' '
COPYGROUP INDD=INDD,OUTDD=OUTDD                                00010000
SELECT MEMBER=(ARCABA)                                          00020000
IEB1013I COPYING FROM PDS INDD=INDD VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I          TO PDS OUTDD=OUTDD VOL=MLD83A
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IGW01557W MEMBER ARCABA NOT COPIED BECAUSE THE NAME ALREADY EXISTS IN
THE OUTPUT DATASET CAUSING A COPY GROUP NO-REPLACE CONFLICT
IGW01558W MEMBER ABA NOT COPIED BECAUSE IT BELONGS TO A GROUP
IN WHICH A PREVIOUS NAME PROCESSED ENCOUNTERED COPY GROUP NO REPLACE CONFLICT
IGW01550I 0 OF 2 SPECIFIED MEMBERS WERE COPIED
IEB1130I A WARNING MESSAGE FROM FAMS PROCESSING APPEARS ABOVE -- DIAGNOSTIC
INFORMATION IS X'2810022D'
IEB147I END OF JOB - 0 WAS HIGHEST SEVERITY CODE
```

Scenario 3

This scenario shows a PDS to PDS copy where the output data set is not empty using COPYGROUP INDD,R OUTDD. SELECT a member that has an alias. Observe that both the member and the alias are copied.

In Example 7-5 on page 110 you find the JCL that was run to show the use of the COPYGROUP statement with the SELECT minor statement to select a member that already

exists in the output PDS. In this example INDD=((INDD,R)) is specified so that the member will be replaced, and the alias will be copied.

Example 7-5 JCL to copy a member that already exists, but with replace specified

```
//MHLRES2C JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
/* example of use of COPYGROUP to contrast with COPYGRP
//S1      EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//INDD    DD DISP=SHR,DSN=SYS1.LINKLIB
//OUTDD   DD DISP=SHR,DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
//SYSIN   DD *
COPYGROUP INDD=((INDD,R)),OUTDD=OUTDD
SELECT MEMBER=(ARCABA)
```

In Example 7-6 on page 110 you find the IEBCOPY messages showing that the alias was copied because the main member was replaced as a result of INDD=((INDD,R)).

Example 7-6 IEBCOPY messages showing that member ARCABA has been replaced and ABA copied

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS    02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2C S1    14:03:11 TUE 17 SEP 2013 PARM=' '
COPYGROUP INDD=((INDD,R)),OUTDD=OUTDD                                00010007
SELECT MEMBER=(ARCABA)                                              00020000
IEB1013I COPYING FROM PDS INDD=INDD VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I          TO PDS OUTDD=OUTDD VOL=MLD83A
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IGW01554I ALIAS ABA OF COPIED PRIMARY ARCABA HAS BEEN COPIED
AND REPLACED
IGW01552I MEMBER ARCABA HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01550I 2 OF 2 SPECIFIED MEMBERS WERE COPIED
IEB147I END OF JOB - 0 WAS HIGHEST SEVERITY CODE
```

Scenario 4

In this scenario a copy of a PDS to PDS where the output data set is not empty. COPY INDD OUTDD. SELECT a member that has an alias is specified. Observe that only the member is copied.

In Example 7-7 on page 110 we show the JCL that was run to show the use of the COPY statement with the SELECT minor statement to select a member that has an alias but we do not want the alias to be copied. In this example INDD=((INDD,R)) was specified so that the member will be replaced.

Example 7-7 JCL to use COPY to copy a member without any alias

```
//MHLRES2C JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
/* example of use of COPYGROUP to contrast with COPYGRP
```



```
//S1      EXEC PGM=IEBCOPY
//SYSPRINT DD  SYSOUT=*
//INDD     DD  DISP=SHR,DSN=SYS1.LINKLIB
//OUTDD    DD  DISP=SHR,DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
//SYSIN    DD  *
COPY      INDD=((INDD,R)),OUTDD=OUTDD
SELECT MEMBER=(IKJEFA00)
```

In Example 7-8 on page 111 you find the IEBCOPY messages showing that only the specified main member was copied.

Example 7-8 IEBCOPY messages showing that COPY copied just the specified member

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS    02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2C S1    14:20:38 TUE 17 SEP 2013 PARM=' '
COPY      INDD=((INDD,R)),OUTDD=OUTDD                                00010008
SELECT MEMBER=(IKJEFA00)                                           00020008
IEB1013I COPYING FROM PDS INDD=INDD  VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I          TO PDS  OUTDD=OUTDD  VOL=MLD83A
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IEB167I FOLLOWING MEMBER(S) COPIED FROM INPUT DATA SET REFERENCED BY INDD
IEB154I IKJEFA00 HAS BEEN SUCCESSFULLY COPIED
IEB1098I 1 OF 1 MEMBERS COPIED FROM INPUT DATA SET REFERENCED BY INDD
IEB144I THERE ARE 747 UNUSED TRACKS IN OUTPUT DATA SET REFERENCED BY OUTDD
IEB149I THERE ARE 9 UNUSED DIRECTORY BLOCKS IN OUTPUT DIRECTORY
IEB147I END OF JOB - 0 WAS HIGHEST SEVERITY CODE
```

Scenario 5

In this scenario we show a copy of a PDS to a PDS where the output data set is not empty using COPYGROUP INDD OUTDD. SELECT the alias of a member is specified. Observe that the alias is not copied because the member is present.

In Example 7-9 on page 111 we show the JCL that was run to show the use of the COPYGROUP statement with the SELECT minor statement to select an alias of a member that already exists in the output.

Example 7-9 JCL to copy alias ACCOUNT

```
//MHLRES2C JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
/* example of use of COPYGROUP to contrast with COPYGRP
//S1      EXEC PGM=IEBCOPY
//SYSPRINT DD  SYSOUT=*
//INDD     DD  DISP=SHR,DSN=SYS1.LINKLIB
//OUTDD    DD  DISP=SHR,DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
//SYSIN    DD  *
COPYGROUP INDD=INDD,OUTDD=OUTDD
SELECT MEMBER=(ACCOUNT)
```

In Example 7-10 on page 112 we show the IEBCOPY messages showing that the specified alias member was not copied because of a NO-REPLACE CONFLICT, which is as expected.

Example 7-10 IEBCOPY messages showing selection of an alias for a member that already exists is not completed

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS    02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2C S1    14:29:59 TUE 17 SEP 2013 PARM=' '
COPYGROUP INDD=INDD,OUTDD=OUTDD                                00010010
SELECT MEMBER=(ACCOUNT)                                         00020011
IEB1013I COPYING FROM PDS  INDD=INDD  VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I          TO PDS  OUTDD=OUTDD  VOL=MLD83A
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IGW01557W MEMBER IKJEFA00 NOT COPIED BECAUSE THE NAME ALREADY EXISTS IN
THE OUTPUT DATASET CAUSING A COPY GROUP NO-REPLACE CONFLICT
IGW01558W MEMBER ACCOUNT  NOT COPIED BECAUSE IT BELONGS TO A GROUP
IN WHICH A PREVIOUS NAME PROCESSED ENCOUNTERED COPY GROUP NO REPLACE CONFLICT
IGW01550I 0 OF 2 SPECIFIED MEMBERS WERE COPIED
IEB1130I A WARNING MESSAGE FROM FAMS PROCESSING APPEARS ABOVE -- DIAGNOSTIC
INFORMATION IS X'2810022D'
IEB147I END OF JOB - 0 WAS HIGHEST SEVERITY CODE
```

Scenario 6

In this scenario we copy PDS to PDS where the output data set is not empty. COPYGROUP INDD,R OUTDD is specified. SELECT the alias of a member. Observe that neither the alias nor the main member is replaced because of a COPYGROUP conflict.

In Example 7-9 on page 111 you find the JCL that was run to show the use of the COPYGROUP statement with the SELECT minor statement to select an alias of a member that already exists in the output. In this case replace the main member name using INDD=((INDD,R)).

Example 7-11 JCL to copy a member alias for a main member that exists, but replace it

```
//MHLRES2C JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
/* example of use of COPYGROUP to contrast with COPYGRP
//S1      EXEC PGM=IEBCOPY
//SYSPRINT DD  SYSOUT=*
//INDD    DD  DISP=SHR,DSN=SYS1.LINKLIB
//OUTDD   DD  DISP=SHR,DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
//SYSIN   DD  *
COPYGROUP INDD=((INDD,R)),OUTDD=OUTDD
SELECT MEMBER=(ACCOUNT)
```

In Example 7-12 on page 113 you find the IEBCOPY messages showing that this COPYGROUP did not complete successfully because of a CONDITIONAL-REPLACE conflict. COPYGROUP is designed to ensure that members of a group are consistent, and that they do not affect members of any other group.

Example 7-12 IEBCOPY messages illustrating CONDITIONAL-REPLACE conflict

```

IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS      02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2C S1   14:56:23 WED 18 SEP 2013 PARM=' '
COPYGROUP INDD=((INDD,R)),OUTDD=OUTDD                                00010011
SELECT MEMBER=(ACCOUNT)                                             00020016
IEB1013I COPYING FROM PDS  INDD=INDD  VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I              TO PDS  OUTDD=OUTDD  VOL=MLD83A
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IGW01559E MEMBER IKJEFA00 NOT COPIED BECAUSE OF COPY GROUP
CONDITIONAL-REPLACE CONFLICT, SINCE AT LEAST ONE NAME IN THIS INPUT
GROUP BELONGS TO A DIFFERENT OUTPUT GROUP
IGW01559E MEMBER ACCOUNT NOT COPIED BECAUSE OF COPY GROUP
CONDITIONAL-REPLACE CONFLICT, SINCE AT LEAST ONE NAME IN THIS INPUT
GROUP BELONGS TO A DIFFERENT OUTPUT GROUP
IGW01550I 0 OF 2 SPECIFIED MEMBERS WERE COPIED
IEB1130W AN ERROR MESSAGE FROM FAMS PROCESSING APPEARS ABOVE -- DIAGNOSTIC
INFORMATION IS X'2810022F'
IEB151I JOB HAS TERMINATED WITH ERROR(S)
IEB147I END OF JOB - 4 WAS HIGHEST SEVERITY CODE

```

The reason for the COPY GROUP CONDITIONAL-REPLACE CONFLICT is the COPY that was done in Scenario 4. We specified to copy member IKJEFA00 but did not also specify alias ACCOUNT, and because we were using COPY there was no automatic inclusion of ACCOUNT.

The listing of the members of the output data set MHLRES2.SYS1.LINKLIB.IEBCOPY1 is shown in Example 7-13 on page 113.

The significant points to note are:

- ▶ Members ABA and the alias ARCABA have the same TTR reference 000121
- ▶ Member ACCOUNT shows that it is an alias of IKJEFA00 and has TTR reference 00031F
- ▶ Member IKJEFA00 shows that it is at TTR 00032C

This effectively means that ACCOUNT and IKJEFA00 are not the same member any longer although they may have the same content. COPYGROUP recognizes this and refuses to do the copy.

Example 7-13 Listing of members in output data set showing TTR inconsistency

EDIT	MHLRES2.SYS1.LINKLIB.IEBCOPY1				Row 0000001 of 0000004			
Command ==>					Scroll ==> CSR			
	Name	Prompt	Alias-of	Size	TTR	AC	AM	RM
_____	ABA		ARCABA	00004140	000121	00	31	24
_____	ACCOUNT		IKJEFA00	00001430	00031F	00	24	24
_____	ARCABA			00004140	000121	00	31	24
_____	IKJEFA00			00001430	00032C	00	24	24
	End							

This situation can not be corrected by using COPYGROUP, it has to be corrected by using COPY and specifying (in this case) both the main member name IKJEFA00 and its alias ACCOUNT.

Scenario 7

In this scenario we copy PDS to PDS where the output data set is not empty using COPY INDD,R OUTDD. SELECT the member and related alias of the member. Observe that neither the alias nor the main member is replaced because of a COPY GROUP conflict.

In Example 7-14 on page 114 you find the JCL that was run to show the use of the COPY statement with the SELECT minor statement to select a member and its alias that already exist in the output data set. In this case we need to replace the main member name and alias so we used INDD=((INDD,R)).

Example 7-14 JCL to COPY member and alias to correct inconsistency

```
//MHLRES2C JOB (999,P0K),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
/* example of use of COPYGROUP to contrast with COPYGRP
//S1      EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//INDD    DD DISP=SHR,DSN=SYS1.LINKLIB
//OUTDD   DD DISP=SHR,DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
//SYSIN   DD *
COPY      INDD=((INDD,R)),OUTDD=OUTDD
SELECT MEMBER=(IKJEFA00)
SELECT MEMBER=(ACCOUNT)
```

In Example 7-15 on page 114 we show the IEBCOPY messages showing that this COPY did complete successfully because both the member name and its alias were specified on the SELECT minor statement.

Example 7-15 IEBCOPY messages showing that both member and alias were copied

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS    02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2C S1    17:34:22 WED 18 SEP 2013 PARM=' '
COPY      INDD=((INDD,R)),OUTDD=OUTDD                                00010000
SELECT MEMBER=(IKJEFA00)                                           00020000
SELECT MEMBER=(ACCOUNT)                                           00030009
IEB1013I COPYING FROM PDS INDD=INDD VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I          TO PDS OUTDD=OUTDD VOL=MLD83A
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IEB167I FOLLOWING MEMBER(S) COPIED FROM INPUT DATA SET REFERENCED BY INDD
IEB154I ACCOUNT HAS BEEN SUCCESSFULLY COPIED
IEB154I IKJEFA00 HAS BEEN SUCCESSFULLY COPIED
IEB1098I 2 OF 2 MEMBERS COPIED FROM INPUT DATA SET REFERENCED BY INDD
IEB144I THERE ARE 745 UNUSED TRACKS IN OUTPUT DATA SET REFERENCED BY OUTDD
IEB149I THERE ARE 9 UNUSED DIRECTORY BLOCKS IN OUTPUT DIRECTORY
IEB147I END OF JOB - 0 WAS HIGHEST SEVERITY CODE
```

The reason for the COPY GROUP CONDITIONAL-REPLACE CONFLICT is the COPY that was done in Scenario 4. We specified to copy member IKJEFA00 but did not also specify alias ACCOUNT, and because we were using COPY there was no automatic inclusion of ACCOUNT.

The listing of the members of the output data set MHLRES2.SYS1.LINKLIB.IEBCOPY1 is shown in Example 7-16 on page 115.

The significant points to note are:

- ▶ Members ABA and the alias ARCABA have the same TTR reference 000121 as before.
- ▶ Member ACCOUNT shows that it is an alias of IKJEFA00 and has TTR reference 000406.
- ▶ Member IKJEFA00 shows that it is at TTR 000406.

This effectively means that ACCOUNT and IKJEFA00 are now the same member and are located at a new location in the output data set.

Example 7-16 Listing of members in output data set showing TTR consistency within groups

BROWSE		MHLRES2.SYS1.LINKLIB.IEBCOPY1				Row 0000001 of 0000004			
Command ==>						Scroll ==> CSR			
	Name	Prompt	Alias-of	Size	TTR	AC	AM	RM	
_____	ABA		ARCABA	00004140	000121	00	31	24	
_____	ACCOUNT		IKJEFA00	00001430	000406	00	24	24	
_____	ARCABA			00004140	000121	00	31	24	
_____	IKJEFA00			00001430	000406	00	24	24	

We have illustrated the value of using COPYGROUP to ensure that members and their aliases remain consistent.

Scenario 8

Using the COYGROUP statement demonstrate the use of SELECT minor statement with wildcard character to specify the primary member(s) to be copied. Expect to see alias names included when their primary member is specified for inclusion.

SYS1.LINKLIB is the source data set.

There are a number of members that start with ANTX, and a number that start with IKJ. Select these with ANTX*, and I%J* respectively

In Example 7-17 on page 115 you find the JCL that was run to show the use of the COPYGROUP statement with the SELECT minor statement to select two sets of members:

- ▶ With prefix ANTX and their aliases.
- ▶ With prefix IxK (where x is any character in position two) and their aliases.

The members may already exist in the output data set so we used INDD=((INDD,R)) so that they will get replaced.

Example 7-17 JCL to run COPYGROUP with SELECT statements with wild cards

```
//MHLRES2C JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
/* example of use of COPYGROUP to contrast with COPYGRP
//S1      EXEC PGM=IEBCOPY
```

```
//SYSPRINT DD SYSOUT=*
//INDD DD DISP=SHR,DSN=SYS1.LINKLIB
//OUTDD DD DISP=SHR,DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
//SYSIN DD *
COPYGROUP INDD=((INDD,R)),OUTDD=OUTDD
SELECT MEMBER=(ANTX*)
SELECT MEMBER=(I%J*)
```

In Example 7-18 on page 116 we show the IEBCOPY messages resulting from the member name SELECT specifications. In particular we see that there are many alias names that are automatically selected even though their names do not match the specified name selection pattern. We noted that 75 members and aliases were copied.

Example 7-18 IEBCOPY messages showing main members selected using wild cards

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS PAGE 1
IEB1135I IEBCOPY FMID HDZ2210 SERVICE LEVEL UA70502 DATED 20130821 DFSMS
02.01.00 z/OS 02.01.00 HBB7790 CPU 2827
IEB1035I MHLRES2C S1 20:25:25 WED 18 SEP 2013 PARM=' '
COPYGROUP INDD=((INDD,R)),OUTDD=OUTDD 00010000
SELECT MEMBER=(ANTX*) 00020017
SELECT MEMBER=(I%J*) 00030017
IEB1013I COPYING FROM PDS INDD=INDD VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I TO PDS OUTDD=OUTDD VOL=MLDA39
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IGW01264I TOTAL PRIMARY NAMES: 3047, FILTER PATTERN MATCHES: 37
IGW01554I ALIAS ACCOUNT OF COPIED PRIMARY IKJEFA00 HAS BEEN COPIED
AND REPLACED
IGW01552I MEMBER ANTXCRCM HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER ANTICAL HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER ANTITSO HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01554I ALIAS CDELPAR OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
AND REPLACED
IGW01553I ALIAS CDELPATH OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
IGW01554I ALIAS CESTPAIR OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
AND REPLACED
IGW01554I ALIAS CESTPATH OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
AND REPLACED
IGW01554I ALIAS CGROUP OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
AND REPLACED
IGW01554I ALIAS CQUERY OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
AND REPLACED
IGW01553I ALIAS CRECOVER OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
IGW01553I ALIAS CSUSPEND OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
IGW01553I ALIAS FCESTABL OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
IGW01553I ALIAS FCQUERY OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
IGW01553I ALIAS FCWITHDR OF COPIED PRIMARY ANTITSO HAS BEEN COPIED
IGW01554I ALIAS GETMSG OF COPIED PRIMARY IKJCNUGW HAS BEEN COPIED
AND REPLACED
IGW01552I MEMBER IKJCNANT HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJCNASR HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01552I MEMBER IKJCNUGW HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
```

IGW01554I ALIAS IKJCNMI OF COPIED PRIMARY IKJCNASR HAS BEEN COPIED
AND REPLACED
IGW01554I ALIAS IKJCT467 OF COPIED PRIMARY IKJCT469 HAS BEEN COPIED
AND REPLACED
IGW01552I MEMBER IKJCT469 HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01552I MEMBER IKJEES40 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFA00 HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01552I MEMBER IKJEFA10 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFA20 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFA30 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFA40 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFD38 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFD40 HAS BEEN COPIED AND REPLACED
IGW01554I ALIAS IKJEFF03 OF COPIED PRIMARY IKJEFF04 HAS BEEN COPIED
AND REPLACED

IEBCOPY MESSAGES AND CONTROL STATEMENTS

PAGE 2
IGW01552I MEMBER IKJEFF04 HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01552I MEMBER IKJEFF10 HAS BEEN COPIED AND REPLACED
IGW01554I ALIAS IKJEFF15 OF COPIED PRIMARY IKJEFF04 HAS BEEN COPIED
AND REPLACED
IGW01552I MEMBER IKJEFF18 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFF19 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFF50 HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01552I MEMBER IKJEFF51 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFF53 HAS BEEN COPIED AND REPLACED
IGW01554I ALIAS IKJEFF55 OF COPIED PRIMARY IKJEFF50 HAS BEEN COPIED
AND REPLACED
IGW01552I MEMBER IKJEFF57 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFT25 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJEFXSR HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJHCA00 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJHCM00 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJHCR00 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJLDI00 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJLDI99 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJLHENP HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJLHENU HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJPRMSG HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJPRM01 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJPRM03 HAS BEEN COPIED AND REPLACED
IGW01552I MEMBER IKJRBBUO HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01552I MEMBER IKJTSEV HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01554I ALIAS RQUERY OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
AND REPLACED
IGW01553I ALIAS RSESSION OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS RVOLUME OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01554I ALIAS TSOENV OF COPIED PRIMARY IKJTSEV HAS BEEN COPIED
AND REPLACED

```

IGW01554I ALIAS UADSREFM OF COPIED PRIMARY IKJRBBUO   HAS BEEN COPIED
AND REPLACED
IGW01553I ALIAS XADD OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01554I ALIAS XADDPAIR OF COPIED PRIMARY ANTXITSO   HAS BEEN COPIED
AND REPLACED
IGW01553I ALIAS XADVANCE OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS XCLIP OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS XCOUPLE OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS XDEL OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS XDELPAIR OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS XEND OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS XQUERY OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS XRCV OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS XRECOVER OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01554I ALIAS XSET OF COPIED PRIMARY ANTXITSO   HAS BEEN COPIED
AND REPLACED
IGW01554I ALIAS XSTART OF COPIED PRIMARY ANTXITSO   HAS BEEN COPIED
AND REPLACED
IGW01553I ALIAS XSTATUS OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01553I ALIAS XSUSPEND OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
IGW01550I 75 OF 75 SPECIFIED MEMBERS WERE COPIED
IEBCOPY MESSAGES AND CONTROL STATEMENTS
PAGE      3
IEB147I END OF JOB - 0 WAS HIGHEST SEVERITY CODE

```

Scenario 9

In this scenario the COPYGROUP statement is used to demonstrate the use of SELECT and EXCLUDE minor statements with wildcard characters to specify the members to be copied. Expect to see alias names excluded when their primary member is specified for exclusion.

SYS1.LINKLIB is the source data set.

There are a number of members that start with ANTX, and a number that start with IKJ.

Select these with ANTX*, and I%J* respectively and check whether any other members with a character other than K in position two of the member name.

Exclude the subset of I%J members that have EFF in positions 4 through 6 of the name

In Example 7-19 on page 119 we show the JCL that was run to show the use of the COPYGROUP statement with the SELECT minor statements to specify two sets of members and use of the EXCLUDE minor statement to exclude members from the second SELECT set.

- ▶ Select members with prefix ANTX and let COPYGROUP pick up their aliases.
- ▶ Select members with prefix IxK (where x is any character in the second position) and let COPYGROUP pick up their aliases.
- ▶ Exclude members with prefix IxJEFF to specify a subset of those with prefix IxK. Let COPYGROUP pick up their aliases and exclude them as well.

Note: The EXCLUDE minor statement is only allowed after a SELECT minor statement.

The members may already exist in the output data set so we used INDD=((INDD,R)) so that they will get replaced.

Example 7-19 JCL showing COPYGROUP with SELECT and EXCLUDE using wild cards

```

000010 //MHLRES2C JOB (999,P0K), 'MHLRES2', CLASS=A, MSGCLASS=T,
000020 //          NOTIFY=&SYSUID, TIME=1440, REGION=6M
000030 /*JOBPARM L=999, SYSAFF=*
000031 /* example of use of COPYGROUP to contrast with COPYGRP
000040 //S1      EXEC PGM=IEBCOPY
000050 //SYSPRINT DD  SYSOUT=*
000051 //INDD    DD  DISP=SHR, DSN=SYS1.LINKLIB
000060 //OUTDD   DD  DISP=SHR, DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
000080 //SYSIN   DD  *
000100 COPYGROUP INDD=((INDD,R)), OUTDD=OUTDD
000200 SELECT MEMBER=(ANTX*)
000300 SELECT MEMBER=(I%J*)
000400 EXCLUDE MEMBER=(I%JEFF*)

```

In Example 7-20 on page 119 you find the IEBCOPY messages resulting from the COPYGROUP major and SELECT and EXCLUDE minor statements. In particular we see that there are many alias names that are automatically selected even though their names do not match the specified name selection pattern.

We noted that 64 member names were copied. This contrasts with the corresponding messages from Scenario 8 as in Example 7-18 on page 116 where 75 members were copied.

The difference is due to the EXCLUDE minor statement that has caused members and aliases to be omitted.

Example 7-20 IEBCOPY messages showing the number of members copied due to EXCLUDE

```

IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS    02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2C S1    20:29:11 WED 18 SEP 2013 PARM=' '
COPYGROUP INDD=((INDD,R)), OUTDD=OUTDD                                00010000
SELECT MEMBER=(ANTX*)                                                00020000
SELECT MEMBER=(I%J*)                                                00030000
EXCLUDE MEMBER=(I%JEFF*)                                            00040018
IEB1013I COPYING FROM PDS  INDD=INDD  VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I          TO PDS  OUTDD=OUTDD  VOL=MLDA39
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IGW01264I TOTAL PRIMARY NAMES: 3047, FILTER PATTERN MATCHES: 29
IGW01554I ALIAS ACCOUNT OF COPIED PRIMARY IKJEFA00  HAS BEEN COPIED
AND REPLACED
IGW01552I MEMBER ANTXCRSM HAS BEEN COPIED  AND REPLACED
IGW01552I MEMBER ANTXICAL HAS BEEN COPIED  AND REPLACED
IGW01552I MEMBER ANTXITSO HAS BEEN COPIED  AND REPLACED
A PRIMARY WITH ALIAS(ES)
IGW01554I ALIAS CDELPAR OF COPIED PRIMARY ANTXITSO  HAS BEEN COPIED
AND REPLACED
IGW01554I ALIAS CDELPATH OF COPIED PRIMARY ANTXITSO  HAS BEEN COPIED
AND REPLACED
IGW01554I ALIAS CESTPAIR OF COPIED PRIMARY ANTXITSO  HAS BEEN COPIED
AND REPLACED

```

IGW01554I ALIAS CESTPATH OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
AND REPLACED

IGW01554I ALIAS CGROUP OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
AND REPLACED

IGW01554I ALIAS CQUERY OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
AND REPLACED

IGW01554I ALIAS CRECOVER OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
AND REPLACED

IGW01554I ALIAS CSUSPEND OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
AND REPLACED

IGW01554I ALIAS FCESTABL OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
AND REPLACED

IGW01554I ALIAS FCQUERY OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
AND REPLACED

IGW01554I ALIAS FCWITHDR OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
AND REPLACED

IGW01554I ALIAS GETMSG OF COPIED PRIMARY IKJCNUGW HAS BEEN COPIED
AND REPLACED

IGW01552I MEMBER IKJCNANT HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJCNASR HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)

IGW01552I MEMBER IKJCNUGW HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)

IGW01554I ALIAS IKJCNMI OF COPIED PRIMARY IKJCNASR HAS BEEN COPIED
AND REPLACED

IGW01554I ALIAS IKJCT467 OF COPIED PRIMARY IKJCT469 HAS BEEN COPIED
AND REPLACED

IGW01552I MEMBER IKJCT469 HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)

IGW01552I MEMBER IKJEES40 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJEFA00 HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)

IGW01552I MEMBER IKJEFA10 HAS BEEN COPIED AND REPLACED

IEBCOPY MESSAGES AND CONTROL STATEMENTS

PAGE 2

IGW01552I MEMBER IKJEFA20 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJEFA30 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJEFA40 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJEFD38 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJEFD40 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJEFT25 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJEFXSR HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJHCA00 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJHCM00 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJHCR00 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJLDI00 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJLDI99 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJLHENP HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJLHENU HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJPRMSG HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJPRM01 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJPRM03 HAS BEEN COPIED AND REPLACED

IGW01552I MEMBER IKJRBBU0 HAS BEEN COPIED AND REPLACED
A PRIMARY WITH ALIAS(ES)

IGW01552I MEMBER IKJTSOEV HAS BEEN COPIED AND REPLACED

A PRIMARY WITH ALIAS(ES)
 IGW01554I ALIAS RQUERY OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS RSESSION OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS RVOLUME OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS TSOENV OF COPIED PRIMARY IKJTSEV HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS UADSREFM OF COPIED PRIMARY IKJRBBUO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XADD OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XADDPAIR OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XADVANCE OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XCLIP OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XCOUPLE OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XDEL OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XDPAIR OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XEND OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XQUERY OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XRCV OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XRECOVER OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XSET OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XSTART OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED

IEBCOPY MESSAGES AND CONTROL STATEMENTS

PAGE 3
 IGW01554I ALIAS XSTATUS OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01554I ALIAS XSUSPEND OF COPIED PRIMARY ANTXITSO HAS BEEN COPIED
 AND REPLACED
 IGW01550I 64 OF 64 SPECIFIED MEMBERS WERE COPIED
 IEB147I END OF JOB - 0 WAS HIGHEST SEVERITY CODE

Scenario 10

In this scenario COPYGRP limitation when working with PDS to PDS copying.

We have illustrated the functions of the COPYGROUP statement when processing PDS to PDS functions. COPYGROUP expanded on COPYGRP functions by including the capability to handle PDS to PDS that were not available with COPYGRP. COPYGRP when used on PDS behaves like COPY; that is it does *not* copy the related members.

In Example 7-21 on page 122 you find the job used to demonstrate that COPYGRP only copies what is specified. In this case member AD was selected which is one alias of the main member name IRRENV00 which was not selected.

Example 7-21 JCL for job to show COPYGRP on PDS not copying the group of data sets

```
//MHLRES2C JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
/* example of use of COPYGRP to contrast with COPYGROUP
//S1      EXEC PGM=IEBCOPY
//SYSPRINT DD  SYSOUT=*
//INDD    DD  DISP=SHR,DSN=SYS1.LINKLIB
//OUTDD   DD  DISP=SHR,DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
//SYSIN   DD  *
COPYGRP  INDD=INDD,OUTDD=OUTDD
SELECT  MEMBER=(AD)
```

In Example 7-22 on page 122 we show that IEBCOPY COPYGRP only copied member AD, which is an alias.

Example 7-22 IEBCOPY messages showing only member AD was copied

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS    02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2C S1    12:33:21 THU 19 SEP 2013 PARM=' '
COPYGRP  INDD=INDD,OUTDD=OUTDD                                00010007
SELECT  MEMBER=(AD)                                           00020007
IEB1013I COPYING FROM PDS  INDD=INDD  VOL=Z21RB1 DSN=SYS1.LINKLIB
IEB1014I          TO PDS  OUTDD=OUTDD  VOL=MLDA39
DSN=MHLRES2.SYS1.LINKLIB.IEBCOPY1
IEB167I FOLLOWING MEMBER(S) COPIED FROM INPUT DATA SET REFERENCED BY INDD
IEB154I AD      HAS BEEN SUCCESSFULLY COPIED
IEB1098I 1 OF 1 MEMBERS COPIED FROM INPUT DATA SET REFERENCED BY INDD
IEB144I THERE ARE 651 UNUSED TRACKS IN OUTPUT DATA SET REFERENCED BY OUTDD
IEB149I THERE ARE 35 UNUSED DIRECTORY BLOCKS IN OUTPUT DIRECTORY
IEB147I END OF JOB - 0 WAS HIGHEST SEVERITY CODE
```

Example 7-23 on page 123 shows the resulting members in the output data set. Alias name AD was copied, and has retained the information that it is an alias of main member of IRRENV00.

Also shown are member names IKJTSOEV and RQUERY. The member name IRRENV00 which should be between those two is missing.

This situation can be corrected by running IEBCOPY with COPYGRP or COPY statement and selecting the main member name IRRENV00 and all the aliases.

Alternatively the COPYGROUP statement can be used to correct the situation more easily but only after the member AD has been deleted. If member AD is not deleted a CONFLICT will occur.

Example 7-23 Listing of the output data set showing member AD as alias of IRRENV00

LIBRARY MHLRES2.SYS1.LINKLIB.IEBCOPY1				Row 0000001 of 0000078			
Command ==>				Scroll ==> CSR			
Name	Prompt	Alias-of	Size	TTR	AC	AM	RM
_ ABA		ARCABA	00004140	000107	00	31	24
_ ACCOUNT		IKJEFA00	00001430	003C16	00	24	24
_ AD		IRRENV00	0004D708	005B08	01	31	24
.							
.							
.							
.							
_ IKJTSOEV			000000D8	004219	00	31	ANY
_ RQUERY		ANTXITSO	00069CD8	004A04	01	31	ANY

Scenarios 1through 10 illustrate the advantage of using the DFSMS V2.1 new COPYGROUP statement.

PDS or PDSE to PDS operations

The original COPYGRP statement does work for PDSE output data sets. The restriction is only for PDS output data sets when COPYGRP behaves if the COPY statement had been specified, subject to the syntax rules for COPYGRP.

Unloading PDS to PDSE data sets to physical sequential data sets

COPYGRP will only work (in group mode) for PDSE to physical sequential (PS). If the input is a PDS then only the name specified on a SELECT statement will be copied.

COPYGROUP will work for PDSE or PDS to PS.

Loading unloaded PDS or PDSE data sets from PS data sets

COPYGRP will only work for PS to PDSE.

COPYGROUP will work for PS to PDS or to PDSE.

Compatibility

The COPYGROUP statement is not available on DFSMS releases prior to DFSMS V2.1.

The COPYGRP statement as provided in DFSMS V2.1 is functionally the same as on releases prior to DFSMS V2.1.

7.2 IEBCOPY user exit capabilities

There are significant enhancements to the user exit capability for IEBCOPY. They are only available when IEBCOPY is invoked from another program. The new exit capability allows:

- ▶ Generation of an input stream for IEBCOPY without using a JCL managed SYSIN stream.
- ▶ Feedback of output to be added to the SYSPRINT stream.

7.2.1 Exit capabilities

There are two exit options, either one or both may be used. We created an example of the use of the control statement exit to supply SYSIN statements.

Control statement exit

The control statement exit may be used to specify statements to be passed as if from SYSIN, and to provide data to be passed to the IEBCOPY SYSPRINT stream.

We created a program (IEBCPYL) to illustrate the use of the control statement exit, while also using the page number setting option that was available prior to DFSMS V2.1.

The assembler source for IEBCPYL is provided in the Appendix at Example C-6 on page 402.

In Example 7-24 on page 124 we show the JCL to run the program IEBCPYL that invokes the IEBCOPY copy program to use the control statement exit. The IEBCPYL program is for illustrative purposes only and issues diagnostic Write-To-Operator (WTO) messages. The JCL is standard for IEBCOPY but note that no SYSIN is present. Normally the absence of a SYSIN JCL statement would cause IEBCOPY to generate an internal COPY from DD SYSUT1 to DD SYSUT2 to pass to the SYSIN stream.

The exit generates the following statements:

- ▶ COPY INDD=SYSUTX,OUTDD=SYSUTY
- ▶ SELECT MEMBER=A

Example 7-24 JCL to invoke IEBCPYL to use the Control Statement Exit capability

```
//MHLRES2E JOB (1234567,COMMENT),MHLRES2,TIME=10,
// MSGCLASS=J,
// REGION=1M,
// MSGLEVEL=1,CLASS=A,
// NOTIFY=&SYSUID
/*JOBPARM S=*
//COPY      EXEC PGM=IEBCPYL,REGION=1M
//STEPLIB   DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.LOAD
//SYSPRINT  DD SYSOUT=*
//SYSUDUMP  DD SYSOUT=*
//SYSUTX    DD DISP=SHR,DSN=SYS1.SIEALNKE
//SYSUTY    DD DISP=(,DELETE),UNIT=SYSALLDA,SPACE=(TRK,(050,1,100)),
//          DCB=(SYS1.SIEALNKE),DSNTYPE=LIBRARY
```

We show the result of running program IEBCPYL in two parts. The first part is the joblog that shows the WTO messages along with the usual SYSOUT. The second part shows the IEBCOPY messages.

The control statement exit invoked by IEBCPYL is called IEBCPYC for this example.

The assembler source for IEBCPYC is provided in Example C-8 on page 404.

In Example 7-25 on page 125 we show the system output from running the IEBCPYL program. The WTO messages are:

- ▶ GOT TO IEBCPYEC - issued each time the IEBCPYC is entered
- ▶ CPLST_CONTROL_INIT - issued when IEBCOPY calls IEBCPYC for initialization
- ▶ CPLST_CONTROL_DATA - n - issued when IEBCOPY calls IEBCPYC for data or to END

Example 7-25 Output resulting from running IEBCPYL (JOBLOG)

```
JES2 JOB LOG -- SYSTEM SC64 -- NODE WTSCPLX2

15.17.46 JOB19260 ---- SUNDAY, 27 OCT 2013 ----
15.17.46 JOB19260 IRRO10I USERID MHLRES2 IS ASSIGNED TO THIS JOB.
15.17.46 JOB19260 ICH70001I MHLRES2 LAST ACCESS AT 15:17:11 ON SUNDAY, OCTOBER
27, 2013
15.17.46 JOB19260 $HASP373 MHLRES2E STARTED - INIT 1 - CLASS A - SYS
SC64
15.17.46 JOB19260 IEF403I MHLRES2E - STARTED - TIME=15.17.46 - ASID=0024 - SC64
15.17.46 JOB19260 +GOT TO IEBCPYEC
15.17.46 JOB19260 +CPLST_CONTROL_INIT
15.17.46 JOB19260 +GOT TO IEBCPYEC
15.17.46 JOB19260 +CPLST_CONTROL_DATA - 1
15.17.46 JOB19260 +GOT TO IEBCPYEC
15.17.46 JOB19260 +CPLST_CONTROL_DATA - 2
15.17.46 JOB19260 +GOT TO IEBCPYEC
15.17.46 JOB19260 +CPLST_CONTROL_DATA - 3
15.17.46 JOB19260 +GOT TO IEBCPYEC
15.17.46 JOB19260 +CPLST_CONTROL_DATA - E
15.17.46 JOB19260 PROGRAM: IEBCPYL IN STEP: COPY - COND CODE: 0000
15.17.46 JOB19260 TOTAL CPU (SECS) BLKS READ/WRITE TOTAL SWAPS
15.17.46 JOB19260 0.07 1.154
15.17.46 JOB19260 IEF404I MHLRES2E - ENDED - TIME=15.17.46 - ASID=0024 - SC64
15.17.46 JOB19260 $HASP395 MHLRES2E ENDED
----- JES2 JOB STATISTICS -----
27 OCT 2013 JOB EXECUTION DATE
13 CARDS READ
111 SYSOUT PRINT RECORDS
0 SYSOUT PUNCH RECORDS
6 SYSOUT SPOOL KBYTES
0.01 MINUTES EXECUTION TIME
1 //MHLRES2E JOB (1234567,COMMENT),MHLRES2,TIME=10,
JOB19260
// MSGCLASS=J,
00020000
// REGION=1M,
00030018
// MSGLEVEL=1,CLASS=A,
00040000
// NOTIFY=&SYSUID
00050000
IEFC653I SUBSTITUTION JCL -
(1234567,COMMENT),MHLRES2,TIME=10,MSGCLASS=J,REGION=1M,MSGLEVEL=1,CLASS=A,
NOTIFY=MHLRES2
2 /*JOBPARM S=*
00060000
3 //COPY EXEC PGM=IEBCPYL,REGION=1M
00070018
4 //STEPLIB DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.LOAD
00080000
5 //SYSPRINT DD SYSOUT=*
00090000
6 //SYSUDUMP DD SYSOUT=*
00100000
```

```

      7 //SYSUTX DD DISP=SHR,DSN=SYS1.SIEALNKE
00110019
      8 //SYSUTY DD DISP=(,DELETE),UNIT=SYSALLDA,SPACE=(TRK,(050,1,100)),
00120019
      //          DCB=(SYS1.SIEALNKE),DSNTYPE=LIBRARY
00130000
STMT NO. MESSAGE
-
      8 IGD01008I SC ACS GETS CONTROL &ACSENVIR=ALLOC
      8 IGD01008I STORCLAS SET TO
ICH70001I MHLRES2 LAST ACCESS AT 15:17:11 ON SUNDAY, OCTOBER 27, 2013
IEF236I ALLOC. FOR MHLRES2E COPY
IEF237I 960D ALLOCATED TO STEPLIB
IEF237I JES2 ALLOCATED TO SYSPRINT
IEF237I JES2 ALLOCATED TO SYSUDUMP
IEF237I 960D ALLOCATED TO SYSUTX
IGD100I A72B ALLOCATED TO DDNAME SYSUTY DATACLAS ( )
GOT TO IEBCPYEC
CPLST_CONTROL_INIT
GOT TO IEBCPYEC
CPLST_CONTROL_DATA - 1
GOT TO IEBCPYEC
CPLST_CONTROL_DATA - 2
GOT TO IEBCPYEC
CPLST_CONTROL_DATA - 3
GOT TO IEBCPYEC
CPLST_CONTROL_DATA - E
IEF142I MHLRES2E COPY - STEP WAS EXECUTED - COND CODE 0000
IEF285I MHLRES2.IEBCOPY.DFSMS21.LOAD KEPT
IEF285I VOL SER NOS= Z21RE1.
IEF285I MHLRES2.MHLRES2E.JOB19260.D0000101.? SYSOUT
IEF285I MHLRES2.MHLRES2E.JOB19260.D0000102.? SYSOUT
IEF285I SYS1.SIEALNKE KEPT
IEF285I VOL SER NOS= Z21RE1.
IEF285I SYS13300.T151746.RA000.MHLRES2E.R0111849 DELETED
IEF285I VOL SER NOS= SBOX76.
*****
* DDNAME : BLKS READ/WRITE *
* STEPLIB 3 *
* SYSPRINT *
* SYSUDUMP *
* SYSUTX 649 *
* SYSUTY 502 *
*****
IEF373I STEP/COPY /START 2013300.1517
IEF032I STEP/COPY /STOP 2013300.1517
CPU: 0 HR 00 MIN 00.07 SEC SRB: 0 HR 00 MIN 00.00 SEC
VIRT: 1024K SYS: 352K EXT: 40K SYS: 50724K
ATB- REAL: 20K SLOTS: 0K
VIRT- ALLOC: 3M SHRD: 0M
IEF375I JOB/MHLRES2E/START 2013300.1517
IEF033I JOB/MHLRES2E/STOP 2013300.1517
CPU: 0 HR 00 MIN 00.07 SEC SRB: 0 HR 00 MIN 00.00 SEC

```

In Example 7-26 on page 127 we show the IEBCOPY output resulting from running program IEBCPYL.

- ▶ The page number has been set to PAGE 4 to illustrate use of the PAGE NUM option.
- ▶ The COPY INDD=SYSUTX,OUTDD=SYSUTY statement has been generated and executed. The copy step failed due to lack of space in the output data set.
- ▶ The COPY INDD=SYSUTX,OUTDD=SYSUTY has been generated again, together with the SELECT MEMBER=A statement. The copy step failed because member A is not in the input data set.

Example 7-26 Output resulting from running IEBCPYL (IEBCOPY)

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    4
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS    02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2E COPY   15:17:46 SUN 27 OCT 2013 PARM=' '
      COPY INDD=SYSUTX,OUTDD=SYSUTY
IEB1013I COPYING FROM PDSE INDD=SYSUTX VOL=Z21RE1 DSN=SYS1.SIEALNKE
IEB1014I          TO PDSE OUTDD=SYSUTY  VOL=SBOX76
DSN=SYS13300.T151746.RA000.MHLRES2E.R0111849
IGW01172T OUT OF SPACE CONDITION ENCOUNTERED DURING MEMBER CREATE
PROCESSING FOR MEMBER IRRSPIM , WITH REASON CODE = X'00000E37'
IGW01173S UNEXPECTED RESULTS FROM AN SMSX SERVICE. RETURN CODE WAS 20
AND REASON CODE WAS X'050BC005'
IGW01550I 0 OF 172 MEMBERS WERE COPIED
IEB1130E A TERMINATING MESSAGE FROM FAMS PROCESSING APPEARS ABOVE -- DIAGNOSTIC
INFORMATION IS X'281D00AD'
      COPY INDD=SYSUTX,OUTDD=SYSUTY
      SELECT MEMBER=A
IEB1013I COPYING FROM PDSE INDD=SYSUTX VOL=Z21RE1 DSN=SYS1.SIEALNKE
IEB1014I          TO PDSE OUTDD=SYSUTY  VOL=SBOX76
DSN=SYS13300.T151746.RA000.MHLRES2E.R0111849
IGW01550I 0 OF 1 SPECIFIED MEMBERS WERE COPIED
IEB177I A      WAS SELECTED BUT NOT FOUND IN ANY INPUT DATA SET
IEB151I JOB HAS TERMINATED WITH ERROR(S)
IEB147I END OF JOB - 8 WAS HIGHEST SEVERITY CODE
```

Member selection exit

The member selection exit may be used to refine the list of members from a data set to be processed. It is invoked by IEBCOPY if requested, and is called by IEBCOPY which passes a data area like the one passed to the control statement exit. Additional data is provided relating to the member list.

Compatibility

The enhanced user exit capability is not provided on systems prior to DFSMS V2.1.

Note: The PARM list that can be passed to DFP utilities is extended to four addresses for IEBCOPY with DFSMS V2.1. The fourth address is for use with the new IEBCOPY exits.

It is possible that some existing programs have three addresses specified but have not indicated that the third address is the last provided. This would not have mattered prior to DFSMS V2.1 since IEBCOPY was only looking for up to three addresses. With DFSMS V2.1 if the third address is not flagged as the last one, IEBCOPY will attempt to process a fourth address with unpredictable results.

7.3 IEBCOPY return code feedback

If IEBCOPY is invoked from another program, feedback is available in DFSMS V2.1 to assist with diagnosis of any problems with the FAMS routines. The information is returned in General Register 0. The return code presented in Register 15 is not new.

The FAMS code is used when processing data sets in PDSE format.

We have written sample code that shows a way to document the special diagnosis information that is available starting with DFSMS V2.1.

In order to test the analysis routine for the data that might be returned in Register 0, the program accepts a PARM value that can be set up to simulate what might be returned in register 0.

The program calls IEBCOPY then presents messages with the results of the analysis.

In Example 7-27 on page 128 we show the JCL to run program IEBCPYF. The program is in data set MHLRES2.IEBCOPY.DFSMS21.LOAD. The source for this program is in Appendix C-1, “IEBCPYF program to display FAMS abend data 1 of 4” on page 398.

As shown in Example 7-27 the PARM is not used because there is a blank before the keyword PARM.

Example 7-27 JCL to run program IEBCPYF

```
//MHLRES2E JOB (1234567,COMMENT),MHLRES2,TIME=10,
// MSGCLASS=J,
// REGION=500K,
// MSGLEVEL=1,CLASS=A,
// NOTIFY=&SYSUID
/*JOBPARM S=*
//COPY      EXEC PGM=IEBCPYF,REGION=500K PARM='TESTREG0:123456789ABCDEF'
//STEPLIB   DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.LOAD
//SYSPRINT  DD SYSOUT=*
//PRINT     DD SYSOUT=*
//SYSUDUMP  DD SYSOUT=*
//SYSUT1    DD DISP=SHR,DSN=SYS1.SIEALNKE
//SYSUT2    DD DISP=(,DELETE),UNIT=SYSAALLDA,SPACE=(TRK,(050,1,100)),
//          DCB=(SYS1.SIEALNKE),DSNTYPE=LIBRARY
//SYSIN     DD DUMMY
```

Example 7-28 shows the messages resulting from running the JCL in Example 7-27 on page 128.

This job was set up so that it would have problems with the IEBCOPY operation, but the problems are not in the not in the FAMS code.

The IEBCOPY process ends with Return code 8, which is also reflected in the messages from IEBCOPY. In this case since the FAMS code did *not* abend, register 0 was set to all zeros.

Example 7-28 Messages from running IEBCOPY sample with no PARM

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS      02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2E COPY  15:48:31 WED 02 OCT 2013 PARM=' '
COPY COPY  INDD=SYSUT1,OUTDD=SYSUT2  GENERATED STATEMENT
IEB1013I COPYING FROM PDSE INDD=SYSUT1 VOL=Z21RE1 DSN=SYS1.SIEALNKE
IEB1014I          TO PDSE OUTDD=SYSUT2  VOL=SBOX82
DSN=SYS13275.T154831.RA000.MHLRES2E.R0100507
IGW01172T OUT OF SPACE CONDITION ENCOUNTERED DURING MEMBER CREATE
PROCESSING FOR MEMBER IRRSPIM , WITH REASON CODE = X'00000E37'
IGW01173S UNEXPECTED RESULTS FROM AN SMSX SERVICE. RETURN CODE WAS 20
AND REASON CODE WAS X'050BC005'
IGW01550I 0 OF 172 MEMBERS WERE COPIED
IEB1130E A TERMINATING MESSAGE FROM FAMS PROCESSING APPEARS ABOVE -- DIAGNOSTIC
INFORMATION IS X'281D00AD'
IEB151I JOB HAS TERMINATED WITH ERROR(S)
IEB147I END OF JOB - 8 WAS HIGHEST SEVERITY CODE
REGISTER CONTENTS AFTER LINK TO IEBCOPY
ON EXIT FROM IEBCOPY R15=0008
ON EXIT FROM IEBCOPY R0=0000000000000000
```

In order to exercise the Register 0 analysis code, the program accepts a PARM of the form PARM='TESTREG0:xxxxxxxxxxxxxx', where the xxxxxxxxxxxxxxxx string can be set to any value.

The JCL used was the same as in Example 7-27 on page 128 but the blank before the PARM= operand was removed so that the PARM takes effect. In Example 7-29 on page 129 we show the effect of running with a PARM.

Example 7-29 Messages from running IEBCOPY program with PARM

```
IEBCOPY MESSAGES AND CONTROL STATEMENTS                                PAGE    1
IEB1135I IEBCOPY  FMID HDZ2210  SERVICE LEVEL UA70502  DATED 20130821 DFSMS
02.01.00 z/OS      02.01.00 HBB7790  CPU 2827
IEB1035I MHLRES2E COPY  15:47:52 WED 02 OCT 2013 PARM=' '
COPY COPY  INDD=SYSUT1,OUTDD=SYSUT2  GENERATED STATEMENT
IEB1013I COPYING FROM PDSE INDD=SYSUT1 VOL=Z21RE1 DSN=SYS1.SIEALNKE
IEB1014I          TO PDSE OUTDD=SYSUT2  VOL=SBOX76
DSN=SYS13275.T154752.RA000.MHLRES2E.R0100506
IGW01172T OUT OF SPACE CONDITION ENCOUNTERED DURING MEMBER CREATE
PROCESSING FOR MEMBER IRRSPIM , WITH REASON CODE = X'00000E37'
IGW01173S UNEXPECTED RESULTS FROM AN SMSX SERVICE. RETURN CODE WAS 20
AND REASON CODE WAS X'050BC005'
IGW01550I 0 OF 172 MEMBERS WERE COPIED
IEB1130E A TERMINATING MESSAGE FROM FAMS PROCESSING APPEARS ABOVE -- DIAGNOSTIC
INFORMATION IS X'281D00AD'
IEB151I JOB HAS TERMINATED WITH ERROR(S)
```

```
IEB147I END OF JOB - 8 WAS HIGHEST SEVERITY CODE
REGISTER CONTENTS AFTER LINK TO IEBCOPY
ON EXIT FROM IEBCOPY R15=0008
ON EXIT FROM IEBCOPY R0=F1F2F3F4F5F6F7F8
*** WARNING - RUNNING WITH TEST PARM
REGISTER 0 CONTENTS ANALYSIS
IEBRN_ABENDED FLAG IS ON INDICATING THAT AN ABEND OCCURRED IN FAMS
IEBRN_ABENDED FLAG IS ON AND ABEND CODE IS: 7F8
IEBRN_ABENDED FLAG IS ON AND REASON CODE IS: F1F2
```

Compatibility

The Register 0 feedback is not formally provided on systems prior to DFSMS V2.1. The contents of register 0 may not have been set explicitly, so although the program could run, any interpretation would be unreliable.

7.4 Open Close and End of volume (OCE) partial release

In this section we describe the OCE partial release enhancements in DFSMS V2.1. First we look at the OCE Partial Release in systems prior to DFSMS V2.1.

7.4.1 Prior to DFSMS V2.1

In systems prior to DFSMS V2.1 unused space at the end of a data set is released under these conditions:

- ▶ Sequential or partitioned data set.
- ▶ RLSE is coded on the DD statement or the management class specifies it.
- ▶ During DFSMSHsm space management or when a program closes the data set that is open for writing.

If the data set has multiple volumes, the space is released only on one volume. Space is not released on subsequent volumes that the data set may have expanded to previously. An exception is for striped data sets where space is released on all volumes if possible.

7.4.2 DFSMS V2.1 enhancement

If a data set is SMS-managed, then all the unused space in the data set on all volumes will be released

- ▶ The Format 1 or 8 Data Set Control Block (DSCB) will remain with no extents.
- ▶ The catalog entry will still show the volume serials.
- ▶ Space is released even if the storage class specifies Guaranteed Space.

7.4.3 Illustration scenarios

We generated an environment to show the different behavior of partial release between a DFSMS V1.13 system and DFSMS V2.1 system. Scenario 1 to scenario 3 are from a DFSMS V1.13 system.

Scenario 1

In this scenario we created a data set specifying that it occupy three volumes.

In Example 7-30 on page 131 we show the JCL used to allocate the sample data set called MHLRES2.TESTREL.PS. This allows use of three volumes for data set MHLRES2.TESTREL.PS. Ten cylinders primary plus 15 extents will be allocated on the first volume then 16 extents on each of the other two volumes, making 48 extents in all.

Example 7-30 JCL to allocate data set MHLRES2.TESTREL.PS on three volumes

```
//MHLRES2H JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
// NOTIFY=&SYSUID,TIME=1440,REGION=6M
//* ALLOCATE SEQUENTIAL DATA SET ON 3 VOLUMES
/*JOBPARM L=999,SYSAFF=*
//TSO      EXEC PGM=IEFBR14
//ALLOC    DD DSN=MHLRES2.TESTREL.PS,DISP=(NEW,CATLG),
//          UNIT=(SYSDA,3),SPACE=(CYL,(10,1)),
//          DCB=(LRECL=80,RECFM=FB,BLKSIZE=160)
```

To illustrate the effect of using a data set to its full extent then show the effect of releasing space after the data set had been used to a much smaller extent we created a program to use the data set. The sample program code is provided in Appendix , “Sample job to initialize data set for OCE Partial Release” on page 425.

The program has two modes

- ▶ Run until the data set was full and all extents used (ending with abend SB37)
- ▶ Run and write only 1000 records

Scenario 2

In Example 7-31 on page 131 we show the JCL to write records to completely fill the data set MHLRES2.TESTREL.PS. Note that the JCL statement containing the RLSE parameter is commented out.

Example 7-31 JCL to write to data set on three volumes

```
//MHLRES2H JOB (1234567,COMMENT),MHLRES2,TIME=10,
// MSGCLASS=J,
// REGION=500K,
// MSGLEVEL=1,CLASS=A,
// NOTIFY=&SYSUID
/*JOBPARM S=*
//COPY      EXEC PGM=GENREC1,REGION=500K PARM='SHORT'
//STEPLIB   DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.LOAD
//SYSPRINT  DD SYSOUT=*
//PRINT     DD SYSOUT=*
//SYSUDUMP  DD SYSOUT=*
//GENDCB    DD DISP=SHR,DSN=MHLRES2.TESTREL.PS
//          SPACE=(TRK,1,RLSE)
//SYSIN     DD DUMMY
```

In Example 7-32 on page 132 we show that the data set MHLRES2.TESTREL.PS has had 57 cylinders allocated in 48 extents.

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To display multiple volumes press Enter or enter Cancel to Exit.

In Example 7-34 on page 133 we show that on the first of the three volumes that data set MHLRES2.TESTREL.PS is allocated on has used 375 tracks. This is 25 cylinders (1 of 10 plus 15 of 1 cylinder each).

Example 7-34 ISPF display showing data set MHLRES2.TESTREL.PS on the first volume

```
Menu Options View Utilities Compilers Help
ssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssss
DSLIS - Data Sets on volume MLD83A                                     Row 22 of 38
Command ==>                                                            Scroll ==> CSR

Command - Enter "/" to select action                                   Tracks %Used    XT
-----
```

MHLRES2.TESTREL.PS	375	100	16
--------------------	-----	-----	----

In Example 7-35 on page 133 we show that on the second of the three volumes that data set MHLRES2.TESTREL.PS is allocated on has used 240 tracks. The 240 tracks is 16 cylinders. The same attributes apply on the third volume which is not shown.

Example 7-35 ISPF display showing data set MHLRES2.TESTREL.PS on the second volume

DSLIST - Data Sets on volume MLDB35		CHARS '.PS' found	
Command ==>		Scroll ==> CSR	
Command - Enter "/" to select action		Tracks	%Used XT
-----		-----	
MHLRES2.TESTREL.PS	240	100	16

Scenario 3

In this scenario we write 1000 records (the number itself is not significant but it is much less than could fit in the full three volume data set. In this case we specify that unused space is to be released by including the JCL statement containing the RLSE statement.

The program PARM specified PARM=SHORT which causes it to write only 1000 records.

In Example 7-36 on page 133 we show the JCL that was used to write 1000 records to the data set MHLRES2.TESTREL.PS.

Example 7-36 JCL to write 1000 records in data set MHLRES2.TESTREL.PS

```
//MHLRES2H JOB (1234567,COMMENT),MHLRES2,TIME=10,  
// MSGCLASS=J,  
// REGION=500K,  
// MSGLEVEL=1,CLASS=A,  
// NOTIFY=&SYSUID  
/*JOBPARM S=*  
//COPY      EXEC PGM=GENREC1,REGION=500K,PARM='SHORT'  
//STEPLIB   DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.LOAD  
//SYSPRINT  DD SYSOUT=*
```

```
//PRINT DD SYSOUT=*
//SYSUDUMP DD SYSOUT=*
//GENDCB DD DISP=SHR,DSN=MHLRES2.TESTREL.PS,
//          SPACE=(TRK,1,RLSE)
//SYSIN DD DUMMY
```

In Example 7-37 on page 134 we show that after writing only 1000 records in the data set MHLRES2.TESTREL.PS on a DD statement that specified the RLSE parameter the overall allocation has reduced from 57 to 33 cylinders. This reduction has occurred on the first volume of the three because the 1000 records were written to only the first volume. There are two volumes with 16 cylinders each still allocated and unused.

Example 7-37 ISPF display showing that data set MHLRES2.TESTREL.PS using 33 CYLs

```
Data Set Information
Command ==>

Data Set Name . . . . : MHLRES2.TESTREL.PS

General Data                                Current Allocation
Management class . . : MCDB22                Allocated cylinders : 33
Storage class . . . : STANDARD                Allocated extents . : 33
Volume serial . . . : MLD83A +
Device type . . . . : 3390
Data class . . . . . : **None**
Organization . . . . : PS
Record format . . . : FB
Record length . . . : 80
Block size . . . . . : 160
1st extent cylinders: 1
Secondary cylinders : 1
Data set name type :
SMS Compressible. . : NO

Current Utilization
Used cylinders . . . : 1
Used extents . . . . : 2

Dates
Creation date . . . : 2013/10/12
Referenced date . . : 2013/10/12
Expiration date . . : ***None***
```

To display multiple volumes press Enter or enter Cancel to Exit.

Scenario 4

In this scenario we repeated scenario 3 but this time running on a DFSMS V2.1 system.

We ran the JCL as in Example 7-36 on page 133 with the only change being to run it on a DFSMS V2.1 system. The ISPPF Data Set Information panel is shown in Example 7-38.

Example 7-38 ISPF display showing data set MHLRES2.TESTREL.PS has reduced to 1 CYL

```
Data Set Information
Command ==>

Data Set Name . . . . : MHLRES2.TESTREL.PS

General Data                                Current Allocation
```

More: +

Management class . . . : MCDB22	Allocated cylinders : 1
Storage class . . . : STANDARD	Allocated extents . . : 3
Volume serial . . . : MLD83A +	
Device type : 3390	
Data class : **None**	
Organization . . . : PS	Current Utilization
Record format . . . : FB	Used cylinders . . . : 1
Record length . . . : 80	Used extents . . . : 1
Block size : 160	
1st extent cylinders: 1	
Secondary cylinders : 1	Dates
Data set name type :	Creation date . . . : 2013/10/12
	Referenced date . . : 2013/10/12
	Expiration date . . : ***None***
SMS Compressible . . : NO	

The potential allocation attributes remain even though there is no space allocated on the additional two volumes.

The data set can be expanded to the three volumes by providing suitable JCL SPACE attributes. The space allocated will depend on what is specified in the run that writes to the data set, not in what was originally allocated.

In Example on page 135 we show the JCL to write to fill data set MHLRES2.TESTREL.PS based on SPACE=(CYL,(10,1)) SPACE attributes which will be added to what was retained in the data set after releasing the space.

Example 7-39 JCL to fill data set MHLRES2.TESTREL.PS based on SPACE=(CYL,(10,1))

```
//MHLRES2H JOB (1234567,COMMENT),MHLRES2,TIME=10,
// MSGCLASS=J,
// REGION=500K,
// MSGLEVEL=1,CLASS=A,
// NOTIFY=&SYSUID
/*JOBPARM S=*
//COPY EXEC PGM=GENREC1,REGION=500K PARM='SHORT'
//STEPLIB DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.LOAD
//SYSPRINT DD SYSOUT=*
//PRINT DD SYSOUT=*
//SYSUDUMP DD SYSOUT=*
//GENDCB DD DISP=SHR,DSN=MHLRES2.TESTREL.PS,
// SPACE=(CYL,(10,1))
//SYSIN DD DUMMY
```

Example 7-40 on page 135 shows that data set MHLRES2.TESTREL.PS has expanded to 46 cylinders allocated across 48 extents which is the limit for three volumes.

Example 7-40 ISPF display showing data set MHLRES2.TESTREL.PS after expansion

Data Set Information

Command ==>

More: +

Data Set Name : MHLRES2.TESTREL.PS

General Data	Current Allocation
Management class . . : MCDB22	Allocated cylinders : 46
Storage class . . . : STANDARD	Allocated extents . . : 48
Volume serial . . . : MLD83A +	
Device type : 3390	
Data class : **None**	Current Utilization
Organization . . . : PS	Used cylinders . . . : 46
Record format . . . : FB	Used extents . . . : 48
Record length . . . : 80	
Block size : 160	Dates
1st extent cylinders: 1	Creation date . . . : 2013/10/12
Secondary cylinders : 1	Referenced date . . : 2013/10/12
Data set name type :	Expiration date . . : ***None***
SMS Compressible . . : NO	

Compatibility

The enhanced space release function is not provided in releases prior to DFSMS V2.1.

A data set that has had space partially released on a pre-DFSMS V2.1 system will have all unused space released if it is written to a DFSMS V2.1 system, even if the new data only goes to the first volume of two or more volumes. That is, as data sets are processed on DFSMS V2.1 their excess space will be released.

7.5 OCE RAS enhancements

There are three Reliability, Availability and Serviceability (RAS) enhancements:

- ▶ S837 RC08 ABEND elimination
- ▶ Comments in SYS1.PARMLIB IEAAPxx member
- ▶ DCBE invalidation message IEC190I

7.5.1 S837 RC08 ABEND elimination

In order to access data on DISK or TAPE, tasks have to set up the necessary definitions. Whether from Job Control Language (JCL) or via dynamic allocation, data sets need to be allocated.

For output data sets, it is also necessary for at least one volume to be specified. This can be specified directly or through the System Managed Storage (SMS) function.

A volume count can be specified in the JCL or via SMS, but it is by default five. Until DFSMS V2.1 if any more than five volumes are required, then the higher number (including the initial five) must be specified.

When access to a data set is defined, a Job File Control Block (JFCB) is allocated in storage, and one of the fields in the JFCB has room for up to five volumes.

If provision for more than five volumes is specified a JFCB Extension is required. If a volume count of more than five is specified in JCL or via dynamic allocation, then the JFCB extension is also created at that time.

Until the DFSMS V2.1 enhancement was implemented, a task that required more than five volumes and that had not been specified, then the task would issue S837 RC08 abend and fail. The task would typically have to be restarted with an update to specify provision for more volumes. DFSMS V2.1 will automatically create a JFCB extension for the additional volumes.

We created a job that required more than five volumes for output without specifying a volume count to demonstrate that the automatic creation of a JFCB extension works.

The JCL for the job is shown in Example 7-41 on page 137. Note that on the TAPE DD statement there is no VOL specification.

Example 7-41 Tape job that requires more than 5 volumes running without a volume count in JCL

```
//MHLRES2D JOB 99990000,UAAALFO,CLASS=A,NOTIFY=&SYSUID
/*JOBPARM S=*
//DFDSS PROC
//DFDSS EXEC PGM=ADRDSSU,REGION=0M
//SYSPRINT DD SYSOUT=A
// PEND
// EXEC DFDSS
//DASD DD DISP=SHR,UNIT=3390,VOL=SER=SERVER1
//TAPE DD UNIT=VT3590,
// DISP=(NEW,PASS),
// DSN=MHLRES2.DSSDUMP.Z21RA1
//SYSIN DD *
        DUMP OUTDDNAME(TAPE) INDD(DASD) ALLEXCP ALLDATA(*) OPTIMIZE(4)
```

The job ran and used 18 volumes. Select parts of the job output are shown in Example 7-42 on page 137.

Example 7-42 Output from TAPE job requiring more than 5 volumes without a volume count specified

```
//MHLRES2D JOB 99990000,UAAALFO,CLASS=A,NOTIFY=&SYSUID
/*JOBPARM S=*
//DFDSS PROC
//DFDSS EXEC PGM=ADRDSSU,REGION=0M
//SYSPRINT DD SYSOUT=A
// PEND
// EXEC DFDSS
//DASD DD DISP=SHR,UNIT=3390,VOL=SER=SERVER1
//TAPE DD UNIT=VT3590,
// DISP=(NEW,PASS),
// DSN=MHLRES2.DSSDUMP.Z21RA1
//SYSIN DD *
        DUMP OUTDDNAME(TAPE) INDD(DASD) ALLEXCP ALLDATA(*) OPTIMIZE(4)
IEC502E RK 0442,VT0027,SL,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC501A M 0442,PRIVAT,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC705I TAPE ON 0442,VT0028,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1,MEDIA5
IEC502E RK 0442,VT0028,SL,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC501A M 0442,PRIVAT,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC705I TAPE ON 0442,VT0029,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1,MEDIA5
IEC502E RK 0442,VT0029,SL,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
```

```

IEC501A M 0442,PRIVAT,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC705I TAPE ON 0442,VT0030,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1,MEDIA5
IEC502E RK 0442,VT0030,SL,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC501A M 0442,PRIVAT,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC705I TAPE ON 0442,VT0031,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1,MEDIA5
IEC502E RK 0442,VT0031,SL,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC501A M 0442,PRIVAT,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC705I TAPE ON 0442,VT0033,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1,MEDIA5
IEC502E RK 0442,VT0033,SL,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC501A M 0442,PRIVAT,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1
IEC705I TAPE ON 0442,VT0034,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1,MEDIA5
IEC502E RK 0442,VT0034,SL,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21RA1

```

.
 .
 Several more IEC705I and IEC502E messages
 and other output, through to

```

..
IEF285I MHLRES2.DSSDUMP.Z21RA1 DELETED
IEF285I VOL SER NOS= VT0024,VT0025,VT0026,VT0027,VT0028,
IEF285I VOL SER NOS= VT0029,VT0030,VT0031,VT0033,VT0034,
IEF285I VOL SER NOS= VT0035,VT0036,VT0037,VT0038,VT0039,
IEF285I VOL SER NOS= VT0040,VT0041,VT0042.
IEF375I JOB/MHLRES2D/START 2013255.2216
IEF033I JOB/MHLRES2D/STOP 2013255.2226
CPU: 0 HR 00 MIN 04.11 SEC SRB: 0 HR 00 MIN 00.64 SEC
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.255 22:16
DUMP OUTDDNAME(TAPE) INDD(DASD) ALLEXCP ALLDATA(*) OPTIMIZE(4) + 0013000
ADMIN
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'DUMP '
ADR109I (R/I)-RI01 (01), 2013.255 22:16:09 INITIAL SCAN OF USER CONTROL STATEME
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.255 22:16:09 EXECUTION BEGINS
ADR006I (001)-STEND(02), 2013.255 22:26:26 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.255 22:26:26 TASK COMPLETED WITH RETURN CODE 0000
ADR012I (SCH)-DSSU (01), 2013.255 22:26:26 DFSMSDSS PROCESSING COMPLETE. HIGHS

```

The IEF285I messages show that 18 volumes were used.

7.5.2 S837 RC08 ABEND in DFSMS V1.13

As an illustration of the consequence of omitting a volume count on an output DD statement when more than five volumes could be required, we ran the same job on a z/OS V1.13 system. The JCL is in Example 7-41 on page 137. The output illustrated in Example 7-43 on page 138 shows that the job failed.

Example 7-43 Output from TAPE job running on z/OS V1.13 showing ABEND S837 RC08

```

utput from TAPE job requiring more than 5 volumes without a volume count specifi
10.36.16 JOB12856 IEC502E RK 042A,VT0051,SL,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.Z21R
10.36.16 JOB12856 *IEC501A M 042A,PRIVAT,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DSSDUMP.
10.36.16 JOB12856 IEC705I TAPE ON 042A,VT0052,SL,COMP,MHLRES2D,DFDSS,MHLRES2.DS
10.36.53 JOB12856 IEC028I 837-08,IFG0554A,MHLRES2D,DFDSS,TAPE,042A,VT0052, 677
677 MHLRES2.DSSDUMP.Z21RA1
10.36.53 JOB12856 - --TIMINGS (MINS.)--
10.36.53 JOB12856 -JOBNAME STEPNAME PROCSTEP RC EXCP CPU SRB CLOCK

```

```

10.36.53 JOB12856 -MHLRES2D          DFDSS          08 46667    .02    .00    3.15
10.36.53 JOB12856 IEF234E K 042A,VT0052,PVT,MHLRES2D
10.36.53 JOB12856 IEF404I MHLRES2D - ENDED - TIME=10.36.53 - ASID=001E - SC63
10.36.53 JOB12856 -MHLRES2D ENDED.  NAME-UAALF0          TOTAL CPU TIME=
10.36.53 JOB12856 $HASP395 MHLRES2D ENDED
----- JES2 JOB STATISTICS -----
  13 SEP 2013 JOB EXECUTION DATE
    13 CARDS READ
    89 SYSOUT PRINT RECORDS
    0 SYSOUT PUNCH RECORDS
    7 SYSOUT SPOOL KBYTES
    3.15 MINUTES EXECUTION TIME
    1 //MHLRES2D JOB 99990000,UAALF0,CLASS=A,NOTIFY=&SYSUID
      /*JOBPARM S=SC63
      IEFC653I SUBSTITUTION JCL - 99990000,UAALF0,CLASS=A,NOTIFY=MHLRES2
    2 //DFDSS PROC
      //DFDSS EXEC PGM=ADRDSSU,REGION=0M
      //SYSPRINT DD SYSOUT=A
      // PEND
    3 // EXEC DFDSS
    4 ++DFDSS PROC
    5 ++DFDSS EXEC PGM=ADRDSSU,REGION=0M
    6 ++SYSPRINT DD SYSOUT=A
    7 //DASD DD DISP=SHR,UNIT=3390,VOL=SER=SERVR1
    8 //TAPE DD UNIT=VT3590,
      // DISP=(NEW,PASS),
      // DSN=MHLRES2.DSSDUMP.Z21RA1
    9 //SYSIN DD *
  STMT NO. MESSAGE
    3 IEFC001I PROCEDURE DFDSS WAS EXPANDED USING INSTREAM PROCEDURE DEFINIT
  ICH70001I MHLRES2 LAST ACCESS AT 23:28:19 ON THURSDAY, SEPTEMBER 12, 2013
  IEF236I ALLOC. FOR MHLRES2D DFDSS
  IEF237I JES2 ALLOCATED TO SYSPRINT
  IEF237I 6619 ALLOCATED TO DASD
  IGD100I 042A ALLOCATED TO DDNAME TAPE          DATACLAS (          )
  IEF237I JES2 ALLOCATED TO SYSIN
  IEC028I 837-08,IFG0554A,MHLRES2D,DFDSS,TAPE,042A,VT0052,
  MHLRES2.DSSDUMP.Z21RA1
  IEF142I MHLRES2D DFDSS - STEP WAS EXECUTED - COND CODE 0008
  IEF285I MHLRES2.MHLRES2D.JOB12856.D0000102.?          SYSOUT
  IEF285I SYS13256.T103344.RA000.MHLRES2D.R0402778      KEPT
  IEF285I VOL SER NOS= SERVR1.
  IEF285I MHLRES2.MHLRES2D.JOB12856.D0000101.?          SYSIN
  IEF373I STEP/DFDSS /START 2013256.1033
  IEF032I STEP/DFDSS /STOP 2013256.1036
          CPU:      0 HR 00 MIN 01.30 SEC   SRB:      0 HR 00 MIN 00.27 SEC
          VIRT: 2172K SYS: 348K EXT: 11732K SYS: 10352K
  IEF285I MHLRES2.DSSDUMP.Z21RA1          DELETED
  IEF285I VOL SER NOS= VT0048,VT0049,VT0050,VT0051,VT0052.
  IEF375I JOB/MHLRES2D/START 2013256.1033
  IEF033I JOB/MHLRES2D/STOP 2013256.1036
          CPU:      0 HR 00 MIN 01.30 SEC   SRB:      0 HR 00 MIN 00.27 SEC
  PAGE 0001      5695-DF175 DFSMSDSS V1R13.0 DATA SET SERVICES      2013.256 10:33
          DUMP OUTDDNAME(TAPE) INDD(DASD) ALLEXCP ALLDATA(*) OPTIMIZE(4) ADMIN 00130004
  ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'DUMP '

```

```

ADR109I (R/I)-RI01 (01), 2013.256 10:33:44 INITIAL SCAN OF USER CONTROL STATEMEN
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.256 10:33:44 EXECUTION BEGINS
ADR049E (001)-STEND(01), 2013.256 10:36:53 DFSMSDSS FUNCTION TASK ABEND RECOVERY
CODE=0008
ADR006I (001)-STEND(02), 2013.256 10:36:53 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.256 10:36:53 TASK COMPLETED WITH RETURN CODE 0008
ADR012I (SCH)-DSSU (01), 2013.256 10:36:53 DFSMSDSS PROCESSING COMPLETE. HIGHEST
TASK      001

```

Compatibility

The enhancement to eliminate the ABEND S837 RC08 is not provided for releases prior to DFSMS V2.1

7.6 IEAAPpx comments

The SYS1.PARMLIB IEAAPpx member is used to specify I/O appendages for special circumstances.

Prior to DFSMS V2.1 it was not possible to add comments to the IEAAPpx member requiring documentation of the reason for particular settings to be held elsewhere.

DFSMS V2.1 has been updated to allow comments in two forms:

- ▶ An asterisk (*) in column one is typically used for comments at the start of the member.
- ▶ Comments anywhere in the member enclosed in the */* comment */* sequence.

An example of comments used in an IEAAP00 sample member is shown in Example 7-44 on page 140.

Example 7-44 IEAAP00 member of SYS1.PARMLIB with comments

```

* MEMBER IEAPPXX OF SYS1.PARMLIB SPECIFIES THE AUTHORIZED
* I/O APPENDAGES THAT ARE TO BE LOADED FOR SPECIAL I/O SITUATIONS.
* A CHANGE WITH DFSMS V2.1 ALLOWS COMMENTS IN THE MEMBER.
* THE COMMENTS MAY BE SPECIFIED BY * IN COLUMN 1
* AS IN THIS PART OF THE MEMBER, TYPICALLY USED FOR A PREFIX FOR
* THE MAIN CONTENTS.
* THE ENTRIES IN THIS MEMBER DENOTE THE SUFFIXES TO MODULES
* WHERE THE FIRST 6 CHARACTERS ARE IGG019
* COMMENTS CAN ALSO BE INSERTED ANYWHERE IN THE MEMBER IF ENCLOSED
* IN THE /* */ SEQUENCE OF CHARACTERS.
/* THIS WOULD TYPICALLY BE USED TO DOCUMENT THE REASON FOR A */
/* PARTICULAR LINE IN THE MEMBER */
SIOAPP Y1,Y2,
EOEAPP X1,Z2,X3,X4,X5,X6,    /* IGG019X3 IS USED FOR EOEAPP */
PCIAPP X3                    /* AND FOR PCIAPP */

```

Compatibility

The enhancement to allow comments in IEAAPpx is not provided for releases prior to DFSMS V2.1

7.7 DCBE invalidation message IEC190I

The Data Control Block Extension (DCBE) is an architected extension to the standard DCB that is required for various data management functions.

The DCBE and related functions can only be used successfully if the linkages to the DCBE are correctly set up.

One of the major uses of the DCBE is to enable DASD EAV volume access. If this form of DASD is not being used there may be no apparent effect caused by the incomplete set up. However, if EAV volumes are implemented at a later stage then problems will occur.

In order to proactively identify and correct these problems, with DFSMS V2.1 a check is made when a data set is opened to verify that the set up is complete.

If the setup is not complete then a version of message IEC190I will be issued. The program issuing this message may continue to work when DASD EAV volumes are not in use, but the warning is issued to allow the program to be corrected.

7.7.1 Maintenance

During z/OS V2.1 testing a number of IBM products have been identified as requiring maintenance.

The following APARs relating to DFSMS V2.1 changes have been identified by searching IBMLINK. There other instances of message IEC190I that relate to other conditions not changes with DFSMS V2.1.

Users may perform a similar search to determine whether there are any additional cases.

z/OS V2.1 by itself and z/OS V2.1 related programs

The APARS listed in this section are likely to be incorporated in the delivered system code. They are presented here as examples of problems you may encounter. The APAR number and the description text are listed:

► **OA42701 AMATERSE INVOCATION RESULTS IN MSGIEC190I**

APAR OA42701 is an example of the situation where message IEC190I is issued, but the program continues to run.

► **OA42694 ABEND0C4 IFG0554P**

APAR OA42694 is an example of a situation where message IEC190I is issued, but the using task abends. The presence of the IEC190I provides valuable information for program diagnosis.

► **OA43037 IRRDBU00 MSGIEC190I**

► **OA43000 IEC190I DURING RECALL OF PDSE DATA SET**

OA42406 OCE FIX ROLLUP FOR HDZ2210

APARs OA42701

The content of the text for APAR OA42701 is shown in Example 7-45 on page 142.

If this situation arises it is an indication that the DCB/DCBE structure is incomplete. This may not cause problems with traditional DASD volumes but could cause problems with EAV DASD volumes.

Example 7-45 APAR OA42701 - message issued and program continues

APAR Identifier OA42701 Last Changed 13/09/04
AMATERSE INVOCATION RESULTS IN MSGIEC190I

Symptom	IN INCORROUT	Status	CLOSED PER
Severity	3	Date Closed	13/07/26
Component	5752SC112	Duplicate of	
Reported Release	790	Fixed Release	999
Component Name	SVA UTILITIES	Special Notice	
Current Target Date	..13/08/15	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 790 : UA70061 available 13/08/07 (F308)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

Under certain conditions executing an AMATERSE with the PACK option will result in a successful execution, but MSGIEC190I xxxxxx,STEP,SYSUT3 INVALID DCBE: STORAGE NOT ADDRESSABLE is issued.

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All users of AMATERSE on release HBB7790.      *
*****
* PROBLEM DESCRIPTION: AMATERSE MSGIEC190I INVALID DCBE:        *
*                        STORAGE NOT ADDRESSABLE                  *
*****
* RECOMMENDATION:                                                *
```

While PACK/ UNPACKing a PDS data set, AMATERSE may successfully complete, but MSGIEC190I is issued. Also, AMATERSE fails to use system determined blocksize for PACK/SPACK output.

PROBLEM CONCLUSION:

AMATERSE's PACK and UNPACK processing is updated to correctly set the relevant DCB data, to prevent the IEC190I message. Also, AMATERSE's processing is changed to allow System Determined Blocksize for the PACK/SPACK output data set.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: AMADECS AMAENCS

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APARs OA42694

The content of the text for APAR OA42694 is shown at Example 7-46 on page 143. This problem is one that needs to be corrected.

Example 7-46 APAR OA42694 - message issued and invoking task fails

APAR Identifier	OA42694	Last Changed	13/08/02
ABENDOC4 IFG0554P			
Symptom	AB ABENDOC4	Status	CLOSED PER
Severity	3	Date Closed	13/07/01
Component	5695DF107	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	DATA MGMT SUPPO	Special Notice	
Current Target Date	..13/07/20	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 210 : UA69759 available 13/07/30 (F307)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:
0c4 in IFG0554P incorrectly accessing a 24bit ucb address
as a 31bit address with a dirty high order byte.

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: All. *

* PROBLEM DESCRIPTION: Abend0c4, invalid IEC190I *

* RECOMMENDATION: *

Abend0c4 out of IFG0554P due to dirty high order UCB address.
Also invalid IEC190I INVALID DCBE: EXCP DCB WITHOUT A
FOUNDATION EXTENSION.

PROBLEM CONCLUSION:
Abend0C4 and invalid IEC190I will no longer occur.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IFG019RA IFG0193A IFG0554P

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

Program products

Fixes for IBM Program products will not normally be delivered as part of the z/OS V2.1 code, so fixes for these situations will have to be addressed specifically. Here we have listed errors related to message IEC190I and the associated description that should be investigated.

- ▶ PM94936 z/OS 2.1 Compatibility fixes.
- ▶ IO18269 MSGIEC190I WHEN MODIFYING THE TARGET VOLUME ON THE VOLUMES AND STORAGE CLASSES PANEL OF Z/OSMF SOFTWARE DEPLOYMENT PLUGIN.

- PM96778 MSG IEC190I RECEIVED RUNNING IBM QMF™ WITH Z/OS 2.1

Non-IBM program products

The term Independent Software Vendor (ISV) is used to describe vendors that create products to use IBM software and hardware. There is often a close relationship between IBM and certain ISVs. IBM collects information about compatibility with its operating systems and the ISVs. It is primarily the responsibility of the end users to obtain the latest information from the vendors.

Compatibility

The enhancement to message IEC190I is not provided in releases prior to DFSMS V2.1. However, in prior releases where the task is not affected the invalidation of the DCBE is still taking place. If this results in a set of control blocks that are incompatible with the DASD access intended, then there may be an abend.

7.8 XTIOt HealthCheck

The SYS1.PARMLIB DEVSUPxx member allows specification of the NON_VSAM_XTIOt=YES option. This option allows application programs to exploit options on dynamic allocation with BSAM, BPAM and QSAM and affects EXCP.

This is not a new feature but the default is NON_VSAM_XTIOt=NO and as a consequence may be overlooked

7.8.1 XTIOt HealthCheck enablement

DFSMS V2.1 introduces a check to verify whether NON_VSAM_XTIOt=YES is set or not. This check is automatically included in the health check process and requires no set up.

7.8.2 XTIOt HealthCheck verification

Once the health check process has run, typically immediately after an IPL, the results will be available from SDSF.

1. To view the status of this check, access SDSF and enter the **CK** command. The result will be a list of all the checks. To locate the XTIOt check issue the **SOt NAME** command.
2. Scroll down the list to find OCE_XTIOt_CHECK
3. Select this entry with **S** in column 1

.An example of output that shows that NON_VSAM_XTIOt=YES has not been set is shown in Example 7-47 on page 145.

Example 7-47 Example of XTIOt HealthCheck result

```
CHECK(IBMOCE,OCE_XTIOt_CHECK)
SYSPLEX:  SANDBOX  SYSTEM: SC64
START TIME: 09/09/2013 10:34:43.397162
CHECK DATE: 20110410  CHECK SEVERITY: LOW
```

* Low Severity Exception *

IECH0101E OPEN macro support for XTIOt, uncaptured UCBs and DSAB

above the line is not enabled for non-VSAM. IBM recommends setting NON_VSAM_XTIOT=YES in the DEVSUPxx member of PARMLIB.

Explanation: IBM suggests setting NON_VSAM_XTIOT=YES in the DEVSUPxx member of PARMLIB to decrease the chances of running out of virtual storage when allocating and concurrently opening many sequential and partitioned data sets.

System Action: The system continues processing.

7.8.3 XTIOT HealthCheck implementation considerations

Care must be taken when deciding to set NON_VSAM_XTIOT=YES because control blocks get relocated from 24 bit addressing to 31 bit addressing. Whether or not all program and application code can support this change depends on the way they have been written.

The XTIOT HealthCheck is new with DFSMS V2.1 and may prompt users to enable the function. This applies to releases prior to DFSMS V2.1 because NON_VSAM_XTIOT=YES is not new with DFSMS V2.1.

To check IBM product compatibility and recommended maintenance, search IBMLINK using keyword XTIOT.

Refer to “APAR OA42667 XTIOT use (also applicable to DFSMS V1.13)” on page 271.

It is advisable to check with the vendors of non-IBM code as to whether or not their code can handle NON_VSAM_XTIOT=YES. If a test system is available that can be used to test all applications, then the DEVSUPxx update can be made, and the tests run.

If a problem arises typically (but not exclusively) a system ABENDS0C4 is issued. If this does occur, the DEVSUPxx member can be updated with NON_VSAM_XTIOT=NO and the command SET DEVSUP=xx can be issued to reset the option dynamically.

Compatibility

The enhanced HealthCheck for the use of NON_VSAM_XTIOT=YES is not provided in release prior to DFSMS V2.1.



DFSMSDss enhancements

This chapter provides an overview of the features and function in DFSMSDss included in z/OS V2.1. The following enhancements are provided:

- ▶ DFSMSDss data set change indicator reset with RESTORE
- ▶ DEBUG option SMSMSG
- ▶ RESTORE command
- ▶ zFS change activity support

8.1 DFSMSdss data set change indicator reset with RESTORE

DASD volumes have a Volume Table of Contents (VTOC) in which control information for all data sets that are on the volume. This control information is stored in the Data Set Control Block (DSCB). There are several DSCB Formats. For the purposes of this section we are only concerned with Formats 1 and 8. These are the primary DSCBs for data sets on the original DASD configured volumes, and for DASD EAV volumes respectively. The specific entry that relates to the information here is the data-set-changed or data set changed indicator (DS1DSCHA) bit which is at the same relative offset in both format DSCBs.

DFSMSdss is used on its own or as part of a DFSMSHsm system to make copies of data sets. Different options are available in DFSMSdss for this purpose. Making copies of data sets is usually a housekeeping task and is usually configured to avoid making copies if a copy already exists and the source data set has not changed. The appropriate setting of the DS1DSCHA bit is essential to that function.

When a task accesses a data set, the DS1DSCHA bit will normally be turned *ON* so that the data set will be included in housekeeping tasks. Once a copy of a data set is taken by DFSMSdss, then it is available to be RESTORED.

After a data set has been RESTORED the setting of the DS1DSCHA bit, for any data set, controls whether new copies will be made or not when housekeeping tasks are run.

If an entire volume is RESTORED the setting of the DS1DSCHA bit for all data sets has to be managed to avoid making new copies of each data set.

8.1.1 DFSMSdss setting of DS1DSCHA prior to DFSMS V2.1

DFSMSdss full volume RESTORE unconditionally resets (turns OFF) DS1DSCHA for each data set it restores. This indicates that until the data set is changed, it will not be selected by any housekeeping process that looks for changed data sets. In some circumstances this process is not granular enough.

8.1.2 DFSMSdss setting of DS1DSCHA options when using DFSMS V2.1

If DFSMSHsm is in use in an installation, care must be taken when using DFSMSdss outside the control of DFSMSHsm. DFSMSHsm normally manages data set copies (backups) and volume DUMPs, and in this process manages the DS1DSCHA bit in the DSCB. If a DFSMSdss process is run that changes the DS1DSCHA bit without DFSMSHsm knowledge the results may be other than expected.

There are DFSMSdss PATCH options that can control the setting of the DS1DSCHA bit, so these must be reviewed to avoid conflict with this enhancement.

DFSMSdss RESET keyword

The DFSMSdss RESET keyword specified when setting up a full volume or a tracks restore (from a full volume) may be used to control the action DFSMSdss is to take.

With DFSMS V2.1 the new options are:

- **RESET(DUMP)**

This is the default. It is a conditional setting depending on what was set at DUMP time. If RESET was specified when the DUMP was run on a DFSMS V2.1 system then the

DS1DSCHA indicators will be turned off. If RESET was *not* specified when the DUMP was run, the DS1DSCHA indicators will not be changed from what they were on the source volume.

- ▶ RESET(YES)

Unconditionally turn DS1DSCHA off regardless of what was set at DUMP time.

- ▶ RESET(NO)

Unconditionally *do not* change DS1DSCHA settings from what they were on the source volume.

Running on DFSMS V2.1 and using a DFSMSdss DUMP from a prior release the options are changed as follows:

- ▶ If RESET(DUMP) if specified then it is treated as if RESET(NO) was specified.
- ▶ If RESET is not specified the default is RESET(NO).

Important: The use of the RESET(DUMP) option described below for RESTORE requires that the DUMP RESET option has been used on a DFSMS V2.1 system

8.1.3 Scenarios of the effects of DUMP options

We ran several jobs to show what the effect of the DUMP RESET and RESTORE RESET(DUMPIYESINO) are.

In order to check what the DS1DSCHA setting is for data sets on a volume, we created a program to read the VTOC and report on up to 20 data sets. The Common VTOC Access Facility (CVAF) returns DSCBs in the order that they are on the volume. Contrast, for example, with ISPF which returns the data set in collating order.

The source of the program is in Appendix C-10, "Source for CVSEQ8D part 1 of 19" on page 407.

Scenario 1: Data set change indicator reset by DFSMSHsm backup

In Example 8-1 on page 149 we show the JCL that was used to run the VTOC listing. The volume to be listed must be specified on the CVAFDD statement.

Example 8-1 JCL to run VTOC listing program showing DS1DSCHA setting

```
//MHLRES2Q JOB (1234567,COMMENT),UAALF0,TIME=10,  
// MSGLEVEL=1,CLASS=A,  
// NOTIFY=&SYSUID  
/*JOBPARM S=*  
//CVSEQ80 PROC  
//RUN      EXEC PGM=CVSEQ8D,REGION=0M  
//STEPLIB DD DISP=SHR,DSN=MHLRES1.EAV.LOAD  
//SYSUDUMP DD SYSOUT=*  
//OUTDD    DD SYSOUT=*  
// PEND  
// EXEC CVSEQ80  
//CVAFDD DD DISP=SHR,UNIT=3390,VOL=SER=SBOX1L
```

In Example 8-2 on page 150 we show the output from the VTOC listing program. This program has code in it to check the status of the EADSCB option since it is important that the environment is correct before using the output information.

The program will list up to 20 entries, but in this case one of the first 20 data sets on the volume is a Format 5 DSCB, so it is not listed.

In the listing the DS1DSCHA flag follows the data set name and is set as follows:

- Y if the bit is on indicating that the data set has changed since last backup.
- N if the bit is off indicating that the data set has not changed since last backup.

As can be seen in the output, of the first 20 data sets on the volume only one has DS1DSCHA set to Y. That is the MHLRES2.CNTL.JCL data set.

Example 8-2 Listing of volume SBOX11 showing DS1DSCHA setting following the dsname

```
CVSEQ8D START OF OUTPUT MESSAGES VOLUME: SBOX1L

CVAFSEQ CALL: EADSCB KEYWORD NOT CODED
CV4EADOK BIT IS NOT SET / EADSCB=NOTOK
RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL
CVSTAT CODE: X"00"    DEC"000"

CVAFSEQ CALL: EADSCB=NOTOK CODED
CV4EADOK BIT IS NOT SET / EADSCB=NOTOK
RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL
CVSTAT CODE: X"00"    DEC"000"

CVAFSEQ CALL: EADSCB=OK CODED
CV4EADOK BIT SET / EADSCB=OK
RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL
CVSTAT CODE: X"00"    DEC"000"

DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:
DSN: SYS1.VTOCIX.SBOX1L                N
DSN: OMVS.SC65.USERS.OLD                N
DSN: SYS1.VVDS.VSBOX1L                N
DSN: OMVS.SC64.USRLOCAL                N
DSN: OMVS.SC63.HERING.TEST.HFS        N
DSN: OMVS.SC63.WEB.DB2V7              N
DSN: MHLRES3.SMSTEST.PSSTRIP2         N
DSN: OMVS.TWSRES6.HFS                 N
DSN: OMVS.TROWELL.HFS                 N
DSN: OMVS.DB8A02S.HFS                 N
DSN: OMVS.D8F202S.HFS.DATA            N
DSN: MHLRES2.CNTL.JCL                 Y
DSN: OMVS.XXXX.AMNTTEST.SC65.ZFS.DATA N
DSN: MHLRES2.SELECT                   N
DSN: OMVS.SC63.WPS5PB1.BS01.CONFIG.ZFS.DATA N
DSN: OMVS.BARI.HFS.DATA               N
END OF DATA REACHED - ALL DATASETS PROCESSED

CVSEQ8D END OF OUTPUT MESSAGES
```

The DS1DSCHA bit is set off when a DFSMSShsm backup of the data set is done. To demonstrate the effect of a DFSMSShsm backup, the HBACKDS ‘MHLRES2.CNTL.JCL’ command was issued.

After receiving a message from DFSMSShsm indicating that the backup had been run, we ran the VTOC listing program using JCL as shown in Example 8-1 on page 149.

In Example 8-3 on page 151 we show the data set listing part of the VTOC listing program showing that the DS1DSCHA bit has been set off for the MHLRES2.CNTL.JCL data set.

Example 8-3 Listing of volume SBOX1I showing DS1DSCHA settings after data set backup

```

DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:
DSN: SYS1.VTOCIX.SBOX1L                      N
DSN: OMVS.SC65.USERS.OLD                      N
DSN: SYS1.VVDS.VSBOX1L                      N
DSN: OMVS.SC64.USRLOCAL                      N
DSN: OMVS.SC63.HERING.TEST.HFS              N
DSN: OMVS.SC63.WEB.DB2V7                    N
DSN: MHLRES3.SMSTEST.PSSTRIP2               N
DSN: OMVS.TWSRES6.HFS                      N
DSN: OMVS.TROWELL.HFS                      N
DSN: OMVS.DB8A02S.HFS                      N
DSN: OMVS.D8F202S.HFS.DATA                 N
DSN: MHLRES2.CNTL.JCL                      N
DSN: OMVS.XXXX.AMNTTEST.SC65.ZFS.DATA       N
DSN: MHLRES2.SELECT                       N
DSN: OMVS.SC63.WPS5PB1.BS01.CONFIG.ZFS.DATA N
DSN: OMVS.BARI.HFS.DATA                   N
END OF DATA REACHED - ALL DATASETS PROCESSED

```

CVSEQ8D END OF OUTPUT MESSAGES

Scenario 2: DFSMSdss DUMP

In order to demonstrate the effect of the DFSMSdss RESTORE RESET options we needed to create a physical DUMP of a volume using the RESET option. We chose volume SBOX1B for this demonstration.

We listed the first 20 data sets on volume SBOX1B using JCL similar that shown in Example 8-1 on page 149. For this scenario we changed the volume on the CVAFFD line from SBOX1L to SBOX1B.

In Example 8-4 on page 151 we show data sets from volume SBOX1B. There are several data sets with the change indicator set to Y.

Example 8-4 Listing of data sets from volume SBOX1B showing DS1DSCHA bit settings are mixed

```

DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:
DSN: SYS1.VTOCIX.SBOX1B                      N
DSN: HFS.SC65.ILMRECON.HFS                  Y
DSN: SYS1.VVDS.VSBOX1B                      N
DSN: OMVS.SC65.XXX.HFS                      Y
DSN: HFS.SC70.DEV                          Y
DSN: HFS.SC64.ETC                          Y
DSN: OMVS.SC63.DOMR5.MAIL3.HFS             Y

```

DSN: OMVS.SC63.USERS.DSSDUMP.ALLDATA	Y
DSN: OMVS.SC63.XML.HFS	Y
DSN: OMVS.HERING.SUBHFS2.TEST	Y
DSN: OMVS.ROGERS.TEST6.SAV	Y
DSN: OMVS.PRADIER.HFS	Y
DSN: OMVS.ROGERS.TEST.SAV.DATA	Y
DSN: OMVS.MORGAR1.HFS	Y
DSN: LUTZ.LOADLIB	Y
DSN: OMVS.EJESWEB.HFS.DATA	Y
DSN: OMVS.MHLRES3.HFS	Y
DSN: MHLEAV1.TEST	N
DSN: OMVS.WATERS.HFS.DATA	Y

END OF DATA REACHED - ALL DATASETS PROCESSED

In Example 8-5 on page 152 we show the job used to create the DFSMSDss DUMP of volume SBOX1B. In this case the significant option is RESET on the DUMP command.

Example 8-5 JCL to run DSS job to DUMP volume SBOX1B

```
//MHLRES2D JOB 99990000,UAA LFO,CLASS=A,NOTIFY=&SYSUID
/*JOBPARM S=*
/* JOB TO DUMP A VOLUME USING THE RESET OPTION SO THAT
/* THE DECISION TO RESET THE CHANGED BIT WHEN RESTORING.
//DFDSS PROC
//DFDSS EXEC PGM=ADRDSSU,REGION=OM
//SYSPRINT DD SYSOUT=A
// PEND
// EXEC DFDSS
//DASD DD DISP=SHR,UNIT=3390,VOL=SER=SBOX1B
//TAPE DD UNIT=VT3590,
// DISP=(NEW,CATLG),
// DSN=MHLRES2.DSSDUMP.SBOX1B
//SYSIN DD *
    DUMP OUTDDNAME(TAPE) INDD(DASD) ALLEXCP ALLDATA(*) OPTIMIZE(4)-
    ADMIN RESET
```

In Example 8-6 on page 152 we show the messages issued by DFSMSDss when running the DUMP with RESET option. The significant message is:

ADR342I (001)-DDTFP(01), DATA SET CHANGE INDICATOR HAS BEEN RESET FOR ALL QUALIFIED DATA SETS ON VOLUME SBOX1B

Example 8-6 DSS messages from running job to DUMP volume SBOX1B

```
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.267 00:45
DUMP OUTDDNAME(TAPE) INDD(DASD) ALLEXCP ALLDATA(*) OPTIMIZE(4)- 00130006
ADMIN RESET 00140006
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'DUMP '
ADR109I (R/I)-RI01 (01), 2013.267 00:45:38 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.267 00:45:38 EXECUTION BEGINS
ADR342I (001)-DDTFP(01), DATA SET CHANGE INDICATOR HAS BEEN RESET FOR ALL
QUALIFIED DATA SETS ON VOLUME SBOX1B
```

```

ADRO06I (001)-STEND(02), 2013.267 00:47:26 EXECUTION ENDS
ADRO13I (001)-CLTSK(01), 2013.267 00:47:26 TASK COMPLETED WITH RETURN CODE 0000
ADRO12I (SCH)-DSSU (01), 2013.267 00:47:26 DFSMSDSS PROCESSING COMPLETE. HIGHEST
RETURN CODE IS 0000

```

We used the VTOC listing program on volume SBOX1B to verify that the DS1DSCHA bit had been reset. In Example 8-7 on page 153 we show the output from the VTOC listing program showing that all DS1DSCHA settings have been RESET to N.

Example 8-7 Listing of data sets from volume SBOX1B showing DS1DSCHA bit settings are all N

```

DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:
DSN: SYS1.VTOCIX.SBOX1B                      N
DSN: HFS.SC65.ILMRECON.HFS                    N
DSN: SYS1.VVDS.VSBOX1B                       N
DSN: OMVS.SC65.XXX.HFS                       N
DSN: HFS.SC70.DEV                            N
DSN: HFS.SC64.ETC                            N
DSN: OMVS.SC63.DOMR5.MAIL3.HFS               N
DSN: OMVS.SC63.USERS.DSSDUMP.ALLDATA         N
DSN: OMVS.SC63.XML.HFS                      N
DSN: OMVS.HERING.SUBHFS2.TEST                N
DSN: OMVS.ROGERS.TEST6.SAV                  N
DSN: OMVS.PRADIER.HFS                       N
DSN: OMVS.ROGERS.TEST.SAV.DATA              N
DSN: OMVS.MORGAR1.HFS                      N
DSN: LUTZ.LOADLIB                           N
DSN: OMVS.EJESWEB.HFS.DATA                  N
DSN: OMVS.MHLRES3.HFS                      N
DSN: MHLEAV1.TEST                          N
DSN: OMVS.WATERS.HFS.DATA                   N
END OF DATA REACHED - ALL DATASETS PROCESSED

```

Scenario 3: RESTORE without RESET(DUMPIYESINO) specified

In this scenario we ran a DFSMSDss RESTORE job without specifying RESET so that the default would take effect.

Since the input DUMP data set was created with RESET specified, the result was expected to be that the target volume data sets had their DS1DSCHA indicator turned off.

In Example 8-8 on page 153 we show the job to RESTORE volume SBOX1B to volume ML9E87. The COPYVOLID statement is required.

Example 8-8 JCL to RESTORE volume SBOX1B to volume ML9E87 with default RESET option

```

//MHLRES2D JOB 99990000,UALF0,CLASS=A,NOTIFY=&SYSUID
/*JOBPARM S=*
//*
//REST EXEC PGM=ADRDSSU,REGION=OM
//SYSPRINT DD SYSOUT=A
//DASD DD UNIT=3390,DISP=OLD,VOL=SER=ML9E87
//TAPE DD DISP=(OLD,KEEP),
// DSN=MHLRES2.DSSDUMP.SBOX1B
//SYSIN DD *
RESTORE INDD(TAPE) OUTDD(DASD) PURGE COPYVOLID

```

/*

In Example 8-9 on page 154 we show the results of the RESTORE. The ADR342I message shows that the data set change indicators have been set to N (off). This is a result of the RESET option having been chosen when the DUMP was created, and that the REST(DUMP) option is the default for the RESTORE.

The ADR320I and ADR344I messages arose as a result of the COPYVOLID on the RESTORE command. Volume ML9E87 has been renamed to SBOX1B. Since it is a duplicate it has been made unavailable, which in z/OS terms means that it is taken offline.

Example 8-9 DSS messages from RESTORE without RESET specified

```
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.267 11:44
RESTORE INDD(TAPE) OUTDD(DASD) PURGE COPYVOLID 00110005
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'RESTORE '
ADR109I (R/I)-RI01 (01), 2013.267 11:44:52 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.267 11:44:52 EXECUTION BEGINS
ADR780I (001)-TDFP (01), THE INPUT DUMP DATA SET BEING PROCESSED IS IN FULL VOLUME
FORMAT AND WAS CREATED BY Z/OS DFSMSDSS VERSION
2 RELEASE 1 MODIFICATION LEVEL 0 ON 2013.267 00:45:38
ADR342I (001)-TDFP (01), DATA SET CHANGE INDICATOR HAS BEEN RESET FOR ALL
QUALIFIED DATA SETS ON VOLUME ML9E87
ADR320I (001)-SBRTN(01), VOLUME SERIAL ML9E87 ON UNIT 9E87 IS CHANGED TO SBOX1B
ADR344I (001)-SBRTN(01), VOLSER ON UCB 9E87 IS A DUPLICATE. VOLUME MADE
UNAVAILABLE
ADR006I (001)-STEND(02), 2013.267 11:46:43 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.267 11:46:43 TASK COMPLETED WITH RETURN CODE 0000
ADR012I (SCH)-DSSU (01), 2013.267 11:46:43 DFSMSDSS PROCESSING COMPLETE. HIGHEST
RETURN CODE IS 0000
```

We ran a job to rename the duplicate SBOX1B back to ML9E87 so that we can put it back online and list the data sets on it.

In Example 8-10 on page 154 we show the job to rename offline volume SBOX1B to ML9E87 using the ICKDSF program REFORMAT command.

Example 8-10 JCL to rename duplicate volume SBOX1B to ML9E87

```
//MHLRES2L JOB (99990000),UAA LFO,CLASS=A,NOTIFY=&SYSUID
/*JOBPARM S=*
// EXEC PGM=ICKDSF,REGION=5012K
//SYSPRINT DD SYSOUT=A
//SYSIN DD *
REFORMAT UNIT(9E87) -
VERIFY(SBOX1B) -
VOLID(ML9E87)
```

The messages from the job in Example 8-10 are shown in Example 8-11.

Example 8-11 ICKDSF messages showing volume rename

```
ICKDSF - MVS/ESA    DEVICE SUPPORT FACILITIES 17.0          TIME: 11:52:04
09/24/13    PAGE    1
```

```
REFORMAT UNIT(9E87) -                                00060003
  VERIFY(SBOX1B) -                                     00070003
VOLID(ML9E87)                                           00080003
```

ICK00700I DEVICE INFORMATION FOR 9E87 IS CURRENTLY AS FOLLOWS:

```
  PHYSICAL DEVICE = 3390
  STORAGE CONTROLLER = 2107
  STORAGE CONTROL DESCRIPTOR = E8
  DEVICE DESCRIPTOR = 0C
  ADDITIONAL DEVICE INFORMATION = 4A00003C
  TRKS/CYL = 15, # PRIMARY CYLS = 10017
```

ICK04000I DEVICE IS IN SIMPLEX STATE

ICK00091I 9E87 NED=002107.900.IBM.75.0000000L3331

ICK03091I EXISTING VOLUME SERIAL READ = SBOX1B

ICK00001I FUNCTION COMPLETED, HIGHEST CONDITION CODE WAS 0

11:52:27 09/24/13

ICK00002I ICKDSF PROCESSING COMPLETE. MAXIMUM CONDITION CODE WAS 0

After the successful rename volume ML9E87 was still offline. The **V 9E87, ONLINE** command was issued to make it available again. We then used the VTOC listing program to show the data sets on the restored volume.

We ran the JCL as in Example 8-1 on page 149 but defining volume ML9E87 in place of SBOX1B. In Example 8-12 on page 155 we show that the data set change indicators are all N (OFF) as expected.

Note that the first data set listed is SYS1.VTOCIX.SBOX1B which came from the original SBOX1B volume.

Example 8-12 Listing of data sets from volume ML9E87 showing DS1DSCHA bit settings are all N

DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:

DSN: SYS1.VTOCIX.SBOX1B	N
DSN: HFS.SC65.ILMRECON.HFS	N
DSN: SYS1.VVDS.VSBOX1B	N
DSN: OMVS.SC65.XXX.HFS	N
DSN: HFS.SC70.DEV	N
DSN: HFS.SC64.ETC	N
DSN: OMVS.SC63.DOMR5.MAIL3.HFS	N
DSN: OMVS.SC63.USERS.DSSDUMP.ALLDATA	N
DSN: OMVS.SC63.XML.HFS	N
DSN: OMVS.HERING.SUBHFS2.TEST	N
DSN: OMVS.ROGERS.TEST6.SAV	N
DSN: OMVS.PRADIER.HFS	N
DSN: OMVS.ROGERS.TEST.SAV.DATA	N
DSN: OMVS.MORGAR1.HFS	N
DSN: LUTZ.LOADLIB	N
DSN: OMVS.EJESWEB.HFS.DATA	N
DSN: OMVS.MHLRES3.HFS	N
DSN: MHLEAV1.TEST	N

DSN: OMVS.WATERS.HFS.DATA
END OF DATA REACHED - ALL DATASETS PROCESSED

N

Scenario 4: RESTORE with RESET(NO) specified

In this scenario we ran a DFSMSDss RESTORE job specifying RESET(NO) so that the RESET setting in the DUMP data set would be overridden.

In Example 8-13 on page 156 we show the job to RESTORE volume SBOX1B to ML9E87 with the RESET(NO) option. This was expected to result in data sets on the restored volume having the data set change indicators set as they were when the volume was dumped.

Example 8-13 JCL to RESTORE volume SBOX1B to volume ML9E87 with RESET (NO) option

```
//MHLRES2D JOB 99990000,UALF0,CLASS=A,NOTIFY=&SYSUID
/*JOBPARM S=*
/*
//REST EXEC PGM=ADRDSSU,REGION=0M
//SYSPRINT DD SYSOUT=A
//DASD DD UNIT=3390,DISP=OLD,VOL=SER=ML9E87
//TAPE DD DISP=(OLD,KEEP),
// DSN=MHLRES2.DSSDUMP.SBOX1B
//SYSIN DD *
    RESTORE INDD(TAPE) OUTDD(DASD) PURGE COPYVOLID RESET(NO)
/*
```

In Example 8-14 on page 156 we show the DFSMSDss messages from running the RESTORE with RESET(NO) specified. There are no messages indicating that the data set change indicators have been reset. The ADR320I and ADR344I are issued as expected.

Example 8-14 DSS messages from RESTORE with RESET(NO) specified

```
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.267 12:05
RESTORE INDD(TAPE) OUTDD(DASD) PURGE COPYVOLID RESET(NO) 00110006
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'RESTORE '
ADR109I (R/I)-RI01 (01), 2013.267 12:05:44 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.267 12:05:44 EXECUTION BEGINS
ADR780I (001)-TDFP (01), THE INPUT DUMP DATA SET BEING PROCESSED IS IN FULL VOLUME
FORMAT AND WAS CREATED BY Z/OS DFSMSDSS VERSION
2 RELEASE 1 MODIFICATION LEVEL 0 ON 2013.267 00:45:38
ADR320I (001)-SBRTN(01), VOLUME SERIAL ML9E87 ON UNIT 9E87 IS CHANGED TO SBOX1B
ADR344I (001)-SBRTN(01), VOLSER ON UCB 9E87 IS A DUPLICATE. VOLUME MADE
UNAVAILABLE
ADR006I (001)-STEND(02), 2013.267 12:07:16 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.267 12:07:16 TASK COMPLETED WITH RETURN CODE 0000
ADR012I (SCH)-DSSU (01), 2013.267 12:07:16 DFSMSDSS PROCESSING COMPLETE. HIGHEST
RETURN CODE IS 0000
```

We ran the same job as in Example 8-10 on page 154 to rename the duplicate volume SBOX1B back to ML9E87.

We then ran the same job as in Example 8-12 on page 155 to list the data sets with the VTOC listing program.

In Example 8-15 on page 157 we show the results of running the CVOL listing program. The list shows that the data set changed indicators are the same as they were before running the DUMP job. This listing can be compared with the one in Example 8-4 on page 151.

Example 8-15 Listing of data sets from volume ML9E87 showing DS1DSCHA bit settings are mixed

DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:

DSN: SYS1.VTOCIX.SBOX1B	N
DSN: HFS.SC65.ILMRECON.HFS	Y
DSN: SYS1.VVDS.VSBOX1B	N
DSN: OMVS.SC65.XXX.HFS	Y
DSN: HFS.SC70.DEV	Y
DSN: HFS.SC64.ETC	Y
DSN: OMVS.SC63.DOMR5.MAIL3.HFS	Y
DSN: OMVS.SC63.USERS.DSSDUMP.ALLDATA	Y
DSN: OMVS.SC63.XML.HFS	Y
DSN: OMVS.HERING.SUBHFS2.TEST	Y
DSN: OMVS.ROGERS.TEST6.SAV	Y
DSN: OMVS.PRADIER.HFS	Y
DSN: OMVS.ROGERS.TEST.SAV.DATA	Y
DSN: OMVS.MORGAR1.HFS	Y
DSN: LUTZ.LOADLIB	Y
DSN: OMVS.EJESWEB.HFS.DATA	Y
DSN: OMVS.MHLRES3.HFS	Y
DSN: MHLEAV1.TEST	N
DSN: OMVS.WATERS.HFS.DATA	Y

END OF DATA REACHED - ALL DATASETS PROCESSED

8.1.4 Security protection over use of RESET with DUMP or RESTORE

Security protection on the use of the RESET option on the DUMP command and on the RESET(YES) option on the RESTORE command is available.

RACF users

Resources are in the FACILITY CLASS. If these resources are not defined then all users have access to the RESET options. If the resources are defined users must have READ access authorization to the RACF resource to use the RESET options.

- ▶ DUMP command RESET option
Read access to the STGADMIN.ADR.DUMP.RESET resource is required.
- ▶ RESTORE command RESET(YES) option
Read access to the STGADMIN.ADR.RESTORE.RESET.YES resource is required.

Other security system users

If using a security system other than RACF it may be necessary to implement the equivalent of the RACF FACILITY profiles before the RESET option can be used.

8.2 DEBUG option SMSMSG

When DFSMSdss allocates data sets it may use the System Managed Storage (SMS) function. SMS applies rules to the allocation of data sets, and may fail a request.

As an aid to problem resolution several commands have a DEBUG option. To facilitate problem analysis when using SMS, the SMSMSG sub-option is provided in DFSMS V2.1. SMSMSG may be used in conjunction with other DEBUG sub operands where appropriate.

When specified, the option DEBUG(SMSMSG) will cause certain SMS messages produced by the control routines to be externalized.

The DEBUG(SMSMSG) option may be used with the following DFSMSdss commands:

- ▶ CONVERTV: Convert existing volumes to and from SMS management.
- ▶ COPY: For logical and physical data sets.
- ▶ RESTORE: For logical and physical data sets.

There may be ACS WRITE statements in the ACS routines. These are not affected by the DFSMSdss SMSMSG option of DEBUG. If coded, they may appear as well as any messages generated by SMSMSG.

Note: The DEBUG(SMSMSG) option depends on DFSMSdss using the ACS routines. If the DFSMSdss BYPASSACS option is specified then DEBUG(SMSMSG) will have no effect.

We provide scenarios to illustrate the use of the DEBUG(SMSMSG) option on the COPY and RESTORE statements.

Scenario 1: DFSMSdss COPY

In Example 8-16 on page 158 we show the job to illustrate use of DEBUG(SMSMSG) with BYPASSACS specified. There are two copy statements in the job, DEBUG(SMSMSG) has to be specified on each.

We specified DEBUG(SMSMSG), but the presence of BYPASSACS resulted in no messages generated.

Example 8-16 JCL to run a DSS COPY job with DEBUG(SMSMSG) and BYPASSACS specified

```
//MHLRES2C JOB (999,P0K),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
//COPY      EXEC PGM=ADRDSSU
//*N        DD VOL=SER=MLD00B,UNIT=3390,DISP=OLD
//OUT        DD VOL=SER=MLDB35,UNIT=3390,DISP=OLD
//OUT2       DD VOL=SER=MLDB35,UNIT=3390,DISP=OLD
//SYSPRINT DD SYSOUT=*
//SYSIN      DD *
COPY DATASET(INCLUDE(MHLRES2.SRCHFOR.LIST)) -
      DEBUG(SMSMSG) -
      OUTDD(OUT) -
      ALLDATA(*) ALLEXCP -
      STORCLAS(HSMFR) RENAMEU(YYY) -
      BYPASSACS(**) ADMIN SHR
COPY DATASET(INCLUDE(MHLRES2.SUPERC.LIST)) -
```



```

DEBUG(SMSMSG)          -
OUTDD(OUT2) -
ALLDATA(*) ALLEXCP -
STORCLAS(HSMFR) RENAMEU(YYY) -
BYPASSACS(**) ADMIN SHR

```

In Example 8-17 on page 159 we show the effect of specifying `DEBUG(SMSMSG)`. No message `ADR803I` is written. However, the `IGD01010I` messages that are from specific `WRITE` statements in the ACS routines do appear.

Example 8-17 DSS messages from COPY job with `DEBUG(SMSMSG)` specified showing no `ADR803I`

```

PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.270 14:38
COPY DATASET(INCLUDE(MHLRES2.SRCHFOR.LIST)) - 00170001
  DEBUG(SMSMSG) - 00170104
  OUTDD(OUT) - 00171000
  ALLDATA(*) ALLEXCP - 00180000
  STORCLAS(HSMFR) RENAMEU(YYY) - 00181000
  BYPASSACS(**) ADMIN SHR 00190000
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'COPY '
COPY DATASET(INCLUDE(MHLRES2.SUPERCL.LIST)) - 00191001
  DEBUG(SMSMSG) - 00191104
  OUTDD(OUT2) - 00192001
  ALLDATA(*) ALLEXCP - 00193001
  STORCLAS(HSMFR) RENAMEU(YYY) - 00194001
  BYPASSACS(**) ADMIN SHR 00195001
ADR101I (R/I)-RI01 (01), TASKID 002 HAS BEEN ASSIGNED TO COMMAND 'COPY '
ADR109I (R/I)-RI01 (01), 2013.270 14:38:02 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.270 14:38:02 EXECUTION BEGINS
ADR709E (001)-VDSS (01), AN ERROR OCCURRED IN THE STORAGE MANAGEMENT SUBSYSTEM WHILE
ALLOCATING DATA SET MHLRES2.SRCHFOR.LIST WITH
NEWNAME YYY.SRCHFOR.LIST. SMS MESSAGES FOLLOW.
IGD01010I SG ACS GETS CONTROL &ACSENVIR=ALLOC
IGD01010I &STORCLAS = HSMFR
IGD01010I &STORGRP = SG1
IGD17206I VOLUME SELECTION HAS FAILED - THERE ARE NOT ENOUGH VOLUMES
WITH SUFFICIENT SPACE FOR DATA SET
YYY.SRCHFOR.LIST
IGD17277I THERE ARE (1) CANDIDATE VOLUMES OF WHICH (1) ARE ENABLED OR QUIESCED
IGD17331I DATA SET YYY.SRCHFOR.LIST COULD
NOT BE ALLOCATED. PREFERRED FAST REPLICATION WAS SPECIFIED
BY THE CALLER.
IGD17290I THERE WERE 1 CANDIDATE STORAGE GROUPS OF WHICH THE FIRST 1
WERE ELIGIBLE FOR VOLUME SELECTION.
THE CANDIDATE STORAGE GROUPS WERE:SG1
IGD17279I 1 VOLUMES WERE REJECTED BECAUSE THEY WERE NOT ON THE INCLUDE LIST
DR809I (001)-VDSS (01), ADDITIONAL DIAGNOSTIC DATA FOR PRECEDING MESSAGE:
  SC=HSMFR MC=MCDB22 DC=
  REQPRI=0000000020BLK REQSEC=0000004096BLK REQVOLS=01
DR801I (001)-DDDS (01), 2013.270 14:38:02 DATA SET FILTERING IS COMPLETE. 1 OF 1 DATA SETS
WERE SELECTED: 0 FAILED SERIALIZATION
AND 0 FAILED FOR OTHER REASONS
DR455W (001)-DDDS (02), THE FOLLOWING DATA SETS WERE NOT SUCCESSFULLY PROCESSED
  MHLRES2.SRCHFOR.LIST
DR006I (001)-STEND(02), 2013.270 14:38:02 EXECUTION ENDS
DR013I (001)-CLTSK(01), 2013.270 14:38:02 TASK COMPLETED WITH RETURN CODE 0008

```

```

DR016I (002)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (002)-STEND(01), 2013.270 14:38:02 EXECUTION BEGINS
ADR405E (002)-DYNA (02), DYNAMIC ALLOCATION OF VOLUME MLD30C FAILED. ERROR CODE 0218.
INFORMATION CODE 0000
ADR283W (002)-FILTC(01), DATA SET MHLRES2.SUPERC.LIST WAS NOT SELECTED, 2
PAGE 0002 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.270 14:38
ADR383W (002)-DDDS (01), DATA SET MHLRES2.SUPERC.LIST NOT SELECTED
ADR455W (002)-DDDS (03), THE FOLLOWING DATA SETS WERE NOT SUCCESSFULLY PROCESSED
      MHLRES2.SUPERC.LIST
ADR470W (002)-DDDS (04), NO DATA SETS SELECTED FOR PROCESSING
ADR006I (002)-STEND(02), 2013.270 14:38:02 EXECUTION ENDS
ADR013I (002)-CLTSK(01), 2013.270 14:38:02 TASK COMPLETED WITH RETURN CODE 0008
ADR012I (SCH)-DSSU (01), 2013.270 14:38:02 DFSMSDSS PROCESSING COMPLETE. HIGHEST RETURN
CODE IS 0008 FROM:
      TASK 001
      TASK 002

```

In Example 8-18 on page 160 we show the effect of coding message ADR803I on a command that *does not* have the BYPASSACS option.

We see that the ADR803I message indicates that the STORCLAS was found to not be NULL. It was set to HSMFR as specified in the COPY statement.

The IGD01010I message also reports that variable &STORCLAS is set to HSMFR.

The significance, and usefulness, of the ADR803I message is that it shows the internal processing that resulted in the STORCLAS setting. If the DFSMSdss processing had not worked due to storage class problems, then the indication that the HSMFR storage class was being used might lead to what the problem was.

We see that in this example there are several problems due to other situations, but they do not result in any further ADR803I messages.

Example 8-18 DSS messages from COPY job with DEBUG(SMSMSG) and no BYPASSACS specified

```

PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.270 14:43
COPY DATASET(INCLUDE(MHLRES2.SRCHFOR.LIST)) - 00170001
  DEBUG(SMSMSG) - 00170104
  OUTDD(OUT) - 00171000
  ALLDATA(*) ALLEXCP - 00180000
  STORCLAS(HSMFR) RENAMEU(YYY) - 00181000
  ADMIN SHR 00190005
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'COPY '
COPY DATASET(INCLUDE(MHLRES2.SUPERC.LIST)) - 00191001
  DEBUG(SMSMSG) - 00191104
  OUTDD(OUT2) - 00192001
  ALLDATA(*) ALLEXCP - 00193001
  STORCLAS(HSMFR) RENAMEU(YYY) - 00194001
  ADMIN SHR 00195005
ADR101I (R/I)-RI01 (01), TASKID 002 HAS BEEN ASSIGNED TO COMMAND 'COPY '
ADR109I (R/I)-RI01 (01), 2013.270 14:43:28 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.270 14:43:28 EXECUTION BEGINS
ADR803I (001)-ACS (01), SMS DIAGNOSTIC MESSAGES:
IGD01008I STORCLAS NOT NULL. SET TO HSMFR
ADR709E (001)-VDSS (01), AN ERROR OCCURRED IN THE STORAGE MANAGEMENT SUBSYSTEM WHILE
ALLOCATING DATA SET MHLRES2.SRCHFOR.LIST WITH

```

```

NEWNAME YYY.SRCHFOR.LIST. SMS MESSAGES FOLLOW.
SG ACS GETS CONTROL &ACSENVIR=ALLOC
IGD01010I &STORCLAS = HSMFR
IGD01010I &STORGRP = SG1
IGD17206I VOLUME SELECTION HAS FAILED - THERE ARE NOT ENOUGH VOLUMES
WITH SUFFICIENT SPACE FOR DATA SET
YYY.SRCHFOR.LIST
IGD17277I THERE ARE (1) CANDIDATE VOLUMES OF WHICH (1) ARE ENABLED OR QUIESCED
IGD17331I DATA SET YYY.SRCHFOR.LIST COULD
NOT BE ALLOCATED. PREFERRED FAST REPLICATION WAS SPECIFIED
BY THE CALLER.
IGD17290I THERE WERE 1 CANDIDATE STORAGE GROUPS OF WHICH THE FIRST 1
WERE ELIGIBLE FOR VOLUME SELECTION.
THE CANDIDATE STORAGE GROUPS WERE:SG1
IGD17279I 1 VOLUMES WERE REJECTED BECAUSE THEY WERE NOT ON THE INCLUDE LIST
ADR809I (001)-VDSS (01), ADDITIONAL DIAGNOSTIC DATA FOR PRECEDING MESSAGE:
      SC=HSMFR MC=MCDB22 DC=
      REQPRI=0000000020BLK REQSEC=0000004096BLK REQVOLS=01
ADR801I (001)-DDDS (01), 2013.270 14:43:28 DATA SET FILTERING IS COMPLETE. 1 OF 1 DATA SETS
WERE SELECTED: 0 FAILED SERIALIZATION
      AND 0 FAILED FOR OTHER REASONS
ADR455W (001)-DDDS (02), THE FOLLOWING DATA SETS WERE NOT SUCCESSFULLY PROCESSED
      MHLRES2.SRCHFOR.LIST
ADR006I (001)-STEND(02), 2013.270 14:43:28 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.270 14:43:28 TASK COMPLETED WITH RETURN CODE 0008
ADR016I (002)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (002)-STEND(01), 2013.270 14:43:28 EXECUTION BEGINS
PAGE 0002 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.270 14:43
ADR405E (002)-DYNA (02), DYNAMIC ALLOCATION OF VOLUME MLD30C FAILED. ERROR CODE 0218.
INFORMATION CODE 0000
ADR283W (002)-FILTC(01), DATA SET MHLRES2.SUPERC.LIST WAS NOT SELECTED, 2
ADR383W (002)-DDDS (01), DATA SET MHLRES2.SUPERC.LIST NOT SELECTED
ADR455W (002)-DDDS (03), THE FOLLOWING DATA SETS WERE NOT SUCCESSFULLY PROCESSED
      MHLRES2.SUPERC.LIST
ADR470W (002)-DDDS (04), NO DATA SETS SELECTED FOR PROCESSING
ADR006I (002)-STEND(02), 2013.270 14:43:28 EXECUTION ENDS
ADR013I (002)-CLTSK(01), 2013.270 14:43:28 TASK COMPLETED WITH RETURN CODE 0008
ADR012I (SCH)-DSSU (01), 2013.270 14:43:28 DFSMSDSS PROCESSING COMPLETE. HIGHEST RETURN
CODE IS 0008 FROM:
      TASK 001
      TASK 002

```

Scenario 2: DFSMSdss RESTORE

When DFSMSdss is used to RESTORE a data set and the operation requires space to be allocated, then the ACS routines would be used to select a suitable location.

The DEBUG(SMSMSG) option is expected to provide information on what the ACS chose to do. Without SMSMSG the storage class selected for the allocation might not have been as selected, and to find out what was used it would be necessary to list the allocation.

In Example 8-19 on page 162 we show the job to run DFSMSdss RESTORE with DEBUG(SMSMSG) specified. The job restores the HSM.BCDS data set with a rename to HSM.BCDS.TEST2 which does not exist on the selected output volume.

Example 8-19 JCL to run DSS RESTORE using DEBUG(SCSMMSG) with new allocation

```
//MHLRES2D JOB 99990000,UALF0,CLASS=A,NOTIFY=&SYSUID
/*JOBPARM S=*
/*
//REST EXEC PGM=ADDRSSU,REGION=0M
//SYSPRINT DD SYSOUT=A
//DASD DD UNIT=3390,DISP=OLD,VOL=SER=SBOX1B
//TAPE DD DISP=(OLD,KEEP),
// DSN=MHLRES2.DUMP.HSM.BCDS
//SYSIN DD *
    RESTORE INDD(TAPE) OUTDD(DASD) -
        DEBUG(SMSMSG) -
        REPLACEUNCONDITIONAL -
        DATASET(INCLUDE(HSM.BCDS)) -
        RENAMEU((HSM.BCDS,MHLRES2.HSM.BCDS.TEST2))
/*
```

In Example 8-20 on page 162 we show the DFSMSdss messages resulting from RESTORE with DEBUG(SCSMMSG).

The significant message is IDR803I which shows IGD01008I STORCLAS NOT NULL. SET TO SCRLS. This information would not have been shown if DEBUG(SCSMMSG) had not been set.

Example 8-20 DSS messages resulting from RESTORE with DEBUG(SMSMSG)

```
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.270 18:06
    RESTORE INDD(TAPE) OUTDD(DASD) - 00110000
        DEBUG(SMSMSG) - 00110110
        REPLACEUNCONDITIONAL - 00110200
        DATASET(INCLUDE(HSM.BCDS)) - 00111009
        RENAMEU((HSM.BCDS,MHLRES2.HSM.BCDS.TEST2)) 00112011
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'RESTORE '
ADR109I (R/I)-RI01 (01), 2013.270 18:06:04 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.270 18:06:04 EXECUTION BEGINS
ADR780I (001)-TDDS (01), THE INPUT DUMP DATA SET BEING PROCESSED IS IN PHYSICAL DATA SET
FORMAT AND WAS CREATED BY Z/OS DFSMSDSS
    VERSION 2 RELEASE 1 MODIFICATION LEVEL 0 ON 2013.268 19:31:03
ADR803I (001)-ACS (01), SMS DIAGNOSTIC MESSAGES:
IGD01008I STORCLAS NOT NULL. SET TO SCRLS
ADR395I (001)-SB105(01), DATA SET HSM.BCDS.DATA ALLOCATED WITH NEWNAME
MHLRES2.HSM.BCDS.TEST2.DATA, ON VOLUME(S): SBOX1B
ADR395I (001)-SB105(02), DATA SET HSM.BCDS.INDEX ALLOCATED WITH NEWNAME
MHLRES2.HSM.BCDS.TEST2.INDEX, ON VOLUME(S): SBOX1B
ADR418I (001)-SB105(01), THE FOLLOWING COMPONENTS FOR CLUSTER MHLRES2. ON SBOX1B MAY HAVE
TO BE CATALOGED IN CATALOG UCAT.VSBOX01
    COMPONENT MHLRES2.HSM.BCDS.TEST2.DATA
    COMPONENT MHLRES2.HSM.BCDS.TEST2.INDEX
ADR378I (001)-TDDS (02), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED FROM VOLUME
SBXHS8
    HSM.BCDS HSM.BCDS.DATA HSM.BCDS.INDEX
    HSM.BCDS
```

In Example 8-21 on page 163 we show the results from IDCAMS listing which shows STORAGECLASS SCRLS which corresponds to the ADR803I message as in Example 8-20 on page 162.

By using DEBUG(SMSMSG) for diagnostic purposes the additional overhead of setting up and running the IDCAMS job is avoided.

Example 8-21 IDCAMS display of details of MHLRES2.HSM.BCDS.TEST2

```

IDCAMS  SYSTEM SERVICES                                TIME: 18:10:50
09/27/13      PAGE      1

/* IDCAMS COMMAND */
  LISTCAT ENTRIES(MHLRES2.HSM.BCDS.TEST2) -
    CLUSTER -
    ALL
CLUSTER ----- MHLRES2.HSM.BCDS.TEST2
  IN-CAT --- UCAT.VSBOX01
  HISTORY
  DATASET-OWNER----- (NULL)    CREATION-----2013.270
  RELEASE-----2      EXPIRATION-----0000.000
  SMSDATA
  STORAGECLASS -----SCRLS    MANAGEMENTCLASS--MCNOACT
  DATACLASS -----HSMCDS    LBACKUP ---0000.000.0000
  CA-RECLAIM----- (YES)
  EATTR----- (NULL)
  BWO STATUS-----00000000    BWO TIMESTAMP---00000 00:00:00.0
  BWO----- (NULL)
  RLSDATA
    LOG -----NONE    RECOVERY REQUIRED --(NO)    FRLOG
----- (NULL)
    VSAM QUIESCED ----- (NO)    RLS IN USE ----- (NO)
LOGREPLICATE----- (NO)
  LOGSTREAMID-----NONE
    RECOVERY TIMESTAMP LOCAL-----X'0000000000000000'

```

Compatibility

The DEBUG option is available on releases prior to DFSMS V2.1, but the SMSMSG sub-option is not.

8.3 RESTORE command

There are updates to existing RESTORE command keywords in DFSMS V2.1:

- ▶ The REPLACEUNCONDITIONAL keyword on the RESTORE command now works for physical data sets.
- ▶ The RENAMEUNCONDITIONAL keyword on the RESTORE command, which previously worked only on non-VSAM physical data sets, will now work on VSAM physical data sets, as well.

In this section we look at the keywords and provide scenarios to show how they work.

8.3.1 Scenarios for RESTORE keywords

We ran several jobs to demonstrate the updates to the DFSMSdss RESTORE command REPLACEUNCONDITIONAL and RENAMEUNCONDITIONAL keywords to show the effect of running these jobs in DFSMS V2.1 and in DFSMS V1.13.

We created a PHYSICAL format DFSMSdss DUMP of a non-VSAM, and of a VSAM data set to use later for the DFSMSdss RESTORE jobs.

DFSMSdss DUMP creation jobs

In Example 8-22 on page 164 we show the job to create a PHYSICAL format DUMP of the data set MHLRES2.CNTL.JCL which is a non-VSAM data set. These enhancements to DFSMSdss RESTORE in DFSMS V2.1 relate to the RESTORE from a PHYSICAL DFSMSdss DUMP so the essential parameter is the option PHYSINDDNAME(IN).

The data set MHLRES2.CNTL.JCL was in use so the TOLERATE(ENQF) option was used which specifies that even if the data set is in use (enqueued or serialized), the enqueue failure should be tolerated.

Example 8-22 JCL to create DSS DUMP of non-VSAM data set

```
//MHLRES2D JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
//COPY     EXEC PGM=ADRDSSU
//IN       DD DISP=SHR,DSN=MHLRES2.CNTL.JCL
//OUT      DD DSN=MHLRES2.DUMP.CNTL.JCL,DISP=(NEW,CATLG),
//          SPACE=(TRK,(500,150),RLSE)
//SYSPRINT DD SYSOUT=*
//SYSIN    DD *
DUMP DATASET(INCLUDE(MHLRES2.CNTL.JCL)) -
      OUTDD(OUT) -
      TOLERATE(ENQF) -
      ALLDATA(*) ALLEXCP -
      PHYSINDDNAME(IN)
/*
```

In Example 8-23 on page 164 we show the DFSMSdss messages from the job creating the PHYSICAL format DFSMSdss DUMP data set. There are no messages explicitly confirming that the DUMP data set is in PHYSICAL format.

The return code 04 from this job is generated because of the message ADR411W in Example 8-23 on page 164.

Example 8-23 DSS messages from job to create DSS PHYSICAL format DUMP of non-VSAM data set

```
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.268 15:51
DUMP DATASET(INCLUDE(MHLRES2.CNTL.JCL)) - 00170035
      OUTDD(OUT) - 00171000
      TOLERATE(ENQF) - 00171135
      ALLDATA(*) ALLEXCP - 00172000
      PHYSINDDNAME(IN) 00173035
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'DUMP '
ADR109I (R/I)-RI01 (01), 2013.268 15:51:07 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
```

```

ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.268 15:51:07 EXECUTION BEGINS
ADR411W (001)-FDSDU(02), DATA SET MHLRES2.CNTL.JCL ON VOLUME SBOX1L WAS NOT
SERIALIZED ON REQUEST
ADR378I (001)-DTDS (01), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED FROM
VOLUME SBOX1L
        MHLRES2.CNTL.JCL
ADR006I (001)-STEND(02), 2013.268 15:51:11 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.268 15:51:11 TASK COMPLETED WITH RETURN CODE 0004
ADR012I (SCH)-DSSU (01), 2013.268 15:51:11 DFSMSDSS PROCESSING COMPLETE. HIGHEST
RETURN CODE IS 0004 FROM:
        TASK      001

```

In Example 8-24 on page 165 we show the job to create a PHYSICAL format DUMP of the data set HSM.BCDS which is a VSAM data set. These enhancements to DFSMSdss RESTORE in DFSMS V2.1 relate to the RESTORE from a PHYSICAL DFSMSdss DUMP so the essential parameter is the option PHYSINDDNAME(IN).

In Example 8-22 on page 164 we have specified TOLERATE(ENQF).

Example 8-24 JCL to create DSS DUMP of VSAM data set

```

//MHLRES2D JOB (999,P0K),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
//COPY      EXEC PGM=ADRDSSU
//IN        DD DISP=SHR,DSN=HSM.BCDS
//OUT       DD DSN=MHLRES2.DUMP.HSM.BCDS,DISP=(NEW,CATLG),
//          SPACE=(TRK,(500,150),RLSE)
//SYSPRINT DD SYSOUT=*
//SYSIN     DD *
        DUMP DATASET(INCLUDE(HSM.BCDS)) -
            OUTDD(OUT) -
            TOLERATE(ENQF) -
            ALLDATA(*) ALLEXCP -
            PHYSINDDNAME(IN)
/*

```

In Example 8-23 on page 164 we show the DFSMSdss messages from the job creating the PHYSICAL format DFSMSdss DUMP data set. There are no messages explicitly confirming that the DUMP data set is in PHYSICAL format.

There is an indication that the data set HSM.BCDS has been recognized as a VSAM data set as shown in the ADR730W message:

```

ADR730W (001)-FDSDU(01), CLUSTER HSM.BCDS IS OPEN

```

CLUSTER is a VSAM construct that does not apply to non-VSAM data sets. The return code 04 from this job was generated because of the message:

```

ADR411W (001)-DYNA (01), DATA SET HSM.BCDS IN CATALOG CATALOG.HSM ON
VOLUME SBXHS8 WAS NOT SERIALIZED ON REQUEST

```

Example 8-25 DSS messages from job to create DSS PHYSICAL format DUMP of VSAM data set

```
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.268 19:30
DUMP DATASET(INCLUDE(HSM.BCDS)) - 00170036
  OUTDD(OUT) - 00171000
  TOLERATE(ENQF) - 00171100
  ALLDATA(*) ALLEXCP - 00172000
  PHYSINDDNAME(IN) 00173000
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'DUMP '
ADR109I (R/I)-RI01 (01), 2013.268 19:30:59 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.268 19:30:59 EXECUTION BEGINS
ADR730W (001)-FDSU(01), CLUSTER HSM.BCDS IS OPEN
ADR411W (001)-DYNA (01), DATA SET HSM.BCDS IN CATALOG CATALOG.HSM ON VOLUME SBXHS8
WAS NOT SERIALIZED ON REQUEST
ADR378I (001)-DTDS (01), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED FROM
VOLUME SBXHS8
      HSM.BCDS
ADR006I (001)-STEND(02), 2013.268 19:31:04 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.268 19:31:04 TASK COMPLETED WITH RETURN CODE 0004
ADR012I (SCH)-DSSU (01), 2013.268 19:31:04 DFSMSDSS PROCESSING COMPLETE. HIGHEST
RETURN CODE IS 0004 FROM:
      TASK    001
```

DFSMSdss RESTORE jobs running on DFSMS V2.1

In this part we demonstrate the use of the updated REPLACEUNCONDITIONAL and RENAMEUNCONDITIONAL operands as used on DSS RESTORE commands.

Non-VSAM data set restore with rename

In Example 8-26 on page 166 we show the job to RESTORE the data set MHLRES2.CNTL.JCL as a copy named MHLRES2.CNTL.JCL.REST1. The option REPLACEUNCONDITIONAL is specified so that if the data set MHLRES2.CNTL.JCL.REST1 already exists it will be replaced.

The RENAMEUNCONDITIONAL option is abbreviated to RENAMEU.

Example 8-26 JCL to restore and rename non-VSAM data set

```
//MHLRES2D JOB 99990000,UALF0,CLASS=A,NOTIFY=&SYSUID
/*JOBPARM S=*
/*
//REST EXEC PGM=ADRDSSU,REGION=OM
//SYSPRINT DD SYSOUT=A
//DASD DD UNIT=3390,DISP=OLD,VOL=SER=SBX1B
//TAPE DD DISP=(OLD,KEEP),
// DSN=MHLRES2.DUMP.CNTL.JCL
//SYSIN DD *
  RESTORE INDD(TAPE) OUTDD(DASD) -
    REPLACEUNCONDITIONAL -
    DATASET(INCLUDE(MHLRES2.CNTL.JCL)) -
    RENAMEU((MHLRES2.CNTL.JCL,MHLRES2.CNTL.JCL.REST1))
/*
```

In Example 8-27 on page 167 we show the DSS messages resulting from the RESTORE.

In this case we have confirmation that in the input data set is in PHYSICAL format as shown by message ADR780I. We see that the data set MHLRES2.CNTL.JCL.REST1 was ALLOCATED.

Example 8-27 DSS messages from RESTORE of non-VSAM data set where newname is new data set

```
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.268 16:53
RESTORE INDD(TAPE) OUTDD(DASD) - 00110007
DATASET(INCLUDE(MHLRES2.CNTL.JCL)) - 00111008
RENAMEU((MHLRES2.CNTL.JCL,MHLRES2.CNTL.JCL.REST1)) 00112008
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'RESTORE '
ADR109I (R/I)-RI01 (01), 2013.268 16:53:11 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.268 16:53:11 EXECUTION BEGINS
ADR780I (001)-TDDS (01), THE INPUT DUMP DATA SET BEING PROCESSED IS IN PHYSICAL
DATA SET FORMAT AND WAS CREATED BY Z/OS DFSMSDSS
VERSION 2 RELEASE 1 MODIFICATION LEVEL 0 ON 2013.268
15:51:11
ADR395I (001)-TDPNV(02), DATA SET MHLRES2.CNTL.JCL ALLOCATED WITH NEWNAME
MHLRES2.CNTL.JCL.REST1, ON VOLUME(S): SBOX1B
ADR378I (001)-TDDS (01), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED FROM
VOLUME SBOX1L
MHLRES2.CNTL.JCL RESTORED ON
SBOX1B
ADR006I (001)-STEND(02), 2013.268 16:53:11 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.268 16:53:11 TASK COMPLETED WITH RETURN CODE 0000
ADR012I (SCH)-DSSU (01), 2013.268 16:53:11 DFSMSDSS PROCESSING COMPLETE. HIGHEST
RETURN CODE IS 0000
```

We ran the job as shown in Example 8-26 on page 166 again. In Example 8-28 on page 167 we show the DFSMSDss messages resulting from the DFSMSDss RESTORE when the new data set name already exists. In this case we see message ADR442I which indicates that the data set was PREALLOCATED.

In this situation the REPLACEUNCONDITIONAL option ensured that the data set was replaced.

Example 8-28 DSS Messages from RESTORE of non-VSAM data set where newname exists

```
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.268 16:56
RESTORE INDD(TAPE) OUTDD(DASD) - 00110007
REPLACEUNCONDITIONAL - 00110108
DATASET(INCLUDE(MHLRES2.CNTL.JCL)) - 00111008
RENAMEU((MHLRES2.CNTL.JCL,MHLRES2.CNTL.JCL.REST1)) 00112008
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'RESTORE '
ADR109I (R/I)-RI01 (01), 2013.268 16:56:15 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.268 16:56:15 EXECUTION BEGINS
ADR780I (001)-TDDS (01), THE INPUT DUMP DATA SET BEING PROCESSED IS IN PHYSICAL
DATA SET FORMAT AND WAS CREATED BY Z/OS DFSMSDSS
VERSION 2 RELEASE 1 MODIFICATION LEVEL 0 ON 2013.268
15:51:11
```

ADR442I (001)-TDPNV(01), DATA SET MHLRES2.CNTL.JCL PREALLOCATED WITH NEW NAME
 MHLRES2.CNTL.JCL.REST1, ON VOLUME(S): SBOX1B
 ADR378I (001)-TDDS (01), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED FROM
 VOLUME SBOX1L

MHLRES2.CNTL.JCL

RESTORED ON

SBOX1B

ADR006I (001)-STEND(02), 2013.268 16:56:15 EXECUTION ENDS

ADR013I (001)-CLTSK(01), 2013.268 16:56:15 TASK COMPLETED WITH RETURN CODE 0000

ADR012I (SCH)-DSSU (01), 2013.268 16:56:15 DFSMSDSS PROCESSING COMPLETE. HIGHEST
 RETURN CODE IS 0000

VSAM data set restore with rename

In Example 8-29 on page 168 we show the job to restore data set HSM.CDS with rename to MHLRES2.HSM.BCDS. There is no requirement to identify this as a VSAM data set, DFSMSDss determines that from the DFSMSDss DUMP data set.

Example 8-29 JCL to RESTORE VSAM data set HSM.BCDS with rename

```
//MHLRES2D JOB 99990000,UALF0,CLASS=A,NOTIFY=&SYSUID
/*JOBPARM S=*
/*
//REST EXEC PGM=ADRDSSU,REGION=0M
//SYSPRINT DD SYSOUT=A
//DASD DD UNIT=3390,DISP=OLD,VOL=SER=SBOX1B
//TAPE DD DISP=(OLD,KEEP),
// DSN=MHLRES2.DUMP.HSM.BCDS
//SYSIN DD *
    RESTORE INDD(TAPE) OUTDD(DASD) -
        REPLACEUNCONDITIONAL -
        DATASET(INCLUDE(HSM.BCDS)) -
        RENAMEU((HSM.BCDS,MHLRES2.HSM.BCDS))
/*
```

In Example 8-30 on page 168 we show the messages resulting from the RESTORE. In this case since the restored data set was a VSAM data set, additional messages are produced.

DFSMSDss has recognized that there is a DATA and INDEX component in the VSAM data set and has allocated these with the MHLRES2.HSM.BCDS prefix.

DFSMSDss has issued message ADR418I advising that the DATA and INDEX components may have to be cataloged. We found that the data set components MHLRES2.HSM.BCDS.DATA and MHLRES2.HSM.BCDS.INDEX were not cataloged.

Example 8-30 DSS messages resulting from RESTORE of VSAM data set

```
PAGE 0001 5695-DF175 DFSMSDSS V2R01.0 DATA SET SERVICES 2013.268 19:34
    RESTORE INDD(TAPE) OUTDD(DASD) - 00110000
        REPLACEUNCONDITIONAL - 00110100
        DATASET(INCLUDE(HSM.BCDS)) - 00111009
        RENAMEU((HSM.BCDS,MHLRES2.HSM.BCDS)) 00112009
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'RESTORE '
ADR109I (R/I)-RI01 (01), 2013.268 19:34:24 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.268 19:34:24 EXECUTION BEGINS
```

```

ADR780I (001)-TDDS (01), THE INPUT DUMP DATA SET BEING PROCESSED IS IN PHYSICAL DATA SET
FORMAT AND WAS CREATED BY Z/OS DFSMSDSS
          VERSION 2 RELEASE 1 MODIFICATION LEVEL 0 ON 2013.268 19:31:03
ADR395I (001)-SB105(01), DATA SET HSM.BCDS.DATA ALLOCATED WITH NEWNAME
MHLRES2.HSM.BCDS.DATA, ON VOLUME(S): SBOX1B
ADR395I (001)-SB105(02), DATA SET HSM.BCDS.INDEX ALLOCATED WITH NEWNAME
MHLRES2.HSM.BCDS.INDEX, ON VOLUME(S): SBOX1B
ADR418I (001)-SB105(01), THE FOLLOWING COMPONENTS FOR CLUSTER MHLRES2. ON SBOX1B MAY HAVE
TO BE CATALOGED IN CATALOG UCAT.VSBOX01
          COMPONENT MHLRES2.HSM.BCDS.DATA
          COMPONENT MHLRES2.HSM.BCDS.INDEX
ADR378I (001)-TDDS (02), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED FROM VOLUME
SBXHS8
          HSM.BCDS                      HSM.BCDS.DATA
          HSM.BCDS                      HSM.BCDS.INDEX
ADR006I (001)-STEND(02), 2013.268 19:34:25 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.268 19:34:25 TASK COMPLETED WITH RETURN CODE 0000
ADR012I (SCH)-DSSU (01), 2013.268 19:34:25 DFSMSDSS PROCESSING COMPLETE. HIGHEST RETURN
CODE IS 0000

```

In Example 8-31 on page 169 we show the job to create the catalog entries for the VSAM data set components MHLRES2.HSM.BCDS.DATA and MHLRES2.HSM.BCDS.INDEX.

Example 8-31 JCL to catalog VSAM DATA and INDEX components

```

//MHLRES2D JOB (999,POK),'MHLRES2',CLASS=A,MSGCLASS=T,
//          NOTIFY=&SYSUID,TIME=1440,REGION=6M
/*JOBPARM L=999,SYSAFF=*
//IDCAMS EXEC PGM=IDCAMS
//HSMBCDS DD UNIT=3390,VOL=SER=SBOX1B,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
    DEFINE CLUSTER(NAME(MHLRES2.HSM.BCDS) VOLUMES(SBOX1B) -
        RECATALOG          ) -
        DATA(NAME(MHLRES2.HSM.BCDS.DATA) -
            ) -
        INDEX(NAME(MHLRES2.HSM.BCDS.INDEX) -
            )
/*

```

In Example 8-32 on page 169 we show the messages from the IDCAMS job to create the catalog entries for MHLRES2.HSM.BCDS.DATA, MHLRES2.HSM.BCDS.INDEX and for the VSAM cluster name MHLRES2.HSM.BCDS.

When DFSMSDss allocated the DATA and INDEX components on the specified volume it also stored data in the VSAM VOLUME RECORD (VVR) on the volume, and the IDCAMS job then used that information to complete the catalog information.

Example 8-32 IDCAMS messages from the catalog job

```

IDCAMS  SYSTEM SERVICES                      TIME: 19:48:13
09/25/13    PAGE      1

DEFINE CLUSTER(NAME(MHLRES2.HSM.BCDS) VOLUMES(SBOX1B) -      00151007
    RECATALOG          ) -                                00156007
    DATA(NAME(MHLRES2.HSM.BCDS.DATA) -                    00157007

```

```

) - 00158007
INDEX(NAME(MHLRES2.HSM.BCDS.INDEX) - 00159007
) 00159107
IDC0508I DATA ALLOCATION STATUS FOR VOLUME SBOX1B IS 0
IDC0509I INDEX ALLOCATION STATUS FOR VOLUME SBOX1B IS 0
IDC0001I FUNCTION COMPLETED, HIGHEST CONDITION CODE WAS 0

IDC0002I IDCAMS PROCESSING COMPLETE. MAXIMUM CONDITION CODE WAS 0

```

DFSMSdss RESTORE jobs running on DFSMS V1.13

We ran the jobs to DFSMSdss RESTORE of non-VSAM and VSAM data sets on DFSMS V1.13 to demonstrate the effect of using the REPLACEUNCONDITIONAL and RENAMEUNCONDITIONAL keywords when used on down level systems.

In Example 8-33 on page 170 we show the DFSMSdss messages resulting from trying to use the REPLACEUNCONDITIONAL and RENAMEUNCONDITIONAL options when running on DFSMS V1.13 and restoring from a physical format DUMP data set.

DFSMSdss issued message ADR780I which indicates that the DUMP data has been recognized as being in physical format.

DFSMSdss messages ADR392E advises that the data set (using its new name) already exists, and because the REPLACEUNCONDITIONAL option is not supported for physical format DUMP data sets it is an error.

DFSMSdss message ADR415W warns that no processing has occurred.

Example 8-33 DSS Messages from job to restore non-VSAM data set run on DFSMS V1.13

```

PAGE 0001 5695-DF175 DFSMSDSS V1R13.0 DATA SET SERVICES 2013.269 01:29
RESTORE INDD(TAPE) OUTDD(DASD) - 00110007
REPLACEUNCONDITIONAL - 00110108
DATASET(INCLUDE(MHLRES2.CNTL.JCL)) - 00111008
RENAMEU((MHLRES2.CNTL.JCL,MHLRES2.CNTL.JCL.REST1)) 00112008
ADR101I (R/I)-RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'RESTORE '
ADR109I (R/I)-RI01 (01), 2013.269 01:29:11 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.269 01:29:11 EXECUTION BEGINS
ADR780I (001)-TDDS (01), THE INPUT DUMP DATA SET BEING PROCESSED IS IN PHYSICAL
DATA SET FORMAT AND WAS CREATED BY DFSMSDSS VERSION
2 RELEASE 1 MODIFICATION LEVEL 0 ON 2013.268 15:51:11
ADR392E (001)-TDPNV(01), MHLRES2.CNTL.JCL EXISTS ON SBOX1B WITH NEWNAME
MHLRES2.CNTL.JCL.REST1
ADR382W (001)-TDDS (01), THE FOLLOWING DATA SETS WERE NOT PROCESSED FROM VOLUME
SBOX1L DUE TO ERRORS
MHLRES2.CNTL.JCL
ADR415W (001)-TDDS (01), NO DATA SETS WERE COPIED, DUMPED, OR RESTORED FROM
LOGICALVOLUME SBOX1L
ADR415W (001)-TDDS (02), NO DATA SETS WERE COPIED, DUMPED, OR RESTORED FROM ANY
VOLUME
ADR006I (001)-STEND(02), 2013.269 01:29:11 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.269 01:29:11 TASK COMPLETED WITH RETURN CODE 0008

```

ADR012I (SCH)-DSSU (01), 2013.269 01:29:11 DFSMSDSS PROCESSING COMPLETE. HIGHEST RETURN CODE IS 0008 FROM:

TASK 001

In Example 8-34 on page 171 we show the DFSMSdss messages resulting from running a job on DFSMS V1.13 with RENAMEUNCONDITIONAL specified.

- ▶ DFSMSdss issued message ADR780I that indicates that the input data set is in DFSMSdss DUMP physical format.
- ▶ DFSMSdss issued message ADR332E advising that VSAM CLUSTER HSM.BCDS was not processed.
- ▶ DFSMSdss issued messages ADR382W and ADR415W which expands on the ADR332E message.

Example 8-34 DSS messages from job to restore VSAM data set run on DFSMS V1.13

```
PAGE 0001 5695-DF175 DFSMSDSS V1R13.0 DATA SET SERVICES 2013.268 19:58
  RESTORE INDD(TAPE) OUTDD(DASD) -                00110000
    REPLACEUNCONDITIONAL -                00110100
    TOLERATE(ENQF) -                00110210
    DATASET(INCLUDE(HSM.BCDS)) -                00111000
    RENAMEU((HSM.BCDS,MHLRES2.HSM.BCDS13))        00112010
ADR101I (R/I)-RI01 (01), TASKID001 HAS BEEN ASSIGNED TO COMMAND 'RESTORE '
ADR109I (R/I)-RI01 (01), 2013.268 19:58:58 INITIAL SCAN OF USER CONTROL STATEMENTS
COMPLETED
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001)-STEND(01), 2013.268 19:58:58 EXECUTION BEGINS
ADR780I (001)-TDDS (01), THE INPUT DUMP DATA SET BEING PROCESSED IS IN PHYSICAL
DATA SET FORMAT AND WAS CREATED BY DFSMSDSS VERSION
      2 RELEASE 1 MODIFICATION LEVEL 0 ON 2013.268 19:31:03
ADR411W (001)-DYNA (01), DATA SET HSM.BCDS IN CATALOG CATALOG.HSM ON VOLUME SBX1B
WAS NOT SERIALIZED ON REQUEST
ADR332E (001)-TDDS (01), CLUSTER HSM.BCDS IN CATALOG CATALOG.HSM NOT PROCESSED.
PHYSICAL DATA SET OPERATION DOES NOT SUPPORT RENAME
      OF VSAM DATA SETS
ADR382W (001)-TDDS (02), THE FOLLOWING DATA SETS WERE NOT PROCESSED FROM VOLUME
SBXHS8 DUE TO ERRORS
      HSM.BCDS
ADR415W (001)-TDDS (01), NO DATA SETS WERE COPIED, DUMPED, OR RESTORED FROM
LOGICALVOLUME SBXHS8
ADR415W (001)-TDDS (02), NO DATA SETS WERE COPIED, DUMPED, OR RESTORED FROM ANY
VOLUME
ADR006I (001)-STEND(02), 2013.268 19:59:02 EXECUTION ENDS
ADR013I (001)-CLTSK(01), 2013.268 19:59:02 TASK COMPLETED WITH RETURN CODE 0008
ADR012I (SCH)-DSSU (01), 2013.268 19:59:02 DFSMSDSS PROCESSING COMPLETE. HIGHEST
RETURN CODE IS 0008 FROM:
```

8.4 zFS change activity support

The zFS change activity support is an enhancement that addresses a difference in the way zFS files have to be handled compared to HFS files.

This support does not require any user interaction, the enhancement that it provides allows zFS files to be handles in a similar manner to other data sets are described in “DFSMSdss data set change indicator reset with RESTORE” on page 148.



System Data Mover enhancements

This chapter introduces a change in DFSMS V2.1 addressing a performance issue in System Data Mover (SDM). SDM primary volumes are now allowed to be offline when volumes are added in a XRC session. Prior to this enhancement, the start of a session had to wait for all volumes to come online delaying a fast creation of an XRC session.

These are the enhancements that we discuss in this chapter:

- ▶ XRC offline volumes
- ▶ Migration and coexistence considerations

9.1 XRC offline volumes

Adding volumes in a XRC session with the XSTART command to restart a session or issuing an XADDPAIR command to resume suspended volumes will now allow the primary volumes in an XRC session to be offline. Previously these actions would require the primary volumes to be online and would delay the resume of a suspended session for several minutes. Having to wait while the volumes were varied online impacts the recovery point objective (RPO) time.

Having the primary volumes in offline status can also remove the risk in having primary volumes known to other systems in online status, and potentially receive writes from these systems. In addition, being able to have XRC volumes offline, will also enable moving XRC volumes to Alternate Channel Subsets, where the current restriction is that these cannot be online.

The first time you initialize a XRC session, you will still have to have all primary volumes online, but once the session is initialized, you can vary these offline.

Trying to add an offline volume to a new XRC session will fail as shown in Example 9-1 on page 174.

Example 9-1 Error message displayed if XRC session is initialized with primary volume offline.

```
ANTA5107E XADDPAIR FAILED FOR VOLUME PAIR(ML91AA,ML9BAF) FOR
SESSION(ITS001), RC=464 REAS=418
```

To see how an offline primary volume appears in a XRC query on a session that was resumed by XSTART or XADDPAIR, we put device address 91AA with volser ML91AA offline and then issued the **TSO XQUERY DATASET(XRC.XQUERY) VOLUME(ALL)** command.

The output from this command is shown in Example 9-2 on page 174.

Example 9-2 Output from XQUERY command having put device 91AA offline

```
ANTQ8200I XQUERY STARTED FOR SESSION(ITS001) ASNAME(ANTAS001)
ANTQ8202I XQUERY VOLUME REPORT - 003
ANTQ8211I PRIM SEC ERROR SYNCH
ANTQ8212I VOL VOL LEVEL % STA CMD OP -----TIMESTAMP-----
ANTQ8203I -----
ANTQ8213I ML91AA ML9BAF SESSION DUP
ANTQ8213I ML91A9 XRCUTL -- UTL
ANTQ8213I ML92AB ML98AE SESSION DUP
ANTQ8213I ML92A9 XRCUTL -- UTL
ANTQ8213I ML93A4 XRCUTL -- UTL
ANTQ8213I ML93A7 ML99AB SESSION DUP
ANTQ8213I ML94AC XRCUTL -- UTL
ANTQ8213I ML94AD ML9AA9 SESSION DUP
ANTQ8213I ML96AB XRCUTL -- UTL
ANTQ8213I ML96AC ML9CAF SESSION DUP
ANTQ8238I TOTAL=10 DUP=5 CPY=0 PND=0 SUS=0 SEQ=0 UTL=5
ANTQ8260I PAV=0
ANTQ8231I DATA CONSISTENT(NO_TIME_AVAILABLE) IDLE(06:13:58.5)
ANTQ8240I DATA EXPOSURE(NO_TIME_AVAILABLE)
ANTQ8232I SESSIONTYPE(XRC) ERRORLEVEL(SESSION) HLQ(SDM)
ANTQ8233I DEFAULT TIMEOUT(STORAGE_CONTROL_DEFAULT)
ANTQ8201I XQUERY VOLUME REPORT COMPLETE FOR SESSION(ITS001)
```

As displayed above, the volser ML91AA still appears in the output listing with the volser identification and not just with the UCB address. This is exactly how the primary volumes will appear in online status.

A new **XQUERY CONFIGURATION** command with the STATUS keyword will have options OFL or ONL. Using this command you will be able to identify either offline or online volumes depending on option setting.

Issuing the command: **XQUERY ITS001 DATASET(XRC.XQUERY) DISP(SHR) CONFIGURATION VOLUME(ALL) STATUS(OFL)** will give you all offline volumes as shown in Example 9-3 on page 175.

Example 9-3 Display of all XRC primary in offline status.

```

ANTQ8200I XQUERY STARTED FOR SESSION(ITS001) ASNAME(ANTAS001)
ANTQ8202I XQUERY CONFIGURATION REPORT - 003
ANTQ8281I -----PRIMARY----- -----SECONDARY-----
ANTQ8284I          SC SC
ANTQ8282I SSID SN ID DVCN CCA VOLSER  SSID DVCN CCA VOLSER
ANTQ8203I -----
ANTQ8283I 8943 -- 01 91AA 2A  ML91AA  8957 9BAF 2F  ML9BAF
ANTQ8206I 1 VOLUME(S) MEET REQUESTED SPECIFICATION
ANTQ8238I TOTAL=10  DUP=5  CPY=0  PND=0  SUS=0  SEQ=0  UTL=5
ANTQ8260I PAV=0
ANTQ8231I DATA CONSISTENT(NO_TIME_AVAILABLE)  IDLE(00:37:26.2)
ANTQ8240I DATA EXPOSURE(NO_TIME_AVAILABLE)
ANTQ8232I SESSIONTYPE(XRC)  ERRORLEVEL(VOLUME)  HLQ(SDM)
ANTQ8201I XQUERY CONFIGURATION REPORT COMPLETE FOR SESSION(ITS001)

```

Note: Utility volumes and secondary volumes will still have to remain

9.2 Migration and coexistence considerations

XRC primary volumes should be defined as coming offline at IPL by default, even though the initial XADDPAIR requires the volumes to be online. After a successful ADDPAIR you can vary the primary volumes offline and leave them in this status.

It is also suggested to remove all unnecessary **VARY ONLINE** commands from scripts related to the previous requirement of having these volumes online.

If you are using any performance or capacity tool on this environment, you may have to move the task collecting information to an application LPAR that has the volumes online.

9.2.1 Additional support

XRC will now listen for ENF 31 and 32, preventing a dynamic activation from removing (or modifying) devices that are in the XRC configuration.

A UCB must be pinned prior to invoking IOS offline device service (IOSODS) macro and pinned while in use. This prevents a device from being deleted by a dynamic activate while the device is in use. The data mover listener will go through the list of devices to be deleted, and if any are devices 'owned' by XRC, the device will be pinned by the listen routine.

Documentation

Refer to the manuals *z/OS DFSMS Advanced Copy Services*, SC35-0428 and *z/OS MVS System Messages Volume 1 (ABA - AOM)*, SA22-7631 for more detail on setting up XRC.



SMS exploiting hardware enhancements

In this chapter we look at features and function in z/OS V2.1 that take advantage of hardware functions and attributes in their implementation.

These are the enhancements that we discuss in this chapter:

- ▶ Cluster and extent pool exploitation
- ▶ DFSMSHsm storage tiers
- ▶ JES3 SMS tape support
- ▶ DADSM CVAf/DEVICE support

10.1 Cluster and extent pool exploitation

Prior to DFSMS V2.1 volume selection will try to allocate or extend a multivolume data set in the same Storage Facility Image (SFI), when the storage class accessibility attribute is set to Continuous or Continuous Preferred.

A SFI can also be translated as a storage control unit. With new support in DFSMS V2.1, DFSMS volume selection will have awareness of cluster and extent pool boundaries in the IBM System Storage® DS8000. This will enable DFSMS to select volumes according to accessibility on fast copy replications and assure optimum performance on data striping.

Cluster and extent pools will have a close relationship in the DS8000, as Cluster 0 (also called server 0) and Cluster 1 (called server 1) will have one or more dedicated extent pools assigned to either even LCUs or odd LCUs. The extent pools will consist of a number of extents depending on the physical capacity installed. For mainframe storage, an extent will be 1113 cylinders (3390 mod 1). These extents can be put together forming logical volumes from 3390 mod 1 up to the EAV size of 1 TB.

10.1.1 Volume selection for multi-volume data sets

When allocating a new multi-volume data set or extending an existing one, DFSMS volume selection will now select a volume in the same cluster (and extent pools assigned to this cluster) provided the ACCESSIBILITY parameter in the storage class is set to Continuous or Continuous Preferred. If a new allocation cannot occur within the same cluster, the allocation will be done within the same SFI (across the two clusters).

A notification will be sent to the joblog if your preference is not met as shown in Example 10-1.

Example 10-1 Example of message if Accessibility requirement is not met

```
IGD17395I DATA SET MHLRES1.SFI.TEST001
WAS NOT ALLOCATED IN THE SAME STORAGE FACILITY IMAGE
BECAUSE SELECTED VOLUMES HAD HIGHER PREFERENCE VALUE
```

Prior to DFSMS V2.1, DFSMS issued IGD17395I if a multi-volume data set was not allocated or extended to volumes in the same SFI. DFSMS issues a new variant of IGD17395I if a multi-volume data set was allocated or extended to volumes in the same SFI, but not in the same cluster.

Volume Selection for DFSMSdss Fast Replication operation

When copying data with a Fast Replication option, the target volume will now be selected within the same cluster if possible. If DFSMS fails to do so, allocation will be able to occur within the same SFI.

Volume Selection for striped data sets

In releases prior to DFSMS V2.1 volume selection would try to distribute striped data sets across LCUs. For striped data sets the support will now enable DFSMS volume selection to spread the allocation (stripes) over multiple extent pools. This will assure that different volumes are used and the individual stripes have no performance impact on one another. When extending a striped data set to a new volume, DFSMS volume selection will prefer extent pools not already assigned to active stripes.

If DFSMS fails to be able to allocate stripes across extent pools, the allocation will happen according to the previous implementation (across LCUs).

10.2 DFSMShsm storage tiers

DFSMShsm Primary Space Management is the function to maintain your primary disk as related to volumes and data sets. Only the most recently created and referenced data is typically retained on primary disk. If no longer needed, data is either deleted based on policy or migrated off the disk to tape. Space release is done along with extent consolidation on the individual data sets to optimize the environment.

Primary disk can either be DFSMS managed volumes or non-DFSMS managed volumes added to DFSMShsm through the ADDVOL command.

10.2.1 Primary Space Management

Primary Space Management is initiated by the AUTOMIGRATE=YES option on the storage group definition for the DFSMS managed volumes, or by command for the non-DFSMS managed volumes. DFSMShsm automigration will happen in two phases.

Phase 1

The first phase will include functions, where no data is being moved. Examples are deletion of temporary and expired data sets, partial release, and Fast Subsequent Migration.

All data set actions are controlled by the management class settings. The management class settings determine if release, delete or migration has to be done. In the first phase migration is limited to Fast Subsequent Migration.

Phase 2

The second phase consists of migrating eligible data sets until the low threshold is reached on the storage group. This may have been accomplished by phase 1 and extent reduction. Extent reduction ignores low threshold, as it usually does not actually free space on the volume. It combines extents so that the data set can continue to grow without running out of extents.

Note: If deletion of expired data sets, Fast Subsequent Migration, and space reduction of the remaining data sets achieves the specified free space, phase 2 will never start.

Migration will continue until the low water mark is reached or there are no more candidates to migrate. Eligible data sets are put onto the migration queue in a priority order based on the size and the age of the data set and are processed accordingly.

10.2.2 DFSMShsm V2.1 Storage Tier basics

The new Storage Tier function in DFSMShsm will support moving data from the DFSMShsm managed primary volumes to the most optimum hardware tier Solid State Device (SSD), normal response time disk (NRT) (also referred to as Enterprise disk) or long response time disk (also called nearline or SATA disk) and ultimately to tape. Movement of data will happen as a class transition as part of the Primary Space Management function outlined previously.

To take advantage of DFSMSHsm Storage Tier function, the environment has to be set up at a storage group level according to the different types of hardware tiers. Basically, each tier will correspond to one or more DFSMS storage groups. Data will be directed to a specified tier as per the storage group requirement.

DFSMSHsm will have no awareness of the specific hardware types, only about the storage groups representing the tier. Class transition as used today for object data will support moving data between the tiers (storage groups) or assure that a recall happens to the correct tier. This support applies for zFS, IBM DB2® and CICS where the Serialization Exit provides special processing support for each of these.

Implementing Storage Tier will require some planning as to how to fit it into your current environment. Storage Tiering implementation most likely has a different approach than separating data in the current storage groups. Storage group distribution of data is most commonly based on type of data and requirements for backup and other requirements for separation. Separation requirements include having production separated from development data or isolation of system data, batch data, data base data and temporary files.

You may have an environment already optimized with the fewest possible storage groups and you will probably not want to add too many to support DFSMSHsm Storage Tier. You may only have Enterprise disk (normal response time disk) - and then you would stay with your current setup.

You might also have nearline disk (long response disk) or plan to implement these along with nearline disk. In this case implement tiering of your primary disk in two tiers. If this would be possible by adding just one storage group for the nearline tier depends on your setup. Consider if you would put all your lower priority primary data in one storage group without taking into account the type of data (for example, database data or sequential data).

The intention of implementing Storage Tier is to assure that your data is always on the most optimum level as to performance and cost. In the graphical presentation in Figure 10-1 you see an example of two primary disk Storage Tiers (Enterprise and Nearline disk). Data will stay 90 days on the most expensive media, and then move to Nearline disk with a lower cost per GB before data will be migrated to tape after 366 days.

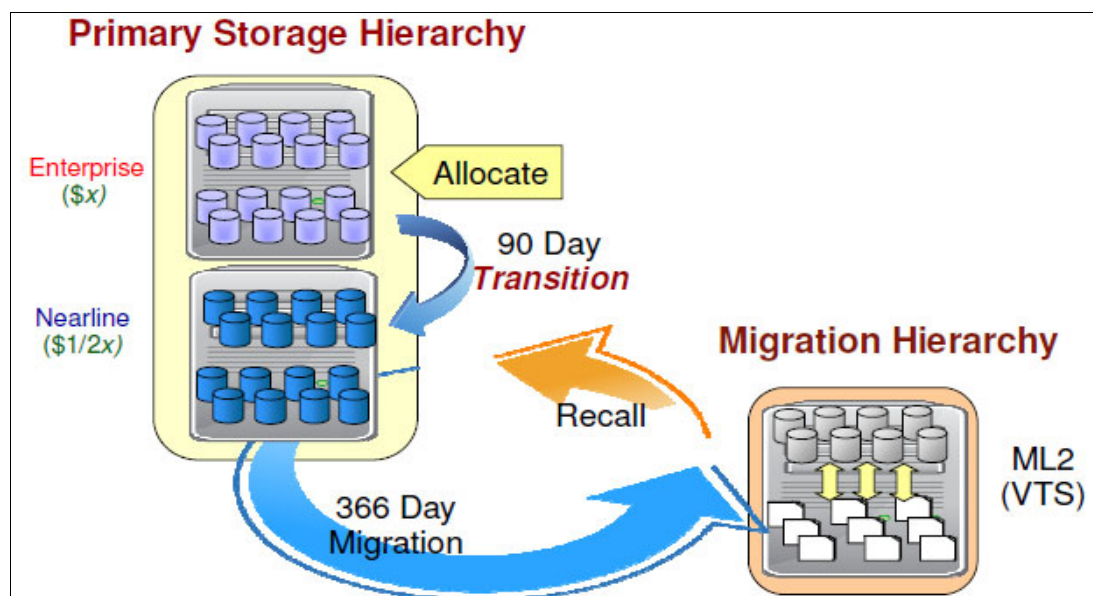


Figure 10-1 Storage Tier concept with two tiers

10.2.3 Storage Tier process

A new flag that will enable Storage Tier function per default will be defined in the MCVT. This new function will be active during primary space management by default with DFSMS V2.1. To summarize the functions in primary space management with the new Storage Tier process:

- ▶ Delete expired data sets
- ▶ Release unused space and free up extents
- ▶ Process data sets in scope for transition
 - Move data sets eligible for class transition to a different storage tier if storage class or storage group changed
 - Change management class only if required
- ▶ Move migration candidates from Primary disk to ML1 or tape

Storage group priority setting

A new priority setting on the DFSMS storage group application will decide the priority sequence in which storage groups are processed. The value can be in the range from 1 to 100 with 50 being the default (1 the lowest, while 100 is the highest priority). This value has been added to make it possible to prioritize the sequence in which your storage groups will be processed during class transition. This assures that the most important storage groups are processed first and data offloaded to make the storage group ready to receive new allocations.

The Processing Priority field in the storage group application in ISMF is shown in Example 10-2.

Example 10-2 New storage group processing priority parameter

POOL STORAGE GROUP ALTER Page 2 of 2
Command ==>

SCDS Name : SYS1.SMS.MHLRES3.SCDs
Storage Group Name : BIGSTUFF

To ALTER Storage Group, Specify:

Allocation/migration Threshold :	High	99	(1-100)	Low . . 1	(0-99)
Alloc/Migr Threshold Track-Managed:	High	99	(1-100)	Low . . 1	(0-99)
Guaranteed Backup Frequency				(1 to 9999 or NOLIMIT)	
BreakPointValue				(0-65520 or blank)	
Processing Priority		50		(1-100)	

You distinguish between your storage groups as to class transition using this to manage the sequence in which these storage groups are serviced by DFSMSHsm class transition.

Class transition processing

Before the migration eligibility checking, DFSMSHsm will determine if the data set is eligible for a class transition based on the class transition settings in the management class and DSCB information in VTOC.

DFSMSHsm has an interface to the ACS routines already. In DFSMS V2.1 a new ACS environment of SPMGCLTR is defined to support class transitions. This is similar to the ones used for class transition on OAM object data.

If the data set is eligible for transition, DFSMSHsm will invoke the DFSMS ACS routines with ACS environment SPMGCLTR to determine if there will be new management class, storage class and storage group values assigned. If a data set is going to have a new management class and keep the existing storage class and an existing storage group, DFSMSHsm will invoke alter of the management class only.

If the storage class and the storage group are subject to change, and a class transition is going to happen, DFSMSHsm will place a Migration Data Set Queue Element (MDQE) at the top of the migration queue. Priority will be the highest, not as it is normally based on data set size and age.

Class transition will happen in the second phase of primary space management. The individual MDQEs will also have a priority order based on type of transition. Preference order will be as follows:

1. All MDQEs that are candidates for both extent reduction and transition will be done first without checking storage group threshold.
2. All MDQEs that are candidates for extent reduction and migration will be processed next. This is done without checking the storage group threshold as extent reduction is mandatory. Data sets will just be migrated as extent reduction will happen on recall.
3. Next in line are the MDQEs that are candidate for class transition and migration. Storage group low threshold will be checked. If utilization is above the threshold, these data sets will be migrated.
4. Last in line, the MDQEs that are candidates for class transition only will be processed, but only if storage group low threshold is exceeded.

Besides the overall priority explained above, data sets will be copied based on size and age of the data set within the initial criteria selected. Copy will stop when the DFSMSHsm low threshold is reached or if target storage group is out of capacity.

The reason behind putting transitions at the top of the queue is that the transitions may meet the need for offloading data from the individual volume (having met the low threshold for this volume). As DFSMSHsm selects each data set for migration, it transfers control to the spacemanagement exit (ARCMDEXT) if the exit is installed and enabled.

Transition rules as to migration versus transition based on eligibility and the return code from the migration exit can be summarized as shown in Table 10-1.

Table 10-1 Transition rules: migration versus class transition.

Eligibility status	Migration Exit RC=0 (Allow Migration)	Migration Exit RC=8 (Disable Migration)	Migration Exit RC=52 (Disable Migration and Class Transition)
Eligible for migration only	Migration	Data set processing fails with RC=45, reason 92	Data set processing fails with RC=45, reason 92
Eligible for migration and class transition	Migration	Class transition	Data set processing fails with RC=45, reason 92

Eligibility status	Migration Exit RC=0 (Allow Migration)	Migration Exit RC=8 (Disable Migration)	Migration Exit RC=52 (Disable Migration and Class Transition)
Eligible for class transition only	Class transition	Class transition	Data set processing fails with RC=45, reason 92

Transition Copy Technique

For the actual class transition move of the individual data set, DFSMSHsm will call DFSMSdss for all data set types. DFSMSHsm will obtain the class Transition Copy Technique from the management class. The new class transition field in the ISMF management class application is shown in Figure 10-2.

MANAGEMENT CLASS ALTER	Page 5 of 6
Command ==>	
SCDS Name : SYS1.SMS.MHLRES3.SCDS	
Management Class Name : MCSATA	
To ALTER Management Class, Specify:	
Class Transition Criteria	
Transition Copy Technique STD	(FRP, FRR, STD, PMP or PMR)
Serialization Error Exit NONE	(DB2, CICS, ZFS, EXIT or NONE)

Figure 10-2 Transition Copy Technique options on management class

The Transition Copy Technique will decide how the copy is going to happen (Fast Replication or normal I/O processing) and the type of action on failure (managed by preferred or required option). Possible settings on the Transition Copy Technique are:

- ▶ FRP: Fast replication preferred
- ▶ FRR: Fast replication required
- ▶ STD: Standard
- ▶ PMP: Flashcopy to PPRC primary (preserve mirror preferred)
- ▶ PMR: Flashcopy to PPRC primary (preserve mirror required)

DFSMSdss performs the transition copy with the appropriate copy option and will give back control to DFSMSHsm at the end of the copy. DFSMSHsm will issue a message on completion. For a copy using Fast Replication this will be handled differently by DFSMSHsm. If there is not a valid backup, the request will be handled as below.

- ▶ Request for transition will fail if FASTREPLICATION option is REQUIRED.
- ▶ If FASTREPLICATION option is PREFERRED, the copy will be performed using standard I/O.

If there is a valid current backup, all replication requests will be processed as normal.

In a PPRC environment, the FCTOPPRCPRIARY and RESERVEMIRROR replication keywords may also be used with the PREFERRED or REQUIRED options. If option PREFERRED is set, standard IO operation will be used if FCTOPPRCPRIARY PRESERVEMIRROR cannot happen using Fast Replication copy. For a Fast Replication copy

with the REQUIRED keyword set, Fast Replication using FCTOPPRCPPRIMARY PRESERVEMIRROR will fail if this is not possible for some reason.

Class transition messages

The message issued by DFSMSHsm after a successful transition of a data set is ARC0734I, displaying the result of the transition is shown in Example 10-3.

Example 10-3 Message displayed when class transition has happened to a data set

```
ARC0522I SPACE MANAGEMENT STARTING ON VOLUME ML921C(SMS) AT 06:39:21 ON 2013/10/11, SYSTEM SC64
ARC0734I ACTION=CLASS-TR FRVOL=ML9B14 TOVOL=MLD83A TRACKS=          4 RC=    0, REASON=    0, AGE=
7,
DSN=MHLRES1.SFI.TEST001
```

By default, failing messages are not issued for data sets that fail a transition because exclusive serialization could not be obtained. It is recommended that these data sets should be analyzed by post processing the FSR records with failing return code RC68 and reason code if desired. If the messages should be needed, you can issue the DFSMSHsm patch in Example 10-4, and have the messages displayed.

Example 10-4 Patch enabling RC68 indicating serialization problem.

```
PATCH .MGCB.+EF BITS(...1....)
```

Last Successful Class Transition Date

To support class transition, a new field was created in the NVR/VVR called *Last Successful Class Transition Date* (LSCTD).

The Last Successful Class Transition Date (LSCTD) field is set and used by DFSMSHsm to manage class transitions. DFSMSHsm will use this field to determine if a data set has already been transitioned and should not be reprocessed. The LSCTD, if it exists, can be displayed for a particular data set with the **LISTCAT** command as shown in Example 10-5.

Example 10-5 LISTCAT of data set having a non zero value in Last Transition:

```
NONVSAM ----- STORTIER.M01.S01.D01.N01.PSFB
IN-CAT --- STRTRFVT.USERCAT
HISTORY
  DATASET-OWNER----- (NULL)      CREATION-----2012.001
  RELEASE-----2      EXPIRATION-----0000.000
  ACCOUNT-INFO----- (NULL)
SMSDATA
  STORAGECLASS ---SCLASS22      MANAGEMENTCLASS-MCLASS01
  DATACLASS -----DCCLASS01    LBACKUP ---2012.001.0701
LAST TRANSITION-2012.013
VOLUMES
  VOLSER-----LSMS12      DEVTYPE-----X'3010200F'
ASSOCIATIONS----- (NULL)
ATTRIBUTES
```

LSCTD field is also available via DCOLLECT output.

When a data set is successfully transitioned, LSCTD is set to zero. This occurs when the management class was changed, or set to the current date if the management class was not changed. DFSMSHsm will attempt to transition a data set if it has met the transition criteria and the LSCTD is zero.

Criteria on management class deciding class transition

The new Storage Tier class transition is based on criteria in the management class already being used for Object Management. Your criteria will depend on your environment (tiers in scope and data availability requirements) and requirement when to move data to the next tier or off to tape. There are three options, you have to consider using when implementing storage tiering:

- ▶ **TIME SINCE CREATION**

Time given in years, months or days, since creation that must have elapsed before transition can happen. A value of 9999/12/31 will cause transition to not happen at all.

- ▶ **TIME SINCE LAST USE**

Time given in years, months or days, since the last reference of data that must have elapsed before transition can happen.

- ▶ **PERIODIC**

A class transition can be scheduled to happen at a determined sequence monthly, quarterly or yearly.

You can only set one of the options per management class. If your data has individual requirements, each group of data must have a transition management class assigned that meets the specific requirement for each group as related to transition.

Recall of data sets in a class transition environment

Data sets may miss class transition while being migrated. Imagine, a data set is on ENTERPRISE disk, when being migrated. During the time the data set is migrated, it may meet the criteria set for a class transition to NEARLINE disk. Based on the creation date and the last referenced date stored in the MCD record, DFSMSHsm is able to determine if a data set missed a transition. If so, DFSMSHsm will run the ACS routines with the SPMGCLTR code and have any new, changed constructs returned. And in this case the data set will end up on NEARLINE disk.

The constructs and candidate storage groups will be handed over to DFSMSdss to have the recall occur to the correct tier.

Note: For RECOVER, ARECOVER and FRRECOV a class transition change will not be verified.

Reporting on class transition

The FSR records will be updated to contain information on the new transition function. The REPORT command will have a new parameter FUNCTION(TRANSITION) that will extract reporting on class transition statistics only.

If REPORT is used without a keyword for a specific FUNCTION, all FUNCTIONS are listed including the new transition function. Example 10-6 shows the class transition report only.

Example 10-6 Example of REPORT on class transition function

```
NUMBER  -----READ-----  -----WRITTEN-----  -----REQUESTS-----  AVERAGE  -----AVERAGE TIME-----
HSM FUNCTION DATASETS TRK/BLK    K-BYTES  TRK/BLK K-BYTES SYSTEM USER FAILED AGE  QUEUED WAIT  PROCESS TOTAL

CLASS TRANSITION
PRIMARY - PRIMARY 0000003  00000053 000001347K 00000034000001368K 000000 00003 00000 0000 0000  00000  00000 00000
-DFSMSHSM STATISTICS REPORT ----- AT 21:23:27 ON 2013/09/29 FOR SYSTEM=SC64
```

Using the REPORT function with SUMMARY sub parameter, the report will limit to the information as displayed in Example 10-7.

Example 10-7 REPORT function on class transition using the SUMMARY sub parameter

Data set class transition = 00000003 requested, 00000000 failed

A new ARC0277I message will now appear on the **QUERY STATISTICS** command. This display will show the number of data sets moved by class transition, the number of failed transitions, and the number of tracks and bytes moved. Example 10-8 shows the output of the command.

Example 10-8 Example of output from a QUERY STATISTICS command

```
ARC0277I DS TRANSITIONED=00000000, DS TRANSITION
ARC0277I (CONT.) FAIL=000, TRKS TRANSITIONED=00000000,  BYTES
ARC0277I (CONT.) TRANSITIONED=000000000
```

Class transition in an IM and ODM environment

Class transition was designed to only happen in the primary space management cycle. However, if volumes and their corresponding data sets are only part of an on-demand migration (ODM) or interval migration (IM) migration type, class transition will not happen for these volumes.

An additional option has been added to support this scenario. Class transition will, through a **SETSYS** command, be able to occur in an environment with ODM or IM active. An example of this command is shown in Example 10-9.

Example 10-9 Example of class transition SETSYS command in an ODM or IM environment

```
SETSYS CLASSTRANSITION(EVENTDRIVENMIGRATION(Y|N SERIALIZATIONEXIT(Y|N)))
```

You can abbreviate the SETSYS parameters shown above as shown in Example 10-10.

Example 10-10 Abbreviated form of CLASSTRANSITION

```
SETSYS CLTR(EDM(Y|N SERL(Y|N)))
```

The default for EVENTDRIVENMIGRATION is Y (YES). With a setting of N (NO), class transition will not occur in an ODM or IM environment.

The sub-parameter SERIALIZATIONEXIT will decide, whether the user data set serialization error exit during event driven migration should be used or not.

Note these additional defaults that will apply to EVENTDRIVENMIGRATION setting:

- ▶ If EDM is not specified, or if EDM is specified without keywords, the default is Y for EDM and for SERIALIZATIONEXIT it is N.
- ▶ If EDM is set to Y and there is no setting on SERIALIZATIONEXIT, then the default for this will be N.

Serialization support

If the serialization exit is called through class transition, the Serialization Error Exit field in the management class decides which action should be taken. Possible values for this field are:

- ▶ DB2
- ▶ CICS

- ZFS
- EXIT (call to user exit)
- NONE

If the call is for a DB2 table and SERIALIZATION(Y) is set, DB2 will be called and asked to close the table, if possible. The same is the case for the other applications listed, just different procedures. For a ZFS data set this will be an unmount as an example. If the data set close is not successful, transition of this particular data set will not happen.

If the setting is NONE, the data set will not be transitioned if an exclusive enqueue fails. Class transition will continue with other candidates.

Figure 10-3 is new the management class panel in ISMF, where the Serialization Error Exit setting is displayed and updated.

MANAGEMENT CLASS ALTER	Page 5 of 6
Command ==>	
SCDS Name : SYS1.SMS.MHLRES3.SCDS	
Management Class Name : MCSATA	
To ALTER Management Class, Specify:	
Class Transition Criteria	
Transition Copy Technique	STD (FRP, FRR, STD, PMP or PMR)
Serialization Error Exit	NONE (DB2, CICS, ZFS, EXIT or NONE)

Figure 10-3 Management panel for updating class transition serialization option.

As this setting is application oriented, each application should have a management class assigned with a setting, that matches this specific application (DB2, CICS, or ZFS).

Ensure that only DB2, CICS or zFS data sets are assigned to a management class with a Serialization Exit specified to one of these types

Getting started with Storage Tier

Migrating to DFSMS V2.1, class transition is enabled by default, but you will need a few additional steps to have all requirements in place. This section will outline the steps you have to go through.

Start identifying your needs and the benefit of doing class transition. Based on your environment and available tiers, determine what data should be included. You may be in a position, where new hardware will be installed introducing new tiers or you just want to take advantage of the DFSMSHsm Storage Tier function on your current multi tier environment.

In ISMF you will have to complete these preparation tasks:

- Create one or more management classes to set the (individual) criteria for transitioning data between tiers. Criteria you will have to look into is Time Since Creation, Time Since Last Use or Periodic for the transition criteria. The Transition Copy Technique for the data mover is also important.
- Create at least one storage class with the accessibility option you have settled on (C,P, S or N). This setting works in conjunction with the management class option Transition Copy Technique.

- Create one or more target storage groups that match your tiers.
- Create transition code in the ACS routines, that will capture the data sets in scope and direct these to the correct target tiers.

Also consider your need for having Class Transition occur for IM or ODM managed data by deciding the settings for EVENTDRIVENMIGRATION and SERIALIZATION as discussed earlier in this chapter.

Having created the constructs for Class Transition, the ACS routines will need to have code similarly to the example in Example 10-11 added. The ACS routine code will assure capture of the right data at the appropriate time moving these to the appropriate defined tiers.

Example 10-11 covers the management class ACS routine code, that enters the SPMGCLTR environment called by DFSMSHsm. A candidate data set is found, it may only have the management class changed, or it may also be transitioned depending on criteria being met.

Example 10-11 ACS routine code that will do transition on management class

```

IF &ACSENVIR = 'SPMGCLTR' THEN
    SELECT (&MGMTCLAS)
        WHEN ('MCNORMA') SET &MGMTCLAS = 'MCSATA'
        OTHERWISE       SET &MGMTCLAS = &MGMTCLAS
    END

```

In Example 10-12 the storage class routine is invoked and assigns a storage class designed for NEARLINE disk (SATA).

Example 10-12 Storage class ACS routine code invoking Class Transition

```

IF &ACSENVIR = 'SPMGCLTR' THEN
    SELECT (&STORCLAS)

        WHEN ('ENTERPRISE') SET &STORCLAS = 'SCSATA'
        OTHERWISE          SET &STORCLAS = &STORCLAS
    END
ELSE ...

```

For the actual transition to occur, data will have to be moved in this case from ENTERPRISE disk to Nearline. A new storage group covering nearline (SATA) disk is assigned and DFSMSdss is called to do the actual transfer of the data set from the current tier (storage group) to the new tier.

Example 10-13 shows the ACS routine code that will support such a transition.

Example 10-13 Storage group ACS routine supporting class transition to new tier

```

IF &ACSENVIR = 'SPMGCLTR' THEN
    SELECT (&STORCLAS)

        WHEN ('SCSATA') SET &STORGRP = 'SGSATA'
        OTHERWISE       SET &STORGRP = &STORGRP
    END
ELSE ...

```

Data will now have moved to a new tier (storage group). The new management class will determine the time data will reside on this tier before eventually being migrated to tape.

Exceptions from class transition

When FORCENONSMS is used in a RECALL command, this will happen as in pre-DFSMS V2.1 releases. The data set will be recalled to a non-SMS managed volume.

For extent reduction, DFSMSHsm will migrate the data set and immediately schedule a recall to the same volume (with same storage class and management class as when the data set was migrated). ACS routines will not be invoked in this case.

Common Recall Queue has been changed to have an indicator for Class Transition to be used for a recall.

New or changed messages

This section lists the new and changed messages in support of class transition.

ARC0277I message

The new ARC0277I message is now issued for QUERY STATISTICS (see Example 10-8). The message provides the data sets moved by class transition, the number of failed transitions, the number of tracks moved and the number of bytes moved.

ARC0278I message

The new ARC0278I message is additionally issued to display EVENTDRIVENMIGRATION and SERIALIZATIONEXIT values as the result of QUERY SETSYS command.

ARC1271I message

The new ARC1271I CLASS TRANSITION FAILED message is issued when IDCAMS return code is given in the ARC0734I message. These reason codes may be included:

- ▶ 9999 Abend during ALTER command processing.
- ▶ 9998 Abend during obtain Last Successful Class Transition Date
- ▶ 9997 Abend during Last Successful Class Transition Date setting

ARC0734I message

ARC0734I message with the new ACTION=CLASS-TR value after class transition processing is issued at the end of the class transition.

ARC0784I message

ARC0784I message with the new LAST SUCCESSFUL CLASS TRANSITION DATE value is issued when:

- ▶ Class transition is disabled (MCVTF_CLTR=OFF) and Last Successful Class Transition Date (LSTD) is not zero
- ▶ The data set is recalled as in prior to DFSMS V2.1 if the LSCTD exists.

ARC1280I message

ARC1280I message with the new CLASS TRANSITION value is issued when DFSMSHsm attempted class transition, but determined that the data set needed to be backed up.

Coexistence support

The following APARS are required on systems prior to DFSMS V2.1.

- ▶ OA36576: DFSMSHsm coexistence support for Storage Tier

Because of the changes in the record sizes in DFSMS V2.1, lower level systems accessing those records should be sure to install the DFSMSHsm PTF for OA36576 on each of the systems in the sysplex.

- ▶ OA37582: DFSMSdss coexistence support for Storage Tier

This APAR provides support for data set COPY and RESTORE to detect data sets that have a last class transition date in the SMS sub-cell of a VVR.

For status on general status on APARS related to Storage Tier see Appendix A, “APARs to be reviewed for DFSMS V2.1” on page 241.

Documentation

For more details on DFSMSHsm Storage Tier, consult the documentation listed below.

- ▶ *z/OS DFSMS Installation Exits*, SC23-6850
- ▶ *z/OS DFSMSHsm Storage Administration*, SC23-6871
- ▶ *z/OS DFSMSdss Storage Administration*, SC23-6868
- ▶ *z/OS DFSMSHsm Diagnosis*, GC52-1387
- ▶ *z/OS DFSMSrmm Reporting*, SC23-6875
- ▶ *z/OS DFSMSHsm Implementation and Customization Guide*, SC23-6869
- ▶ *z/OS MVS System Messages, Volume 2 (ARC-ASA)*, SA22-7632

10.3 JES3 SMS tape support

This new TS7700 support for JES3 will be available with the release of DFSMS V2.1. For JES2 the Device Allocation Assistance (DAA) support was introduced in release 1.5 of the TS7700 virtual engine in December 2008. Scratch Allocation Assistance (SAA) came with TS7700 2.0 release in May 2011.

To be able to take advantage of the new support all systems in the JES3 plex must be at the DFSMS V2.1 level. JES3 can be at an earlier level, but you need to make changes to the INISH deck to enable the function.

10.3.1 Device Allocation Assistance (private/specific mounts)

Release 1.5 of the TS7700 code introduced Device Allocation Assistance (DAA). Device Allocation Assistance is a function that allows the host to query the TS7700 Virtualization Engine to determine which clusters should be preferred for a private (specific) mount request. DAA returns an ordered allocation list of Subsystems (logical control units with 16 devices per control unit image). The list is built in, and presented in preferred cluster order. This is to maximize local cache hits and minimize remote cache hits, but also at the same time balance the workload across a multi cluster grid. With this support, scratch allocations were still randomized across eligible libraries and devices.

10.3.2 Scratch Allocation Assistance (Scratch Mounts)

Scratch Allocation Assistance (SAA) was introduced with the TS7700 2.0 code. SAA will select the cluster for scratch mount allocation based on management class. In a mixed environment consisting of both TS7720 and TS7740 users start to request allocations from specific workloads like DFSMSHsm to go to TS7720 for better response time. A new

management class option has been introduced enabling the user to direct scratch mounts to a specific cluster. If no cluster is available standard MVS allocation will be used (WTOR, WAITHOLD, WAITNOH, CANCEL). With 256 drives available this will hardly be a problem in normal operation.

Figure 10-4 illustrates a configuration where two different types of workload are directed to a specific cluster (TS7740 or TS7720). One management class will point to cluster 0 (TS7720) and another to cluster 1 (TS7740). Cluster 2 is used for a remote application.

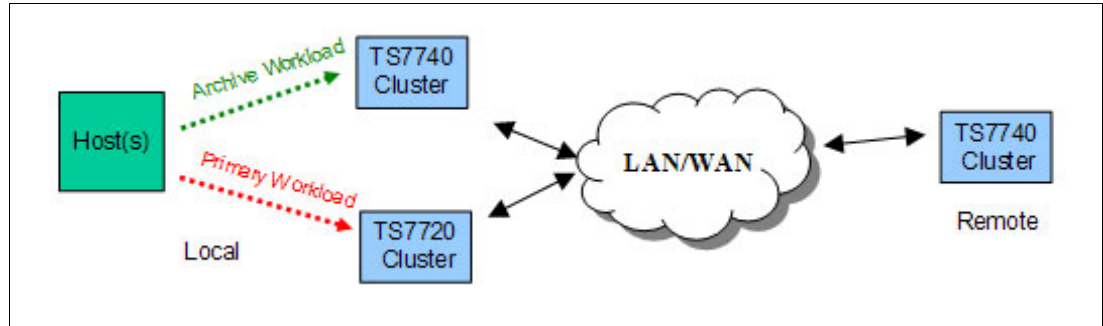


Figure 10-4 Example of three cluster grid where workload is split through management class setting

When the allocation request is issued for cluster 0 or cluster 1, only the candidate clusters LCUs and devices will be returned. MVS Allocation will select a device from this list without any awareness of cluster.

10.3.3 Implementing TS7700 Allocation Assistance for JES3

All systems in a JES3 plex must migrate to DFSMS V2.1 before implementing the TS7700 Allocation Assistance.

To enable the support, a new keyword JES3_ALLOC_ASSIST=YES/NO in DEVSUPxx member in SYS1.PARMLIB has to be added.

If option YES is set the Allocation Assistance support will be enabled for use with JES3. Before enabling the support, refer to *z/OS DFSMS OAM Planning, Installation, and Storage Administration Guide For Tape Libraries*, SC23-6867 for more detail.

To enable this feature without an IPL, issue the **SET DEVSUP=xx** command.

A new esoteric (library-specific distributed name) is introduced to support Allocation Assistance support. The new esoteric has to be added in the JES3 INISH deck.

Make sure that you have the outboard management classes in place that point to the specific libraries. Define this in the Add/copy Management Classes panel on the TS7700 as shown in Figure 10-5 on page 192. Only those clusters specified through the assigned management class are considered for the scratch mount request.

Add/Copy Management Classes

Management class settings

Name:

Secondary pool:

Retain copy mode: ☐

Description:

Management class settings for each cluster

Clusters	Copy Mode	Options
"IBMTSK00[0]" (#16190)	Deferred	Scratch mount candidate: ☒
"IBMTSS00[1]" (#16189)	Deferred	Scratch mount candidate: ☒

OK Cancel

Figure 10-5 Display of management class definition in TS7700

Additionally, the Scratch and Device Allocation Assistance support must be explicitly enabled on the TS7700 through the LIBRARY REQUEST command. See Example 10-14.

Example 10-14 LIBRARY REQUEST commands that will enable Allocation support in TS7700

```
LIBRARY REQUEST,composite-libraryname,SETTING,DEVALLOC,SCRATCH,ENABLE
LIBRARY REQUEST,composite-libraryname,SETTING,DEVALLOC,PRIVATE,ENABLE
```

Note that the default for Scratch Allocation support is DISABLED, while for PRIVATE mounts, the default is ENABLED.

Consider defining the allocation preference in the ALLOCxx member as shown in Example 10-15.

Example 10-15 Setting the Library preference across multiple TS7700 libraries

```
TAPELIB_PREF(EQUAL|BYDEVICES)
```

This setting specifies the policy for balancing non-specific tape library requests (for example, scratch tape requests) across multiple tape libraries. The description of the TAPELIB_PREF parameters are:

► **EQUAL**

Indicates that for non-specific tape library requests, all tape libraries must be treated as equal, and receive an equal share of the requests.

► **BYDEVICES**

Indicates that non-specific tape library requests must be balanced across all tape libraries according to the number of online tape devices in the tape library. Tape libraries with more online tape devices will receive more non-specific tape requests than libraries with fewer online devices when all devices have the same attributes.

JES3 allocation requirements

In a JES3 environment the setup consists of one processor called the Global Processor, while the remaining processors are called local processors. The Global processors manage most of

the JES3 processing like allocating resources to jobs before this is sent to the Local Processor for execution.

Depending on individual user setup, converter and interpreter processing may happen in the Global Processor or in one of the Local Processors. Furthermore, the processor in which the converter and interpreter processing happens may not be the same as the executing processor. It is suggested, that all tape devices in the IBM Tape Library are connected to all processors in a JES3 plex to have these working on the same candidate list.

JES3 esoterics

The existing esoterics used in a JES3 environment for system managed tape libraries and IBM Tape Libraries will continue to be used. The JES3 esoterics are listed below.

- ▶ complex-wide name
- ▶ library-specific name
- ▶ library-specific device name
- ▶ complex-wide device type

The new esoteric *library-specific distributed name* will have an eight character string consisting of LDX followed by a five-digit library number for the distributed library. This will only be used, if DAA or SAA support is enabled.

JES3 INISH deck

The JES3 INISH deck has to be updated if you migrate to the new TS7700 Allocation assistance support for JES3. The existing device statements covering, for example two libraries, have to be split to be able to point to a specific library. An example of how this is done is in Example 10-16.

Example 10-16 Example of JES3 device statements in INISH deck before and after update

```
DEVICE,XTYPE=(CLB12345,CA),XUNIT=(1100,*ALL,,OFF),NUMDEV=512
```

line above splits into >>>>

```
DEVICE,XTYPE=(DLB10001,CA),XUNIT=(1100,*ALL,,OFF),NUMDEV=256
DEVICE,XTYPE=(DLB10002,CA),XUNIT=(1200,*ALL,,OFF),NUMDEV=256
```

The SETNAME statements has to have the new esoteric library-specific distributed name added to the existing esoterics as shown in Example 10-17.

Example 10-17 Example of JES3 SETNAME statements updated with the new esoteric.

```
SETNAME,XTYPE=DLB10001,NAMES=(LDGW3495,LDG12345,LDG3490E,LDE12345,LDX10001)
SETNAME,XTYPE=DLB10002,NAMES=(LDGW3495,LDG12345,LDG3490E,LDE12345,LDX10002)
```

One of the factors used by JES3 in selecting devices for volume mounting is the ADDRSORT parameter on the SETPARAM initialization statement. This parameter specifies that devices are either allocated in the same order as the DEVICE statement defining them (ADDRSORT=NO) or allocated by the order of their device numbers in ascending order (ADDRSORT=YES the default).

Documentation

For further detail, consult the documentation listed the below.

- ▶ IBM TS7700 Virtualization Engine customer information center
<http://publib.boulder.ibm.com/infocenter/ts7700/cust/index.jsp>
- ▶ z/OS Host Command Line Request User's Guide Version 3.0
[http://www.ibm.com/support/techdocs/atsmastr.nsf/5cb5ed706d254a8186256c71006d2e0a/6234eaf422f7d6b78625733d006582a7/\\$FILE/White%20Paper%20-%20TS7700%20Library%20Request%20Command%20V31.pdf](http://www.ibm.com/support/techdocs/atsmastr.nsf/5cb5ed706d254a8186256c71006d2e0a/6234eaf422f7d6b78625733d006582a7/$FILE/White%20Paper%20-%20TS7700%20Library%20Request%20Command%20V31.pdf)
- ▶ *z/OS DFSMS Object Access Method Planning, Installation, and Storage Administration Guide for Tape Libraries*, SC23-6867

10.4 DADSM CVAF/DEVICE support

DFSMS V2.1 provides DADSM/CVAF improvements as discussed in this section.

10.4.1 Rebuilding of VTOC index in VTOC full scenario

Prior to DFSMS V2.1 it was not possible to rebuild a VTOC index once the corresponding VTOC was full. Consequently you had to go through a somewhat time consuming and disruptive process of extending and even moving the VTOC to accomplish this.

With DFSMS V2.1 it is possible for the CVAF convert routine to return VTOC records with free space so that these do not need to be written to disk. Rebuild of the INDEX will be able to occur without the need for deleting data sets. Less time will be spent on problem determination and impact on the system will be reduced.

The ICK01322I message will be set when the conversion has detected that there is room to expand the VTOC physically on the volume as shown in Example 10-18.

Prerequisites for this function are ICKDSF R17 and PM36724.

Example 10-18 New message when VTOC is successfully expanded.

```
ICK01322I THE VTOC HAS BEEN EXPANDED TO ***** TRACKS
```

10.4.2 Reclaim of orphaned FORMAT 3 DSCBs in DADSM convert routine

This enhancement in DADSM convert routine identifies orphaned DSCBs during conversion from INDEXED to OS VTOC. The convert routine is called to build free space chains, but will now additionally identify orphaned FORMAT3 DSCBs and also convert these to FORMAT 0, creating more free DSCBs in the VTOC through this.

10.4.3 CVAFFILT macro uses the multi-DSCB read interface of CVAF

The CVAFFILT macro, with its underlying services is provided to obtain information about the data sets on an indexed or non-indexed VTOC. CVAFFILT has been modified in DFSMS V2.1 to use the multi-DSCB Read Interface making it possible to read all DSCBs for one data set into CVAFDIR in one operation. The most important functions of CVAFDIR are:

- ▶ Reads or write one or more DSCBs by specifying the name of the data set they represent.
- ▶ Reads or write one or more DSCBs by specifying their addresses.

10.4.4 PARTREL macro enhancement

The PARTREL macro is called at data set close, at restart processing, during DFSMSHsm space management processing, or when directly called. PARTREL would not be attempted in some cases where module IGG020D1 determined that there were not enough format 0 DSCBs available to update the free space chain on an OS VTOC volume. In DFSMS V2.1 it is modified to account for all FORMAT 0 DSCBs, including those released in PARTREL processing.

10.4.5 Erase On Scratch enhancement

The purpose of this enhancement is to replace existing channel program used to erase tracks with Locate Record with Erase operation code. The existing Erase On Scratch (EOS) algorithm builds a Format Write R0 channel program to erase one track at a time. The main objective for this change has been to improve performance and to position for the future.

Basically the new EOS algorithm uses an LR Erase operation code and specifies 1 to N tracks. Specifying a range of tracks in a single channel program will increase throughput.



DFSMSHsm enhancements

This chapter describes the various enhancements, functions and features introduced with DFSMSHsm V2.1. The DFSMSHsm functions described in this chapter are:

- ▶ Exploiting Storage Tiers in space management
- ▶ Extend number of tape volumes for migrated or backed up data sets
- ▶ Migration subtasking throughput enhancements
- ▶ GDG support for PDSE
- ▶ DFSMSHsm Fast Replication enhancements
- ▶ Reliability, availability, serviceability and usability improvements.

11.1 Exploiting Storage Tiers in space management

Prior to z/OS V2.1, DFSMSHsm treated all data in Level 0 (L0) as being in one single tier in the overall storage hierarchy. There were no policies to enable automated data movement within that L0 tier. In z/OS V2.1, DFSMSHsm is enhanced to move data from one class of devices to another within the L0 hierarchy. DFSMSHsm V2.1 implements class transition, when a data set migrates from a ML0 volume to a ML0 volume.

11.1.1 Storage Tiers migration concept

DFSMS V2.1 provides a policy-based automation for moving data sets within the ML0 volumes including active (opened) data sets.

The reason you would want to migrate data sets among ML0 3390 volumes is that these volumes are not the same as far as performance is concerned based on the disk technology. For example, in a DS8870 the 3390 volumes may be associated with physical disks, such as:

- ▶ Fast solid state devices (SSD)
- ▶ Medium fast enterprise-class drives, that can be full disk encryption or not
- ▶ Slower Nearline SAS disk, that also can be full disk encryption or not
- ▶ Slower SATA drives

Each layer of disk technology is called a tier. The association of a 3390 volume with the disks in the array of a tier is done through the DS8000 hardware console (DS storage manager).

The association of data sets and 3390 volumes at creation (or recall) time is done through assigning an SMS storage group (SG) to the data set. The SMS construct has the 3390 volumes candidates to contain the data sets assigned to this SG.

Concerning those tiers, to have the best performance for your business, two automatic mechanisms must be in place:

- ▶ To avoid bad performance (meaning higher I/O disconnect time) caused by 3390 data set volumes hot spots. It means data set extents with high I/O activity being located at 3390 associated with a slow tier. The solution is to use the DS8000 easy tier function. It has the automated capability of moving hot spots 3390 data set extents from a slower tier to a faster tier, such as SSD.
- ▶ To avoid bad performance (meaning higher I/O disconnect time) for 3390 volumes data sets with a not so high I/O activity (not a hot spot participant) but with key I/O operations for the system such as catalogs, RACF data base, JES2 checkpoint, important customer data sets, and so on. The solution is to allocate initially such data sets at ML0 3390 volumes associated with faster tiers and ask DFSMSHsm to an later “class transition” (in this case, we do not use the word migration). Class transition means to move such data sets to ML0 3390 volumes associated with slower tiers, after some time of no use. As expected, the MVS catalog is updated as a part of the class transition movement.

As an example, a user may tell DFSMSHsm to use data set class transition from an 3390 volume associated with a SSD tier to a 3390 volume associated with a SATA disk tier after 5 days of no use, and then becoming eligible for migration to ML2 after 30 days of continuous inactivity. Refer to Figure 11-1, where three tiers are pictured.

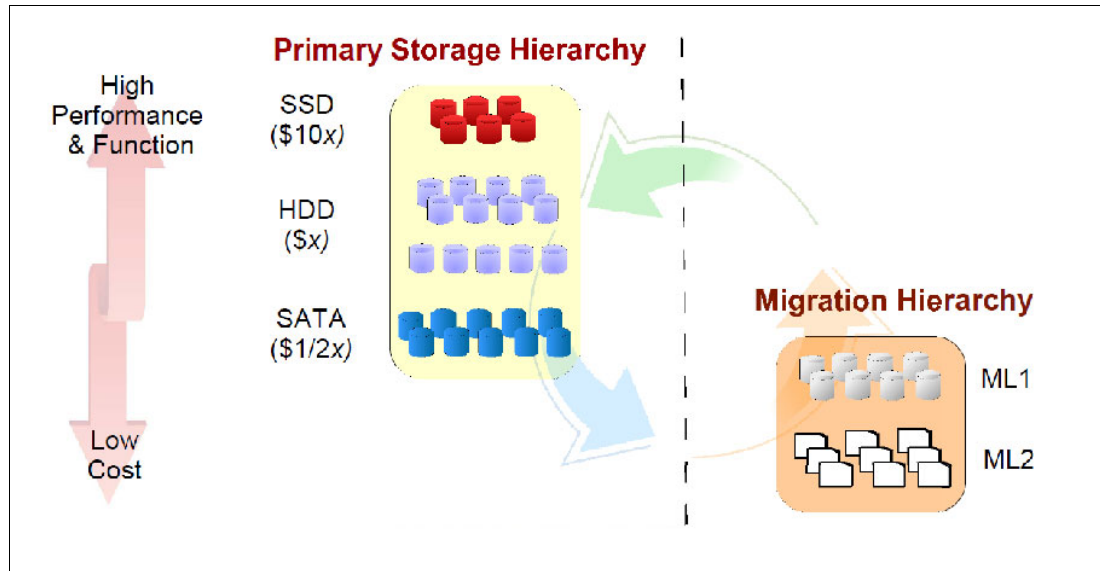


Figure 11-1 ML0 volumes with three tiers

The definition of DFSMSHsm space management function has been updated:

“DFSMSHsm using policy-based automation ensures that 3390 volumes within the primary storage hierarchy (ML0) have enough free space for new data sets and to ensure that data sets are stored at the lowest acceptable tier (performance wise) in the storage hierarchy.”

With DFSMS V2.1, we have the following functions for the DFSMSHsm space management:

- Automatic volume space management

- Primary space management

It is executed on each requested DFSMSHsm-managed volume at a specified time of day. It has two phases:

- Phase one deletes expired and temporary data sets, releases unused over allocated space.
 - Phase two migrates data sets (under control of the data set management class) optionally first from ML0 to ML0 (class transition), and second from ML0 to ML1 or to ML2. Migrations and optional class transition occur until the volume reaches its low threshold. This phase is executed if the first phase does not free enough space on the volume, that is, reaches the low threshold.

- Interval space migration

Ensures on an hourly basis that a specified amount of space is available on DFSMSHsm-managed ML0 volumes. It also has the two same phases of primary space management optionally including class transitioning operations. Interval space management only runs against storage groups that have this function enabled through the storage group SMS construct.

- On-demand migration

Introduced in DFSMS V1.13, it performs space management on a volume, as soon as it goes over high threshold. It is an optional replacement for on-the-hour interval space migration processing. It has the same two phases that includes optionally class transition, as well.

- Automatic secondary space management.

Secondary space migration moves data sets (under control of the data set management class) from ML1 to ML2 volumes and processes the migration cleanup function. Secondary space management is usually run before primary space management.

The SETSYS command may indicate if event driven migration (on demand migration) and interval migration should process class transitions. Refer to , “Class transition in an IM and ODM environment” on page 186 for more information about this SETSYS command.

Due to the introduction of class transition, some installations may need to adjust the number of DFSMSHsm tasks executing space management to keep the same performance.

The advantages of class transition a data set (compared with migrating) are:

- ▶ Keep the data set at the business performance required tier
- ▶ Data set is always available to the program task, that is, no need for a recall delay
- ▶ The class transition maybe applied to an opened data set under certain circumstances
- ▶ Increase the number of days that the data set must be unreferenced before migrating directly to ML2.
- ▶ Class transition operation is faster and less resource consuming because it may use the DS8000 FlashCopy function. However, such function is only applied between volumes of the same physical DS8000. As a reminder, the previous releases of DFSMSHsm already exploit FlashCopy for back up purposes.

The data sets that would benefit from class transition are:

- ▶ Data sets currently not eligible for migration because they always need to be immediately accessible. Or in other words, the recall delay is unacceptable.
- ▶ Data sets that are eligible for migration today, but there would be a benefit to keep them online for a longer period of time, as the ones, that goes through cycles of activity and inactivity. In this case with class transition, you may avoid the data set going through many migration and recall iterations.

For the complete discussion of Storage Tiers in DFSMSHsm V2.1 and how to implement it refer to “DFSMSHsm storage tiers” on page 179 and “DFSMS storage tiers” on page 100.

11.2 Extend number of tape volumes for migrated or backed up data sets

DFSMSHsm migration and backup data sets can span up to a maximum of 40 tape volumes. Since the size of a virtual tape is limited to 6 GB, DFSMSHsm cannot migrate or back up data sets larger than 600 GB to virtual tape volumes, assuming a 2.5:1 compaction factor. This fact is now a limitation for some users, mainly the ones using the typically small tape volume sizes configured for virtual tape subsystems, such as VTS.

11.2.1 Using and invocation

The solution is to extend the maximum number of volumes that a migration or backup tape data set can span from 40 to 254 volumes.

The tape recycle function is also enhanced to process connected sets of up to 254 volumes. The recycle function provides the capability of moving the valid data sets out from the original

tapes and consolidates the data on another tape. This makes one tape with all unexpired data sets and leaving the recycled tapes available for reuse.

There is no user option to implement this new function in DFSMSHsm V2.1. DFSMSHsm automatically spans up to 254 tape volumes during data set migration, backup, and recycle.

FSR records have been extended from a maximum of 1260 bytes in length to 6396 bytes in order to contain up to 508 volume serials (254 input and 254 output).

- ▶ FSRs that list more than 144 tape volumes truncates when written to the DFSMSHsm LOG log data sets. The existing 2048 fixed logical record length (RECL) prevents FSRs with more than 144 volumes from being written to the DFSMSHsm log data sets in their entirety.
- ▶ Truncated FSRs affects ARCPRLOG and ARCPEDIT output. These are IBM supplied programs to list the DFSMSHsm log data set.
- ▶ ARCPRLOG output displays only the portion of each FSR that was written to the log.
- ▶ When RECYCLE volumes are truncated, ARCPRLOG and ARCPEDIT output will display "TOVOL=*****".

11.3 Migration subtasking throughput enhancements

DFSMSHsm migrates a data set in several steps:

- ▶ Preprocessing: This includes eligibility checking, enqueue for serialization, CDS updates, and catalog update
- ▶ Data movement: Moving ML0 data sets to ML0 (class transition), or to ML1 or to ML2
- ▶ Postprocessing: This includes dequeue, CDS Updates, catalog update, and scratch

The target of this enhancement in DFSMSHsm V2.1 is to increase the parallelism of data set migration, and consequently increase the data set migration throughput.

11.3.1 Subtasking prior to DFSMSHsm V2.1

Prior to DFSMS V2.1 multitasking allows the parallel processing of multiple data sets migration (each one in each migration task). However, data set migration internal operations such as pre-processing, data movement, and postprocessing of each data set during are still performed sequentially. Figure 11-2 on page 202, shows an example of three data sets being migrated to ML2 (tapes) by one migration task. Only one task is shown for ease of explanation.

However, if the migration is to tapes, there is only one concurrent data movement to tape device per migration task, as shown in Figure 11-2 on page 202.

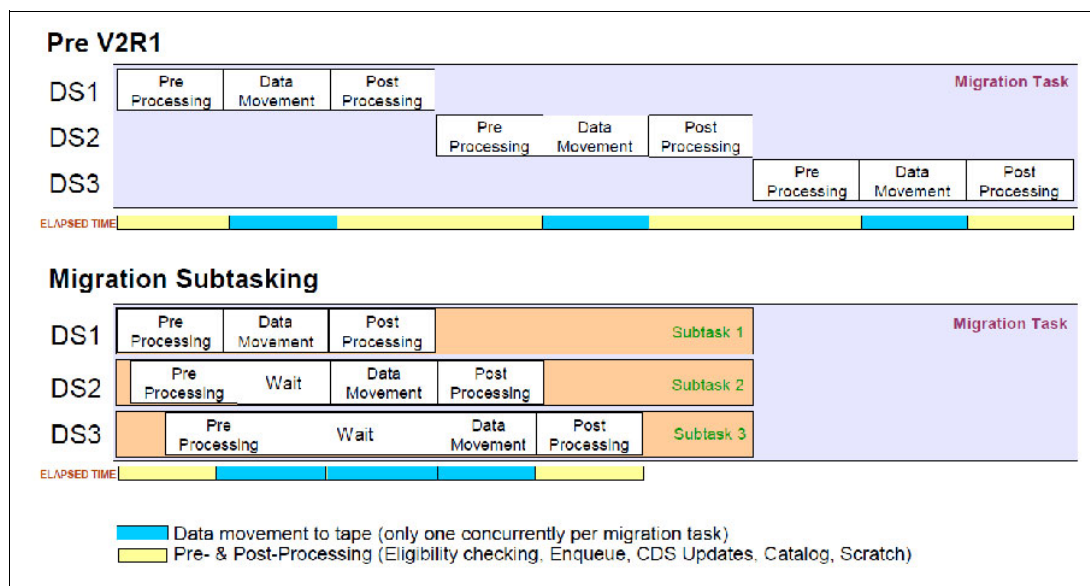


Figure 11-2 Migration throughput enhancements due to multitask

By implementing subtask migration, the aggregate throughput of all the migration tasks is improved.

11.3.2 Migration subtasks implementation

Two new keywords of the SETSYS command are added in DFSMSHsm V2.1:

- **MIGRATIONSUBTASKS(YES | NO)**

With the YES option, DFSMSHsm runs multiple subtasks concurrently under each migration task for primary space management, on-demand migration, and interval migration. The number of these subtasks per migration task is decided internally by DFSMSHsm.

- **ADDITIONALSUBTASKS(nn)**

This is a sub-parameter of the MIGRATION SUBTASKS parameter. It allows you to dynamically change the number of additional subtasks that DFSMSHsm can define, running under each migration task. These additional subtasks add to the number of subtasks that DFSMSHsm already uses when the MIGRATIONSUBTASKS parameter is specified.

Note that the total number of subtasks (per task) in a moment can vary, and it is lower, if a large value is specified with the maximum former migration tasks (MAXMIGRATIONTASKS) parameter. Conversely, the total migration subtasks used is higher, up to 15, if a smaller value is specified in the MAXMIGRATIONTASKS parameter.

11.4 GDG support for PDSE

DFSMSHsm V2.1 is updated so that it can support PDSEs that are generation data sets. Generation data sets defined with DSNTYPE=LIBRARY will be PDSEs. As a result, DFSMSHsm will invoke DFSMSdss to dump and restore those data sets. Each of the DFSMSHsm functions provide this support:

- ▶ Backup
- ▶ Recover
- ▶ Migration
- ▶ Recall
- ▶ Aggregate backup
- ▶ Aggregate recover

Systems prior to z/OS V2R1 which are running DFSMSHsm ABACKUP and ARECOVER will detect SMS-managed PDSE GDS in the ALLOCATE list and fail the command.

DFSMSHsm migration, recall, backup, and recovery functions will invoke DFSMSdss. The following existing messages were updated in both DFSMSHsm and DFSMSdss:

- ▶ ARC1001I dsn MIGRATE FAILED, RC=0087, REAS=0006
- ▶ ARC1287I A DISCREPANCY WAS FOUND IN THE DATA SET VTOC ENTRY
- ▶ ARC1001I dsn BACKDS FAILED, RC=0087, REAS=0006
- ▶ ARC1387I A DISCREPANCY WAS FOUND IN THE DATA SET VTOC ENTRY
- ▶ ARC6172E DATA SET dsn IS NOT SUPPORTED IN AN ALLOCATE LIST FOR AGGREGATE GROUP group_name
- ▶ ADR285E DATA SET dsn WAS NOT PROCESSED BECAUSE pre-allocated DATA SET TYPE IS NOT SUPPORTED IN THIS RELEASE, 15
- ▶ ADR778E DATA SET dsn WAS NOT SELECTED BECAUSE THE DATA SET TYPE IS NOT SUPPORTED IN THIS RELEASE, 20

11.5 DFSMSHsm Fast Replication enhancements

DFSMSHsm may use fast replication for backing up data sets. Fast replication uses, through DFSMSdss, the FlashCopy function of DS8000 controllers. FlashCopy is point-in-time implementation that gives the appearance of an almost instantaneous data set (or volume) copy.

The DFSMSHsm V2.1 fast replication enhancements are:

- ▶ DFSMSdss physical data set COPY and RESTORE commands
- ▶ DFSMSHsm fast replication and physical data set recovery functionality

11.5.1 Concept review

Here, we introduce certain basic concepts.

- ▶ Consistency group is designed to guarantee the sequence of the multiple I/O operations connected to a non atomic write along a process of Metro Mirror or FlashCopy, when a rolling over disaster is unfolding. A non atomic write has several sequenced I/O operations, such as an update in place (read, update in memory and writes back). Consistency group guarantees (for integrity purpose) that the sequence of the I/O operation associated with dependent write are the same in the primary and in the secondary volume.
- ▶ Let us refresh the concept of restore and recovery in DFSMSHsm. The data set level recovery is the opposite action of backup and the volume level restore is the opposite action of dump. By the way, DFSMSHsm does not automatically recover or restore a data

set, if it becomes damaged. The recover and restore process is driven by manual commands.

Fast replication uses volume-level fast replication to create backup versions for sets of storage groups. You can define a set of storage groups through the SMS copy pool construct. The DFSMSHsm FRBACKUP command creates a fast replication backup version for each volume in every storage group defined within a copy pool.

11.5.2 Implementation and use

Here we discuss considerations for the implementation and use of the Fast Replication enhancements in DFSMS V2.1.

► DFSMSdss physical data set COPY and RESTORE commands

Here, we have two sub items:

- DS RENAME and REPLACE for COPY and RESTORE commands

Prior to DFSMSHsm V2R1, DFSMSdss only allows the rename of non VSAM data sets during physical DS COPY and RESTORE.

In DFSMS V2.1, a NEWNAME(newwsname) parameter has been added to the FRRECOV command. It allows DFSMSHsm to use a new, VSAM fully-qualified data set name for the recovered backup version or dump copy. It does not apply to VSAM alternate Index cluster. This function gives users access to production data set and restored data set without having to restore entire volume. The benefit of such improvement would allow users to recover a data set to another name for analysis without replacing the production data. This function is not implemented for VSAM data set restore operations.

- Physical DS alternate SMS volume

Prior to DFSMSHsm V2.1, DFSMSdss physical DS COPY and RESTORE commands only support one output volume. Volume selection is not an option. If the target volume does not have enough space, the operation fails, like in the scenarios:

- COPY FULL from source to target, COPY DS from target to source
- DUMP FULL from target to tape, RESTORE DS from tape to source

Space might no longer be available on source if data set size changed pre allocated target might have had to be scratched/reallocated.

DFSMSHsm at V2R1 will no longer be restricted to recovering data sets back to the original volumes. If DFSMSHsm is not able to recover a data set to the original volumes, it instead selects the most eligible volumes with the most free space within the storage group. There are no changes to the DFSMSHsm commands.

This enhancement is implemented by adding of a new user interaction module (UIM) exit:

- Physical DS Alternate SMS Volume (EIOPTION 30)
- Added to ADREID0 mapping
- Allows program to pass a list of volumes that DFSMSdss can use to attempt to allocate data set on during physical DS copy and restore, when necessary

This exit is only called when original target volume could not be used.

► DFSMSHsm fast replication and physical data set recovery functionality

Here, we have three sub items:

- FlashCopy consistency group

When application data set resides on multiple volumes, data-consistent copy might need to be created across multiple volumes for FRBACKUP command. Prior to DFSMS V2R2, DFSMSHsm does not have this support, although DFSMSdss has already implemented it.

At DFSMSHsm V2R1 fast replication backup supports FlashCopy consistency groups. It freezes the source volumes causing subsequent writes to be held. After FlashCopy has completed, the held I/O is resumed.

Note: The consistency group must be formed and completed within a (default 2-minute) time window, therefore it is intended for backing up small LOG copy pools.

Specify whether the FlashCopy consistency group option is required to be used when the copy pool is backed up to disk with the new SMS copy pool settings:

- FlashCopy Consistency Group (Y or N)

Y - yes, perform consistency group FlashCopy

N (default) - no, do not perform consistency group FlashCopy.

If you specify Yes, then DFSMSHsm FRBACKUP function invokes DFSMSdss specifying:

- FCFREEZE on the COPY commands. "Freeze" places the source volume in an extended long busy state (suspended). All subsequent writes to the source volume are held. Associated dependent writes to any volumes also cannot complete.
- CGCREATED ACCVOL FCCGVERIFY command (in DFSMSdss) to thaw the source volumes after all volumes have been copied. The previously held I/O resumes. FCCGVERIFY validates the consistency of the copy.

DFSMSdss accepts multiple volume serial numbers on the FCCGVERIFY keyword of the CGCREATED command. The keyword will accept up to 255 volume serial numbers.

If the FlashCopy consistency group function fails, the FRBACKUP command is terminated and the new or in-process copy pool backup version is invalidated. An ARC1806E message is issued in the console:

ARC1806E FAST REPLICATION BACKUP HAS FAILED FOR COPY POOL cpname,
RC=90

RC90 means that a data-consistent copy pool backup version could not be created successfully. The FlashCopy consistency group timer might have expired or reset before FRBACKUP finished processing all the volumes

The FlashCopy consistency group option can be used in combination with other FlashCopy options, as:

- LIST COPYPOOL(cpname) command can be used to display (at FCCG=Y/N field) and determine whether the copy pool backup version was created using the FlashCopy consistency group option. Refer to Figure 11-3 on page 206 for get the output of such command.

LISTCOPYPOOL command output

```
COPYPOOL=CP6
ALLOWPPRCP FRB=PN FRR=PN
VERSION VTOCENQ DATE TIME FASTREPLICATIONSTATE DUMPSTATE
001 Y 2012/08/06 11:03:50 RECOVERABLE NONE
TOKEN(C)=C"
TOKEN(H)=X"
TOTAL NUM OF VOLUMES=00009,INCREMENTAL=N,CATINFO=Y,FCFRR=N,RECOVERYINCOMPLETE=N,FCCG=Y
SGNAME SOURCE - TARGET SOURCE - TARGET SOURCE - TARGET SOURCE - TARGET
SGRP1 SRC01B - TGT01B SRC02B - TGT02B
SGRP2 SRC03B - TGT05B SRC04B - TGT06B SRC05B - TGT07B
SGRP6 SRC31B - TGT31B SRC32B - TGT32B SRC33B - TGT33B SRC34B - TGT34B
```

Figure 11-3 LISTCOPYPOOL command output

Now, data-consistent copy can be created across multiple volumes. For example, DB2 customers can use this option on the LOG copy pool for a consistent, non-fuzzy backup of the logs and BSDS.

- Fast replication data set recovery to different volumes

Prior to DFSMSHsm V2R1, fast replication data set recovery must recover data sets to the original volumes where they resided at the time of backup. In order to achieve that, DFSMSHsm currently captures catalog information at the time of the fast replication backup in order to recover deleted and moved data sets back to the original volumes. However, the recovery might fail for reasons such as insufficient space.

The solution is to remove the fast replication recovery restriction that data sets must recover back to the original volumes, to implement such, there are no changes to the fast replication commands.

Now, at DFSMSHsm V2R1, customers are able to recover data sets to different volumes than the original volumes.

DFSMSHsm will first attempt to recover the data set to the original volumes.

If the initial allocation attempt failed, DFSMSHsm performs SMS volume selection within the primary storage group and retry data set recovery.

- Fast replication data set recovery with rename

Prior to DFSMSHsm V2R1, fast replication data set recovery does not allow data sets to be recovered to a new name,

Also, the RECOVER command currently does not allow VSAM data sets to be renamed when restoring from dump.

The solution at DFSMSHsm V2R1 is to enhance:

- FRBACKUP command to allow data set to be recovered to a new name
- RECOVER command to remove the restriction that VSAM data set must be recovered to the original name when restoring from dump

A new optional parameter: NEWNAME(newdsname) introduces the new name. If a data set exists with the same name as the new data set name you are specifying, you must specify the existing REPLACE parameter to replace the existing data set

The benefit of such improvement would allow users to recover a data set to another name for analysis without replacing the production data.

11.6 Reliability, availability, serviceability and usability improvements

Since the very beginning there is an non stop effort on the z/OS platform for improving reliability, availability and serviceability (RAS). A list of the improvements for DFSMSHsm at V2.1 are:

- ▶ DFSMSHsm no longer capture UCBs into below the 16-M line
- ▶ Automatic recycle retry for a tape takeaway
- ▶ Automatic recycle retry for a duplex error
- ▶ QUERY ACTIVE(TCBADDRESS) command improvements
- ▶ Enhance ARC0936I with catalog return/reason code
- ▶ Update ARCHRCAL macro to use different TCB
- ▶ SMSVSAM server termination handling

11.6.1 DFSMSHsm no longer capture UCBs into below the 16-M line.

Unit Control Block (UCB) is a control block used by the MVS component Input Output Supervisor (IOS) to represent a device or an additional alias address of a device, if you implement Parallel Access Volume (PAV). All UCBs are located above the 16-M line in ESQA. They can be:

- Captured UCBs, although located above the 16 M line, they are also mapped and accessible by AMODE24 programs for compatibility needs. This is possible through the implementation of the IBM z/Architecture® concept of shared pages, that is, two virtual pages pointing to the same real storage frame.
- Non capture UCBs, as defined in an HCD. They are not mapped below the 16 M line.

Because captured UCBs consumes virtual storage below the 16-M line, large customers face periodic 878-Abends in the DFSMSHsm address space. DFSMSHsm has increased the available virtual storage below 16 M line by not capturing UCBs below the line anymore.

11.6.2 Automatic recycle retry for a tape takeaway

DFSMSHsm V2R1 can automatically generate a new Recycle command for the same original target tape, when this Recycle must terminate due to:

- ▶ Takeaway process. A takeaway happens when a recall request needs the tape which is currently being written to by a space management task as migration to ML2, or
- ▶ When the recycle input tape is in use by another DFSMSHsm task

Prior to DFSMSHsm V2R1 space management task closes the ML2 cartridge after the takeaway. The space management task will then request a new scratch tape, and the partially filled ML2 tape is not reused to append any more data to it. This can result in many cartridges being only partially full and becoming candidates for subsequent recycles.

Automatically retrying a recycle is managed by the new option at SETSYS command:

SETSYS RECYCLETAKEAWAYRETRY(YES MAXRETRYATTEMPTS(nn) DELAY(mmmm) | NO)

- ▶ RECYCLETAKEAWAYRETRY is an optional parameter used to generate a new Recycle command for the same original tape when the original Recycle must terminate due to the takeaway process or when the input tape is in use by the other DFSMSHsm task.
- ▶ MAXRETRYATTEMPTS(nn) sets the maximum number of recycle retry attempts
- ▶ DELAY(mmmm) sets the delay in seconds between recycle attempts

Range and defaults:

- nn -the value can be 1 – 99. The default value is 12.
- mmmm - the value can be 1 – 9999 The default value is 300.

11.6.3 Automatic recycle retry for a duplex error

Prior to DFSMSHsm V2R1, when an alternate tape of a duplex copy pair fails to be created, DFSMSHsm generates an automatic request to create a tape copy of the original. At DFSMSHsm V2R1, when such error occurs, an option is available to generate a RECYCLE command instead of TAPECOPY command to create a new original and alternate tape.

This can be done through a new SETSYS option added to the Duplex Tape function. When an error occurs on the duplex alternate tape, the alternate tape is demounted and discarded, DFSMSHsm continues to write to the original tape. When DFSMSHsm has completed writing the original tape, a TCN record for Recycle is created and written to OCDS. DFSMSHsm starts RECYCLE of the original tape to create a new duplex pair of tapes.

The new option CONTINUE (RECYCLE) creates a new duplex pair of tapes via the Recycle command after the original finishes:

SETSYS DUPLEX (BACKUP (Y ERRORALTERNATE(CONTINUE(TAPECOPY| RECYCLE)) | N) MIGRATION(Y ERRORALTERNATE(CONTINUE(TAPECOPY| RECYCLE) | MARKFULL) | N))

11.6.4 QUERY ACTIVE(TCBADDRESS) command improvements

At DFSMSHsm V2R1 the QUERY ACTIVE(TCBADDRESS) command is improved to display on top of the TCB address for data movement tasks, to show: the tape volser, device address and task name. It is very clear that just the TCB address is not enough for task identification. This helps users with correctly identifying which TCB should be cancelled. Users often know which tape drive that is in error, but do not know how to relate that to a particular DFSMSHsm TCB.

11.6.5 Enhance ARC0936I with catalog return and reason code

DFSMSHsm ARC0936I message now contains additional diagnostic information. When FAST VVDS ACCESS interface returns RC12, REAS98, DFSMSHsm uses the information in FVPPROB field to put it into SUBREAS field of the ARC0936I message.

11.6.6 Update ARCHRCAL macro to use different TCB

The ARCHRCAL macro allows customer programs to recall a migrated data set. The syntax of the ARCHRCAL macro is changed to use a different TCB. A new TCBTOKEN=CURRENT

parameter is added to the syntax. This parameter is provided to support changes in BCP for batch job recall processing.

11.6.7 SMSVSAM server termination handling

This support greatly improves the usability and robustness of DFSMSHsm in regards to its response to SMSVSAM server errors. When an SMSVSAM server error occurs, DFSMSHsm detects the error, and quiesce all DFSMSHsm CDS I/O activity. Once the SMSVSAM server initializes, DFSMSHsm automatically closes and reopens the CDSs and resumes all requests waiting on CDS I/O operations.



DFSMSrmm enhancements

The DFSMSrmm enhancements in z/OS V2.1 DFSMS provide improvements in the areas of usability and maintainability. New function in this release includes the following:

- ▶ SMS management class expiration attributes for tape
- ▶ RETENTIONMETHOD(EXPDT) enhancements
- ▶ DFSMSrmm conversion support

At the end of the chapter you should be able to use the retention specified in your SMS management classes. You should understand how to select different retention methods for your volumes and to exclude some of your data sets from vital record processing.

12.1 DFSMSrmm overview

DFSMSrmm is a z/OS feature. In your enterprise, you probably store and manage removable media in several types of media libraries. For example, in addition to your traditional tape library, a room with tapes, shelves, and drives, you might have several automated, virtual, and manual tape libraries. You probably also have both on-site libraries and off-site storage locations, also known as vaults or stores.

With DFSMSrmm, you can manage your removable media as one enterprise-wide library across systems and SYSPLEXes. DFSMSrmm manages your installation's tape volumes and the data sets on those volumes. DFSMSrmm also manages the shelves where volumes reside in all locations except in automated tape libraries.

DFSMSrmm manages all tape media, such as cartridge system tapes and 3420 reels, as well as other removable media you define to it. For example, DFSMSrmm can record the shelf location for optical disks and track their vital record status. It does not manage the objects on optical disks.

12.2 SMS management class expiration attributes for tape

You can now set the expiration date in DFSMSrmm for a tape data set with a DFSMS management class (MC). When you enable use of management class attributes by DFSMSrmm, the management class expiration attributes (except the management class Expiration attribute Retention limit) are retrieved by DFSMSrmm during OPEN for output and used to set the expiration date for the tape data set, and also to set the LASTREF extra days in the tape data set record on retention method EXPDT managed volumes. Regardless of whether the expiration attributes are retrieved from the management class, from a DFSMSrmm default parmlib option, or from an installation exit, an expiration date is calculated and will be used to manage expiration. Any attributes needed to continue management of retention, such as *days non-usage* are bound to the data set record in the DFSMSrmm CDS, depending on the retention method. This is a one-time action, thus avoiding any overhead of repeating the policy decisions as part of inventory management.

At OPEN for input the management class attributes are not considered for processing. At OPEN for output with disposition MOD the management class attributes are not considered for processing. The enablement in DFSMSrmm provides options to use or not use these management class attributes for all volumes. It provides an option for VRSEL managed volumes to exclude the management class attribute *Expire after Date/Days*. This last option is recommended if it is desired that the processing of VRSEL managed volumes not change with DFSMSrmm V2.1.

Management class attributes for tape management

You can use ACS routines to assign a management class to a data set to allow DFSMSrmm to:

- Retrieve and use the management class attributes relevant for tape management.

DFSMSrmm calls ACS routines directly to provide support for non-system-managed tape if you have enabled it with the EDGRMMxx parmlib option SMSACS. This DFSMSrmm processing enables you to use an SMS management class to manage system-managed and non-system-managed the usage of the management class attributes. The SMS management class attributes that can be retrieved are the expiration attributes *Expire after Days Non-usage* and *Expire after Date/Days*. They are bound to the DFSMSrmm data set attributes EXPDT and LASTREF when the data set is first written.

- Match a DFSMSRmm data set vital record specification to the management class name.
This processing applies only when the VRSEL retention method is used. DFSMSRmm calls ACS routines directly to provide support for non-system-managed tape, if you have enabled it with the EDGRMMxx parmlib option SMSACS. This DFSMSRmm processing enables you to use an SMS management class to manage system-managed and non-system-managed tape data sets.

Using SMS ACS processing you have an automated way to set the expiration date and LASTREF extra days of a data set from the management class expiration attributes 'Expire after Date/Days' and 'Expire after Days Non-usage'. You can tailor the use of the management class attributes by DFSMSRmm with the EDGRMMxx parmlib option MCATTR.

Using SMS ACS processing you can consolidate policy assignment decisions in a single place whether you use system-managed tape or not. You can use SMS ACS routines to assign management class for your data sets instead of using vital record specification management values assigned by the EDG_EXIT100 exit. You can assign a management policy by name to either a non-system-managed or a system-managed tape data set.

If the use of the management class attributes is enabled by the EDGRMMxx parmlib OPTION MCATTR, you can use the management class Expiration Attributes (Expire after Date/Days and Expire after Days Non-usage) to set the data set EXPDT and LASTREF attributes, where appropriate.

You can exploit the SMS management class attributes for both system-managed and non-system-managed volumes. For non-system-managed volumes, you must set the EDGRMMxx parmlib option SMSACS(YES).

12.2.1 Using SMS management class in a system-managed tape environment

To use the SMS management class expiration attributes you have to change the MCATTR option in you EDGRMMnn parmlib member from the default MCATTR(NONE) to ALL or VRSELXDI to enable the use of DFSMS management class expiration attributes.

Update EDGRMMnn parmlib member

In Figure 12-1 you can see the MCATTR and RETENTIONMETHOD settings in the EDGRMMnn parmlib member to use DFSMS management class expiration attributes.

OPTION	OPMODE(P)	/* protect mode	*/ -
	ACCOUNTING(J)	/* Account information	*/ -
	BACKUPPROC(RMMBKUP)	/* backup procedure	*/ -
	BLP(RMM)	/* bypass label process	*/ -
	CATRETPD(12)	/* catalog retention	*/ -
...			
	MAXRETPD(NOLIMIT)	/* maximum retention	*/ -
	MCATTR(VRSELXDI)	/* management class use	*/ -
...			
	RETENTIONMETHOD(VRSEL)	/* VRSEL retain process	*/ -
	RETPD(0)	/* default retention	*/ -
...			
	SMSACS(YES)	/* SMSACS option	*/ -
...			

Figure 12-1 Sample EDGRMMnn parmlib member

The following values apply to Figure 12-1 on page 213:

MAXRETPD	Specifies the maximum retention period that a user can request for data sets on volumes. The following values are valid:
NOLIMIT	Specify NOLIMIT to use the dates 99365 or 99366 which mean to never expire. If the calculated date is 31 December 1999, the expiration date 1 January 2000 is used. The default is MAXRETPD(NOLIMIT).
nnnnn	Specify a value between 0 and 93000 days. When a value between 0 and 93000 days is specified, the value is added to the current
MCATTR	Specifies whether DFSMSrmm should be enabled to use of DFSMS management class expiration attributes that apply to tape management. The following values are valid:
ALL	The use of all applicable management class attributes is enabled. The management class attributes are exploited as they are appropriate, depending on the retention method. The management class attribute Retention Limit is not applied.
NONE	No management class attributes are used. This is the default.
VRSELXDI	The use of management class attributes is enabled. The management class attributes are exploited as they are appropriate, depending on the retention method. The exception is the Expire after Date/Days attribute, which is ignored if the data set is on a volume managed by the VRSEL retention method. The management class Retention Limit attribute is not applied. VRSELXDI is recommended if it is desired that the processing of VRSEL managed volumes is the same as in prior DFSMSrmm releases.
RM	RETENTIONMETHOD can be abbreviated as RM. Use this operand to set the system-wide retention method default for new tape volume sets. New tape volume sets may be created during Open/Close/End-of-Volume (O/C/EOV) processing, or through DFSMSrmm commands. A tape volume set may be a multivolume set, or a single tape volume. The following values are valid:
EXPDT	Specify EXPDT to set the default retention method for new tape volume sets to be based on EXPDT. Data sets and volumes managed by this retention method are never processed by VRSEL inventory management. When you specify the EXPDT retention method the DFSMSrmm inventory management EXPROC processing always attempts to return volumes to scratch on the same run as the volume is released (this is as if the SCRATCHIMMEDIATE attribute is set for the volume). DFSMSrmm maintains a consistent expiration date and time for all data set records of a multivolume data set, unless the volume set is retained by first file. EXPDT can be specified either as EXPDT or EXPDT(options). The available options are:
LASTREF	LASTREF specifies the default for the data set record LASTREF attribute. The LASTREF attribute specifies the number of days that the data set will be retained after the data set was last referenced by a read or write operation. LASTREF applies only to data sets on volumes managed by the EXPDT retention method. The parameter <i>extra_days</i> is a decimal number between 0 and 93000. The value must not exceed the

	maximum retention period (MAXRETPD) specified in the EDGRMMxx parmlib member. An extra_days value of 0 has the same effect as the NOLASTREF operand. When a data set is added to DFSMSrmm on a volume managed by the EXPDT retention method and neither LASTREF nor NOLASTREF are specified for the data set, then DFSMSrmm uses the default LASTREF value. DFSMSrmm uses the data set LASTREF value to determine the data set expiration date. The extra days are added to the date of last reference. The data set expiration date is set to the maximum of the date calculated using data set LASTREF value and the date resulting from applying the EXPDT, RETPD, or default RETPD. Any reference to the data set by a read or write operation will change the expiration date.
NOLASTREF	NOLASTREF is the default setting for the data set record LASTREF attribute. If neither LASTREF nor NOLASTREF are specified in parmlib, NOLASTREF is used by default. NOLASTREF specifies that DFSMSrmm does not consider the data set last reference date when determining the data set expiration date. NOLASTREF applies only to data sets on volumes managed by the EXPDT retention method.
RETAINBY	RETAINBY specifies how DFSMSrmm is to retain volumes or multivolume sets that are managed by the EXPDT retention method. In a multivolume set, RETAINBY is assigned only to the first volume in a multivolume sequence. All other volumes added to the set assume the same RETAINBY. It has several valid values described in “Using RETAINBY option” on page 224.
VRSEL	Specify VRSEL to set the default retention method for new tape volume sets to be VRSEL. This option enables DFSMSrmm inventory management to attempt to match data sets and volumes to vital record specifications, and if a match is found, to determine if the data set or volume is to be retained by VRS. The VRSEL retention method is controlled by all the other VRS related options in parmlib including OPTION RETAINBY MOVEBY. The default is RETENTIONMETHOD(VRSEL). For volumes managed by the VRSEL retention method, use the RETAINBY option to obtain a similar function.
RETPD	Specifies the default retention period for all new data sets on volumes. Specify a value between 0 and 93000 days. The specified value is added to the current date to determine the expiration date. Select a default retention for parmlib RETPD that is a small value to ensure that all tape data created outside the service levels is released as soon as possible. The MAXRETPD value you specify in the parmlib limits the calculated expiration date. The default is RETPD(5).
SMSACS	Specify this operand to control whether DFSMSrmm calls SMS ACS processing to enable use of storage group and management class values with DFSMSrmm for non-system managed data. The following values are valid:
NO	Specify NO to prevent DFSMSrmm from calling the SMS ACS processing to obtain management class and storage group names. DFSMSrmm system-based scratch pooling, and scratch pooling

and VRS management values based on the EDG_EXIT100 installation exit are used. The default is SMSACS(NO).

YES

Specify YES to enable DFSMSrmm calls to the SMS ACS processing to obtain management class and storage group names. If values are returned by the SMS ACS routines the values are used instead of the DFSMSrmm and EDG_EXIT100 decisions.

If there is no expiration date in the JFCB or management class, DFSMSrmm uses the EDGRMMxx RETPD value to calculate the new expiration date. If the RETPD value allows the volume to be retained longer, DFSMSrmm uses that date to update the volume's expiration date.

Figure 12-2 shows you the result of the LISTCONTROL OPTION subcommand if you have the default MCATTR option setting.

```

System options:
PARMLIB Suffix = 02           Operating mode = P
Control data set name = RMM.CONTROL.DSET
Journal file data set name = RMM.JOURNAL.DSET
Journal threshold = 75%           Journal transaction = NO
Catalog SYSID = Notset
Scratch procedure name = EDGXPROC
Backup procedure name = EDGCDSBK
IPL date check = N           Date format = J           RACF support = N
SMF audit = 0           SMF security = 42           CDS id = SC70
MAXHOLD value = 100           Lines per page = 54           System ID = SC70
BLP = RMM           TVEXT purge = RELEASE Notify = N
                        days = 0
Uncatalog = Y           Message case = M
MASTER overwrite= USER           Accounting = J
Disp DD name = DISPDD           Disp msg ID = EDG4054I
PREACS = NO           SMSACS = YES           CMDAUTH OWNER = NO
Reuse bin = CONFIRMMOVE           CMDAUTH DSN = YES
Local tasks = 10           Media name = 3480

Retention period: Default = 0           Maximum = NOLIMIT
                  Catalog = 6           hours
Use of Management Class Attributes = VRSELXDI
Retention method: Default = VRSEL

RM(VRSEL) defaults:           RM(EXPDT) defaults:
Retain by = SET           Retain by = VOLUME
Move By = SET
VRS selection = NEW           LASTREF extra days = 0
VRS change = INFO
VRSMIN action = INFO
VRSMIN count = 1
VRS job name = 2
GDG duplicate = BUMP
GDG cycle by = GENERATION

```

Figure 12-2 EDGRMMnn parmlib member without use MC attributes

12.2.2 Update your tape related SMS management classes

You have to specify your expiration settings in the SMS management classes related for tape. Figure 12-3 shows an SMS management class used for tape data sets with specific expiration date settings.

MANAGEMENT CLASS ALTER		Page 1 of 6
Command ==>		
SCDS Name : SYS1.SMS.MHLRES3.SCDS		
Management Class Name : NSRMMC		
To ALTER Management Class, Specify:		
Description ==> SPECIAL MANAGEMENT CLASS TO SUPPORT DFSMSRMM SMS ACS SUPPORT ==>		
Expiration Attributes		
Expire after Days Non-usage . . 10	(1 to 93000 or NOLIMIT)	
Expire after Date/Days 20	(0 to 93000, yyyy/mm/dd or NOLIMIT)	
Retention Limit 30	(0 to 93000 or NOLIMIT)	
Use ENTER to Perform Verification; Use DOWN Command to View next Panel;		

Figure 12-3 Sample SMS management class expiration setting

The following values apply to Figure 12-3 on page 217:

EXPIRE AFTER DAYS NON-USAGE

This attribute is used to specify when each data set in this management class will expire. The data sets will expire if they are not used within the number of days you specify. If you do not want the data sets to expire, specify NOLIMIT.

This attribute is valid only if the retention period (RETPD) or expiration date (EXPDT) is not specified or is not derived from the data class.

The *retention* attributes override the *migration* attributes. The following values are valid:

- | | |
|-------------------|---|
| 1 to 93000 | Specify any value between 1 to 5 digits. |
| NOLIMIT | Specify this attribute if you want to ignore the expiration time. |

EXPIRE AFTER DATE/DAYS

To specify either the expiration date for data sets or objects in this management class or the number of days before data sets or objects expire, beginning with the creation date. This attribute is valid only if the retention period (RETPD) or expiration date (EXPDT) is not specified or is not derived from the data class. The retention attributes override the migration attributes. The following values are valid:

- | | |
|-------------------|--|
| yyyy/mm/dd | Specify four digits that specify a year, the month as 01 to 12, the day as 01 to 31. |
| NOLIMIT | Specify if you want the data set to not expire. |

0 to 93000	One to five digits to specify the number of days.
RETENTION LIMIT	To control what a user or data class can specify for retention period (RETPD) or expiration date (EXPDT) during allocation. The following values are valid:
0	Do not use the RETPD and EXPDT that the user or data class specified.
1 to 93000	Use this value only if the RETPD or EXPDT is more than this value.
NOLIMIT	Do not set a limit to the allowable RETPD or EXPDT.

Important: If both EXPIRE AFTER DAYS NON-USAGE and EXPIRE AFTER DATE/DAYS specify NOLIMIT, the data sets or objects never expire. If either field has a value of NOLIMIT and the other field specifies an expiration date or the number of days until expiration, the data sets or objects expire at that specified time. If both fields specify an expiration date or the number of days until expiration, the data sets or objects expire on the later date.

The use of the management class is illustrated further in this chapter. We provide examples that show you what is happened with the different DFSMSrmm PARMLIB settings.

12.2.3 SMS management class in a non-system-managed tape environment

For non-system managed tape, DFSMSrmm calls the ACS routines to obtain a management class. The management class is used in place of the vital record specification management value assigned by the EDG_EXIT100 installation exit. When a management class name is assigned using ACS routines, the EDG_EXIT100 installation exit is not called for a vital record specification management value. The decision to call the EDG_EXIT100 installation exit is made each time a new data set is created on a tape based on whether a construct is assigned by ACS processing. You have the flexibility to identify one request to be handled by ACS and the next request to be handled by the EDG_EXIT100 installation exit. For compatibility, DFSMSrmm passes the vital record specification management value that is determined by the EDG_EXIT100 installation exit by using the pre-ACS interface in the MSPOLICY variable. You might use the vital record specification management value in the MSPOLICY variable as the base for management class assignment for system managed tape. Even when you use SMS ACS support to assign management class names, you can have separate policies for retention and movement by using a primary data set name vital record specification and a secondary management class vital record specification. Use the ACS routine to assign the management class as the secondary vital record specification and the DFSMSrmm data set name vital record specification to assign the primary vital record specification.

Note: You can still use the EDG_EXIT100 installation exit to check for either EXPDT= or ACCODE= specifying special values and override them to ensure correct retention processing by DFSMSrmm.

Management class assignment

In addition to the description in section “Using SMS management class in a system-managed tape environment” on page 213 you must enable this by specifying the OPTION SMSACS in your EDGRMM parmlib member.

To assign a management class, you must have the SMS subsystem active and have a valid SMS configuration.

You use the ACS routines to process the special calls that DFSMSrmm makes to the SMS subsystem for ACS processing. DFSMSrmm requests that the management class routine is run. The environment variable is set to RMMVRS so that you can differentiate allocation requests for system-managed data sets from requests by DFSMSrmm for a management class name. When DFSMSrmm calls the ACS routines with the &ACSENVIR variable set to either RMMPOOL or RMMVRS, the storage class (&STORCLAS variable) is set to the word USERMM.

DFSMSrmm calls the ACS routines to request the assignment of storage group and management class names for non-system managed tape data sets. Table 12-1 lists the read-only variables that are set for DFSMSrmm requests.

Table 12-1 SMS read-only variables set by DFSMSrmm

Variables			
&ACCT_JOB	&ACCT_STEP	&ACSENVIR	&DD
&DSN	&DSORG	&DSTYPE	&EXPDT
&FILENUM	&GROUP	&HLQ	&JOB
&LABEL	&LIBNAME	&LLQ	&NQUAL
&PGM	&STORGRP	&SYSNAME	&SYSPLEX
&UNIT	&USER	&XMODE	

Note: The SMS pre-ACS exit variables &MSPOOL, &MSPOLICY, &MSDEST, and &MSPARM read-only variables can be used by a tape management system only.

The following values apply to Table 12-1 on page 219:

&ACCT_JOB	The accounting information from the JOB statement. The maximum value is 142 characters.
&ACCT_STEP	The accounting information from the STEP statement. The maximum value is 142 characters.
&ACSENVIR	The environment on which the ACS routine was invoked. The following values are valid:
RECALL	For data set recall operations.
RECOVER	For data set recover operations.
RENAME	For data set alter rename operations.
RMMPOOL	For DFSMSrmm requests for a storage group name.
RMMVRS	For DFSMSrmm requests for a management class name.
CONVERT	For data set convert in place operations.
ALLOC	For new data set allocations. This is the default.
STORE	OSREQ object store environment.
CHANGE	OSREQ object change environment.
CTRANS	OSMC object class transition environment.
other	Installation exit can set its own value before re-invoking ACS Max value: 8 characters

&DD	DDNAME in the DD statement of the data set.
&DSN	The name of the data set or collection for which ACS processing is taking place. For VSAM data sets, only the cluster name is passed to the ACS routine, the component names are not. If the data set has an absolute or relative generation number, it is stripped from &DSN. The generation number is the low-level qualifier of the data set name.
&DSORG	The data set organization. The following are valid values:
PS	Physical sequential
PO	Partitioned
VS	VSAM organization
DA	BDAM organization
null	No value specified
&DSTYPE	The data set type. The following are valid values:
GDS	One generation data set of a generation data group, or any data set allocated with a relative generation number (such as A.B.C(+1)) or an absolute generation number (such as A.B.C.G0000V00).
PERM	Standard permanent data sets.
TEMP	Temporary data sets.
null	None of the above.
&EXPDT	The expiration date in the form of YYYYDDD where YYYY is the four-digit number for the year. The maximum allowable value for yyyy is 9799. DDD is the three-digit number for the day of the year from 1 to 366. Note that expiration dates of 99365 and 99366 are considered NEVER-SCRATCH dates.
&FILENUM	The value of the FILENUM ACS read-only variable. This variable corresponds to the data set sequence number on the JCL LABEL parameter. The default is 1. This field is optional.
&GROUP	The RACF-defined group to which you are connected, or the group specified in the GROUP keyword on the JCL JOB statement. If the environment is recall or recover, &GROUP is set only if the requester of the recall or recover is not a DFSMSHsm authorized user. When DFSMSHsm invokes the ACS routines, &GROUP is the group associated with &USER.
&HLQ	The high-level qualifier of the data set or collection name.
&JOB	The job name, the started task name, or the TSO/E user ID from the JOB statement, depending on the execution mode (&XMODE).
&LABEL	The value of the LABEL ACS read-only variable. This variable corresponds to the label field of the JCL LABEL parameter. Allowable values are NL, AL, SL, NSL, SUL, AUL, BLP, LTM or blank. The default is IBM Standard Label. This field is optional.
&LIBNAME	The name for the LIBNAME ACS read-only variable, can contain a 1 to 8 character tape library name. This field is optional.
&LLQ	The low-level (last) qualifier of the data set or collection name.
&MSPDEST	The destination, specified in data set name format, for a tape management system-driven tape allocation. This value is specified through the AMS pre-ACS installation exit. The data set name format

lets you specify a sequence of destinations to be identified, where each qualifier is a specific destination. For example, a data set vaulted first at location OUTD and then sent to OLTS can have an MSPDEST of 'OUTD.OLTS'. The actual values depend on the support provided by your tape management system.

&MSPARM	Additional information related to a tape management system-driven tape allocation. This is a variable length field that can be indexed. The value is specified through an external exit.
&MSPOLICY	The name of a management policy associated with tape data for a tape management system-driven allocation. You can use the DFSMSrmm EDGUX100 installation exit to set MSPOLICY to a VRS management value name. You can also set the value of this variable using the SMS pre-ACS installation exit, or allow your tape management system to set it using the pre-ACS installation exit.
&MSPOOL	A tape pool name associated with the data set being allocated. In a system-managed tape environment with scratch pool support, you can use this variable to specify a default storage group, where the tape storage group is equivalent to the tape pool specified in the variable. If you use the DFSMSrmm EDGUX100 installation exit, you can set this variable to the pool name or prefix determined by the DFSMSrmm scratch pool processing. This variable can also be set through the pre-ACS installation exit.
&NQUAL	The number of qualifiers in the data set or collection name.
&PGM	The name of the program the system is running.
&SYSNAME	Specifies the system name of the system on which the ACS routine is executing. This field is optional.
&SYSPLEX	Specifies the Parallel Sysplex name of the system on which the ACS routine is executing. This field is optional.
&UNIT	IBM-supplied or installation defined generic name for a device type (for example, 3380, SYSDA). A slash (/) preceding a four digit number represents a unit address (for example, /3380). When you allocate a tape data set with DISP=MOD, and no unit information is specified in the JCL, this variable is blank and SMS might attempt to manage the tape data set as a DASD-resident data set.
&USER	The user ID of the person allocating the data set. When DFSMSShsm invokes the ACS routines, &USER is either the requestor of the recall or recover, or the user ID of the DFSMSShsm address space. If the environment is recall or recover, the variable is set only if the requestor of the recall or recover is not a DFSMSShsm authorized user.
&XMODE	The execution mode in which the data set is being allocated. The following are valid values:
BATCH	Batch execution mode
TSO	TSO execution mode
TASK	A started address space

Set up your SMS management class routine to ask for the ACSENVIR RMMVRS environment and select a management class set up for tape data sets. Example 12-1 is a sample management class routine for managing non-system-managed volumes. In this sample all

data set names starting with RMM, MHLRES7 or SCHLUM will be assigned a management class of NSRMMMC. All other tape data sets will become the management class catalog, except if a special meaning expiration date is selected.

Example 12-1 Sample management class routine for managing non-system-managed volumes

```

PROC 1 MGMTCLAS
...
FILTLIST RMMDSN INCLUDE(RMM.**, MHLRES7.**, SCHLUM.** ) /* $02 */
...
IF &MGMTCLAS ^= ' ' THEN DO /* IF THE DATA SET ALREADY */
    EXIT /* HAS A MANAGEMENT CLASS */
        /* USE IT */
END /*
/*****
/* RMM POLICY FILTER LISTS */
/*****
IF &ACSENVIR = 'RMMPOOL' THEN EXIT /* $01 */
...
IF &ACSENVIR = 'RMMVRS' THEN DO /* $02 */
/*****
/* RMM VRS ASSIGNMENT DECISIONS */
/*****
    SELECT /* $02 */
        WHEN (&RETPD=1990000 OR /* $02 */
            &RETPD=1999000) /* $02 */
            SET &MGMTCLAS = 'MC99000' /* $02 */
        WHEN (&RETPD=1990001) /* $02 */
            SET &MGMTCLAS = 'MC90001' /* $02 */
        ...
        WHEN (&RETPD=1990365) /* $02 */
            SET &MGMTCLAS = 'MC90365' /* $02 */
        WHEN (&RETPD=1998001) /* $02 */
            SET &MGMTCLAS = 'MC98001' /* $02 */
        ...
        WHEN (&RETPD=1998365) /* $02 */
            SET &MGMTCLAS = 'MC98365' /* $02 */
        WHEN (&RETPD=1999001) /* $02 */
            SET &MGMTCLAS = 'MC99001' /* $02 */
        ...
        WHEN (&RETPD=1999364) /* $02 */
            SET &MGMTCLAS = 'MC98364' /* $02 */
        WHEN (&HLQ=&RMMDSN) /* $03 */
            SET &MGMTCLAS = 'NSRMMMC' /* $03 */
        OTHERWISE SET &MGMTCLAS = 'CATALOG' /* $02 */
    END /* $02 */
    EXIT /* $02 */
END /* $02 */
...
END

```

12.3 RETENTIONMETHOD(EXPDT) enhancements

With DFSMSrmm V2.1 there are two new attributes added to the RETENTIONMETHOD expiration data setting. They are listed here.

LASTREF	Specifies the default for the data set record LASTREF attribute. The LASTREF attribute specifies the number of days that the data set will be retained after the data set was last referenced by a read or write operation. LASTREF applies only to data sets on volumes managed by the EXPDT retention method.
----------------	---

RETAINBY	Specifies how DFSMSrmm is to retain volumes or multivolume sets that are managed by the EXPDT retention method.
-----------------	---

These new options apply only to the EXPDT retention method, not to the VRSEL expiration method. The processing of volume sets managed by the VRSEL retention method is unchanged. The use of the new RETENTIONMETHOD settings are shown in Figure 12-4.

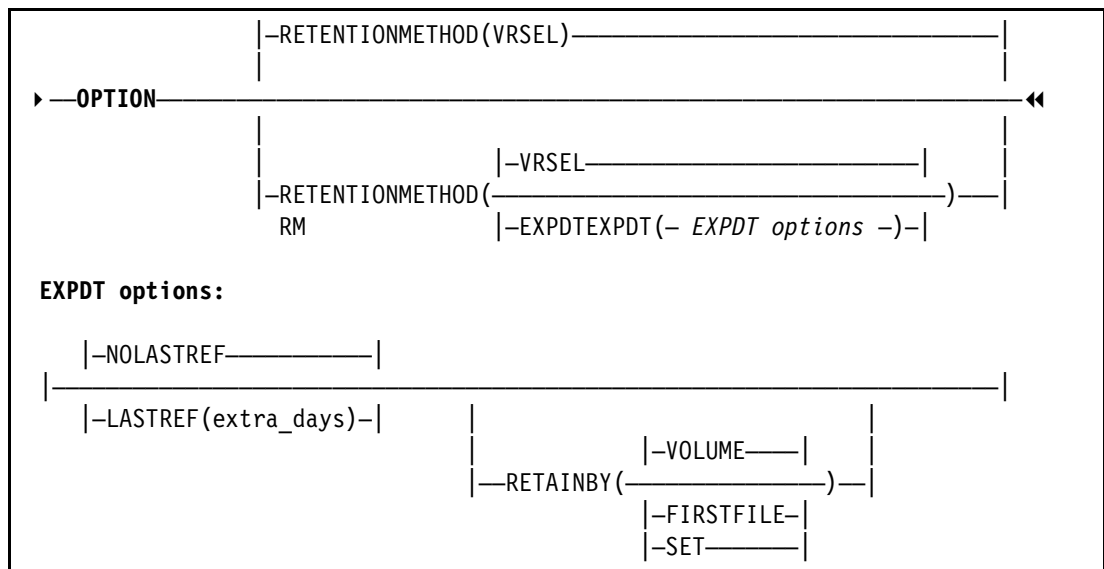


Figure 12-4 EDGRMMnn RETENTIONMETHOD operand

12.3.1 Using LASTREF option

LASTREF extra days is added to the data set record for data sets on volumes managed by the EXPDT retention method. Then, the installation can manage its data based on the number of days since the data set was last read or written on volumes managed by the EXPDT retention method.

The LASTREF extra days apply to data sets on RM(EXPDT) managed volumes. LASTREF extra days cannot be specified for data sets on VRSEL-managed volumes. Extra days means the number of days that the data set will be retained after the data set was last referenced.

DFSMSrmm uses the LASTREF extra days to evaluate the data set expiration date. The extra days are added to the date of last reference. The data set expiration date is set to the maximum of the date calculated using LASTREF extra days, and the date from applying the EXPDT RETPD or default RETPD. Any reference to the data set by a write or read operation will re-determine the expiration date.

The LASTREF extra days option can be set only for data sets on volumes managed by the EXPDT retention method.

You can set the LASTREF extra days retention option in several ways:

- ▶ By the management class, when the use of the management class attributes is enabled by the EDGRMMxx member option MCATTR and there is a value in Expiry after days non-usage in the management class. The advantage is that you do not need to define the function to each data set explicitly through the JCL, data class, TSO DFSMSrmm command, or using the EDGRMMxx member. With management class assignment, we implement automation in the process. Refer to “Update your tape related SMS management classes” on page 217 to see how to implement LASTREF extra days retention option in an SMS management class.
- ▶ By DFSMSrmm ADDDATASET subcommand, when the data set record is created.
- ▶ By DFSMSrmm CHANGEDATASET subcommand any time after the data set record has been created.

If the volume set is retained by SET or VOLUME the LastReferenceDays data set attribute will be kept equal for all files of a multi-volume data set. The latest LastReferenceDays update to a single file in a multivolume data set is propagated to all files that belong to the same multi-volume data set. However, for volume set retained by FIRSTFILE the LastReferenceDays data set attribute will be set but not equalized across the multi-volume data set because the expiration date depends only of the first file of the first volume, and its LastReferenceDays.

Additional information on the EXPDT options LASTREF and NOLASTREF can be found in “Using SMS management class in a system-managed tape environment” on page 213.

12.3.2 Using RETAINBY option

EXPDT retention method now allows retention to be based on a single volume or volume set, or on a controlling first file. You now have the option of retaining volumes with the EXPDT retention method based on a single volume or volume set, or on a controlling first file. Use the RETAINBY keyword of the RETENTIONMETHOD option to specify one of the following retention methods:

VOLUME	DFSMSrmm expires volumes in a multi volume set at the volume level. Each volume in the set has its own expiration date. Each file on a volume can increment the volume expiration date. RETAINBY(VOLUME) is the default value.
SET	DFSMSrmm expires volumes in a multi volume set at the volume set level. All volumes in the set have the same expiration date, which is the maximum expiration date of all volumes in the set (except if changed by the CV command). Any file on any volume of the set can increment the volume expiration date.
FIRSTFILE	DFSMSrmm expires volumes in a multi volume set at the volume set level. All volumes in the set have the same expiration date, which is the expiration date of the first file in the volume set (here a single volume is treated as a volume set with only one volume in it). The volumes will be released to scratch in the same run of DFSMSrmm inventory management.

12.3.3 Using EDGUX100 exit to specifying the retention method to be used

You can use the EDGUX100 installation exit to set the retention method to be used for new tape data sets. When you create a new tape volume set, or rewrite an existing set from the first file you can override the system default retention method. Example 12-2 is a sample EDGUX100 exit.

Example 12-2 Sample EDGUX100 retention method selection table

```
.....
*****
* Table to match job/data set/program name combinations for
* setting the Retention Method (RM).
* Entries are 8-byte jobname, 8-byte program name, and 44-byte
* dataset name. If an entry is matched, the retention method is set.
* Partial generic entries are supported - '%' may be used to
* represent a single non-blank character, and '*' is supported
* at the end of an entry only, to indicate a generic prefix.
* Some sample commented out entries are included. If you decide to
* change the format of this table, remember to update the RMENT
* DSECT accordingly.
*****

      SPACE 1
EDGUX100 CSECT
RMTAB   DS      OF          START OF RM TABLE
      SPACE 1
      DC      CL8'*'          JOBNAME
      DC      CL44'*'          DATA SET NAME
      DC      CL8'*'          PROGRAM NAME
      DC      AL1(PL100_RM_VRSEL) RETENTION METHOD VRSEL
      DC      XL3'00'          RESERVED
      SPACE 1
      DC      CL8'*'          JOBNAME
      DC      CL44'MHLRES7.*'    DATA SET NAME
      DC      CL8'*'          PROGRAM NAME
      DC      AL1(PL100_RM_EXPDT) RETENTION METHOD VRSEL
      DC      XL3'00'          RESERVED
      SPACE 1
      DC      CL8'*'          JOBNAME
      DC      CL44'RMM.*'        DATA SET NAME
      DC      CL8'*'          PROGRAM NAME
      DC      AL1(PL100_RM_EXPDT) RETENTION METHOD EXPDT
      DC      XL3'00'          RESERVED
      SPACE 1
      DC      CL8'RM END'        END OF RM TABLE MARKER
*****

* Table to match job/data set/program name combinations for
* setting the VRSELEXCLUDE attribute (VX).
* Entries are 8-byte jobname, 8-byte program name, and 44-byte
* dataset name. If an entry is matched, the VRSELEXCLUDE
* attribute is set.
* Partial generic entries are supported - '%' may be used to
* represent a single non-blank character, and '*' is supported
* at the end of an entry only, to indicate a generic prefix.
* Some sample commented out entries are included. If you decide to
* change the format of this table, remember to update the RMENT
```

```

* DSECT accordingly.
*****
VXTAB    DS      OF                      START OF VX TABLE          §NSC
        SPACE 1                          §NSC
        DC      CL8'*'                   JOBNAME                    §NSC
        DC      CL44'HSM.*'              DATA SET NAME            §NSC
        DC      CL8'*'                   PROGRAM NAME                §NSC
        SPACE 1                          §NSC
        DC      CL8'*'                   JOBNAME                    §NSC
        DC      CL44'OAM.*'              DATA SET NAME            §NSC
        DC      CL8'*'                   PROGRAM NAME                §NSC
        SPACE 1                          §NSC
        DC      CL8'VX END'              END OF VX TABLE MARKER    §NSC

*-----
* DSECT to map RMTAB table entries
*-----
RMENT    DSECT ,                        map of a RMTAB table entry    §NSC
RMJOB    DS      CL8                    jobname                      §NSC
RMDSN    DS      CL44                   data set name                §NSC
RMNAME    DS      CL8                    program name                §NSC
RMMETHOD DS      FL1                   retention method              §NSC
          DS      XL3                   reserved                      §NSC
RMENTL    EQU    *-RMENT                length of an entry           §NSC
          SPACE 1                      §NSC

*-----
* DSECT to map VXTAB table entries
*-----
VXENT    DSECT ,                        map of a VXTAB table entry    §NSC
VXJOB    DS      CL8                    jobname                      §NSC
VXDSN    DS      CL44                   data set name                §NSC
VXNAME    DS      CL8                    program name                §NSC
VXENTL    EQU    *-VXENT                length of an entry           §NSC
          EJECT ,                      §NSC

...

```

The following values apply to Example 12-2 on page 225:

RMTAB	All data set names ‘*’ will be retained by the VRSEL retention except the data set names starting with ‘MHLRES7.*’ and ‘RMM.*’
VXTAB	All data set names starting with ‘HSM.*’ and ‘OAM.*’ are excluded from VRSEL processing as they are created or rewritten.

Important: The order in which the table entries are listed is important because the exit scans the table until it finds the first entry where the job name, data set name and program name masks match the current request. You can change the priority of matching by changing the order of the table entries.

12.3.4 Show new retention method settings

The DFSMSRmm Volumes panel displays the values for RetentionMethod (RM), RetainBy and Set Retained in the Ret column as shown in Figure 12-5.

DFSMSrmm Volumes (Page 1 of 2)									
Command ==>									
Row 1 to 13 of 18									
Scroll ==> CSR									
Enter HELP or PF1 for the list of available line commands									
Use the RIGHT command to view other data columns									
	Volume		Assigned	Expir./			Dest-	T Data	
S	serial	Owner	date	Retn. date	Ret	Status	Location	ination	R sets
---	-----	-----	-----	-----	---	-----	-----	-----	-----
	THS000	MHLRES7	2013/273	2014/008	V	VRS	LIB2		N 1
	THS002	MHLRES7	2013/273	2014/008	V	VRS	LIB2		N 1
	THS001	MHLRES7	2013/273	2013/363	EV	MASTER	LIB2		N 1
	THS005	MHLRES7	2013/273	2013/363	ES	MASTER	LIB2		N 1
	THS010	MHLRES7	2013/273	2013/363	ES	MASTER	LIB2		N 1
	THS012	MHLRES7	2013/273	2013/273	EV	MASTER	LIB2		N 1
	THS017	MHLRES7	2013/274	2013/294	EV	MASTER	LIB2		N 1
	THS021	MHLRES7	2013/273	2014/008	ES	MASTER	LIB2		N 1
	THS022	MHLRES7	2013/270	2013/293	ES	MASTER	LIB2		N 1
	THS024	MHLRES7	2013/270	2013/363	EV	MASTER	LIB2		N 1
	THS025	MHLRES7	2013/270	2013/363	EV	MASTER	LIB2		N 1
	THS027	MHLRES7	2013/270	2013/363	EV	MASTER	LIB2		N 1

Figure 12-5 DFSMSrmm Volumes panel example

Each subcolumn is one character wide. The retention method is displayed in the first subcolumn, RetainBy in the second, Set Retained in the third. RetainBy is only provided for RM(EXPDT) volumes, Set Retained is only provided for RM(VRSEL) volumes. The possible values are listed here.

- Retention Method
 - E indicates Retention Method EXPDT
 - V indicates Retention Method VRSEL
- RetainBy
 - V indicates RetainBy value VOLUME
 - S indicates RetainBy value SET
 - F indicates RetainBy value FIRSTFILE
- Set Retained
 - Y indicates Volume is Set Retained
 - blank indicates Volume is not Set Retained

12.3.5 ADDDATASET and CHANGEDATASET subcommand enhancements

Figure 12-6. shows the new operands of the RMM ADDDATASET and CHANGEDATASET subcommands.

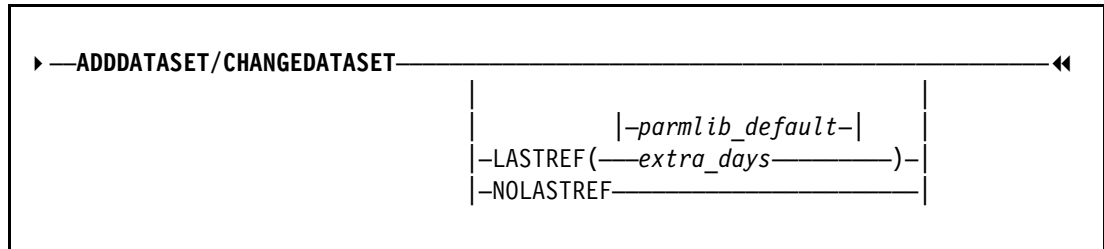


Figure 12-6 ADDDATASET and CHANGEDATASET subcommands

The following values apply to Figure 12-6:

LASTREF

Specifies the number of days that the data set will be retained after the data set was last referenced. LASTREF(extra_days) applies only to data sets on EXPDT-managed volumes. LASTREF cannot be specified for data sets on VRSEL-managed volumes.

extra_days is a decimal number between 0 and 93000. The value must not exceed the maximum retention period (MAXRETPD) specified in the DFSMSrmm EDGRMMxx parmlib member. An extra_days value of 0 has the same effect as using NOLASTREF. DFSMSrmm uses the LASTREF extra days to evaluate the data set expiration date. The extra days are added to the date of last reference. The data set expiration date is set to the maximum of the date calculated using LASTREF extra days, and the date resulting from applying the EXPDT, RETPD, or default RETPD. Any reference to the data set by a read or write operation will redetermine the expiration date.

If neither LASTREF nor NOLASTREF is specified for a new data set, DFSMSrmm uses the LASTREF default value specified in OPTION RETENTIONMETHOD(EXPDT) command in the parmlib member EDGRMMxx.

NOLASTREF

Specifies that DFSMSrmm does not consider the data set last reference date when evaluating the data set expiration date.

NOLASTREF applies only to data sets on volumes managed by the EXPDT retention method. NOLASTREF cannot be specified for data sets on volumes managed by the VRSEL retention method.

When a file is added to a multivolume data set, the LASTREF or NOLASTREF attribute is copied from the preceding file. For a volume set retained by VOLUME or SET DFSMSrmm ensures that the LASTREF(extra_days) or NOLASTREF data set attribute is the same for all files of a multivolume data set. For a volume set retained by FIRSTFILE, no additional processing is performed to keep the LASTREF extra days attribute consistent across the multivolume data set, because the expiration date depends only on the LASTREF extra days attribute of the first file of the first volume.

If neither NOLASTREF nor LASTREF is specified for a new data set, DFSMSrmm uses the LASTREF default value specified by the OPTION RETENTIONMETHOD(EXPDT) command in the parmlib member EDGRMMxx.

12.3.6 ADDVOLUME and CHANGEVOLUME subcommand enhancements

Figure 12-7 shows the new operands of the RMM ADDDATASET and CHANGEDATASET subcommands.

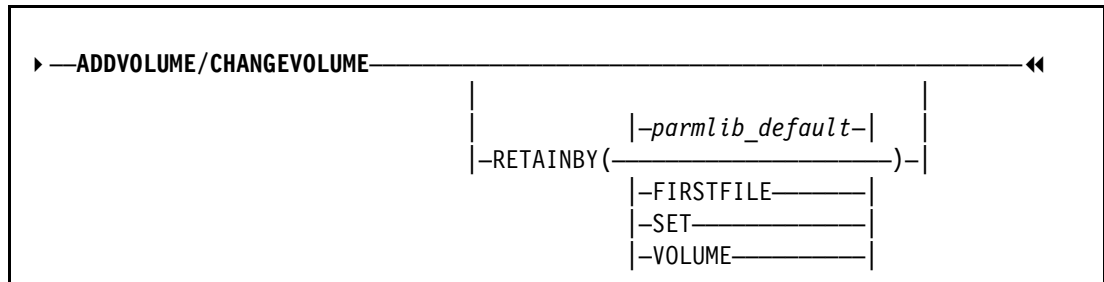


Figure 12-7 *ADDDATASET* and *CHANGEDATASET* subcommands

The following values apply to Figure 12-7:

RETAINBY	Specifies how DFSMSrmm is to retain an EXPDT-retained volume or multi-volume set:
FIRSTFILE	The expiration date of the first file is used to determine the expiration date of a single volume or a multi-volume set. All volumes in a multi-volume set will have exactly the same expiration date and will be released to scratch in the same run of DFSMSrmm inventory management. Additional data sets added later to a volume or to a multi-volume set can have different expiration dates that are independent of the volume expiration date.
SET	DFSMSrmm uses the highest expiration date of all volumes in the set. All volumes in the set will have exactly the same expiration date and will be released to scratch on the same run of DFSMSrmm inventory management.
VOLUME	The expiration date of the volume is considered for each volume separately and each file on a volume can increment the volume expiration date.

The RETAINBY operand cannot be specified for a volume managed by the VRSEL retention method. Use the RETAINBY operand only for volume sets that use the EXPDT retention method.

When a RETAINBY value is defined for a non-scratch volume, it is not overridden to the default during OPEN output processing, but can be changed using CHANGEVOLUME subcommand.

If RETAINBY is omitted, the EXPDT(RETAINBY) value specified in parmlib is used as the default value.

12.3.7 New REXX exec variables

Table 12-2 lists the new variables in RMM subcommand order that you can use in your REXX execs.

Table 12-2 New REXX exec variables

Variable Name	Abbreviation of Subcommands	Contents	Format
EDG@EXRB	LC OPT, LV VOL	RETENTIONMETHOD EXPDT RETAINBY	Nine characters
EDG@LRED	LC OPT, LD	Last reference extra days	Numeric: 0 - 93000
EDG@MCAT	LC OPT	SMS management class attributes	ALL, NONE VRSELXDI
EDG@RMSB	LV VOL	Retention method set by	Up to 10 characters. One of: CMD CMD_DEF CNVT EXPORT_DEF INERS_DEF LASTREF LCS_DEF OCE_DEF OCE_EXIT UNDEFINED
EDG@XDSB	LV VOL, LD	Expiration date set by	10 characters: blank (not set) CMD CMD_DEF CMD_VOLCAT CNVT EXPORT LASTREF LCS LCS_DEF OCE_DEF OCE_EXIT OCE_JFCB OCE_MAX OCE_MC OCE_VOLCAT TVEXTPURGE

12.3.8 Using RMM TSO subcommands with system REXX

DFSMSrmm displays the ID of the user that caused the most recent change to a record. If the most recent change was made by DFSMSrmm processing the ID starts with an asterisk (*). Internal IDs include these values:

*CAT	Updates due to catalog status changes
*HKP	Updates due to inventory management
*MIM	Updates due to SARS MIM message interception
*OAM	Updates due to system-managed tape support
*OCE	Updates due to Open/Close/EOV support
*UT	Updates due to execution of EDGUTIL
*WTO	Updates due to message interception

12.4 Testing MCATTR and RETENTIONMETHOD settings

In this section we use actual jobs to provide an overview of how the different MCATTR and RETENTIONMETHOD settings work. We used the same JCL to create four data sets in both an SMS managed environment and a non-SMS managed environment. As you can see in the JCL in Figure 12-8, we created data sets both with and without a retention date set in the JCL.

```
//STEP01 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=RMM.ADDONS.CNTL(IEBGENER),DISP=SHR
//SYSUT2 DD DSN=RMM.TEST.RM.VRSEL.MCATTR.SETTINGS,DISP=(,KEEP),
// UNIT=ATL3,LABEL=(1,SL),MGMTCLAS=NSRMMMC,RETPD=90
//SYSIN DD DUMMY
//STEP01A EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=RMM.ADDONS.CNTL(IEBGENER),DISP=SHR
//SYSUT2 DD DSN=RMM.TEST.RM.VRSEL.MCATTR.SETTINGS,DISP=(,KEEP),
// UNIT=ATL3,LABEL=(1,SL),MGMTCLAS=NSRMMMC RETPD=90
//SYSIN DD DUMMY
//STEP02 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=RMM.ADDONS.CNTL(IEBGENER),DISP=SHR
//SYSUT2 DD DSN=RMM.RM.VRSEL.MCATTR.ALL,DISP=(,KEEP),
// UNIT=VT3590,LABEL=(1,SL),MGMTCLAS=NSRMMMC,RETPD=90
//SYSIN DD DUMMY
//STEP02A EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=RMM.ADDONS.CNTL(IEBGENER),DISP=SHR
//SYSUT2 DD DSN=RMM.RM.VRSEL.MCATTR.ALL,DISP=(,KEEP),
// UNIT=VT3590,LABEL=(1,SL),MGMTCLAS=NSRMMMC
//SYSIN DD DUMMY
```

Figure 12-8 Sample JCL used to test the different settings

To test the vital record processing as well, we have specified a DSNAME VRS matching the data sets as shown in Figure 12-9 on page 232.

```

                                DFSMSrmm Display Data Set VRS
Command ==>

Data set mask . : 'RMM.**'
Job name mask . :

Count . . . . : 100
Delay . . . . : 0    Days
Location . . . . : CURRENT
Number in location : 100
Priority . . . . : 0

Next VRS in chain . :
Chain using . . . :

Owner . . . . . : MHLRES7
Description . . :
Last reference  : 2013/275    18:10:43 ( YYYY/DDD HH:MM:SS )

                                More:      +
                                GDG . . : NO

Retention type . . . . . : DAYS
While cataloged . . . . . : NO
Until expired . . . . . : NO

Release options:
Expiry date ignore . . . . : NO
Scratch immediate . . . . : NO

```

Figure 12-9 Sample DSNAME VRS

The following values apply to Figure 12-9 on page 232:

Count	Specifies how many days or how many cycles of a data set should be retained in the location specified in the Location field. How the cycles or days are counted is determined by the Retention type field. For example, if the retention type is BYDAYC the count specifies the number of cycles retained and each cycle is all data sets created on the same day.
Retention type	Specifies how data sets are retained as vital records. The retention type specifies how each instance of a data set is managed and moved while under the control of this VRS. Possible values for the Retention type field are listed here.
CYCLES	Each occurrence of a data set is managed as a separate cycle.
BYDAYC	All data set occurrences created on the same date are managed as a cycle.
DAYS	Retention is based on days since creation.
REFDAYS	Retention is based on days since last referenced.
XTRDAYS	Is only valid on NAME type VRSs. Retention is based on days since the NAME VRS gained control of the data set.

12.4.1 MCATTR(VRSELXDI) and RETENTIONMETHOD(VRSEL)

Table 12-3 shows you the result of the allocation of the data sets in Figure 12-8 on page 231 if you have specified MCATTR(VRSELXDI), SMSACS(NO) and RETENTIONMETHOD(VRSEL) in your EDGRMMnn parmlib member.

Table 12-3 Retention method VRSEL and MCATTR VRSELXDI

Data set allocated	1	2	3	4
Retention Method	VRSEL	VRSEL	VRSEL	VRSEL
MCATTR	VRSELXDI	VRSELXDI	VRSELXDI	VRSELXDI
SMSACS	NO	NO	NO	NO
JCL EXPDT/RETPD	90	default	90	default
SMS managed Volume	yes	yes	no	no
Volume information	THS004	THS021	VT0058	VT0059
Assign date	2013/273	2013/273	2013/273	2013/273
Expiration date Set by	2013/363 OCE_JFCB	2013/273 OCE_DEF	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
Retention Method Set by	VRSEL LCS_DEF	VRSEL LCS_DEF	VRSEL OCE_DEF	VRSEL OCE_DEF
EXPDT Retain by				
Retention date after VRS	2014/008	2014/008	2014/008	2014/008
Data set information				
Creation date	2013/273	2013/273	2013/273	2013/273
Expiration Date Set by	2013/363 OCE_JFCB	2013/273 OCE_DEF	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
LASTREF extra days	0	0	0	0
VRS selected	yes	yes	yes	yes

12.4.2 MCATTR(ALL) and RETENTIONMETHOD(VRSEL)

Table 12-3 shows you the result of the allocation of the data sets in Figure 12-8 on page 231 if you have specified MCATTR(ALL), SMSACS(NO) and RETENTIONMETHOD(VRSEL) in your EDGRMMnn parm lib member.

Table 12-4 Retention method VRSEL and MCATTR VRSELXDI

Data set allocation	1	2	3	4
Retention Method	VRSEL	VRSEL	VRSEL	VRSEL
MCATTR	ALL	ALL	ALL	ALL
SMSACS	NO	NO	NO	NO
JCL EXPDT/RETPD	90	default	90	default
SMS managed Volume	yes	yes	no	no
Volume information	THS011	THS000	VT0060	VT0061
Assign date	2013/273	2013/273	2013/273	2013/273

Data set allocation	1	2	3	4
Expiration date Set by	2013/363 OCE_JFCB	2013/293 OCE_MC	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
Retention Method Set by	VRSEL LCS_DEF	VRSEL LCS_DEF	VRSEL OCE_DEF	VRSEL OCE_DEF
EXPDT Retain by				
Retention date after VRS	2014/008	2014/008	2014/008	2014/008
Data set information				
Creation date	2013/273	2013/273	2013/273	2013/273
Expiration Date Set by	2013/363 OCE_JFCB	2013/293 OCE_MC	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
LASTREF extra days	0	0	0	0
VRS selected	yes	yes	yes	yes

12.4.3 MCATTR(NONE) and RETENTIONMETHOD(VRSEL)

Table 12-3 shows you the result of the allocation of the data sets in Figure 12-8 on page 231 if you have specified MCATTR(NONE), SMSACS(NO) and RETENTIONMETHOD(VRSEL) in your EDGRMMnn parmlib member.

Table 12-5 Retention method VRSEL and MCATTR VRSELXDI

Data set allocation	1	2	3	4
Retention Method	VRSEL	VRSEL	VRSEL	VRSEL
MCATTR	NONE	NONE	NONE	NONE
SMSACS	NO	NO	NO	NO
JCL EXPDT/RETPD	90	default	90	default
SMS managed Volume	yes	yes	no	no
Volume information	THS002	THS016	VT0062	VT0063
Assign date	2013/273	2013/273	2013/273	2013/273
Expiration date Set by	2013/363 OCE_JFCB	2013/293 OCE_MC	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
Retention Method Set by	VRSEL LCS_DEF	VRSEL LCS_DEF	VRSEL OCE_DEF	VRSEL OCE_DEF
EXPDT Retain by				
Retention date after VRS	2014/008	2014/008	2014/008	2014/008
Data set information				

Data set allocation	1	2	3	4
Creation date	2013/273	2013/273	2013/273	2013/273
Expiration Date Set by	2013/363 OCE_JFCB	2013/293 OCE_DEF	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
LASTREF extra days	0	0	0	0
VRS selected	yes	yes	yes	yes

12.4.4 MCATTR(VRSELXDI) and RETENTIONMETHOD(EXPDT)

Table 12-3 shows you the result of the allocation of the data sets in Figure 12-8 on page 231 if you have specified MCATTR(ALL), SMSACS(NO) and RETENTIONMETHOD(VRSEL) in your EDGRMMnn parmlib member.

Table 12-6 Retention method VRSEL and MCATTR VRSELXDI

Data set allocation	1	2	3	4
Retention Method	EXPDT	EXPDT	EXPDT	EXPDT
MCATTR	VRSELXDI	VRSELXDI	VRSELXDI	VRSELXDI
SMSACS	NO	NO	NO	NO
JCL EXPDT/RETPD	90	default	90	default
SMS managed Volume	yes	yes	no	no
Volume information	THS025	THS027	VT0066	VT0067
Assign date	2013/273	2013/273	2013/273	2013/273
Expiration date Set by	2013/363 OCE_JFCB	2013/293 OCE_MC	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
Retention Method Set by	EXPDT LCS_DEF	EXPDT LCS_DEF	EXPDT OCE_DEF	EXPDT OCE_DEF
EXPDT Retain by	VOLUME	VOLUME	VOLUME	VOLUME
Retention date after VRS				
Data set information				
Creation date	2013/273	2013/273	2013/273	2013/273
Expiration Date Set by	2013/363 OCE_JFCB	2013/293 OCE_MC	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
LASTREF extra days	10	10	0	0
VRS selected	no	no	no	no

12.4.5 MCATTR(ALL) and RETENTIONMETHOD(EXPDT)

Table 12-3 shows you the result of the allocation of the data sets in Figure 12-8 on page 231 if you have specified MCATTR(ALL), SMSACS(NO) and RETENTIONMETHOD(VRSEL) in your EDGRMMnn parmlib member.

Table 12-7 Retention method VRSEL and MCATTR VRSELXDI

Data set allocation	1	2	3	4
Retention Method	EXPDT	EXPDT	EXPDT	EXPDT
MCATTR	ALL	ALL	ALL	ALL
SMSACS	NO	NO	NO	NO
JCL EXPDT/RETPD	90	default	90	default
SMS managed Volume	yes	yes	no	no
Volume information	THS025	THS027	VT0066	VT0067
Assign date	2013/273	2013/273	2013/273	2013/273
Expiration date Set by	2013/363 OCE_JFCB	2013/293 OCE_MC	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
Retention Method Set by	EXPDT LCS_DEF	EXPDT LCS_DEF	VRSEL OCE_DEF	VRSEL OCE_DEF
EXPDT Retain by				
Retention date after VRS				
Data set information				
Creation date	2013/273	2013/273	2013/273	2013/273
Expiration Date Set by	2013/363 OCE_JFCB	2013/293 OCE_DEF	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
LASTREF extra days	10	10	0	0
VRS selected	no	no	no	no

12.4.6 MCATTR(NONE) and RETENTIONMETHOD(EXPDT)

Table 12-3 shows you the result of the allocation of the data sets in Figure 12-8 on page 231 if you have specified MCATTR(NON), SMSACS(NO) and RETENTIONMETHOD(VRSEL) in your EDGRMMnn parmlib member.

Table 12-8 Retention method VRSEL and MCATTR VRSELXDI

Data set allocation	1	2	3	4
Retention Method	EXPDT	EXPDT	EXPDT	EXPDT
MCATTR	ALL	ALL	ALL	ALL
SMSACS	NO	NO	NO	NO

Data set allocation	1	2	3	4
JCL EXPDT/RETPD	90	default	90	default
SMS managed Volume	yes	yes	no	no
Volume information	THS010	THS012	VT0068	VT0069
Assign date	2013/273	2013/273	2013/273	2013/273
Expiration date Set by	2013/363 OCE_JFCB	2013/273 OCE_DEF	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
Retention Method Set by	EXPDT LCS_DEF	EXPDT LCS_DEF	VRSEL OCE_DEF	VRSEL OCE_DEF
EXPDT Retain by	VOLUME	VOLUME		
Retention date after VRS				
Data set information				
Creation date	2013/273	2013/273	2013/273	2013/273
Expiration Date Set by	2013/363 OCE_JFCB	2013/273 OCE_DEF	2013/363 OCE_JFCB	2013/273 OCE_DEF
Original expiration date	2013/363		2013/363	
LASTREF extra days	0	0	0	0
VRS selected	no	no	no	no

12.4.7 MCATTR(ALL) with RETENTIONMETHOD(EXPDT) and SMSACS(YES)

Table 12-3 shows you the result of the allocation of the data sets in Figure 12-8 on page 231 if you have specified MCATTR(ALL), SMSACS(YES) and RETENTIONMETHOD(VRSEL) in your EDGRMMnn parmlib member. In this case the non-SMS managed tape volumes will get the SMS management class retention attributes as well.

Table 12-9 Retention method VRSEL and MCATTR VRSELXDI

Data set allocation	3	4
Retention Method	EXPDT	EXPDT
MCATTR	ALL	ALL
SMSACS	NO	NO
JCL EXPDT/RETPD	90	default
SMS managed volume	no	no
Volume information	VT0072	VT0073
Assign date	2013/274	2013/274
Expiration date Set by	2013/364 OCE_JFCB	2013/294 OCE_MC
Original expiration date	2013/364	

Data set allocation	3	4
Retention Method Set by	EXPDT OCE_DEF	EXPDT OCE_DEF
EXPDT Retain by	VOLUME	VOLUME
Retention date after VRS		
Data set information		
Creation date	2013/274	2013/274
Expiration Date Set by	2013/364 OCE_JFCB	2013/364 OCE_MC
Original expiration date	2013/363	
LASTREF extra days	10	10
VRS selected	no	no

12.5 DFSMSrmm conversion support

Historically, DFSMSrmm provides conversion programs to simplify conversion from ISV tape management programs to DFSMSrmm. Beginning with z/OS V2.1, the DFSMSrmm conversion samples and documentation have been moved from SYS1.SAMPLIB to the “as-is z/OS downloads” web page as a zipped sequential XMIT file of a PDS. It can be found at:

<http://www-03.ibm.com/systems/z/os/zos/tools/downloads/index.html#asis>

Figure 12-10 shows you the z/OS DFSMSrmm conversion part of the “as-is z/OS downloads”.

z/OS DFSMSrmm Conversion Support

Beginning with z/OS release V2R1, some DFSMSrmm samples and modules have been removed from SYS1.SAMPLIB and SYS1.LINKLIB, including the documentation in EDGCMM01

- The documentation can now be found here in PDF format: [Data Extraction Programs Users Guide for Conversion to DFSMSrmm.pdf](#)
- To generate the removed modules, from old V1R13 sources, the JCL in this JCL sample [can be used](#)

z/OS DFSMSrmm Customization Wizard

This wizard is designed to help you set up DFSMSrmm by selecting a standard setup that uses DFSMSrmm default values or a customized setup that you tailor for your installation. It builds the EDGRMMxx parmlib member, provides JCL jobs to setup RACF for DFSMSrmm, and jobs to allocate and initialize the datasets needed by DFSMSrmm. Editions of this wizard are available for:

- [z/OS V1R12](#)
- [z/OS V1R11](#)

Figure 12-10 DFSMSrmm conversion samples

12.5.1 Conversion samples provided

In this section we list the z/OS DFSMSrmm conversion samples that are still shipped with the product. At this time the provided samples support conversions to DFSMSrmm from CA-1, TLMS, Control-M/Tape up to release 7.0, ZARA/Automedia up to release 1.7, TAPE2000 and ICF user catalogs.

Load module samples

The load module samples shipped with the product are listed here.

EDGCDYNM	Source for TLMS extract program
EDGCSVDS	Source for CA-1 extract program
EDGC5BIN	Source for CA-1 5.0 and above extract program
EDGC5LDR	Source for CA-1 5.0 and above extract program
EDGCRFMT	Source for CA-1 5.0 and above extract program
EDGCSRDS	Source for CA-1 extract program
EDGCVRSX	Sample EDGUX100 exit
EDGCVRSG	Macro used for EDGCVRSX/F
EDGCVRSF	Part of sample EDGUX100 table
EDGCVRSE	Part of sample EDGUX100 table
EDGCVRSL	Sample to load EDGUX100 table
EDGC1PRM	Parameters extract program from TMS parmlib
EDGC5UJL	Convert data sets using unique Julian date
EDGCSPCC	CA-1 special character conversion
EDGC1PPL	Maps the TMS parameter area

REXX exec samples

The REXX exec samples shipped with the product and a brief description are listed here.

EDGRACS	For TMS to create MC ACS statements
EDGRCSR	For TMS 5.0 and above pool conversion
EDGCVTC0	To create LISTCAT commands by using the LISTCAT UCAT command
EDGCVTC4	To create ADDRACK, ADDVOLUME and CHANGEVOLUME commands to build a RMM CDS from the ICF-CATALOG input
EDGCVTC5	To create additional ADDVOLUME and CHANGEVOLUME commands
EDGCVTC6	To create CHANGEVOLUME commands to build the volume chains
EDGRT2EX	For TAPE2000 extract exec
EDGRCTT1	To create the extended Control-M/Tape record
EDGRCTT2	For Control-M/Tape media data base (MDB) extract
EDGRCTT3	For Control-M/Tape Rules extract
EDGRCZA1	To modify the unloaded ZARA/AutoMedia records for future processing
EDGRCZA2	To create L-, D-, K-, and O-Records for the EDGCNVT conversion program
EDGRCZA3	To create K-Records out of a ZARA/Automedia expiration report

EDGCLMS Sample conversion CLIST

JCL samples

The JCL samples shipped with the product are listed here.

EDGJYPRE	To compile and link EDGCDYNM
EDGJDYNM	To execute EDGCDYNM
EDGJ5PRE	To compile and link the provided load module samples
EDGJSRDS	To execute EDGCSRDS and EDGRACS
EDGJ5BIN	To execute EDGC5BIN
EDGJ5LDR	To execute EDGC5LDR and reformat the RDS and VPDD
EDGJCSCR	To execute exec EDGRCSCR
EDGJVTC1	To convert from manual management part 1
EDGJVTC2	To convert from manual management part 2
EDGJVTC3	To convert from manual management part 3
EDGJT2EX	To execute EDGRT2EX TAPE2000 conversion exec
EDGJCTT1	To execute EDGRCTT1 CONTROL-M/Tape conversion exec
EDGJCTT2	To execute EDGRCTT2 CONTROL-M/Tape conversion exec
EDGJCTT3	To execute EDGRCTT3 CONTROL-M/Tape conversion exec
EDGJCZA1	To execute EDGRCZA1 ASG-ZARA/AutoMedia conversion exec
EDGJCZA2	To execute EDGRCZA2 ASG-ZARA/AutoMedia conversion exec
EDGJCZA3	To execute EDGRCZA3 ASG-ZARA/AutoMedia conversion exec
EDGJSVDS	To execute EDGCSVDS and reformat the RDS and VPDD
EDGCVRSP	Procedure to run EDGCVRSL



APARs to be reviewed for DFSMS V2.1

The APARs listed in this appendix should be reviewed for impact to your installation prior to implementing z/OS V2.1 DFSMS.

Where APAR text is shown it is at time of writing of this IBM Redbooks publication. The current version should be checked for updates. When an APAR has had a PTF shipped, and a problem is then found, the PTF will be marked PE, and also the original APAR may be updated to show the fixing APAR numbers.

There two APAR lists in this IBM Redbooks publication. One for fixes that should be applied to DFSMS V2.1 before it is used, and one that should be applied to DFSMS V1.13 or DFSMS V1.12 before DFSMS V2.1 is used.

The lists that follow do not include every APAR that may be required, but is representative. If the PTFS for the listed APARS are applied then many other APARS will also be included.

This information in this Appendix is for DFSMS V2.1. The corresponding list for DFSMS V1.13 and V1.12 can be found at Appendix B, “APARs to be reviewed for DFSMS V1.13 or DFSMS V1.12” on page 309.

DFSMS V2.1 suggested APARS

The APARs in this appendix are divided into two sections. There are informational APARs and then APARs that correct errors.

Information APARS

There a number of Information APARS that provide additional information, or provide links to other Information APARS.

APAR II14670 LISTCAT LEVEL change in DFSMS V2.1

Example A-1 contains the cover letter for APAR II14670.

Example A-1 II14670

APAR Identifier	II14670	Last Changed	13/09/27
LISTCAT LEVEL CHANGE IN 2.1			

Symptom	IN INFO	Status	INTRAN
Severity	3	Date Closed	
Component	INFOV2LIB	Duplicate of	
Reported Release	001	Fixed Release	
Component Name V2 LIB INFO ITE		Special Notice	
Current Target Date ..		Flags	
SCP			
Platform			

Status Detail: Not Available

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

The LISTCAT LEVEL function has been corrected in release 2.1 to report on entries matching the level specified where in prior releases a LISTCAT LEVEL may return a non-zero return code if no entries matching the level with additonal qualifiers were found.

For example:

```
DEFINE GDG MY.GDG.BASE
(no GDSs defined)
```

@ 1.13

LISTCAT LVL(MY.GDG.BASE)

IDC3012I ENTRY MY.GDG.BASE NOT FOUND
IDC3007I ** VSAM CATALOG RETURN-CODE IS 8
IDC1566I ** MY.GDG.BASE NOT LISTED

IDC0001I FUNCTION COMPLETED, HIGHEST CONDITION CODE WAS 4

@ 2.1

LISTCAT LVL(MY.GDG.BASE)

GDG BASE ----- MY.GDG.BASE
IN-CAT --- MY.USER.CAT

LISTING FROM CATALOG --

THE NUMBER OF ENTRIES PROCESSED WAS:

AIX -----0
ALIAS -----0
CLUSTER -----0
DATA -----0
GDG -----1

IDC0001I FUNCTION COMPLETED, HIGHEST CONDITION CODE WAS 0

In 2.1, the behavior of LISTCAT LEVEL has been updated to mimic that expected of ISPF 3.4 which returns entries matching the key of the level specified as well as any keys with additional qualifiers.

II14250 explains past enhancements to the LISTCAT LEVEL command

CATKEYS: CAT2013 IDC2013 IDCLISTC IDCNEW IDCOUTPUT CATINFO
IDCINFO

LOCAL FIX:

Use the IDCNOGFL interface or change the LISTCAT LEVEL(A.B.C) to a LISTCAT ENT(A.B.C.*) for comparable results to 1.13 and below.

APAR II14708 PDSE maintenance for DFSMS V2.1

Example A-2 contains the cover letter for APAR II14708.

Example A-2 II14708

APAR Identifier II14708 Last Changed 13/05/30
ZOS DFSMS 2.1 HDZ2210 CURRENT PDSE MAINTENANCE

Symptom DD DOC	Status INTRAN
Severity 3	Date Closed
Component INFOV2LIB	Duplicate of

Reported Release	001	Fixed Release	
Component Name	V2 LIB INFO ITE	Special Notice	
Current Target Date	..	Flags	
SCP			
Platform			

Status Detail: Not Available

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

This informational APAR contains a current list of DFSMS HDZ2210 apars/ptfs for PDSE which includes both HIPER and non-HIPER maintenance.

Compid: 5695DF115 Fmid: HDZ2210

DFSMS HDZ2210 will be found on z/OS 2.1 systems.

We encourage you to install all current maintenance and refer frequently to PSP upgrade ZOSV2R1 subset DFSMS for HIPER maintenance.

.
 Reference II14255 for current PTFs for HDZ1180.
 Reference II14403 for current PTFs for HDZ1190.
 Reference II14459 for current PTFs for HDZ1A10.
 Reference II14519 for current PTFs for HDZ1B10.
 Reference II14580 for current PTFs for HDZ1C10.
 Reference II14632 for current PTFs for HDZ1D10

```
*****
* PTFs for DFSMS HDZ2210 ( R210 ) *
*****
```

LOCAL FIX:

none

Fix APARs

There are fixes required for pre-DFSMS V2.1 systems. Some of these are conditioning fixes that must be installed and in use on all systems in a SYSPLEX *before* any DFSMS V2.1 system is added to the SYSPLEX. See "APARs to be reviewed for DFSMS V1.13 or DFSMS V1.12" on page 309.

Note: In some cases there are no PTFs for the listed APARs, and some may have been superseded. The current listing from IBMLINK should be checked for current status and PTFs.

APAR OA42081 NFS applicable to DFSMS V1.13

Example A-3 contains the cover letter for APAR OA42081.

Example A-3 OA42081

APAR Identifier OA42081 Last Changed 13/07/02
REGR TESTS FOR #176 D1N IN SAF MODE WITH CHECKLIST' /HFS/
<NOSAF> SETTING FAILED:

Symptom IN INCORROUT Status CLOSED PER
Severity 3 Date Closed 13/06/07
Component 5695DF121 Duplicate of
Reported Release 21N Fixed Release 999
Component Name NETWORK FILE SY Special Notice
Current Target Date ..13/06/28 Flags
SCP
Platform

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release C1N : UA69454 available 13/06/26 (F306)
Release D1N : UA69455 available 13/06/26 (F306)
Release 21N : UA69456 available 13/06/26 (F306)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

Regr tests for #176 D1N in SAF mode with checklist' /hfs/
<nosaf> setting failed:

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: All NFSS users. *

* PROBLEM DESCRIPTION: The function v4_access(returns *
* invalid result in SAFEXP mode with the *
* export entry /hfs/ <nosaf>. *

* RECOMMENDATION: *

The local variable mode in h4_access(is not initialized if SAF mode and bypassaf flag in MB is set.

PROBLEM CONCLUSION:

Add MB->bypassaf check in h4_access(.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS:

GFS44UMC

SRLS:

NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42239 PDSE DFSMS V2.1 roll up

Example A-4 contains the cover letter for APAR OA42239.

Example A-4 OA42239

APAR Identifier	OA42239	Last Changed	13/07/02
SMS V2R1 ROLLUP APAR			
Symptom	IN INCORROUT	Status	CLOSED PER
Severity	4	Date Closed	13/05/24
Component	5695DF101	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	STORAGE MGMT SU	Special Notice	
Current Target Date ..		Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for

shipment.

PE PTF List:

PTF List:

Release 210 : UA69253 available 13/06/22 (F306)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

SMS V2R1 ROLLUP APAR FIXES

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: HDZ2210 USERS WHO UPGRADE THEIR SYSTEMS TO *
*                   DFSMS HDZ2210 DRIVER                      *
*****
* PROBLEM DESCRIPTION: THIS IS THE SMS V2R1 ROLLUP APAR. THIS *
*                   APAR CONTAINS THE z/OS V2R1 FIXES FOR:    *
*                   ----- *
*                   ENHANCEMENT: ADD A NEW MESSAGE VARIANT   *
*                   FOR IGD068I TO BE ISSUED WHEN            *
*                   AN OPERATOR ATTEMPTS TO                   *
*                   ACTIVATE A COMMUNICATION DATA            *
*                   SET OR AN ACTIVE CONFIGURATION            *
*                   DATA SET BY SPECIFYING IT IN              *
*                   THE IGDSMSxx PARMLIB MEMBER TO            *
*                   ENHANCE USABILITY AND                     *
*                   SERVICEABILITY.                            *
*                   ENHANCEMENT: ADD CODES TO DISREGARD THE   *
*                   NEW IGDSMSxx PARMLIB PARAMETER            *
*                   HONOR_DSNTYPE_PDSE WHEN THERE             *
*                   IS NO DSNTYPE SPECIFIED ON                 *
*                   EITHER JCL OR DATA CLASS.                *
*                   OA42065: SMS ALLOWS COPY POOL BACKUP       *
*                   STORAGE GROUP NAME TO START               *
*                   WITH A NON-ALPHABETIC                     *
*                   CHARACTER.                                  *
*                   OA42227: ABEND0B0 RC08 ATTEMPTING TO       *
*                   EXTEND A DATA SET TO A NEW               *
*                   VOLUME. JFCBX TOKEN IS ZEROES.           *
*                   OA41461: OVERALLOCATION OF PS EXTENDED     *
*                   FORMAT DATASET WHEN ALLOCATING            *
*                   IN BLOCKS AND USING DSS AND/OR            *
*                   HSM RESTORE/RECALL.                        *
*****
* RECOMMENDATION: *
*****
```

SUMMARY: THIS IS A SMS V2R1 ROLLUP APAR FOR OA42065, OA42227, OA41461. PLEASE SEE THE SUMMARIES FOR THOSE APARS.

PROBLEM CONCLUSION:

THIS IS THE SMS V2R1 ROLLUP APAR FOR OA42065, OA42227, OA41461. PLEASE SEE PROBLEM CONCLUSIONS FOR THOSE APARS.

THE NEW MESSAGE VARIANT FOR IGD068I SHOULD BE AS FOLLOWING:

IGD068I cds_type dsname SPECIFIED IS THE SAME AS THE ACTIVE
cds_type - ACTIVE cds_type IS STILL IN EFFECT

Explanation: An operator attempts to activate the active control data set or a communication data set by specifying it in the IGDSMSxx PARMLIB member, but that is the same as the currently active one.

In the message text:

cds_type

The configuration data set types:

ACDS - Active Configuration Data Set

COMMDS - Communication Data Set

dsname

The data set name

System action: Validation processing rejected the SMS request. The currently active configuration data sets are still in effect . The other PARMLIB parameters specified are still processed.

Operator response: Contact the system programmer.

System programmer response: No further action if there is no need to change the system configuration. Otherwise, specify a different configuration data set.

Source: Storage Management Subsystem (SMS)

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IGDCSDTV IGDICMGC IGDMCSMT IGDOPCDM IGDVTSCM
IGDVTSCR IGDVTSC1 IGDVTSDP

SRLS: SA38067500

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42283 PDSE

Example A-5 contains the cover letter for APAR OA42283.

Example A-5 OA42283

APAR Identifier OA42675 Last Changed 13/09/26
ABENDOC4 PIC38 IGWIRADB+OCB0 @ HDZ2210

Symptom AB ABENDOC4	Status CLOSED PER
Severity 3	Date Closed 13/07/03
Component 5695DF115	Duplicate of
Reported Release 210	Fixed Release 999
Component Name EXTENDED DATA S	Special Notice HIPER
Current Target Date ..13/08/15	Flags
SCP	
Platform	DATALOSS

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 210 : UA69767 available 13/07/11 (F307)

Parent APAR: OA42051

Child APAR list:

ERROR DESCRIPTION:

ABENDOF4 PIC39 CSECT=IGWIRADB+OCB0 @ HDZ2210 base
ABENDOC1 IGWIRPAD+0B4E @ HDZ2210 base
ABENDOC1 IGWIRPAD+17D4 @ HDZ2210 base
ABENDOC1 IGWIRPND+1848 @ HDZ2210 base
ABENDOF4 IGWIRPAD RSN1419A084 1419A084

LOCAL FIX:

None

PROBLEM SUMMARY:

* USERS AFFECTED: PDSE which has been updated extensively on *

```

*               a release lower than 2.1 may               *
*               have a sparse index.  When 2.1 updates the  *
*               the index may be broken,                    *
*****
* PROBLEM DESCRIPTION: See users affected.                  *
*****
* RECOMMENDATION:                                           *
*****
Modify PDSE processing to position to the correct place in the
index before attempting to remove empty pages,

```

PROBLEM CONCLUSION:
See Problem Summary

TEMPORARY FIX:
* HIPER *

COMMENTS:

MODULES/MACROS: IGWIRCMT IGWIRDIS IGWIRICR IGWIRIEM IGWISTAM

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42300 PDSE

Example A-6 contains the cover letter for APAR OA4230.

Example A-6 OA42300

APAR Identifier	OA42300	Last Changed	13/07/02
PERFORMANCE DEGRADATION IN RLS			
Symptom	PF PERFM	Status	CLOSED PER
Severity	3	Date Closed	13/05/30
Component	5695DF122	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name VSAM REC LEV SH		Special Notice	HIPER

Current Target Date ..13/08/01 Flags
SCP PERFORMANCE
Platform

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:
Release 210 : UA69307 available 13/06/04 (F306)

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:
A degradation of RLS performance is detected when processing clusters opened in rls, record-level sharing.

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: ALL VSAM RECORD-LEVEL SHARING, RLS USERS *
* WITH RELEASE HDZ2210. *

* PROBLEM DESCRIPTION: PERFORMANCE DEGRADATION WHEN PROCESSING *
* RLS REQUESTS INCLUDING GET, PUT, ERASE. *

* RECOMMENDATION: *

PERFORMANCE DEGRADATION DURING RLS REQUEST PROCESSING.

PROBLEM CONCLUSION:
RLS REQUEST PERFORMANCE IS CORRECTED.
KEYWORDS: RLSPSP/K

TEMPORARY FIX:

* HIPER *

COMMENTS:

MODULES/MACROS:
IDAVRR10 IDAV193S IGWLJ020 IGWMRT3 IGWSDRDM IGWSDRDS IGWSDSVT
IGWSDWRM IGWSDWRS IGWSRDPD IGWSSCAN IGWSSCN2 IGWSSDCN IGWSSNOM

IGWSSSCS IGW8RBID IGW8RCID IGW8RRES

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42347 DFSMSHsm

Example A-7 contains the cover letter for APAR OA42347.

Example A-7 OA42347

APAR Identifier	OA42347	Last Changed	13/08/02
DFSMSHSM MIGRATES BEYOND STORAGE GROUP MIGRATION LOW THRESHOLD DURING SPACE MANAGEMENT.			
Symptom	IN INCORROUT	Status	CLOSED PER
Severity	4	Date Closed	13/06/25
Component	5695DF170	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	DFSMSHSM, ISMF	Special Notice	ATTENTION
Current Target Date	..13/08/16	Flags	
SCP			
Platform		PERVASIVE	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:
Release 210 : UA69550 available 13/07/10 (F307)

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:
DFSMSHsm migrates beyond storage group MIGR LOW threshold during space management.

LOCAL FIX:
N/A

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All users of DFSMSHsm Space Management      *
*                   functions on z/OS V2R1.                    *
*****
* PROBLEM DESCRIPTION: DFSMSHsm migrates beyond storage group *
*                   migration low threshold during Space      *
*                   Management.                                *
*****
* RECOMMENDATION:                                             *
*****
DFSMSHsm migrates beyond storage group migration low threshold
during Space Management.
```

PROBLEM CONCLUSION:

DFSMSHsm has been modified to correct this problem.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: ARCSMPMQ

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42358 PDSE

Example A-8 is the controlling APAR for the PDSE member generations update. There are many other APARS that constitute the whole collection.

Example A-8 OA42358

This is the APAR for SPE Line Item PDSE Member generations

Local fix

Problem summary

* USERS AFFECTED: All PDSE users in z/OS V2R1 *

* PROBLEM DESCRIPTION: PDSE is being enhanced to allow a user *
* to save a fixed number of previous *
* generations of each member, the number *
* of generations for each member is *
* specified when the data set is created. *
* A new generation of each member will be *
* saved whenever a member is replaced or *
* deleted, PDF will provide services to *
* view saved generations of members. *

* This feature will work only on PDSE *
* Version 2 datasets. *

* Toleration APAR for this project is *
* OA43433 (Install the PTFs for this APAR *
* in R12 and R13 to avoid abnormal errors *
* when using a PDSE (V2 and MAXGENS > 0)) *
* in the lower releases. *

* PDF is enhanced to access generations *
* and identify data sets which have *
* generations. (OA42247 and OA42248) *

* New functions *
* DESERV FUNC=GET_G *
* DESERV FUNC=GET_ALL_G *

* New PARMS for FIND Macro: *
* G - Find a member generation *

*

*

* New PARMS for STOW Macro: *
* DG - delete a member generation *
* RG - replace an old member generation. *
* RECOVERG - Recover a member generation. *

* For further details please read the *
* latest refresh of the manuals to know *
* more about this new function. *

* RECOMMENDATION: *

Provide new functions for PDSEs (See Description)

APAR OA42390 PDSE

Example A-9 contains the cover letter for APAR OA42390.

Example A-9 OA42390

APAR Identifier OA42390 Last Changed 13/07/02
PDF REPORTS NO MEMBERS FOR NON-EMPTY PDSE ON HDZ2210

Symptom	IN INCORROUT	Status	CLOSED PER
Severity	2	Date Closed	13/06/11
Component	5695DF115	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	EXTENDED DATA S	Special Notice	HIPER
Current Target Date	..13/06/30	Flags	
SCP			
Platform			DATALOSS

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 210 : UA69489 available 13/06/26 (F306)

Parent APAR:

Child APAR list: OA42573

ERROR DESCRIPTION:

Existing PDSE appears to have no members. Read of directory for PDSE will return no member data for certain data sets.

LOCAL FIX:

You may be able to bypass this by adding a member to the PDSE with a very low key '#' for example.
The problem is related to an empty directory block and can be resolved by adding a record to that block.

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: Users access PDSEs updated on systems below *
*                  the HDZ2210 level may find the PDSE seems to *
*                  be empty on HDZ2210. The problem is most    *
*                  likely on PDSEs where the members with the *
*                  first member names are deleted.             *
*****
* PROBLEM DESCRIPTION: See Users affected.                     *
*****
* RECOMMENDATION:                                              *
*****
```

Add code to the PDSE Index manager to tolerate empty Directory Blocks.

PROBLEM CONCLUSION:
See Problem Summary.

TEMPORARY FIX:
HIPER AA42390

ADD MEMBER WITH VERY LOW KEY '# ' TO THE PDSE

COMMENTS:

MODULES/MACROS:
IGWDABDN IGWDACN2 IGWDADCD IGWDADPD IGWDARD1 IGWDDCR3 IGWDDDSF
IGWDLALR IGWDLCLS IGWDLDES IGWDPDMP IGWDRLLDR

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42406 OCE Fix roll up

Example A-10 contains the cover letter for APAR OA42406.

Example A-10 OA42406

APAR Identifier	OA42406	Last Changed	13/08/02
OCE FIX ROLLUP FOR HDZ2210			
Symptom	IN INCORROUT	Status	CLOSED PER
Severity	3	Date Closed	13/06/25
Component	5695DF107	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	DATA MGMT SUPPO	Special Notice	
Current Target Date	..13/08/30	Flags	

SCP
Platform

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:
Release 210 : UA69690 available 13/07/30 (F307)

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:
Fixes PCK in IFG019TR. Also adds ALIAS name in SMF14/15. Also fixes an invalid IEC190I EXCP DCB without a foundation extension..

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: This is a rollup APAR for HDZ2210. *

* PROBLEM DESCRIPTION: This APAR fixes 3 problems and *
* adds 1 additional function. *

* RECOMMENDATION: *

3 problems fixed:
1. Extraneous characters in IEC999I DEBADDR= message.
2. PCK in IFG019TR.
3. IEC190I invalid DCBE: EXCP DCB without foundation ext
is invalidly issued.
Additional function:
1. Alias data set name added to SMF14 Additional Data Set
characteristics section (type 5).

PROBLEM CONCLUSION:
See problem summary.
SMF14 Additional data set characteristics section (type 5):

New bit in SMF14: Byte SMF14BFG0:
SMF14ALS EQU X'10' ALIAS NAME PRESENT

New Alias name 44 character field:
SMF14ALI ALIAS DATA SET NAME

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IFGSMF14 IFGOTCOA IFG019RA IFG019TR IFG0202H
IFG0202I

SRLS: SA22763025

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42488 IEBCOPY user exit

Example A-11 contains the cover letter for APAR OA42488.

Example A-11 OA42488

APAR Identifier OA42488 Last Changed 13/07/02
IEB1169T USING IEBCPARM AND IEBCPLST

Symptom MS MSGIEB1169T	Status CLOSED PER
Severity 3	Date Closed 13/06/12
Component 5695DF102	Duplicate of
Reported Release 210	Fixed Release 999
Component Name BASE ACCESS MET	Special Notice
Current Target Date ..13/07/31	Flags
SCP	
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 210 : UA69515 available 13/06/26 (F306)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:
IEBCPARM and IEBCPLST are not presently included SYS1.MACLIB

LOCAL FIX:
n/a

PROBLEM SUMMARY:

* USERS AFFECTED: Users wishing to use the IEBCOPY user exits *
* shipped in HDZ2210. *

* PROBLEM DESCRIPTION: The required macros were not shipped. *
* They should have been in SYS1.MACLIB. *

* RECOMMENDATION: *

The required macros were not shipped.

PROBLEM CONCLUSION:
The macros are shipped.

The description of the new IEBCOPY user exits is incomplete.

These IEBCPLST fields are expected to be updated by the user exit:

CPLST_PARMLIST_RC
CPLST_PARMLIST_USR
CPLST_PARMLIST_MAXRETC
CPLST_PARMLIST_MAXRSNC
CPLST_CONTROL_OFLAGS
CPLST_CONTROL_DATA_ADDR
CPLST_CONTROL_DATA_LEN
CPLST_MEMBER_OFLAGS
CPLST_MEMBER_RC
CPLST_MEMBER_NEWNAME_ADDR

If the control statement user exit gives a return code 4, it signals IEBCOPY to process all currently pending control statements. Otherwise, IEBCOPY will automatically execute the "in progress step" whenever it receives (from the exit a new "IEBCOPY command" (such as COPY, COPYGRP, etc. control statement.

A control statement user exit may choose to never use return code 4 and completely depend on the standard IEBCOPY processing rules, but it must always return an RC=32 to signal an "End of control statement input data".

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS:
IEBCPARM IEBCPLST

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42540 PDSE binder

Example A-12 contains the cover letter for APAR OA42540.

Example A-12 OA42540

APAR Identifier OA42540 Last Changed 13/08/02
ABENDOF4 RC24 RSN141AA7FE IGWIRPND+ODE2 @ OA42051

Symptom AB ABENDxxx	Status CLOSED PER
Severity 3	Date Closed 13/06/27
Component 5695DF115	Duplicate of
Reported Release 210	Fixed Release 999
Component Name EXTENDED DATA S	Special Notice HIPER
Current Target Date ..13/09/15	Flags
SCP	
Platform	DATALOSS

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 210 : UA69724 available 13/07/11 (F307)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

HSM back up of a PDSE data set on z/OS 2.1 LPAR failed with
ABENDOF4 RC24 RSN141AA7FE IGWIRPND+ODE2 @ OA42051.

RSN141AA7FE equates IMF_DirStorePageBad1 indicating that the front and rear compression data for LONG_NAME_PROCESS_RECORD is not what is expected.

LOCAL FIX:
None

PROBLEM SUMMARY:

* USERS AFFECTED: PDSE binder users. *

* PROBLEM DESCRIPTION: Abend0F4 when accessing PDSEs contain *
* long alias names. *

* RECOMMENDATION: *

Long alias names that were not compressed were being
uncompressed unnecessarily.

PROBLEM CONCLUSION:
The index manager has been updated to handle non compressed records correctly.

TEMPORARY FIX:

* HIPER *

INSTALL ++APAR AA42540.

COMMENTS:

MODULES/MACROS: IGWDAALN IGWDABDN IGWDACRN IGWDDCR2 IGWDDCR3
IGWIRCMT IGWIRPAD IGWIRPND

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42541 Catalog

Example A-13 contains the cover letter for APAR OA42541.

Example A-13 OA42541

APAR Identifier OA42541 Last Changed 13/10/02
ABEND S130 IN IDA0200T AFTER INSTALLING PTFS FOR OA36916

Symptom IN INCORROUT Status CLOSED PER
Severity 2 Date Closed 13/08/02
Component 5695DF106 Duplicate of
Reported Release D10 Fixed Release 999
Component Name DFSMS VSAM Special Notice PE
Current Target Date ..13/08/31 Flags
SCP
Platform

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List: UA68644 UA68643

PTF List:

Release A10 : PTF not available yet
Release B10 : PTF not available yet
Release C10 : UA70130 available 13/08/28 (F308)
Release D10 : UA70131 available 13/08/28 (F308)
Release 210 : UA70146 available 13/08/28 (F308)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

An abend S130 may occur in IDA0200T at offset X'3600' after installing the PTFS for OA36916. The error is a dequeue for a resource that is not held.

The dequeue will show a request to dequeue SYSVSAM with a catalog name as the resource and ending in a "N".

LOCAL FIX:

Remove the PTFS for OA36916

PEX/ UA68641

PEX/ UA68642

PROBLEM SUMMARY:

* USERS AFFECTED: All z/OS 1.10 VSAM users and above who *
* OPEN and CLOSE a catalog as a normal *
* VSAM Data Set. *

```

* PROBLEM DESCRIPTION: While closing a user catalog that has *
*                      been opened with multiple ACBs, an *
*                      ABEND S130 will result. *
*****
* RECOMMENDATION: Apply PTF. *
*****
VSAM CLOSE processing was not correctly verifying
the last close for a user catalog.

```

PROBLEM CONCLUSION:
VSAM CLOSE processing was modified to correctly
determine the last close for a user catalog.
KEYWORDS: ZOS0201C/K

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IDA0200T

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42543 XTIOT Health Checker

Example A-14 contains the cover letter for APAR OA42543.

Example A-14 OA42543

APAR Identifier	OA42543	Last Changed	13/08/02
MSGIECH0101E IECH0101E HEALTH CHECKER			
MSGIECH0100I IECH0100I			
Symptom		MS IECH0101E	
Severity	2	Status	CLOSED PER
Component	5695DF107	Date Closed	13/07/02
Reported Release	210	Duplicate of	
Component Name DATA MGMT SUPPO		Fixed Release	999
Current Target Date ..13/08/30		Special Notice	ATTENTION
SCP		Flags	
Platform		PERVASIVE	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 210 : UA69765 available 13/07/31 (F307)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

When the system is dynamically updated via the MVS command SET DEVSUP=xx, the following message is issued even though NON_VSAM_XTIOT=YES is specified in SYS1.PARMLIB(DEVSUPxx):

IECH0101E OPEN macro support for XTIOT, uncaptured UCBs and DSAB above the line is not enabled for non-VSAM. IBM recommends setting NON_VSAM_XTIOT=YES in the DEVSUPxx member of PARMLIB.

When this situation occurs ensure the following:

- 1) Check SYS1.PARMLIB(DEVSUPxx) to ensure commas are used for multi-line statements. Without the commas, only one line will be read.
- 2) If you are using multiple DEVSUPxx members then ensure NON_VSAM_XTIOT=YES is specified in all members.

If DEVSUPxx is updated successfully then a dynamic update from SET DEVSUP=xx will issue message IEA253I:

IEA253I DEVSUP XTIOT FOR NON-VSAM IS SUPPORTED

!!!NOTE: A dynamic update of DEVSUPxx will NOT update health checker. Health Checker will need to be refreshed by implementing one of the following:

- Restarting health checker via commands:
F HZSPROC,STOP
S HZSPROC
- Go to SDSF CK panel selecting R and E (refresh)
- F HZSPROC,RUN,CHECK=(IBMOCE,OCE_XTIOT_CHECK)
- F HZSPROC,REFRESH,CHECK=(IBMOCE,OCE_XTIOT_CHECK)

After Health Checker is refreshed, IECH0100I will be issued instead of IECH0101E. After this APAR is implemented, message IECH0100I will move from an "exception" to an "information" message class.

LOCAL FIX:

n/a

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All.                                     *
*****
* PROBLEM DESCRIPTION: Incorrect message class.           *
*****
* RECOMMENDATION:                                         *
```

The original submitter text was based on the incorrect assumption that updating DEVSUPxx keyword: NON_VSAM_XTIOT followed by system command: SET DEVSUP=xx would immediately be reflected in the Health Checker display for the OCE_XTIOT_CHECK.

After updating DEVSUPxx and issuing command SET DEVSUP=xx the following commands will reexecute the health checker:

```
F HZSPROC,RUN,CHECK=(IBMOCE,OCE_XTIOT_CHECK)
F HZSPROC,REFRESH,CHECK=(IBMOCE,OCE_XTIOT_CHECK)
```

This APAR however,does change the message class for IECH0100I to "information" from "exception".

PROBLEM CONCLUSION:

IECH0100I issued when NON_VSAM_XTIOT=YES is specified and activated in DEVSUPxx has been changed to message class "Information".

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IFGHCM01 IFGHCOC1 IFGHCOC2

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42551 RLS DSS

Example A-15 contains the cover letter for APAR OA42551.

Example A-15 OA42551

APAR Identifier OA42551 Last Changed 13/10/02
ADR952E ADR957I DURING LOGICAL DUMP OF USERCATALOG WITH DSS DUE
TO A FAILURE IN RLS DURING QUIESCE PROCESS 13/06/14 PTF PECHANGE

Symptom MS MSGADR952E Status CLOSED PER
Severity 2 Date Closed 13/06/21
Component 5695DF122 Duplicate of
Reported Release D10 Fixed Release 999
Component Name VSAM REC LEV SH Special Notice PE
Current Target Date ..13/07/31 Flags
SCP
Platform

Status Detail: SHIPMENT - Packaged solution is available for
shipment.

PE PTF List: UA68643 UA68644

PTF List:

Release A10 : PTF not available yet
Release B10 : PTF not available yet
Release C10 : UA69623 available 13/07/23 (F307)
Release D10 : UA69624 available 13/07/23 (F307)
Release 210 : UA69625 available 13/07/23 (F307)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

A logical dump of a usercat defined with a large number
of extents (approximatly 30 and more) will fail with DSS
messages:

ADR952E (001)-DTDSC(01), THE IDAQDMP MACRO FAILED DURING QUIESCE
PROCESSING FOR FOR CLUSTER dsname WITH RETURN CODE
00000008 AND REASON CODE 61FF0001

This issue occurs after application of OA36422 and OA36403

Problem is due to workarea shortage during RLS locate request
for catalog information.

ADDITIONAL KEYWORDS:

RLSPSP/K
SMSVSAM RLS VSAMRLS

LOCAL FIX:

You can use DSS patch to circumvent the RLS quiesce processing described in DFSMSdss Storage Administration
1.14.30 Bypassing RLS processing (OW32817)
PEX/ UA68641
PEX/ UA68642

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All Catalog users who issue DSS DUMP          *
*                   command to backup user catalog data sets.    *
*****
* PROBLEM DESCRIPTION: When performing the DSS DUMP function     *
*                   to backup user catalog data sets with        *
*                   about 30 extents, the following error        *
*                   is generated:                                  *
*                   ADR952E (001)-DTDSC(01), THE IDAQDMP         *
*                   MACRO FAILED DURING QUIESCE PROCESSING       *
*                   FOR CLUSTER dsname WITH RETURN CODE         *
*                   00000008 AND REASON CODE 61FF0001           *
*****
* RECOMMENDATION:                                                *
*****
If a user catalog data set has approximately 30 or more extents,
any attempt to back it up using the DSS DUMP will failed with
ADR952E error message.
```

PROBLEM CONCLUSION:

The error is due to the work area being too small to handle a user catalog with many extents. The fix will allocate a larger work area and retry the operation when detecting the work area too small error.
KEYWORDS: RLSPSP/K ZOS0201C/K

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IDAVQCAT

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42553 SDM

Example A-16 contains the cover letter for APAR OA42553.

Example A-16 OA42553

APAR Identifier OA42553 Last Changed 13/10/03
AUTHORIZATION CHECKING FOR ANTRQST STATESAVE CALLERS

Symptom	IN INCORROUT	Status	CLOSED PER
Severity	3	Date Closed	13/08/20
Component	5695DF117	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	SYSTEM DATA MOV	Special Notice	
Current Target Date	..13/10/31	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 210 : UA70479 available 13/09/18 (F309)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

Change authorization checking for callers of
ANTRQST STATESAVE

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: Users of ANTRQST API. *
*****
* PROBLEM DESCRIPTION: A return code 7010 can occur indicating *
*                       that a caller is not authorized even if *
*                       the caller is in an authorized key and *
*                       in supervisor state. *
*****
* RECOMMENDATION: *
*****
```

PROBLEM CONCLUSION:
The problem has been corrected.
KEYWORDS: SDMAPI/K

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS:
ANTCDPI

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42654 SDM

Example A-17 contains the cover letter for APAR OA42654.

Example A-17 OA42654

APAR Identifier OA42654 Last Changed 13/09/04
HANG IN SMSVSAM BETWEEN LOCK ALTER AND LOCK GRANT PROCESSING

Symptom IN INCORROUT	Status CLOSED PER
Severity 2	Date Closed 13/08/01
Component 5695DF122	Duplicate of
Reported Release 210	Fixed Release 999
Component Name VSAM REC LEV SH	Special Notice HIPER
Current Target Date ..13/08/15	Flags
SCP	FUNCTIONLOSS
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:
Release 210 : UA70040 available 13/08/08 (F308)

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:

Hang encountered between two SMS threads attempting to process the same lock. D SMS,SMSVSAM,DIAG(C) display shows contention building on the RHT latch:

```
.
IGW343I VSAM RLS DIAG STATUS (V.01)
|----RESOURCE---| |----- WAITER -----| |--HOLDER---| ELAPSED
TYPE          ID   JOB NAME ASID  TASK  ASID  TASK  TIME
-----
LATCH      7F5B93F8 CICS2ACA 01A3 00000000 000A 70C2DOC0 00:12:45
LATCH      7F5B93F8 CICS2ACB 01A2 00000000 000A 70C2DOC0 00:13:15
LATCH      7F5B93F8 CICS2ACC 01A6 00000000 000A 70C2DOC0 00:13:16
LATCH      7F5B93F8 CICS2ACA 01A3 00000000 000A 70C2DOC0 00:13:20
LATCH      7F5B93F8 CICS2ACA 01A3 00000000 000A 70C2DOC0 00:13:20
```

.
The holder is a thread attempting to ALTER the Component2 lock and is waiting for the request to be granted. A waiter is also processing the same lock attempting to grant it, but is waiting on the RHT latch held by the first thread.

.
Additional Problems

During rebuild a cancel of CICS regions can orphan GRSLatch and Sidb latch causing the REBUILD to hang

Additional keywords:

RLSHANG SMSVSAM IGWLN17 IGWLN77 IGWLN19 IGWLN78
DIWA COMP2 RHT SPLIT
RLS_GRS_LATCH_SET

LOCAL FIX:

Restart SMSVSAM

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All VSAM RLS users.                                     *
*****
* PROBLEM DESCRIPTION: Hang in SMSVSAM between lock alter and             *
*                               Lock grant processing.                       *
*****
* RECOMMENDATION:                                                         *
*****
```

Problem #1.

Problem occurs if 3 or more jobs are competing for a component2 lock. Job A is the holder and Job C (same system) waits for the

lock. Job A is about to release the lock and Job B (from other system) tries to obtain it. Due to an un-initialized parameter that SMSVSAM uses for lock processing, Job A may wait for Job B to complete before it can grant the lock to the waiter (Job C), resulting in a deadlock situation.

Problem #2.

Cancelling a job during record lock request processing can hang due to internal routine error not releasing some of the internal latches.

PROBLEM CONCLUSION:

Problem #1.

The code has been modified to initialize the parameter correctly to prevent the deadlock situation.

Problem #2.

The code has been modified to release all the internal latches if cancel occurs during record lock request processing.

KEYWORDS: RLSPSP/K

TEMPORARY FIX:

* HIPER *

COMMENTS:

MODULES/MACROS:

IDAVRCLO IGWLN12 IGWLN17 IGWLN19 IGWLN20 IGWLN36 IGWLN40
IGWLN60 IGWLN77 IGWLN78 IGWLN10

SRLS:

NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42667 XTLOT use (also applicable to DFSMS V1.13)

Example A-18 contains the cover letter for APAR OA42667.

Example A-18 OA42667

APAR Identifier OA42267 Last Changed 13/10/01
IEC999I IFG0194D ABENDOC1 ABENDOC4 NON_VSAM_XTIOT DEVSUPXX
TRKCALC RC04 IECOSCR1

Symptom AB ABENDOC4 Status CLOSED PER
Severity 2 Date Closed 13/09/16
Component 5695DF107 Duplicate of
Reported Release D10 Fixed Release 999
Component Name DATA MGMT SUPPO Special Notice
Current Target Date ..13/09/30 Flags
SCP
Platform

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release D10 : UA70761 available 13/10/01 (1000)
Release 210 : UA70762 available 13/10/01 (1000)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

When keyword NON_VSAM_XTIOT is set to YES in PARMLIB member DEVSUPxx, IFG0194D +x'186A' UA68824 makes a TRKCALC call without LOC=ANY and the UCB is a 31-bit address pulled directly from the XTIOT. TRKCALC module IECOSCR1 takes an ABENDOC4 as a result. IFG0194D may also take an intentional ABENDOC1 to header IFG0194D TRKCALC_ERROR_SEE_R15 ERROR when register 15 is set to RC04. The joblog will show an IEC999I IFG0194D message.

LOCAL FIX:

Set NON_VSAM_XTIOT to NO in PARMLIB member DEVSUPxx

PROBLEM SUMMARY:

* USERS AFFECTED: All. *

* PROBLEM DESCRIPTION: Abend0c1 in IFG0194D. *

* RECOMMENDATION: *

IEC999I IFG0194D,POFFOR,POFFOR,TRKCALC ERROR SEE R15
due to an intentional abend0c1 after a TRKCALC failure.
TRKCALC is passed a 31bit UCB address without including
keyword: LOC=ANY in the macro invocation. OPEN is processing
with an XTIOT containing 31bit UCB addresses.

PROBLEM CONCLUSION:
IFG0194D now issues TRKCALC with the LOC=ANY keyword.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IFG0194D

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42675 PDSE

Example A-19 contains the cover letter for APAR OA42675.

Example A-19 OA42675

APAR Identifier OA42675 Last Changed 13/09/26
ABENDOC4 PIC38 IGWIRADB+OCB0 @ HDZ2210

Symptom AB ABENDOC4	Status CLOSED PER
Severity 3	Date Closed 13/07/03
Component 5695DF115	Duplicate of
Reported Release 210	Fixed Release 999
Component Name EXTENDED DATA S	Special Notice HIPER
Current Target Date ..13/08/15	Flags
SCP	
Platform	DATALOSS

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:
Release 210 : UA69767 available 13/07/11 (F307)

Parent APAR: OA42051
Child APAR list:

ERROR DESCRIPTION:
ABENDOF4 PIC39 CSECT=IGWIRRRDB+OCB0 @ HDZ2210 base
ABENDOC1 IGWIRPAD+0B4E @ HDZ2210 base
ABENDOC1 IGWIRPAD+17D4 @ HDZ2210 base
ABENDOC1 IGWIRPND+1848 @ HDZ2210 base
ABENDOF4 IGWIRPAD RSN1419A084 1419A084

LOCAL FIX:
None

PROBLEM SUMMARY:

* USERS AFFECTED: PDSE which has been updated extensively on *
* a release lower than 2.1 may *
* have a sparse index. When 2.1 updates the *
* the index may be broken, *

* PROBLEM DESCRIPTION: See users affected. *

* RECOMMENDATION: *

Modify PDSE processing to position to the correct place in the
index before attempting to remove empty pages,

PROBLEM CONCLUSION:
See Problem Summary

TEMPORARY FIX:
* HIPER *

COMMENTS:

MODULES/MACROS: IGWIRCMT IGWIRDIS IGWIRICR IGWIRIEM IGWISTAM

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42986 NFS applies to DFSMS V1.13 as well

Example A-20 contains the cover letter for APAR OA42986.

Example A-20 OA42986

APAR Identifier OA42986 Last Changed 13/10/03
ZNFSC VN_TRUNC SOC4 DUE TO UNINITIALIZED VARIABLE

Symptom AB ABENDOC4	Status CLOSED PER
Severity 3	Date Closed 13/08/22
Component 5695DF121	Duplicate of
Reported Release 21N	Fixed Release 999
Component Name NETWORK FILE SY	Special Notice
Current Target Date ..13/09/13	Flags
SCP	
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release C1N : UA70539 available 13/09/04 (F309)
Release D1N : UA70540 available 13/09/04 (F309)
Release 21N : UA70541 available 13/09/04 (F309)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

zNFSC vn_trunc SOC4 due to uninitialized variable

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: Users of zOS NFS Client *
*****
* PROBLEM DESCRIPTION: OC4 abend in vn_trunc when accessing a *
*                      file in an NFSv4 mount *
*****
* RECOMMENDATION: *
*****
```

An uninitialized variable led to attempting to write to incorrect memory, leading to an OC4abend.

PROBLEM CONCLUSION:

Properly initializing the pRes variable in the vn_trunc NFSv4 path eliminated an OC4abend.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS:

GFSCVNAC

SRLS:

NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42846 Single striped, multi-volume sam tailored compressed data set

Example A-21 contains the cover letter for APAR OA42846.

Example A-21 OA42846

APAR Identifier OA42846 Last Changed 13/09/12
APAR OA41459 CAN CAUSE LOOP IN IFG0232D

Symptom LP LOOP	Status CLOSED PER
Severity 2	Date Closed 13/08/05
Component 5695DF107	Duplicate of
Reported Release D10	Fixed Release 999
Component Name DATA MGMT SUPPO	Special Notice PE
Current Target Date ..13/09/15	Flags
SCP	
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List: UA68273 UA68272 UA68824 UA68823

PTF List:

Release C10 : UA70205 available 13/09/12 (1000)
Release D10 : UA70204 available 13/09/12 (1000)
Release 210 : UA70203 available 13/09/12 (1000)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

APAR OA41459 can cause a LOOP in IFG0232D when there are fewer than 5 used physical blocks on the last volume for tailored compressed data sets during CLOSE TYPE=T (TCLOSE), this LOOP will hold ENQ on SYSZTIOT.

LOCAL FIX:

remove PTF for OA41459

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: users of single striped, multi-volume sam      *
*                  tailored compressed data sets.                  *
*****
* PROBLEM DESCRIPTION: users of single striped, multi-volume      *
*                  sam tailored compressed data sets may          *
*                  see loops when close type=t leave is           *
*                  issued when open for input or issued           *
*                  twice when open for output on the last         *
*                  volume and there is very little data           *
*                  (less than five physical blocks) on the        *
*                  current volume.                                  *
*****
* RECOMMENDATION:                                                  *
*****
users of single striped, multi-volume sam tailored compressed
data sets may see loops when close type=t leave is issued.
```

PROBLEM CONCLUSION:

Change code so that the count of physical blocks used by the tailored dictionary are ignored for the second and subsequent volumes of a single striped, multi-volume data set.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IFG0232D

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42891 PDSE

Example A-22 contains the cover letter for APAR OA42891.

Example A-22 OA42891

APAR Identifier	OA42891	Last Changed	13/10/03
IEC036I 002-94 MAXIMUM ALLOWABLE NUMBER OF MEMBERS OF PDSE EXCEEDED			
Symptom	AB ABEND	Status	CLOSED PER
Severity	3	Date Closed	13/08/13
Component	5695DF115	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	EXTENDED DATA S	Special Notice	
Current Target Date	..13/10/31	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 210 : UA70333 available 13/09/04 (F309)

Parent APAR: OA40985

Child APAR list:

ERROR DESCRIPTION:

ABEND002 RC94 received when adding to a pdse that has less than the maximum Number of members allowed for a PDSE as displayed by ISPF 3.4 Info - I .

S002 rsn94 msgIEC036I

LOCAL FIX:
IEBCOPY PDSE to a newly allocated PDSE

PROBLEM SUMMARY:

* USERS AFFECTED: Users who create new PDSE members. *

* PROBLEM DESCRIPTION: An Abend002 RC94 occurs when adding a *
* new member to a PDSE *

* RECOMMENDATION: *

Prevent Member counts from going negative

PROBLEM CONCLUSION:
A total member count was invalid and misrepresented that the
maximun count had been reached.
The code has been modified to prevent the member counts from
reaching an invalid range

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS:
IGWIRCMT

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42904 z/HPF

Example A-23 contains the cover letter for APAR OA42904.

Example A-23 OA42904

APAR Identifier OA42904 Last Changed 13/08/02

EXPOSURE IN Z/HPF IN R2.1: CHECK FOR FCX_FMTWRITE TO
IFG0555H

Symptom	IN INCORROUT	Status	CLOSED PER
Severity	3	Date Closed	13/07/29
Component	5695DF107	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	DATA MGMT SUPPO	Special Notice	HIPER
Current Target Date	..13/08/30	Flags	
SCP			
Platform		DATALOSS	

Status Detail: SHIPMENT - Packaged solution is available for
shipment.

PE PTF List:

PTF List:

Release 210 : UA70011 available 13/07/31 (F307)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

Check for FCX_FmtWrite to IFG0555H

LOCAL FIX:

none

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All users of SAM z/HPF writing multi-volume *
*                  data sets.                                   *
*****
* PROBLEM DESCRIPTION: Data may be lost when writing a SAM      *
*                  z/HPF data set if the control unit is      *
*                  not at the latest EC level.                 *
*****
* RECOMMENDATION:                                             *
*****
Data may be lost when writing SAM z/HPF data if the control unit
microcode is not at the latest level.
```

PROBLEM CONCLUSION:

Added check to verify the correct control unit microcode level
before writing with z/HPF.

TEMPORARY FIX:

* HIPER *

COMMENTS:

MODULES/MACROS:
IFG0555H

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42935 IEBCOPY

Example A-24 contains the cover letter for APAR OA42935.

Example A-24 OA42935

APAR Identifier OA42935 Last Changed 13/10/03
ABEND C03 UNLOADING AN EMPTY PDSE USING IEBCOPY

Symptom AB ABENDC03	Status CLOSED PER
Severity 2	Date Closed 13/08/21
Component 5695DF114	Duplicate of
Reported Release 210	Fixed Release 999
Component Name DFSMS UTILITIES	Special Notice
Current Target Date ..13/09/16	Flags
SCP	
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:
Release 210 : UA70502 available 13/09/12 (F309)

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:

When attempting to do an UNLOAD of an Empty PDSE using IEBCOPY an Abend C03 occurs.

Errors received resemble:

```
IEC999I IFGOTCOA,IFGOTCOB,TEST2    ,UNLOAD  ,
      DEB ADDR=8C2394-1,DSN = xxxxx.xxxxxx
```

SYSTEM COMPLETION CODE=C03

When the PDSE to be unloaded is empty, module IEBCFAMS is not closing the output DCB, which causes the C03 abend during task termination, since the DCB has been previously freed.

The SYSTRACE shows this storage was freed up as part of a large area by IEBVTM1 + x'COE'.

LOCAL FIX:

n/a

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: Users trying to unload an empty PDSE using *
*                  IEBCOPY.                                   *
*****
* PROBLEM DESCRIPTION: Users trying to unload an empty PDSE *
*                  using IEBCOPY will see an abend C03      *
*                  during step termination.                  *
*****
* RECOMMENDATION:                                           *
*****
Abend C03 unloading an empty PDSE using IEBCOPY.
```

PROBLEM CONCLUSION:

Removed an asterisk that commented out a call to close the output data set.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS:

IEBCFAMS

SRLS:

NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA43003 VSAM RLS

Example A-25 contains the cover letter for APAR OA43003. DFSMS V1.13 equivalent APAR is OA42947.

Example A-25 OA43003

APAR Identifier OA43003 Last Changed 13/09/19
POSSIBLE INCORRECT RBA RETURNED AFTER ESDS PUT UPDATE OR DELETE

Symptom IN INCORROUT	Status CLOSED PER
Severity 3	Date Closed 13/09/19
Component 5695DF106	Duplicate of
Reported Release 210	Fixed Release 999
Component Name DFSMS VSAM	Special Notice
Current Target Date ..13/10/31	Flags
SCP	
Platform	

Status Detail: TESTPACKAGING - Packaged solution is being tested.

PE PTF List:

PTF List:

Release 210 : PTF not available yet

Parent APAR: OA42947

Child APAR list:

ERROR DESCRIPTION:

For ESDS processing, it is possible for the RBA to be corrupted by application programs after it's initially passed in correctly. An application program specifies the correct RBA for a GET UPDATE request, VSAM RLS processes the GET UPDATE with the correct RBA successfully, then the application somehow corrupts the RBA, possibly by mistake, followed by a PUT UPDATE request. RLS also processes the PUT UPDATE successfully since RBA is not

required. However, the incorrect RBA is returned at the end of the processing.

LOCAL FIX:

Avoid corrupting the RBA between requests.

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: VSAM users running with non-keyed GET, PUT *
*                  and ERASE processing.                      *
*****
* PROBLEM DESCRIPTION: A user application can corrupt the    *
*                  input RBA value after a GET UPDate        *
*                  request and the subsequent PUT or ERASE   *
*                  request can return with an incorrect      *
*                  RBA value.                                 *
*****
* RECOMMENDATION: APPLY PTF                                  *
```

A user application specifies a correct RBA value for the GET UPDate request, then the application corrupts the RBA value. Since VSAM is already positioned to the record, the next PUT or ERASE request can run successfully but the returned RBA value in RPL area is still corrupted.

PROBLEM CONCLUSION:

VSAM will save the input RBA value for non-keyed GET request and the subsequent PUT or ERASE request will restore the RBA value in RPL before returning to user application.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IDA019R1

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA43112 VSAM

Example A-26 contains the cover letter for APAR OA43112.

Example A-26 OA43112

APAR Identifier OA43112 Last Changed 13/09/23
IDA019RV+05A0 FEEDBACK CODE: 010804F8, IDA019RZ+61A PLH STACK
ERROR

Symptom IN INCORROUT	Status OPEN
Severity 3	Date Closed
Component 5695DF106	Duplicate of
Reported Release 210	Fixed Release
Component Name DFSMS VSAM	Special Notice
Current Target Date ..13/12/31	Flags
SCP	
Platform	

Status Detail: DESIGN/CODE - APAR solution is being designed
and coded.

PE PTF List:

PTF List:

Parent APAR: OA40707

Child APAR list:

ERROR DESCRIPTION:

VSAM produces dump titled:

VSAM DYNAMIC RPL DUMP - IDA019RV+05A0 FEEDBACK CODE: 010804F8.

The dump shows (in the first entry in the PLH stack) that IDA019RZ+61A was attempting to return using a zero stack pointer and so he had no module to return to. This produced the RPL error code and dump.

The application is using LSR buffer pools, Asynchronous VSAM calls, with many TCBs multitasking, and it is very busy. The call that received the error was an Asynchronous Direct PUT to a cluster with an AIX.

The nature of this problem is very timing dependent and, we would expect, rarely seen. The sequence leading to the error can be seen by closely reviewing entries in the MVS SYSTRACE for this call, processing at the same time from the same TCB for other asynchronous calls. This current call had gone to VSAM for the PUT, the IO was started, but we then were interrupted by a timer interrupt (EXT 1005) in IDA019RZ+57E (IDA019RZ is base 1.12 code). At this point the MVS supervisor saves the regs for this Ext, interrupt in the TCB.

Other work gets done under this same TCB, as long as it is only IRB related work, or stems from IRB work, such as previously started VSAM IO for this TCB. Before we get to redispach this TCB after the EXT. interrupt PSW, the PLH gets reused for AIX processing for another request. This is all under this same TCB same. The AIX request processes, the PLH stack is reused and goes to zero as the AIX request completes. We then dispatch the Externally interrupted work in IDA019RZ+57E using the regs saved in the TCB. He expects the PLH stack to be as he left it, when the EXT. interrupt occurred, but the stack is zeros. He then gets rpl feedback code 010804F8 when he tries to "return" at IDA019RZ+61A.

=====

This is a sysroute of OA40707 for 2.1 only. The design of the serviceability enhancement for OA40707 was restricted by the storage available in the footsteps. This is a more complete version and will include a timestamp and other information.

=====

LOCAL FIX:
none

APAR OA43128 DFSMSrmm applicable to DFSMS V2.1 and V1.13

Example A-27 contains the cover letter for APAR OA43128.

Example A-27 OA43128

APAR Identifier OA43128 Last Changed 13/10/03
GARBLED TEXT ON JAPANESE PANEL EDGP@LIB

Symptom IN INCORROUT	Status CLOSED PER
Severity 3	Date Closed 13/08/27
Component 5695DF186	Duplicate of
Reported Release 21K	Fixed Release 999
Component Name DFSMSRMM	Special Notice
Current Target Date ..13/10/17	Flags
SCP	
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release C1K	: UA70581 available 13/09/28 (F309)
Release D1K	: UA70582 available 13/09/28 (F309)
Release 21K	: UA70583 available 13/09/28 (F309)

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:
Unreadable characters appear on Japanese panel EDGP@LIB,
shipped with z/OS V2.1 (R21K) PTF/APAR UA69568 / OA42135.
Additional Keywords:
DFSMSrmm DFRMM RMM RMMCJH KANJI

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: All DFSMSrmm users using KANJI panels. *

* PROBLEM DESCRIPTION: There is a problem with the Kanji panel *
* EDGP@LIB: *
* - in release V2R1 the panel contains *
* garbled text; *
* - in releases V1R12, V1R13 the panel *
* is missing. *

* RECOMMENDATION: Apply the applicable PTS(s). *

PROBLEM CONCLUSION:
The problem has been fixed.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS:
EDGP@LIB TDGP@LIB

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA43153 VSAM

Example A-28 contains the cover letter for APAR OA43153.

Example A-28 OA43153

APAR Identifier OA43153 Last Changed 13/09/27
PLH "ROUND ROBIN" PROBLEM UNDER R210.

Symptom IN INCORROUT	Status OPEN
Severity 2	Date Closed
Component 5695DF106	Duplicate of
Reported Release 210	Fixed Release
Component Name DFSMS VSAM	Special Notice
Current Target Date ..13/12/01	Flags
SCP	
Platform	

Status Detail: REVIEW - APAR solution is being reviewed.

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

PLH "round robin" problem under r210.

Under r210 a POINT returns RPL FEEDBACK = x'93080010', and the following GET returns x'00000000' along with the 1st record in data set.

Under previous releases the POINT returned x'93080010' and the following GET returned x'58080058' and no record.

LOCAL FIX:

If a POINT request is rejected, failed it should not be used as the basis for a following GET NEXT (sequential GET). Since the POINT request failed, if it is used with a subsequent GET the results are unpredictable.

APAR OA43162 DSS Storage Tier

Example A-29 contains the cover letter for APAR OA43162.

Example A-29 OA43162

APAR Identifier OA43162 Last Changed 13/09/25
COPY USING STORAGE TIER, WITH DB2 DATABASE, LDS, ISGQUERY
RECEIVES ADR564E WITH RC4 AND RSN051B0405

Symptom MS MSGADR564E	Status OPEN
Severity 2	Date Closed
Component 5695DF175	Duplicate of
Reported Release 210	Fixed Release
Component Name DFSMSDSS, ISMF	Special Notice
Current Target Date ..	Flags
SCP	
Platform	

Status Detail: DESIGN/CODE - APAR solution is being designed
and coded.

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

Customer running Storage Tier through HSM, copy of a DB2
database, which is a VSAM LDS, customer will receive the
following message:

ADR564E (001)-DDFLT(01), ISGQUERY ISSUED A RETURN CODE OF
00000004 AND REASON CODE OF 051B0405 FOR DATA SET xxxxxxxx

ISGQUERY returns information indicating with return code x'4',
which indicated the ISGQUERY succeeded, but a warning is issued.
For reason code xxxx0405, the AnswerAreaFull indicator has been
returned. Either the answer area for the query is too small, or
a RESUMETOKEN was not used correctly.

Additional info: STORTIERS/K

LOCAL FIX:

N/A

APAR OA43169 DFSMShsm

Example A-24 contains the cover letter for APAR OA43169.

Example A-30 OA43169

APAR Identifier	OA43169	Last Changed	13/10/03
ABEND878 OUT-OF-STORAGE CONDITION DUE TO ORPHANED X'11F8' BYTE			
ALLOCATIONS IN SUBPOOL 0 KEY 8.		13/09/10 PTF PEREMOVE	
Symptom	AB ABEND878	Status	CLOSED PER
Severity	2	Date Closed	13/09/10
Component	5695DF170	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	DFSMSHSM, ISMF	Special Notice	HIPER
Current Target Date	..13/09/27	Flags	
SCP		FUNCTIONLOSS	
Platform		PERVASIVE	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release 210 : UA70684 available 13/09/24 (F309)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

S878 ABEND occurs because of orphaned X'11F8' byte allocations in SPO Key8. These allocations will exhaust user region above the 16M line and then proceed to exhaust it below the 16M line. In IPCS, " IP VERBX VSMDATA 'NOG SUM' " will show thousands of the following orphaned storage obtains:

DQE:	Addr 007C3000	Size	2000
FQE:	Addr 007C41F8	Size	E08

LOCAL FIX:

Determine size of storage leak by seeing how long HSM was up prior to the S878 ABEND. Schedule regular restarts of HSM accordingly.

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All z/OS DFSMSHsm release hdz2210 users.          *
*****
* PROBLEM DESCRIPTION: Customer is experiencing ABEND878            *
*                        during DFSMSHsm space management.           *
*****
```

* RECOMMENDATION: *

AB878 is occurring during DFSMSHsm space management. Getmained storage is not being freed.

PROBLEM CONCLUSION:

Apply the PTF for this APAR to resolve this problem.

TEMPORARY FIX:

* HIPER *

COMMENTS:

MODULES/MACROS: ARCATTEC ARCMDSUV ARCMSTAI ARCMVDS

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

Apply ++APAR temporary fix when available. Estimated availability date is September 9, 2013.

MESSAGE TO SUBMITTER:

APAR OA43191 RLS

Example A-31 contains the cover letter for APAR OA43191.

Example A-31 OA43191

APAR Identifier OA43191 Last Changed 13/09/30
HANG DURING REBUILD / REALLOCATE OF SMSVSAM LOCK STRUCTURE
RLS_GRS_LATCH_SET

Symptom WS WAIT	Status OPEN
Severity 2	Date Closed
Component 5695DF122	Duplicate of
Reported Release 210	Fixed Release
Component Name VSAM REC LEV SH	Special Notice
Current Target Date ..14/03/31	Flags
SCP	
Platform	

Status Detail: REVIEW - APAR solution is being reviewed.

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

A hang can be seen during a REBUILD or REALLOCATE of the SMSVSAM lock structure. The hang is seen when the lock structure is in SIMPLEX state. A large amount of contention will be reported on RLS_GRS_LATCH_SET with several holders and requestors for SHARED state and 1 requestor (Rebuild) for exclusive.

Additional Keyword:

RLS_GRS_LATCH_SET
NAME=SMSVSAM/RLS_GRS_LATCH_SET
RLSHANG
RLSPSP/K
SMSVSAM RLS VSAMRLS DF122

LOCAL FIX:

Restart SMSVSAM

APAR OA43195 Catalog

Example A-32 contains the cover letter for APAR OA43195.

Example A-32 OA43195

APAR Identifier OA43195 Last Changed 13/09/26
BAD RB ERROR 246-24 IN IGG0CLXA AT 2.1

Symptom IN INCORROUT	Status OPEN
Severity 2	Date Closed
Component 5695DF105	Duplicate of
Reported Release 210	Fixed Release
Component Name ICF CATALOG & I	Special Notice
Current Target Date ..13/12/31	Flags
SCP	
Platform	

Status Detail: DESIGN/CODE - APAR solution is being designed and coded.

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

APAR OA36157 introduced a new error to capture a doc when control is improperly transferred between user address spaces and CAS.

In 2.1, new code was introduced in support of RLS Catalogs and related commands. One such code path will issue an informational message to inform a user of any suspended catalogs (IEC367W message) that are waiting for SMSVSAM to become available. It is possible, however, for this timer code path to be invoked for terminated catalogs as well as suspended catalogs. This causes the timer pop code to be erroneously called for terminated catalogs, leading to unexpected IRB interrupts, and 246-24 errors.

LOCAL FIX:

n/a

APAR OA43198 PDSE

Example A-33 contains the cover letter for APAR OA43198.

Example A-33 OA43198

APAR Identifier	OA43198	Last Changed	13/09/02
ADD PDSE DATA SET NAME IN EREP SYMPTOM RECORD FOR ABEND0F4 RC24			
IGWIRADB RSN141FA7FD			
Symptom	IN INCORROUT	Status	OPEN
Severity	2	Date Closed	
Component	5695DF115	Duplicate of	
Reported Release	210	Fixed Release	
Component Name	EXTENDED DATA S	Special Notice	
Current Target Date	..13/12/31	Flags	
SCP			
Platform			

Status Detail: ANALYSIS - APAR is being investigated or debugged.

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

The intent of this apar is to externalize the pdse data set name in the erep symptom record (symrec) for the ABEND0F4 141FA7FD detected from module IGWIRADB to assist in the identification of the PDSE data set involved with the abend.

LOCAL FIX:

APAR OA43214 PDSE

Example A-34 contains the cover letter for APAR OA43214.

Example A-34 OA43214

APAR Identifier	OA43214	Last Changed	13/09/23
ABENDOC1 IGWIRMT COPYING INTO A PDSE DATA SET WITH A SPARSE ND NAME DIRECTORY INDEX			
Symptom	AB ABEND	Status	CLOSED PER
Severity	3	Date Closed	13/09/23
Component	5695DF115	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	EXTENDED DATA S	Special Notice	
Current Target Date	..13/12/31	Flags	
SCP			
Platform			

Status Detail: TESTPACKAGING - Packaged solution is being tested.

PE PTF List:

PTF List:

Release 210 : PTF not available yet

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

IEBCOPY to copy a PDSE to a PDSE data set with sparse ND index results in following errors :

IGW01173S RC36 RSN06105AF8

ABENDOC1 out of IGWIRMT+92EC @ UA69767

LOCAL FIX:

PROBLEM SUMMARY:

```
*****  
* USERS AFFECTED: PDSE which have been process on system levels below HDZ2210 and have very empty indices, can cause HDZ2210 to fail when attempting to add members to the PDSE. The emptiness of the PDSE indices can be determined by running the IEBPDSE utility against the data set.  
* The error is most likely when the ND free space is greater than 75%.  
* +IGW700I PDSE Directory Validation Successf  
* +DSN:SYSPLEX.T1#04031.PDSE01  
* +ADPages:164 IXRecords:10056  
* +ADPagesInCore:11 ADPagesRead:153  
* +ADTreeLevels:3  
* +NDPages:3506 IXRecords:4000  
* +NDPagesInCore:1 NDPagesRead:3505  
* +NDTreeLevels:3  
* +AD ND Tree Nodes:4000  
* +ADPercentFree:43 NDPercentFree:96<-----  
* +Version:1  
*  
*  
*  
*  
*  
*  
*  
*  
*  
*  
*****  
* PROBLEM DESCRIPTION: See Users Affected.  
*****  
* RECOMMENDATION:  
*****  
PDSE processing will be changed to correctly handle empty directory blocks and almost empty blocks.
```

PROBLEM CONCLUSION:

See Problem Summary.

TEMPORARY FIX:

COMMENTS:

```
MODULES/MACROS:  IGWIRCMT IGWIRPAD IGWIRPND
```

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA43220 DSS Storage Tier

Example A-35 contains the cover letter for APAR OA43220.

Example A-35 OA43220

APAR Identifier OA43220 Last Changed 13/09/25
MSGADR560E DB2 CAF CONNECT FUNCTION RC08 RSN00F30012 ISSUED WHEN
DOING CLASS TRANSITION IN HSM PSM TO TIER2 OF A DB2 TABLE SPACE

Symptom MS MSGadr560e	Status OPEN
Severity 2	Date Closed
Component 5695DF175	Duplicate of
Reported Release 210	Fixed Release
Component Name DFSMSDSS, ISMF	Special Notice
Current Target Date ..	Flags
SCP	
Platform	

Status Detail: DESIGN/CODE - APAR solution is being designed
and coded.

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

Four systems exist in the sysplex all running DB2 which has a table space in use on each system in the sysplex. HSM primary space management starts and begins processing class transition to move a DB2 table space to tier2 dasd. This fails with msgadr560e (001)-DDFLT(01), DB2 CAF CONNECT FUNCTION ISSUED A RETURN CODE OF 00000008 AND REASON CODE OF 00F30012 FOR SUBSYSTEM DBX3
additional info: STORTIERS/K

LOCAL FIX:
n/a

APAR OA43221 DSS Storage Tier

Example A-36 contains the cover letter for APAR OA43221.

Example A-36 OA43221

APAR Identifier OA43221 Last Changed 13/09/27
ABENDOF4 IN IGWLNRI0 + 6E4 AFTER APPLICATION OF OA42654

Symptom AB ABENDOF4	Status CLOSED PER
Severity 3	Date Closed 13/09/27
Component 5695DF122	Duplicate of
Reported Release 210	Fixed Release 999
Component Name VSAM REC LEV SH	Special Notice HIPER
Current Target Date ..13/12/31	Flags
SCP	FUNCTIONLOSS
Platform	

Status Detail: TESTPACKAGING - Packaged solution is being tested.

PE PTF List:

PTF List:

Release 210 : PTF not available yet
Release 210 : Relief is available in the form of: CALL IBM

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

After applying fix for OA42654, testing received ABENDOF4 in
IGWLNRI0+06E4 due to all latches not being owned.

Symptom string:

COMPID=DF122,CSECT=IGWLNRI0+06E4,DATE=07/24/13,MAINTID=OA42654
,ABND=0F4,RC=00000024,RSN=66F32050

LOCAL FIX:
None

PROBLEM SUMMARY:

* USERS AFFECTED: All VSAM RLS users. *

```

* PROBLEM DESCRIPTION: ABENDOF4 in IGWLNRI0 at offset X'6E4' *
*                      after application of OA42654          *
*****
* RECOMMENDATION:                                           *
*****
Cancelling a job waiting for a lock request can cause an abend
due to an internal routine having erroneously released some of
the internal latches which were not owned.

```

```

PROBLEM CONCLUSION:
IGWLNRI0 is modified to release all the internal latches
only if they are owned.
KEYWORDS: RLSPSP/K

```

```

TEMPORARY FIX:
*****
* HIPER *
*****

```

COMMENTS:

MODULES/MACROS: IGWLNRI0

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA43221 PDSE

Example A-37 contains the cover letter for APAR OA43434.

Example A-37 OA43434

```

APAR Identifier ..... OA43234      Last Changed ..... 13/09/18
ABENDOF4 RSN5E061100 IGWFB120 BTREE INVALID TOPNODE

```

```

Symptom ..... AB ABEND              Status ..... CLOSED PER
Severity ..... 3                     Date Closed ..... 13/09/10
Component ..... 5695DF115            Duplicate of .....
Reported Release ..... 210            Fixed Release ..... 999

```

Component Name EXTENDED DATA S Special Notice
Current Target Date ..13/12/31 Flags
SCP
Platform

Status Detail: TESTPACKAGING - Packaged solution is being tested.

PE PTF List:

PTF List:
Release 210 : PTF not available yet

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:
TITLE=COMPID=DF115,CSECT=IGWFB120+2838,DATE=03/12/13,MAINTID=
NONE ,ABND=0F4,RC=00000024,RSN=5E061100

Subcomponent: PDSE B-Tree Services
Module name : IGWFB120
Reason : BTree_InvalidTopNode

Module flow : IGWFB125 <- IGWIRLRU <- IGWIRLRT

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: Abend can occur if the PDSE LRU returns a *
* page buffer and the block is reassigned *
* immediately. *

* PROBLEM DESCRIPTION: See Users affected. *

* RECOMMENDATION: *

Correct logic to save next block in btree before the block may
be deleted.

PROBLEM CONCLUSION:
See Summary.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IGWIRLRU

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA43234 PDSE

Example A-38 contains the cover letter for APAR OA43234.

Example A-38 OA43234

APAR Identifier OA43234 Last Changed 13/09/18
ABENDOF4 RSN5E061100 IGWFB120 BTREE INVALID TOPNODE

Symptom AB ABEND	Status CLOSED PER
Severity 3	Date Closed 13/09/10
Component 5695DF115	Duplicate of
Reported Release 210	Fixed Release 999
Component Name EXTENDED DATA S	Special Notice
Current Target Date ..13/12/31	Flags
SCP	
Platform	

Status Detail: TESTPACKAGING - Packaged solution is being tested.

PE PTF List:

APAR OA43417 DSS Storage Tier

Example A-34 contains the cover letter for APAR OA43417.

Example A-39 OA43417

APAR Identifier OA43417 Last Changed 13/09/25
MSGADR567E RC67 RSN105 EXCI ERROR WHEN ATTEMPTING COPY A CICS
RLS DATA SETS USING STORAGE TIER PROCESSING.

Symptom MS MSGadr567e	Status OPEN
Severity 2	Date Closed
Component 5695DF175	Duplicate of
Reported Release 210	Fixed Release
Component Name DFSMSDSS, ISMF	Special Notice

Current Target Date .. Flags
SCP
Platform

Status Detail: DESIGN/CODE - APAR solution is being designed
and coded.

PE PTF List:

PTF List:

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:
Processing CICS RLS data set using storage tiers to copy the
data set receives the following error:
ADR568I (001)-EXCIC(01), INVOCATION OF CICS INTERFACES BEGIN
=====DFSMSdss EXCI Client Program (ADREXCIC)=====
* Input Parameters: CICS CLOS APPLID=" "
* Linking to CICS EXCI server routine ADREXCIS
* The EXEC CICS LINK PROGRAM request failed. Return codes are:
* RESP=00000058 RESP2=000001AD. ABCODE:
*=====End of DFSMSdss EXCI Client Program
(ADREXCIC)=====*
ADR567E (001)-DDFLT(01), INVOCATION OF EXCI CICS INTERFACES FOR
DATA

SET RLSTEST.VF01D.BANKACCT FAILED WITH RETURN CODE 00000008

AND REASON CODE 00000067

DIAGNOSTIC CODE IS 0C000105

Error is due to an invalid applid being used by ADRDSOCS.
additional info: STORTIERS/K

LOCAL FIX:
none

APAR OA43418 DSS Storage Tier

Example A-40 contains the cover letter for APAR OA43418.

Example A-40 OA43418

APAR Identifier OA43418 Last Changed 13/09/25
LOOP IN ADRDSOC +X'D04' THROUGH X'D8E' TRYING TO LOOP THROUGH
THE ANSWER AREA RETURNED BY GRS.

Symptom	LP LOOP	Status	OPEN
Severity	2	Date Closed	
Component	5695DF175	Duplicate of	
Reported Release	210	Fixed Release	
Component Name	DFSMSDSS, ISMF	Special Notice	
Current Target Date ..		Flags	
SCP			
Platform			

Status Detail: DESIGN/CODE - APAR solution is being designed and coded.

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

Using Storage tier processing in DFSMSHsm to process CICS non-RLS data sets appears to hang. Analysis of the dump shows we are in a loop in ADRDSOC from +x'd04' through x'D8e' while processing the data returned in the answer area (ISGYQUAAR) by GRS.

additional information: STORTIERS/K

LOCAL FIX:

none

APAR OA43430 DFSMSdfp

Example A-41 contains the cover letter for APAR OA43430.

Example A-41 OA43430

APAR Identifier	OA43430	Last Changed	13/09/20
BAD IEC205I MESSAGE AFTER 10 VOLSERS IN Z/OS2.1			

Symptom	MS MSGIEC205I	Status	OPEN
Severity	3	Date Closed	
Component	5695DF107	Duplicate of	
Reported Release	210	Fixed Release	
Component Name	DATA MGMT SUPPO	Special Notice	
Current Target Date ..		Flags	
SCP			
Platform			

Status Detail: DESIGN/CODE - APAR solution is being designed

and coded.

PE PTF List:

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

When creating a Multi-volume data set and not specifying a volume count the IEC205I message will not reflect correctly the volume serial numbers after the 10th volume.

Following is an example of the bad message:

```
IEC205I TAPE,TGAIR0,BACKUP,FILESEQ=1, COMPLETE VOLUME LIST, 917
DSN=TGAIR.M9SG03.DUMP,
VOLS=A00760,A00761,A00762,A00763,A00764,A00765,A00766,A00767,
VOLS=A00768,A00769,3 TIONAL VOLS,          TOTALBLOCKS=454
```

LOCAL FIX:

Provide a volume count in JCL

APAR OA43583 PDSE (also applies to pre-DFSMS V2.1)

Example A-42 contains the cover letter for APAR OA43583.

Example A-42 OA43583

```
APAR Identifier ..... OA43583          Last Changed ..... 13/10/21
ABENDOF4 RC14 RSN25400018 IGWDLCLS+OC32 @ UA68850
```

Symptom AB ABENDOF4	Status CLOSED PER
Severity 2	Date Closed 13/10/21
Component 5695DF115	Duplicate of
Reported Release D10	Fixed Release 999
Component Name EXTENDED DATA S	Special Notice HIPER
Current Target Date ..14/01/15	Flags
SCP	
Platform	DATALOSS

Status Detail: TESTPACKAGING - Packaged solution is being tested.

PE PTF List:

PTF List:

```
Release C10   : PTF not available yet
Release D10   : PTF not available yet
Release 210   : PTF not available yet
```

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

ABEND0F4 RC14 RSN25400018 out of the module IGWDLCLS+0C32 at UA68850 maintenance level.

The reason code RSN25400018 translates to GETMAIN_FAILED.

VSM summary data (IP VERBX VSMDATA 'NOG SUMM') shows that SMSPDSE1 extended private storage is almost exhausted:

LOCAL STORAGE MAP

Extended LSQA/SWA/229/230 Address	80000000 <- Top of Ext. Private 80000000 <- Max Ext. User Region
(Free Extended Storage)	32914000 <- ELSQA Bottom 32908000 <- Ext. User Region Top
Extended User Region	32900000 <- Ext. User Region

An IP VERBX SMSXDATA 'F(POOLS) JOBNAME(SMSPDSE1)' will show an extremely large allocation for the "JCDM DSC POOL FOR LSTB DREFD" pool:

```
*****
Below the Bar Storage
Exts  TotSize      TotCells  InUseCells  SP  Key Description
*****
11315 1,205,002,240  1,131,500  1,131,500  124 50 JCDM DSC POOL
                                FOR LSTB
                                DREFD
```

The build up of the LSTB control structures is the result of PDSE index code that is attempting to join two index pages.

This failure can occur on PDSE data sets that are shared in a sysplex with mixed z/OS levels where at least one LPAR is at the HDZ2210 / z/OS 2.1 level.

LOCAL FIX:

Restart the SMSPDSE1 address space

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: PDSE index records can rarely be broken when *
*                   the data set is updated on a release HDZ2210 *
*                   which had been updated on a lower release.  *
```

```

*****
* PROBLEM DESCRIPTION: See Users Affected.
*****
* RECOMMENDATION:
*****
PDSE index manager will be modified to now join index pages
which have created on HDZ2210 and a lower release.

```

PROBLEM CONCLUSION:
See Problem summary.

```

TEMPORARY FIX:
*****
* HIPER * AA45383
*****

```

COMMENTS:

MODULES/MACROS:
IGWIRCMT IGWIVRJ2

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA43701 SAM EF V2 (also applies to pre-DFSMS V2.1)

Example A-43 contains the cover letter for APAR OA43701.

Example A-43 OA43701

```

APAR Identifier ..... OA43701      Last Changed ..... 13/10/28
VOLUME LABEL MAY BE OVERWRITTEN BY SAM EXTENDED FORMAT VERSION 2
DATA SET - Z/OS 2.1 ONLY

Symptom ..... IN INCORROUT          Status ..... CLOSED PER
Severity ..... 2                     Date Closed ..... 13/10/28
Component ..... 5695DF102            Duplicate of .....
Reported Release ..... 210           Fixed Release ..... 999
Component Name BASE ACCESS MET       Special Notice      HIPER

```

Current Target Date ..13/12/15 Flags
SCP
Platform PERVASIVE DATALOSS

Status Detail: TESTPACKAGING - Packaged solution is being tested.

PE PTF List:

PTF List:
Release C10 : PTF not available yet
Release D10 : PTF not available yet
Release 210 : PTF not available yet

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:

If a multi-volume single striped SAM Extended Format Version 2 data set is allocated using Guaranteed Space and then opened for EXTEND or opened for OUTPUT or OUTIN when allocated DISP=MOD, the volume label on the second to last volume may be overwritten by user data.

LOCAL FIX:

Do not use Version 2 if using Guaranteed Space allocation and either OPEN EXTEND or OPEN OUTPUT/OUTIN and DISP=MOD is used. To bypass this problem you must do one of the following:
1) Specify DSNTYPE=(EXT,1) or DSNTYPE=(EXT) on the JCL
OR
2) Specify EXT_VERSION(1) in IGDSMSxx parmlib member AND do NOT specify DSNTYPE=(EXT,2) on the JCL.

The volume can be recovered by varying it offline to all systems and running the following job.

```
//*****  
/* THIS STEP REBUILDS CYL 0 HEAD 0 TO POINT TO VTOC      */  
/* CHANGE THE VOLID TO THE VOLSER REQUIRED, UNIT TO DEVICE*/  
/* NUMBER AND THE START OF THE VTOC TO THE CYLINDER and */  
/* HEAD where it is  
//*****  
//RFBT      EXEC  PGM=ICKDSF  
//SYSPRINT DD SYSOUT=*  
//SYSIN     DD  *  
  REFORMAT UNIT(dddd) NVFY VTOCPTR(Cyl,HD) PURGE VOLID(vvvvvv)  
/*
```

A successful vary online is a good indication all should be fine.

PROBLEM SUMMARY:

```

* USERS AFFECTED: All users of multi-volume single striped SAM *
*                   extended format data sets allocated using   *
*                   guaranteed space.                             *
*****
* PROBLEM DESCRIPTION: Multi-volume single striped SAM          *
*                   extended format version 2 data sets         *
*                   allocated using guaranteed space may         *
*                   overlay the volume label on the second      *
*                   to last volume if the first OPEN is for     *
*                   EXTEND or for OUTPUT or OUTIN and           *
*                   DISP=MOD is specified. In some cases,      *
*                   such as ISPF browse, OPEN for INPUT or     *
*                   INOUT will position to the volume          *
*                   label. Though version 2 data sets may      *
*                   not be created on releases prior to        *
*                   HDZ2210, this problem could occur if       *
*                   the data set was allocated on HDZ2210      *
*                   and first opened on an earlier release.    *
*****
* RECOMMENDATION:                                             *
*****
Multi-volume single striped SAM extended format version 2 data
sets allocated using guaranteed space may overlay the volume
label on the second to last volume.

```

PROBLEM CONCLUSION:

Changed code to correctly build internal control blocks to only allow access to the data set extents on the last volume.

TEMPORARY FIX:

* HIPER * AA43701

COMMENTS:

MODULES/MACROS:

IGG0193V

SRLS:

NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:



APARs to be reviewed for DFSMS V1.13 or DFSMS V1.12

This APARs listed in this appendix should be reviewed for impact to your installation prior to implementing z/OS 2.1 DFSMS.

Where APAR text is shown it is at time of writing of this IBM Redbooks publication. The current version should be checked for updates. When an APAR has had a PTF shipped, and a problem is then found, the PTF will be marked PE, and also the original APAR may be updated to show the fixing APAR numbers.

There two lists in this Redbooks publication. One for fixes that should be applied to DFSMS V2.1 before it is used, and one that should be applied to DFSMS V1.13 or DFSMS V1.12 before DFSMS V2.1 is used.

The lists that follow do not include every APAR that may be required, but is representative. If the PTFS for the listed APARS are applied then many other APARS will also be included.

This list is for DFSMS V1.13 and DFSMS V1.12.

The corresponding list for DFSMS V2.1 can be found at Appendix A, “APARs to be reviewed for DFSMS V2.1” on page 241

DFSMS suggested and required fixes for pre-DFSMS V2.1

In some cases there are no PTFs for the listed APARs, and some may have been superseded. The current listing from IBMLINK should be checked for current status and PTFs.

Some of the APARs listed in this part are also applicable to DFSMS V2.1.

APAR OA35808 RMM

Example B-1 contains the cover letter for APAR OA45808.

Example B-1 OA35808

APAR Identifier	OA35808	Last Changed	13/06/03
NF - TOLERATION FOR Z/OS DFSMSRMM V2R1			
Symptom		NF NF	Status
Severity		4	CLOSED UR1
Component		5695DF186	Date Closed
Reported Release		A10	13/04/16
Component Name		DFSMSRMM	Duplicate of
Current Target Date ..		13/05/08	Fixed Release
SCP			999
Platform			Special Notice
			ATTENTION
			Flags
			NEW FUNCTION

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	:	UA68801 available 13/05/02 ()
Release B10	:	UA68802 available 13/05/02 ()
Release C10	:	UA68803 available 13/05/03 (F305)
Release D10	:	UA68804 available 13/05/03 (F305)

Parent APAR:

Child APAR list: OA42058

ERROR DESCRIPTION:

Toleration for future new function

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: All users of DFSMSrmm. *

```

* PROBLEM DESCRIPTION: This ARAR provides toleration          *
*                      for z/OS V2R1 DFSMSrmm.                *
*                      ZOS0201C/K                             *
*****

```

```

* RECOMMENDATION: Apply the applicable PTF(s).                *
*****

```

APAR OA35808 provides coexistence support to allow systems at z/OS V1R10 (HDZ1A10) and above to coexist in an RMMplex with a z/OS V2R1 (HDZ2210) level system. This APAR is required on the lower level systems to tolerate new support introduced in z/OS V2R1, or for backout of the new function.

OA35808 recognizes and supports the new data set attribute LASTREF, and a new volume attribute RETAINBY.

Coexistence support pertains an existing set of consistent LASTREF extra_days in formation for a multi-volume data set created or maintained in V2R1 whenever retention information for a file in that set is added or changed in a lower release. New single data set records will get extra_days 0 assigned. An extra_days value of 0 has the same effect as using NOLASTREF. When an existing multi volume data set is enlarged with MOD and a new data set record is added to the RMM CDS because of EOVS then the extra_days value for LASTREF is taken from the existing previous data set record.

Coexistence support pertains an existing volume set of consistent RETAINBY information and consistent volume expiration date for a multi-volume set created or maintained in V2R1 whenever retention information for a file or a volume in that set is added or changed in a lower release. EXPDT equalization for volume sets depending on the RETAINBY attribute will be rolled down. New single volumes will get RETAINBY 0 assigned. This has the same effect as using RETAINBY(VOLUME). When an existing multi volume set is enlarged and a new volume record is added to the RMM CDS because of EOVS then the RETAINBY is taken from the existing previous volume record.

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
 NF - TOLERATION FOR Z/OS DFSMSRMM V2R1
 ZOS0201C/K

There are two new messages in
 z/OS MVS System Messages, Vol 5 (EDG-GFS) SA22-7635

EDG3369I FILE EXPIRATION ATTRIBUTES ACCEPTED BUT IGNORED
 FOR THE VOLUME EXPIRATION BECAUSE VOLUME IS RETAINED BY
 FIRSTFILE

Explanation: You issued an ADDDATASET or a CHANGEDATASET subcommand with a RETPD, EXPDT, or LASTREF operand for a file which is not the first file of the volume set. RMM detects that the data set resides on a volume managed by the EXPDT retention method with the RETAINBY(FIRSTFILE) attribute. The data set will get the new expiration date or LASTREF attribute assigned, but the effective expiration date of the volume is unchanged as long the first file of the volume set is not changed.

System Action: The subcommand ends with return code 4 reason code 278

Operator Response: n/a.

Application Programmer Response: Check the expiration date of the volume. If it is lower than the needed expiration date of the data set, you can enlarge the expiration date of the first file of the volume set.

Source: DFSMSrmm

Source: DFSMSrmm

Detecting Module: EDGMTSO

EDG3372I CHANGE OF EXPIRATION DATE NOT ALLOWED FOR VOLUMES
RETAINED BY FIRST FILE

Explanation: You issued a CHANGEVOLUME subcommand with a RETPD or an EXPDT operand for a volume or volume set defined with RETAINBY(FIRSTFILE) attribute. This is not supported. System action: The subcommand ends with return code 12 reason code 284.

Operator Response: n/a.

Application Programmer Response: To change the expiration date of all volumes of the set you can issue a CHANGEDATASET subcommand for the first file of the volume set. Alternatively the RETAINBY attribute of the volume set can be changed to RETAINBY(VOLUME) or RETAINBY(SET).

Source: DFSMSrmm

Detecting Module: EDGMTSO

MODULES/MACROS: EDGDOC EDGDOCS EDGEXTSY EDGMFIO EDGMTAB
EDGMTSO EDGSOCE EDGSOCS EDGTSO EDGTSORT EDGXMLAP

SRLS: SA22763516 SA22763518 SA22763519 SA22763521

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA36403 RLS

Example B-2 contains the cover letter for APAR OA3643.

Example B-2 OA36403

APAR Identifier	OA36403	Last Changed	13/09/13
NEW FUNCTION			
Symptom	NF NF	Status	CLOSED UR1
Severity	3	Date Closed	13/04/05
Component	5695DF122	Duplicate of	
Reported Release	B10	Fixed Release	999
Component Name	VSAM REC LEV SH	Special Notice	ATTENTION
Current Target Date	..13/04/30	Flags	
SCP			NEW FUNCTION
Platform			

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release A10	:	UA68641 available 13/04/15 (	)
Release B10	:	UA68642 available 13/04/15 (	)
Release C10	:	UA68643 available 13/04/16 (F304	)
Release D10	:	UA68644 available 13/04/16 (F304	)

Parent APAR:

Child APAR list: OA36409 OA36414 OA36422 OA36492 OA36916

ERROR DESCRIPTION:

New function.

LOCAL FIX:

PROBLEM SUMMARY:

 * USERS AFFECTED: ALL RLS users. *

 * PROBLEM DESCRIPTION: RLS Toleration APAR. *

 * RECOMMENDATION: *

 RLS Toleration APAR.
 ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
 RLS Toleration APAR.
 KEYWORDS: RLSPSP/K

MODULES/MACROS: IDAVCBK IDAVCBL2 IDAVCMAT IDAVCUBL IDAVOMAT
 IDAVQCAT IDAVQDMP IDAVQDRV IDAVQENA IDAVQPCP IDAVQSET IDAVQSRL
 IDAVQSXT IDAVQUAC IDAVRRND IDAVRRN2 IDAVRRST IDAVRSSF IDAVUACK
 IDAVXMAT IGWLNIO1 IGWLNII6 IGWLPQYS IGWSDISP IGWSDIS2 IGWSGMSG
 IGWSRTE3 IGW8QEXR

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA36409 RLS

Example B-3 contains the cover letter for APAR OA36409.

Example B-3 OA36409

APAR Identifier	OA36409	Last Changed	13/09/13
NEW FUNCTION			

Symptom	NF NF	Status	CLOSED UR1
Severity	3	Date Closed	13/04/05

Component	5695DF119	Duplicate of	
Reported Release	B10	Fixed Release	999
Component Name	COMM FUNCTION/R	Special Notice	ATTENTION
Current Target Date	..13/04/30	Flags	
SCP			NEW FUNCTION
Platform			

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release A10	:	UA68641 available 13/04/15 (	)
Release B10	:	UA68642 available 13/04/15 (	)
Release C10	:	UA68643 available 13/04/16 (F304	)
Release D10	:	UA68644 available 13/04/16 (F304	)

Parent APAR: OA36403

Child APAR list:

ERROR DESCRIPTION:

New function.

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: ALL RLS users.                                     *
*****
* PROBLEM DESCRIPTION: RLS Toleration APAR.                         *
*****
* RECOMMENDATION:                                                  *
*****
RLS Toleration APAR.
ZOS0201C/K
```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

RLS Toleration APAR.

MODULES/MACROS: INTLPRTS

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA36414 DFSMSHsm RLS

Example B-4 contains the cover letter for APAR OA36414.

Example B-4 OA36414

APAR Identifier	OA36414	Last Changed	13/05/03
NEW FUNCTION - DFSMSHSM TOLERATION COEXISTENCE SUPPORT FOR RLS USER CATALOGS			
Symptom	NF NF	Status	CLOSED UR1
Severity	3	Date Closed	13/04/08
Component	5695DF170	Duplicate of	
Reported Release	B10	Fixed Release	999
Component Name	DFSMSHSM, ISMF	Special Notice	
Current Target Date	..13/07/31	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	:	UA68663	available	13/04/15	(	)
Release B10	:	UA68664	available	13/04/15	(	)
Release C10	:	UA68665	available	13/04/16	(F304	)
Release D10	:	UA68666	available	13/04/16	(F304	)

Parent APAR: OA36403

Child APAR list:

ERROR DESCRIPTION:

DFSMSHsm toleration coexistence support for RLS user catalogs.

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: Users with DFSMSHsm hosts on z/OS V2R1 *
* which share CDSs and ICF catalogs with *
* hosts running at z/OS V1R13, V1R12, V1R11, *
* and/or V1R10 *

* PROBLEM DESCRIPTION: Changes in the RLS ICF user catalog *
* function in z/OS V2R1 require that *
* z/OS V1R10 through V1R13 hosts be *
* updated. This APAR provides toleration *
* support for Record Level Sharing (RLS) *
* user catalogs that are processed *
* using DFSMSHsm. *

* RECOMMENDATION: *

Changes in the RLS ICF user catalog function in z/OS V2R1
require that z/OS V1R13 through V1R10 hosts be updated.

.
Additional keywords:
ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

DFSMSHsm has been modified such that the RLS user catalog
support for z/OS V2R1 can be detected and processed
appropriately.

.
With the toleration/coexistence PTFs installed, DFSMSHsm will
invoke DFSMSdss to support backup and recover of SMS managed
RLS-eligible user catalogs on z/OS pre-V2R1 systems.

.
See DFSMSdss OA36422 for more information.

MODULES/MACROS: ARCBUDS ARCKUCAT ARCNCAAT ARCNVIFY ARWCNT
ARCWDEF C ARCWLSTC

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA36415 RLS

Example B-5 contains the cover letter for APAR OA36415.

Example B-5 OA36415

APAR Identifier OA36415 Last Changed 13/05/03
NEW FUNCTION TOLERATION

Symptom	NF NEWFUNCTION	Status	CLOSED UR1
Severity	4	Date Closed	13/04/10
Component	5695DF122	Duplicate of	
Reported Release	A10	Fixed Release	999
Component Name	VSAM REC LEV SH	Special Notice	
Current Target Date	..13/09/30	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	:	UA68691 available 13/04/15 ()
Release B10	:	UA68692 available 13/04/15 ()
Release C10	:	UA68693 available 13/04/17 (F304)
Release D10	:	UA68694 available 13/04/17 (F304)

Parent APAR:

Child APAR list: OA36443

ERROR DESCRIPTION:

Toleration apar for new function

LOCAL FIX:

No local fix

PROBLEM SUMMARY:

* USERS AFFECTED: All users of z/OS HDZ1A10 and above. *

* PROBLEM DESCRIPTION: New function APAR. *

* RECOMMENDATION: *

New function apar.

ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

New function APAR.

KEYWORDS: RLSPSP/K

MODULES/MACROS: IDAV192A IDAV192B IDAV192C IDAV192F IDAV193A
IDAV194A IDAV200B IDAV200T IDAV201T IDAV202T IDAV203T IDAV231T
IDAV232T IDAV233T IGWMCOLD IGWMPROP IGWSDCD2 IGWSDRDM IGWSDRDS
IGWSDWRM IGWSDWRS IGWSXMSG

SRLS: SA22763714

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA36422 DSS RLS

Example B-6 contains the cover letter for APAR OA36422.

Example B-6 OA36422

APAR Identifier	OA36422	Last Changed	13/06/26
NEW FUNCTION - DFSMSDSS SUPPORT OF RLS USER CATALOGS TOLERATION/ COEXISTENCE			
Symptom	NF NF	Status	CLOSED UR1
Severity	3	Date Closed	13/04/05
Component	5695DF175	Duplicate of	
Reported Release	B10	Fixed Release	999
Component Name	DFSMSDSS, ISMF	Special Notice	ATTENTION
Current Target Date ..		Flags	
SCP			NEW FUNCTION
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10 : UA68646 available 13/04/15 ()
Release B10 : UA68647 available 13/04/15 ()
Release C10 : UA68648 available 13/05/03 (F304)
Release D10 : UA68649 available 13/05/03 (F304)

Parent APAR: OA36403

Child APAR list:

ERROR DESCRIPTION:

New function - Toleration/coexistence for DFSMSdss handling of
RLS user Catalogs.

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: Users of DFSMSdss processing User          *
*                   Catalogs.                                *
*****
* PROBLEM DESCRIPTION: This APAR provides toleration support *
*                   for Record Level Sharing ( RLS )         *
*                   User Catalogs ( BCS ) that are           *
*                   processed using DFSMSdss commands.       *
*****
* RECOMMENDATION:                                           *
```

ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

Users of DFSMSdss attempting to process a User h
in any release prior to zOS V2.1 that is opened in RLS
mode will be failed on lower level systems. Errors that
could be encountered while attempting to process a User
Catalog that is opened in RLS mode in a prior release are
the following:

IEC161I 006-0122

IEC331I 4-86

ADR724E

ADR380E

If a User Catalog is not open in RLS mode on zOS V2.1,

DFSMSdss will successfully process the User Catalog on a lower level system if no errors are encountered.

Documentation changes for OA36422:

The following new reason code Explanation and Application Programmer Response will be added in the z/OS V1.13 MVS System Messages, Vol 1 (ABA-AOM) under existing message ADR380E.

Explanation: 87 The data set is a Catalog open for use with RLS access.

Application Programmer Response: Process the RLS Catalog using z/OS V2R1 or higher.

MODULES/MACROS: ADRCATLG ADRDDFLT ADRDTDSC ADRFMSCT ADRKVDSN
ADRTDDS ADRTDLOG

SRLS: SA22763100

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER: corrected APAR number in responder text.

APAR OA36443 RLS related

Example B-7 contains the cover letter for APAR OA36443.

Example B-7 OA36443

APAR Identifier	OA36443	Last Changed	13/09/13
NEW FUNCTION TOLERATION			
Symptom	NF PERFM	Status	CLOSED UR1
Severity	4	Date Closed	13/04/10
Component	5695DF106	Duplicate of	
Reported Release	A10	Fixed Release	999
Component Name	DFSMS VSAM	Special Notice	
Current Target Date	..13/06/30	Flags	
SCP			
Platform			

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release A10 : UA68691 available 13/04/15 ()
Release B10 : UA68692 available 13/04/15 ()
Release C10 : UA68693 available 13/04/17 (F304)
Release D10 : UA68694 available 13/04/17 (F304)

Parent APAR: OA36415

Child APAR list:

ERROR DESCRIPTION:

Toleration for new funtion

LOCAL FIX:

No local fix

PROBLEM SUMMARY:

* USERS AFFECTED: All users of z/OS HDZ1A10 and above. *

* PROBLEM DESCRIPTION: New function apar. *

* RECOMMENDATION: *

New funtion APAR.
ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

New funtion toleration.

MODULES/MACROS: IDA0A05B IDA0192A IDA0192D IDA0192G IDA0200B
IDA0200T IDA0231B IDA0557E

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA36492 RLS and catalog

Example B-8 contains the cover letter for APAR OA36492.

Note: PTF fixes are in error. See APAR OA42551 and APAR OA42541.

Example B-8 OA36492

APAR Identifier	OA36492	Last Changed	13/08/02
NEW FUNCTION TOLERATION APAR			
Symptom	NF NF	Status	CLOSED UR1
Severity	3	Date Closed	13/04/05
Component	5695DF105	Duplicate of	
Reported Release	A10	Fixed Release	999
Component Name	ICF CATALOG & I	Special Notice	
Current Target Date ..		Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	: UA68641 available 13/04/15 ()
Release B10	: UA68642 available 13/04/15 ()
Release C10	: UA68643 available 13/04/16 (F304)
Release D10	: UA68644 available 13/04/16 (F304)

Parent APAR: OA36403
Child APAR list: OA41517 OA42462

ERROR DESCRIPTION:
New function.

CATKEYS: CAT2012 CATRLS CATNEW

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: New Function *

* PROBLEM DESCRIPTION: TOLERATION SUPPORT for RLS for Catalog. *

```

*                                ZOS0201C/K                                *
*****
* RECOMMENDATION:                                                         *
*****
TOLERATION SUPPORT for RLS for Catalog function.

```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

```

None
**** PE13/06/14 FIX IN ERROR. SEE APAR OA42551  FOR DESCRIPTION
**** PE13/06/14 FIX IN ERROR. SEE APAR OA42541  FOR DESCRIPTION

```

```

MODULES/MACROS:  IGGOCLAH IGGOCLA9 IGGOCLC1 IGGOCLFA IGGOCLFB
IGGOCLFH IGGOCLFK IGGOCLFN IGGOCLFO IGGOCLK0 IGGOCLKV

```

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA36536 PDSE member size

Example B-9 contains the cover letter for APAR OA36563.

Example B-9 OA36536

APAR Identifier	OA36536	Last Changed	13/05/03
COEXISTANCE APAR FOR PDSE			
Symptom	NF NEWFUNCTION	Status	CLOSED UR1
Severity	2	Date Closed	13/04/11
Component	5695DF102	Duplicate of	
Reported Release	A10	Fixed Release	999
Component Name	BASE ACCESS MET	Special Notice	
Current Target Date	..13/05/15	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10 : UA68751 available 13/04/15 ()
Release B10 : UA68752 available 13/04/15 ()
Release C10 : UA68753 available 13/04/16 (F304)
Release D10 : UA68754 available 13/04/16 (F304)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

COEXISTANCE APAR for PDSE.

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All users of PDSEs that have z/OS release *
*                  2.1 installed.                             *
*****
* PROBLEM DESCRIPTION: Compatibility for PDSE.                 *
*****
* RECOMMENDATION:                                             *
*****
Compatibility PTF for PDSE z/OS release 2.1
```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

Compatibility/coexistence PTF for PDSE member size limitation.
ZOS0201C/K

MODULES/MACROS: IGG019BK

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA36576 DFSMShsm Storage Tiers

Example B-10 contains the cover letter for APAR OA36576.

Note: PTFs for this APAR are in error. See APAR OA42529.

Example B-10 OA36576

APAR Identifier OA36576 Last Changed 13/08/14
NEW FUNCTION: Coexistence for Storage Tiers - DFSMShsm Support.

Symptom NF NEW FUNCTION	Status CLOSED UR1
Severity 4	Date Closed 13/04/10
Component 5695DF170	Duplicate of
Reported Release A10	Fixed Release 999
Component Name DFSMSHSM, ISMF	Special Notice ATTENTION
Current Target Date ..13/05/31	Flags
SCP	
Platform	
	XSYSTEM

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10 : UA68714 available 13/04/15 ()
Release B10 : UA68715 available 13/04/15 ()
Release C10 : UA68716 available 13/04/17 (F304)
Release D10 : UA68717 available 13/04/17 (F304)

Parent APAR:

Child APAR list: OA37582

ERROR DESCRIPTION:
NEW FUNCTION

LOCAL FIX:
NEW FUNCTION

PROBLEM SUMMARY:

* USERS AFFECTED: All users of an HSMplex with V2R1 and lower *
* level DFSMSHsm releases. *

* PROBLEM DESCRIPTION: z/OS V1R13, V1R12, V1R11, V1R10 *


```

*           levels of DFSMSHsm need to tolerate      *
*           the DSR, MCD and VSR records expanded   *
*           due to the class transition function     *
*           introduced in V2R1 DFSMSHsm release.     *
*****
* RECOMMENDATION:                                     *
*****
z/OS V1R13, V1R12, V1R11 and V1R10 levels of DFSMSHsm need to
tolerate the DSR, MCD and VSR records expanded due to the class
transition function introduced in V2R1 DFSMSHsm release.

```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

With the new class transition function there is a new type of FSR record. ARCDSCR and ARCVSR records derived from ARCF SR records are expanded to contain the new function information. To avoid record truncation on lower systems, these records must be also expanded and all affected modules recompiled. Work areas using these fields must be expanded as well. To tolerate the expansion of the MCD record, all affected modules must also be recompiled.

Recall.

Recalls will be processed normally if no Last Successful Class Transition Date (LSCTD) exists or the data set Management Class does not contain Class Transition parameters. Recalls will be directed to a V2.1 host using the Common Recall Queue (CRQ) if a CRQ V2R1 host available in the Sysplex and if a LSCTD exists or the data set Management Class does not contain Class Transition parameters. If there is no V2.1 host or CRQ is not established and a data set has gone through a Class Transition rules, then the recalls will fail. The failing message (ARC1179I Reason=7(new)) will be issued to state that the data set should be recalled on a V2.1 system where Class Transition rules can be followed.

DFSMSHsm will not issue a warning message for RECOVER, FRRECOV and ARECOVERs when a LSCTD is lost because backup/recover processing does not lend itself to transitions as space management does.

If a user wants to recall a data set on the pre-V2.1 system then the new MCVTF_SKIP_CLTR flag should be patched ON (PATCH .MCVT.+286 BITS(...1....)). The warning ARC0784I message 'LAST SUCCESSFUL CLASS TRANSITION DATA FOR DATA SET dsname NOT RETAINED DURING THE CLASS TRANSITION' will be issued for the data sets recalled on pre-V2.1 system for which LSCTD exists.

The default value of MCVTF_SKIP_CLTR flag is OFF.

++HOLD(DOC)

V1R10, V1R11, V1R12 and V1R13 MVS System Messages Vol 2(ARC-ASA)
SRL: SA22763218, SA22763219, SA22763220, SA22763221

1. A new reason code RSN07 will be added for RC79 for the
ARC1179I message:

Explanation:

7 - Class Transition rules are not followed for recall
of this data set.

Application Programmer Response:

Use DFSMSHsm V2.1 or higher to recall this data set.

2. ARC0784I message will be extended:

ARC0784I {EXTENDED ATTRIBUTES | LAST SUCCESSFUL CLASS
TRANSITION DATE} FOR DATA SET dsname {WERE | WAS} NOT

RETAINED DURING THE RECALL | RECOVER | ARECOVER

Explanation:

The data set was recalled from a migration copy, but
Class Transition rules are not followed for the recall
of this data set.

z/OS V1R13 DFSMSHsm Data Areas
SRL:GC521083XX

MCDMigration Control Data Set Data Set Record
(Table 84) will be modified:

456(1C8) 392(188) FIXED 4 MCDCSZ_HO High order for MCDSZ.
460(1CC) 396(18C) FIXED 15 MCD_STGR_LENGTH FIXED Storage
group name length
462(1CE) 398(18E) CHAR 30 MCD_STGR_NAME Storage group name
492(1EC) 428(1AC) CHAR 4 MCD_CLTR_DATE CHAR The last class
transition date or zero
496(1F0) 432(1B0) CHAR 24
520(208) 456(1C8) CHARACTER MCDEND End of record.

++HOLD(A0)

SPECIAL CONDITIONS:

AO: MSG=ARC1179I CHANGE=New RSN 07

AO: MSG=ARC0784I NEW=TEXT-INSERT

AO: ENDAO COMMENT='***End of the list of changes
that could affect your automated
operations. The DOC text has
details of the changes.***'

++HOLD MTS

This PTF will not be fully effective on the system it is being applied until the PTF(s) are applied to all systems in the sysplex.

New release

Keyword

z/OS V2R1

ZOS0201C/K

**** PE13/06/13 FIX IN ERROR. SEE APAR OA42529 FOR DESCRIPTION

MODULES/MACROS: ARCASTAI ARCAUDIT ARCAUDMS ARCAXREP ARCAZWC
ARCCP ARCCPQST ARCCPRTN ARCCSTAI ARCCTL ARCCVSR ARCDCOLL
ARCDSTAI ARCESTAI ARCFCRE ARCFDEL ARCFSTAI ARCGDSRV ARCGSTAI
ARCGVDS ARCISTAT ARCJRELP ARCJRPLT ARCJRSEL ARCJSTAE ARCJSTAI
ARCKBLDC ARCKCDSR ARCKDELA ARCKMITA ARCKMIT1 ARCKOTHR ARCKSDSP
ARCKWRIT ARCKWRTT ARCMDSMV ARCMDSUV ARCMLCLN ARCMREC N ARCMSCLN
ARCMMSG ARCMSTAI ARCMVCLN ARCMVDS ARCNCD SW ARCNDEL R ARCNMVR1
ARCNMVR2 ARCNOTR1 ARCNRCHK ARCNRDSI ARCNRDS1 ARCNVRFY ARCNVRF1
ARCNWRIT ARCPRLOG ARCPRQ ARCRCYPQ ARCREUIM ARCRNVDS ARCRPT
ARCRSTR ARCRVDS ARCSMINT ARCUDATA ARCUTIL ARCWSMS ARCXSTAI
ARCYSTAI ARCZMSG S ARCZREAD ARCZRNX T ARCZUPDT

SRLS: SA22763218 GC521083XX SA22763219 SA22763220
SA22763221

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA36916 RLS

Example B-11 contains the cover letter for APAR OA36916.

Example B-11 OA36916

APAR Identifier	OA36916	Last Changed	13/09/13
NEW FUNCTION COMPROUTE OA36403			
Symptom	NF NF	Status	CLOSED UR1
Severity	3	Date Closed	13/04/05
Component	5695DF106	Duplicate of	
Reported Release	A10	Fixed Release	999
Component Name	DFSMS VSAM	Special Notice	
Current Target Date	..13/04/30	Flags	
SCP			
Platform			

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release A10 : UA68641 available 13/04/15 ()
Release B10 : UA68642 available 13/04/15 ()
Release C10 : UA68643 available 13/04/16 (F304)
Release D10 : UA68644 available 13/04/16 (F304)

Parent APAR: OA36403

Child APAR list:

ERROR DESCRIPTION:

New function.

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: ALL RLS users. *

* PROBLEM DESCRIPTION: RLS Toleration APAR. *

* RECOMMENDATION: *

RLS Toleration APAR.

ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

RLS Toleration APAR.

MODULES/MACROS: IDA0192A IDA0192C IDA0192X IDA0200B IDA0200T

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA37336 PDSE in GDG DSS support

Example B-12 contains the cover letter for APAR OA37336.

Example B-12 OA37336

APAR Identifier OA37336 Last Changed 13/05/03
NEW FUNCTION - DFSMSDSS TOLERATION/COEXISTENCE OF PDSE DATA SETS
DEFINED AS A GENERATION DATA SET (GDS).

Symptom NF NEWFUNCTION Status CLOSED UR1
Severity 3 Date Closed 13/04/10
Component 5695DF175 Duplicate of
Reported Release A10 Fixed Release 999
Component Name DFSMSDSS, ISMF Special Notice ATTENTION
Current Target Date .. Flags
SCP NEW FUNCTION
Platform

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10 : UA68720 available 13/04/15 ()
Release B10 : UA68721 available 13/04/15 ()
Release C10 : UA68722 available 13/04/16 (F304)
Release D10 : UA68723 available 13/04/16 (F304)

Parent APAR:

Child APAR list: OA37349

ERROR DESCRIPTION:

TOLERATION/COEXISTENCE for DFSMSdss handling of PDSE
data sets defined as a Generation Data Set (GDS)

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: Users of DFSMSdss COPY, DUMP, RESTORE and *
* CONVERTV. *

* PROBLEM DESCRIPTION: This APAR provides toleration support *
* for PDSE data sets defined as a GDS *
* (Generation Data Set) that are *
* processed using DFSMSdss commands. *

* RECOMMENDATION: *

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
ZOS0201C/K

Users of DFSMSdss attempting to process an SMS-managed PDSE data set defined as a GDS in any release prior to z/OS V2.1 will be failed.

Documentation changes for OW37336:

The following new reason code explanation will be added to the zOS V1.13 MVS System Messages, Vol 1 (ABA-AOM) under existing message ADR285E:

Explanation: 15 The PDSE data set is defined as a generation data set (GDS).

The following new reason code explanation will be added to the zOS V1.13 MVS System Messages, Vol 1 (ABA-AOM) under existing message ADR778E:

Explanation: 20 The PDSE data set is defined as a GDS.

MODULES/MACROS: ADRDDFLT ADRD TDSC ADRFDS DU ADRFDSRL ADRFDSRS
ADRFRLBO ADRKVDSF ADRPCNV S ADRPPRNV ADRPREDS

SRLS: SA22763100

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA37349 PDSE in GDG DFSMSHsm support

Example B-13 contains the cover letter for APAR OA37349.

Example B-13 OA37349

APAR Identifier OA37349 Last Changed 13/05/03
NEW FUNCTION - DFSMSHSM TOLERATION COEXISTENCE FOR PDSE DATA
SETS DEFINED AS GENERATION DATA SETS (GDS)

Symptom NF NEWFUNCTION Status CLOSED UR1
Severity 4 Date Closed 13/04/10
Component 5695DF170 Duplicate of
Reported Release A10 Fixed Release 999
Component Name DFSMSHSM, ISMF Special Notice
Current Target Date ..13/07/31 Flags
SCP
Platform

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10 : UA68728 available 13/04/15 ()
Release B10 : UA68729 available 13/04/15 ()
Release C10 : UA68730 available 13/04/17 (F304)
Release D10 : UA68731 available 13/04/17 (F304)

Parent APAR: OA37336
Child APAR list:

ERROR DESCRIPTION:

DFSMSHsm toleration / coexistence support for PDSE generation data sets (GDS).

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: Users with DFSMSHsm hosts on z/OS V2R1 *
* which share CDSs and process PDSEs that are *
* generation data sets on hosts running at *
* z/OS V1R13, V1R12, V1R11, and/or V1R10 *

* PROBLEM DESCRIPTION: The generation data groups (GDGs) *
* support for PDSE generation data sets *
* (GDSs) in z/OS V2R1 requires z/OS *
* V1R10 through V1R13 hosts to be *
* updated. *

* RECOMMENDATION: *

Changes in the GDG support for PDSE in z/OS V2R1 require that z/OS V1R10 through V1R13 hosts to be updated.

.
Additional keywords:
ZOS0201C/K
DFSMSdss OA37366

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

DFSMSShsm has been modified so that the new PDSE generation data sets can be detected and handled appropriately.

.
With the toleration/coexistence PTF installed, SMS managed PDSE generation data sets created by z/OS V2R1 hosts will be failed by DFSMSShsm or DFSMSdss on z/OS pre-V2R1 hosts with an existing DFSMSShsm message or the new DFSMSdss MSGADR778E RC20.

.
MSGARC1001I dsn MIGRATE FAILED, RC=0087, REAS=0006
MSGARC1287I A DISCREPANCY WAS FOUND IN THE DATA SET VTOC ENTRY

.
MSGARC1001I dsn BACKDS FAILED, RC=0087, REAS=0006
MSGARC1387I A DISCREPANCY WAS FOUND IN THE DATA SET VTOC ENTRY

.
MSGARC6172E DATA SET dsn IS NOT SUPPORTED IN AN ALLOCATE LIST FOR AGGREGATE GROUP group_name

.
MSGADR778E DATA SET dsn WAS NOT SELECTED BECAUSE THE DATA SET TYPE IS NOT SUPPORTED IN THIS RELEASE, 20

.
The following publication updates will be made:
z/OS MVS System Messages, Vol 2 (ARC-ASA), SA22-7632
SRL: SA22763218, SA22763219, SA22763220, SA22763221
Update the Application Programmer Response section and add the following to the examples of unsupported situations for ABACKUP:

- datasetname represents an SMS-managed PDSE generation data set. Use DFSMSShsm V2R1 or higher release to process the data set.

.
Add following text to the ARECOVER section:
Ensure the data sets specified in the ALLOCATE list are acceptable for processing. Some examples of unsupported data sets are:
- datasetname represents an SMS-managed PDSE generation data set. Use DFSMSShsm V2R1 or higher release to process the data set.

MODULES/MACROS: ARCKALLC ARCKBLDC ARCMDSUV ARCNOTR1

SRLS: SA22763218 SA22763219 SA22763220 SA22763221

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA37582 DSS support of DFSMSHsm Storage Tier

Example B-14 contains the cover letter for APAR OA37582.

Example B-14 OA37582

APAR Identifier	OA37582	Last Changed	13/05/03
NEW FUNCTION - DFSMSDSS SUPPORT OF LAST CLASS TRANSITION DATE TOLERATION/COEXISTENCE			
Symptom	NF NEW FUNCTION	Status	CLOSED UR1
Severity	4	Date Closed	13/04/10
Component	5695DF175	Duplicate of	
Reported Release	A10	Fixed Release	999
Component Name	DFSMSDSS, ISMF	Special Notice	ATTENTION
Current Target Date ..		Flags	
SCP			NEW FUNCTION
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	: UA68710 available 13/04/15 ()
Release B10	: UA68711 available 13/04/15 ()
Release C10	: UA68712 available 13/04/16 (F304)
Release D10	: UA68713 available 13/04/16 (F304)

Parent APAR: OA36576

Child APAR list:

ERROR DESCRIPTION:

NEW FUNCTION - Toleration/Coexistence for DFSMSdss handling
of DFSMSHsm Storage Tiers last class transition date

LOCAL FIX:
NEW FUNCTION

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All users of DFSMSdss V1R13, V1R12, V1R11      *
*                  and V1R10 attempting to COPY or RESTORE a      *
*                  data set with a last class transition date     *
*                  from a DFSMShsm storage tier process.          *
*****
* PROBLEM DESCRIPTION: This APAR provides toleration support     *
*                  when attempting to restore or copy a          *
*                  data set that has a last class                 *
*                  transition date from a DFSMShsm                *
*                  storage tier process.                           *
*****
* RECOMMENDATION:                                                *
*****
ZOS0201C/K
```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

z/OS DFSMSdss V1R13, V1R12, V1R11 and V1R10 are modified to restore or copy data that has a last class transition date from a DFSMShsm storage tier process.

The last successful class transition date will be maintained the same except for preserving the date from the source. For these cases, message ADR556W will be issued with reason code of 3 to indicate that the class transition date was lost.

For preallocated data sets that are replaced, the date will not be modified. For preallocated data sets that are scratched and re-allocated, the date will be lost and ADR556W is issued.

When a data set is not replacing a preallocated target, but is being renamed or the management class is changing, then the date is cleared and ADR556W is not issued.

When a data set is not replacing a preallocated target, not being renamed and the management class didn't change, then the message ADR556W is issued indicating the date was lost.

ADR556W is an existing message, but reason code 3 is new. Also, the word "EXTENDED" has been removed from the message text.

ADR556W (ttt)-,mmm(yy), ATTRIBUTES FOR DATA SET dsname

WERE NOT RETAINED DURING {COPY | RESTORE},rsn

Explanation: The data set was copied or restored but some extended attributes were lost for the following reasons (rsn):

- 1 Vendor attributes from the F9 DSCB of the dumped data set were not retained for the target data set because the volume on which it was placed did not support F8/F9 DSCBs.
- 2 Vendor attributes from the F9 DSCB of the dumped or copied data set were not retained for the target data set due to problems updating the target's F9 DSCB.
- 3 The last successful class transition date of the dumped or copied data set was not retained for the target data set because the release of z/OS on which it was processed did not support it.

System action: None.

Application Programmer Response:

1. If the extended attributes are desired, rerun the COPY or RESTORE and specify target volumes or an SMS group that supports F8/F9 DSCBs.
2. An error occurred while reading or writing an F9 DSCB. Retry the restore or copy for the data set.
3. If the last successful class transition date is desired, rerun the COPY or RESTORE on a release that supports it.

MODULES/MACROS: ADRCNVSM ADRCVSAM ADRMSGs ADRPCNVX ADRPCVSX
ADRSB105 ADRTDDS ADRTDLOG ADRTDPNV ADRXVSAM

SRLS: SA22763100

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA38185 EXCP support for zHPF on z/OS V1.12 and V1.13

Example B-15 contains the cover letter for APAR OA38185.

Example B-15 OA38185

APAR Identifier OA38185 Last Changed 13/10/07
NEW FUNCTION APAR - EXCP VIRTUAL SUPPORT FOR ZHPF

Symptom NF FUNCTION Status CLOSED UR1
Severity 3 Date Closed 12/10/04

Component	5752SC1C3	Duplicate of	
Reported Release	770	Fixed Release	999
Component Name	IOS	Special Notice	ATTENTION
Current Target Date ..		Flags	
SCP			NEW FUNCTION
Platform			

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release 770 : UA66846 available 12/10/17 (F210)
 Release 780 : UA66847 available 12/10/17 (F210)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

NEW FUNCTION APAR

LOCAL FIX:

PROBLEM SUMMARY:

```

*****
* USERS AFFECTED: Users at HBB770 and above.                                *
*****
* PROBLEM DESCRIPTION: New function APAR to further enhance                  *
*                       the High Performance FICON for System z              *
*                       (zHPF) facility.                                     *
*                       (D/T2817, D/T2818, D/T2107)                         *
*****
* RECOMMENDATION:                                                            *
*****
New function APAR to further enhance the High Performance
FICON for System z (zHPF) facility.
  
```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

This APAR provides EXCP virtual support for zHPF.

The following Publication updates will be made:

. z/OS MVS System Codes (SA22-7626)

ABEND Code 800

Explanation: During processing of an I/O request, execute channel program (EXCP) processing encountered an error.

...

Code Explanation

01 An error occurred during channel program validity check processing. Either an access protection error or a page fix error occurred when:

* The EXCP IDAW routine tried to access the IDAW using an incorrect storage protection key.

* The EXCP CCW validity check routine tried to access the virtual channel program using an incorrect storage protection key.

* The CCW channel program attempted to execute a restricted operation.

* The EXCP zHPF validity check routine tried to access the virtual channel program using an incorrect storage protection key.

* An error occurred during page-fix or page-unfix processing. A page-fix error can occur if the EXCP processor tries to fix pages that are not assigned to the caller's address space identifier (ASID).

Note: An access protection or page-fix error can occur when there is an error in the channel program. Correct the channel program.

02 System error. Search problem reporting databases for a fix for the problem. If no fix exists, contact the IBM Support Center.

03 An error occurred during page-fix processing for a data extent block (DEB) for an EXCP V=R request.

...

09 The IOBE specified that this is a zHPF channel program request, but zHPF channel programs are only supported for EXCPVR and EXCP virtual requests for non-VIO data sets. They are not supported for VIO data sets or EXCP V=R requests.

...

- OD A zHPF channel program for an EXCP virtual request specified a TCCB that crosses a page boundary.
- OE The number of virtual TIDAWs in a zHPF channel program has exceeded the maximum allowed value (65,536). This can occur due to an error in the channel program. For example:
 - * A virtual TIDAW in the list is linked to a previous virtual TIDAW causing a loop.
 - * An incorrect TIDAL virtual storage address is specified in the TCW.
 - * The last virtual TIDAW in the list does not have the last TIDAW in the list flag turned on.

...

. *-----|

. MVS Data Areas Volume 2 (GA32-0854)

TCCW Translator Fix List(IECDFIX)

Update to FIX Information section:

FIX Map:

OFFSET	HEX	TYPE	LENGTH	NAME (DIM)	DESCRIPTION
=====	=====	=====	=====	=====	=====
(0)		STRUCTURE	0	FIX	
(0)		ADDRESS	4	FIXCHAIN	Fix block chain pointer
(4)		ADDRESS	1	FIXInUse	Number of fix list entries in use - used only by the zHPF translator
(5)		CHARACTER	3	*	Reserved.
...					
(8)		ADDRESS	4	FIXLSTST	Start address of area to be fixed
	1...			FIXCONT	"X'80'" Fix list continuation flag
(C)		ADDRESS	4	FIXLSTEN	End address of area to be fixed
	1...			LASTENT	"X'80'" Last fix entry flag
	1...			FIXLast	"X'80'" Last fix entry flag

...

FIX List Equates

(C) X'8'	0	FIXHL	"FIXLSTST-FIX" Header length
(C) X'8'	0	FIXEL	"FIXLSTEN+L'FIXLSTEN-FIXLSTST" Fix list entry length
(C) X'13'	0	FIXNE	"19" Number of fix list entries- 160 byte block caller
(C) X'1E'	0	FIXNEL	"30" Number of fix list entries- 248 byte block caller
(C) X'F8'	0	FIXBL	"FIXHL+FIXEL*FIXNEL" Size of fix list block

. *-----|

. MVS Data Areas Volume 3 (GA32-0855)

zHPF Channel Program Information Area (IOSDZHPF)

Update to IOSDZHPF Information section:

IOSDZHPF Map:

OFFSET

HEX	TYPE	LENGTH	NAME (DIM)	DESCRIPTION
=====	=====	=====	=====	=====
(0)	STRUCTURE	32	ZHPF_INFO	zHPF Channel Program Information Area
(0)	UNSIGNED	1	ZHPF_VERSION	Version number
(1)	UNSIGNED	1	ZHPF_FLAG1	Capabilities flag 1
1... ..			ZHPF_BIDI	bidirectional data transfer Indicates that all of the online paths for the device support bidirectional data for zHPF I/O requests.
.1.. ..		1	ZHPF_EXCPVR	zHPF is supported for EXCPVR requests
..1.		1	ZHPF_Incorrect_Len	The incorrect length facility is supported by the processor
...1		1	ZHPF_EXCP	zHPF is supported for EXCP virtual requests
.... 1111		*		Reserved
(2)	CHARACTER	2	*	Reserved
(4)	UNSIGNED	4	ZHPF_MAXXFERSIZE	Maximum amount of data (in bytes) that can be

transferred in a single
Transport Control Area
(TCA)

...

. *-----|

. MVS Data Areas Volume 6 (GA32-0858)

Channel Program Scan Exit Parm List/Work Area (IEXCPS)

Update to XCPS Information section:

XCPS Map:

OFFSET	HEX	TYPE	LENGTH	NAME (DIM)	DESCRIPTION
=====	=====	=====	=====	=====	=====
(0)		STRUCTURE	0	CPS	
(0)		CHARACTER	4	CPSCPS	CPS acronym
(4)		BITSTRING	1	CPSENTRY	Entry reason byte
		...1		CPSENSIO	"X'01'" STARTIO entry
		..1.		CPSENIOE	"X'02'" I/O error entry
		..11		CPSENEOE	"X'03'" End of Extent entry
		.1..		CPSENML	"X'04'" Normal-end entry
(5)		CHARACTER	3	CPSRESV1	Reserved
(8)		ADDRESS	4	CPSRQE	EXCP RQE address
(C)		ADDRESS	4	CPSIOSB	EXCP IOSB address
(10)		ADDRESS	4	CPSCPX	CPS extension address
(14)		CHARACTER	12	CPSRESV2	Reserved
(20)		CHARACTER	216	CPSWA	Work area for use by the channel program scan exit, includes prefix CCWs, set to 0's on initial entry
(20)		CHARACTER	104	CPS_THPFRegs	IECVTHPF register save area during zHPF channel program translation
(20)	X'F8'		0	CPSLEN	"*" CPS block length

KEYWORDS: ZHPF/K

MODULES/MACROS: IECDFIX IECDXCPS IECVEXCP IECVEXFR IECVEXPR
IECVBLK IECVTHPF IECVCCW IOSDIOBE IOSDZHPF IOSVZHPF

SRLS: SA22762600 GA32085400 GA32085500 GA32085800

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39530 PDSE V2

Example B-16 contains the cover letter for APAR OA39530.

Example B-16 OA39530

APAR Identifier	OA39530	Last Changed	13/06/03
COEXISTANCE APAR FOR PDSE			
Symptom	NF NEWFUNCTION	Status	CLOSED UR1
Severity	2	Date Closed	13/04/17
Component	5695DF115	Duplicate of	
Reported Release	B10	Fixed Release	999
Component Name	EXTENDED DATA S	Special Notice	
Current Target Date	..13/05/31	Flags	
SCP			
Platform			

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release C10 : UA68849 available 13/05/07 (F305)
Release D10 : UA68850 available 13/05/07 (F305)

Parent APAR:

Child APAR list: OA41790 OA41864

ERROR DESCRIPTION:

Coexistence Apar for PDSE

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: All PDSE users that have the ability to *
* create PDSE V2 (z/OS V2R1) and want to be *
* able to access in lower releases (R12 and *
* R13). *

* PROBLEM DESCRIPTION: This APAR will allow to process PDSE V2 *
 * datasets in Releases 12 and 13. PDSE V2 *
 * datasets are created in V2R1 and above *
 * releases. *

* RECOMMENDATION: *

Modify PDSE code to allow the process of PDSE V2 datasets in R12
 and R13

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

Allow the process of PDSE V2 datasets in lower releases (R12 and
 R13)

ZOS0201C/K - coexistence keyword for z/OS v2.1

MODULES/MACROS: IGWAMAMT IGWAMCVO IGWAMOP0 IGWBIRP1 IGWBIWP1
 IGWBVLP1 IGWBVRP1 IGWDAASU IGWDACND IGWDACN2 IGWDACRD IGWDARD1
 IGWDAV00 IGWDAV10 IGWDAV20 IGWDBHOM IGWDDCR2 IGWDDCR3 IGWDDDSF
 IGWDLCLS IGWDLFAS IGWDLFNS IGWDLGAS IGWDLINI IGWDLISA IGWDLNAS
 IGWDLRFL IGWDPRSP IGWDRRDR IGWDRRD3 IGWDRRRC IGWDRRRX IGWDRSVF
 IGWDRUPR IGWDSRNA IGWDSSTX IGWICONM IGWICPSC IGWICPSG IGWICPSP
 IGWICREM IGWIFCOM IGWIFMLT IGWIGSIB IGWIIDEL IGWIIUW IGWIMAIN
 IGWINEWT IGWINITM IGWIREAD IGWISRCH IGWISTRE IGWIVRJN IGWIVRJ2
 IGWIVRSC IGWIVRSG IGWIVRSP IGWIOIP3

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39551 SAM EF V2 compatibility

Example B-17 contains the cover letter for APAR OA39551.

Example B-17 OA39551

APAR Identifier OA39551 Last Changed 13/06/03
 NEW FUNCTION TOLERATION

Symptom	NF NEW FUNCTION	Status	CLOSED UR1
Severity	3	Date Closed	13/04/17
Component	5695DF107	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	DATA MGMT SUPPO	Special Notice	
Current Target Date	..13/05/15	Flags	
SCP			
Platform			

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release C10 : UA68823 available 13/05/02 (F305)
 Release D10 : UA68824 available 13/05/03 (F305)

Parent APAR:

Child APAR list: 0A39869

ERROR DESCRIPTION:

Coexistence for future new function.

LOCAL FIX:

PROBLEM SUMMARY:

```

*****
* USERS AFFECTED: All users of SAM EF version 2 data sets.      *
*****
* PROBLEM DESCRIPTION: Compatibility support for SAM EF          *
*                        version 2 data sets.  z/OS version 2.1  *
*                        is required to create SAM EF version 2  *
*                        data sets.                               *
*****
* RECOMMENDATION:                                              *
*****
Compatibility support for SAM EF version 2 data sets.
  
```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

Compatibility/coexistence support for SAM EF version 2 data sets. ZOS0201C/K

MODULES/MACROS: IECEQU IECPDINI IEZDEB IFGSMF14 IFGWAX
IFG019RS IFG0193A IFG0194D IFG0195H IFG0196V IFG0202I IFG0232D
IFG0550P IFG0553X IFG0554N IFG0555H

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39618 Catalog

Example B-18 contains the cover letter for APAR OA39618.

Example B-18 OA39618

APAR Identifier OA39618 Last Changed 13/05/03
CO-EXISTENCE APAR FOR NEW FUNCTION

Symptom NF NEWFUNCTION	Status CLOSED UR1
Severity 4	Date Closed 13/04/08
Component 5695DF105	Duplicate of
Reported Release D10	Fixed Release 999
Component Name ICF CATALOG & I	Special Notice
Current Target Date ..13/04/30	Flags
SCP	
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	: UA68659 available 13/04/15 ()
Release B10	: UA68660 available 13/04/15 ()
Release C10	: UA68661 available 13/04/16 (F304)
Release D10	: UA68662 available 13/04/16 (F304)

Parent APAR:

Child APAR list: OA39619 OA39620 OA39621

ERROR DESCRIPTION:

THIS IS A CO-EXISTENCE APAR FOR NEW FUNCTION

CATKEYS: CAT2012 CATNEW

LOCAL FIX:
N/A

PROBLEM SUMMARY:

* USERS AFFECTED: All users of z/OS HDZ1A10 and above. *

* PROBLEM DESCRIPTION: NEW FUNCTION APAR. *

* RECOMMENDATION: *

SUMMARY:
NEW FUNCTION APAR

ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
NEW FUNCTION APAR

MODULES/MACROS: IGGOCLEQ IGGOCLEV

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39619 VSAM

Example B-19 contains the cover letter for APAR OA39619.

Example B-19 OA39619

APAR Identifier OA39619 Last Changed 13/09/13
CO-EXISTENCE APAR FOR NEW FUNCTION

Symptom	NF NEWFUNCTION	Status	CLOSED UR1
Severity	4	Date Closed	13/04/08
Component	5695DF106	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	DFSMS VSAM	Special Notice	
Current Target Date	..13/04/30	Flags	
SCP			
Platform			

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release A10	:	UA68659	available	13/04/15	(		)
Release B10	:	UA68660	available	13/04/15	(		)
Release C10	:	UA68661	available	13/04/16	(F304		)
Release D10	:	UA68662	available	13/04/16	(F304		)

Parent APAR: OA39618

Child APAR list:

ERROR DESCRIPTION:

THIS IS A CO-EXISTENCE APAR FOR NEW FUNCTION

LOCAL FIX:

N/A

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All users of z/OS HDZ1A10 and above.          *
*****
* PROBLEM DESCRIPTION: NEW FUNCTION APAR.                        *
*****
* RECOMMENDATION:                                                *
*****
```

NEW FUNCTION APAR

ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
Co-existence support for new function.

MODULES/MACROS: IDA0192C

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39620 DSS

Example B-20 contains the cover letter for APAR OA39620.

Example B-20 OA39620

APAR Identifier OA39620 Last Changed 13/05/03
CO-EXISTENCE APAR FOR NEW FUNCTION

Symptom NF NEWFUNCTION	Status CLOSED UR1
Severity 4	Date Closed 13/04/11
Component 5695DF175	Duplicate of
Reported Release D10	Fixed Release 999
Component Name DFSMSDSS, ISMF	Special Notice ATTENTION
Current Target Date ..	Flags
SCP	NEW FUNCTION
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	: UA68740 available 13/04/15 ()
Release B10	: UA68741 available 13/04/15 ()
Release C10	: UA68742 available 13/04/16 (F304)
Release D10	: UA68743 available 13/04/16 (F304)

Parent APAR: OA39618
Child APAR list:

ERROR DESCRIPTION:
THIS IS A CO-EXISTENCE APAR FOR NEW FUNCTION

LOCAL FIX:
N/A

PROBLEM SUMMARY:

* USERS AFFECTED: All users of DFSMSdss z/OS HDZ1A10 *
* and above. *

* PROBLEM DESCRIPTION: NEW FUNCTION APAR. *

* RECOMMENDATION: *

ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
NEW FUNCTION APAR

MODULES/MACROS: ADRBLDCN ADRCATLG ADRDDFLT ADRTDSC ADRFDSOU
ADRFDSRL ADRKVDSF ADRPCVSM ADRSB004 ADRTDDS

SRLS: SA22763100

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39621 DFSMSHsm

Example B-21 contains the cover letter for APAR OA39621.

Example B-21 OA39621

APAR Identifier OA39621 Last Changed 13/05/30
NEW FUNCTION: Coexistence for DFSMSHsm Non-SMS VSAM LDS > 4GB
Support

Symptom NF INCORROUT	Status CLOSED UR1
Severity 4	Date Closed 13/04/11
Component 5695DF170	Duplicate of
Reported Release D10	Fixed Release 999
Component Name DFSMSHSM, ISMF	Special Notice ATTENTION
Current Target Date ..13/05/31	Flags
SCP	
Platform	

XSYSTEM

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10 : UA68756 available 13/04/15 ()
Release B10 : UA68757 available 13/04/15 ()
Release C10 : UA68758 available 13/04/17 (F304)
Release D10 : UA68759 available 13/04/17 (F304)

Parent APAR: OA39618
Child APAR list:

ERROR DESCRIPTION:

THIS IS A CO-EXISTENCE APAR FOR NEW FUNCTION

LOCAL FIX:

N/A

PROBLEM SUMMARY:

* USERS AFFECTED: All users of DFSMSHsm in an HSMplex with *
* V2R1 and lower level DFSMSHsm releases. *

* PROBLEM DESCRIPTION: z/OS V1R10, V1R11, V1R12 and V1R13 *
* levels of DFSMSHsm need to fail *
* backup, migration and ABACKUP of *
* non-SMS VSAM LDS with the Extended *
* Addressability(EA) attribute *
* specified. *

* RECOMMENDATION: *

z/OS V1R10, V1R11, V1R12 and V1R13 levels of DFSMSHsm need to
fail backup, migration and ABACKUP of non-SMS VSAM LDS

with the Extended Addressability(EA) attribute specified.

New release	Keyword
z/OS V2R1	ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

On V2R1, non-SMS VSAM LDS's with the Extended Addressability(EA) are fully supported. The z/OS V1R10, V1R11, V1R12 and V1R13 levels of DFSMSHsm need to fail backup, migration and ABACKUP this type of data set.

Non-SMS EA LDSes migrated, backed up or ABACKUPed on V2.1 system will be able to be recalled, recovered or arecovered on V1R10, V1R11, V1R12 and V1R13 levels of DFSMSHsm.

++HOLD(DOC)

V1R10, V1R11, V1R12 and V1R13
MVS System Messages Vol 2 (ARC-ASA)
SRL: SA22763218, SA22763219, SA22763220, SA22763221

1. New reason code RSN67 will be added for RC58
for the ARC1258I message

Explanation:

67 - The data set being migrated is a non-SMS-managed VSAM linear data set which has the Extended Addressable(EA) attribute specified. This data set type is not supported for migration in pre-V2R1 systems.

Application Programmer Response:

Use DFSMSHsm V2R1 or higher to migrate this data set.

2. New reason code RSN67 will be added for RC56
for the ARC1356I message

Explanation:

67 - The data set being backed up is a non-SMS-managed VSAM linear data set which has the Extended Addressable(EA) attribute specified. This data set type is not supported for backup in pre-V2R1 systems.

Application Programmer Response:

Use DFSMSHsm V2R1 or higher to backup this data set.

3. New reason code RSN67 will be added for RC58
for the ARC1358I message

Explanation:

67 - The data set being backed up is a non-SMS-managed

VSAM linear data set which has the Extended Addressable(EA) attribute specified. This data set type is not supported for backup in pre-V2R1 systems.

Application Programmer Response:

Use DFSMSHsm V2R1 or higher to backup this data set.

4. ARC6172E message - a new text will be added to the Application Programmer Response section.

Application Programmer Response: For ABACKUP, ensure

that all data sets specified in the INCLUDE, ALLOCATE, or ACCOMPANY lists are acceptable for processing.

Some examples of unsupported situations are:

- data-set-name represents a GDG base.
- The data set organization is unsupported.
- The data set BLOCKSIZE or LRECL is invalid or larger than supported by the ABARS release being executed.
- data-set-name does not reside on a supported device.
- data-set-name is in the ACCOMPANY list, but is migrated.
- data-set-name is a z/FS data set.
- Non-VSAM data-set-name data set has Format 8 DSCB.
- Non-SMS VSAM linear data set which has the Extended Addressable(EA)attribute specified. Use DFSMSHsm V2R1 or higher to perform ABACKUP for this data set.

New release	Keyword
z/OS V2R1	ZOS0201C/K

MODULES/MACROS: ARCBDSMP ARCKALLC ARCKBLDC ARCKFILT ARCKFIL1
ARCKFIND ARCKGDGR ARCKPAG1 ARCKUCAT ARCNDCLR ARCNFIND ARCNLGDG
ARCNVIFY ARCNVRFY ARCVVSC ARWCLOC ARWCNT ARWCTL ARCWLOC
ARCWLOC ARWLSTC

SRLS: SA22763218 SA22763219 SA22763220 SA22763221

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39869 SAM EF V2

Example B-22 contains the cover letter for APAR OA39869.

Example B-22 OA39869

APAR Identifier	OA39869	Last Changed	13/06/03
NEW FUNCTION - TOLERATION			
Symptom	NF NEW FUNCTION	Status	CLOSED UR1
Severity	3	Date Closed	13/04/17
Component	5695DF102	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	BASE ACCESS MET	Special Notice	
Current Target Date	..13/05/15	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release C10 : UA68823 available 13/05/02 (F305)
Release D10 : UA68824 available 13/05/03 (F305)

Parent APAR: OA39551

Child APAR list: OA39870 OA39871 OA39872 OA39873 OA39875 OA40259
OA40477

ERROR DESCRIPTION:

Coexistence for future new function.

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All users of SAM EF version 2 data sets.      *
*****
* PROBLEM DESCRIPTION: Compatibility support for SAM EF          *
*                      version 2 data sets.  z/OS version 2.1    *
*                      is required to create SAM EF version 2    *
*                      data sets.                                *
*****
* RECOMMENDATION:                                              *
*****
Compatibility support for SAM EF version 2 data sets.
```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

Compatibility/coexistence support for SAM EF version 2 data sets. ZOS0201C/K

MODULES/MACROS: IGGESDR0 IGGESDR1 IGGESDR2 IGGESDR3 IGG019BW
IGG019VX IGG019V7 IGG019V8 IGG019V9 IGG0191N IGG0193V IGG0193W
IGG0196I IGG0201V IGG055V1

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39871 SAM EF V2

Example B-23 contains the cover letter for APAR OA39871.

Example B-23 OA39871

APAR Identifier	OA39871	Last Changed	13/05/03
NEW FUNCTION TOLERATION			
Symptom	NF NEW FUNCTION	Status	CLOSED UR1
Severity	3	Date Closed	13/04/10
Component	5695DF103	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	ACCESS METHOD S	Special Notice	
Current Target Date	..13/06/30	Flags	
SCP			
Platform			

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release C10 : UA68732 available 13/04/16 (F304)
Release D10 : UA68733 available 13/04/16 (F304)

Parent APAR: OA39869
Child APAR list:

ERROR DESCRIPTION:
Coexistence for future new function.

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: All users of SAM EF version 2 data sets *
* created on a z/OS V2R1 or above system. *

* PROBLEM DESCRIPTION: Compatibility support for SAM EF *
* version 2 data sets. z/OS version 2.1 *
* is required to create SAM EF version 2 *
* data sets. *

* RECOMMENDATION: *

Summary: Compatibility support for SAM EF version 2 data sets.
FIXCAT KEYWORD: ZOS0201C/K
This APAR COREQs OA39872.

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
Compatibility/coexistence support for SAM EF version 2 data
sets. FIXCAT KEYWORD: ZOS0201C/K

MODULES/MACROS: IDCDC01 IDCDC02 IDCDC03 IDCDOUT IDCLC01
IDCLC02 IDCLC03 IDCLC04 IDCLC05 IDCLC06 IDCLC07 IDCLC08
IDCLC09 IDCLC10 IDCRP01 IDCTSLC0

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39872 Catalog

Example B-24 contains the cover letter for APAR OA39872.

Example B-24 OA39872

APAR Identifier	OA39872	Last Changed	13/05/03
NEW FUNCTION TOLERATION			
Symptom	NF NEW FUNCTION	Status	CLOSED UR1
Severity	3	Date Closed	13/04/10
Component	5695DF105	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	ICF CATALOG & I	Special Notice	
Current Target Date	..13/09/30	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release C10 : UA68732 available 13/04/16 (F304)
Release D10 : UA68733 available 13/04/16 (F304)

Parent APAR: OA39869

Child APAR list:

ERROR DESCRIPTION:

Coexistence for future new function.

CATKEYS: CAT2013 CATNEW

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All users sharing systems with z/OSV2R1      *
*                   or above.                                     *
*****
* PROBLEM DESCRIPTION: Toleration for new function in z/OS      *
*                   V2R1.                                         *
*****
* RECOMMENDATION:                                               *
```

Summary: FIXCAT KEYWORD: ZOS0201C/K
This APAR COREQs OA39871.

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
Apply maintenance if sharing with a system using z/OS V2R1 or
above.

MODULES/MACROS: IGGOCLE0 IGGOCLFE IGGOCLKN

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA39873 SAM EF V2 DSS

Example B-25 contains the cover letter for APAR OA39873.

Example B-25 OA39873

APAR Identifier	OA39873	Last Changed	13/06/03
NEW FUNCTION - DFSMSDSS SUPPORT FOR VERSION 2 EXTENDED FORMAT SEQUENTIAL DATA SETS TOLERATION/COEXISTENCE			
Symptom	NF NEW FUNCTION	Status	CLOSED UR1
Severity	3	Date Closed	13/04/17
Component	5695DF175	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	DFSMSDSS, ISMF	Special Notice	ATTENTION
Current Target Date ..		Flags	
SCP			NEW FUNCTION
Platform			

Status Detail: SHIPMENT - Packaged solution is available for
shipment.

PE PTF List:

PTF List:

Release C10 : UA68831 available 13/05/02 (F305)

Release D10 : UA68832 available 13/05/03 (F305)

Parent APAR: OA39869

Child APAR list:

ERROR DESCRIPTION:

NEW FUNCTION - TOLERATION/COEXISTENCE for DFSMSdss handling
of version 2 extended format sequential data sets.

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: Users of DFSMSdss copy and restore          *
*                  processing single striped multi-volume      *
*                  extended format sequential data sets.       *
*                                                              *
*****
* PROBLEM DESCRIPTION: This APAR provides toleration support  *
*                  for single striped multi-volume            *
*                  extended format sequential Version 2       *
*                  data sets.                                  *
*****
* RECOMMENDATION:                                             *
```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

DFSMSdss will be modified to process version 2
extended format single striped multi-volume sequential
data sets during copy, dump, and restore. Processing
version 2 extended format single striped data sets
that were defined in z/OS V2.1 in a prior release
may result in the version type to be converted. Whether
the data set is converted or not the data set will be
successfully copied, dumped, and restored if no errors
are encountered.

During logical copy and restore if the target has to

be allocated it will result in the target data set not defined as a versions 2 type. If the target is preallocated the version type of the target will be preserved. If the target has to be scratched and reallocated the version type will not be preserved.

During physical data set copy and restore, the target version type will be the version type of the source data set.

ZOS0201C/K

MODULES/MACROS: ADRCNVSM ADRDALOC ADRTDSC ADREXTND ADRFDSO
ADRFDSRL ADRFILTC ADRFRLBO ADRMPPAM ADMRRFM ADRPROTD ADRPSECM
ADRPSEDM ADRPSERM ADRSBRTN ADRSB105 ADRSB106 ADRSB107 ADRSB108
ADRTDDS ADRTDEXT ADRTDLOG ADRTDNVS ADRTDPNV ADRTDPVL ADRTDRF1
ADRTDUNL ADRTDVSM

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA40252 DFSMSHsm Fast Replication

Example B-26 contains the cover letter for APAR OA40252

Example B-26 OA40252

APAR Identifier	OA40252	Last Changed	13/05/03
NEW FUNCTION - DFSMSHSM TOLERATION COEXISTENCE SUPPORT FOR FLASHCOPY CONSISTENCY GROUP AND CAPTURE CATALOG INFO FUNCTIONS			
Symptom	NF NEW FUNCTION	Status	CLOSED UR1
Severity	4	Date Closed	13/04/11
Component	5695DF170	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	DFSMSHSM, ISMF	Special Notice	ATTENTION
Current Target Date	..13/05/31	Flags	
SCP			
Platform			

XSYSTEM

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10 : UA68734 available 13/04/15 ()
Release B10 : UA68735 available 13/04/15 ()
Release C10 : UA68736 available 13/04/17 (F304)
Release D10 : UA68737 available 13/04/17 (F304)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

This APAR provides DFSMSHsm toleration coexistence support for FlashCopy Consistency Group support and catalog information data set (CIDS) compatibility.

LOCAL FIX:

PROBLEM SUMMARY:

* USERS AFFECTED: DFSMSHsm fast replication users of *
* - FlashCopy consistency group function, or *
* - Capture Catalog Information REQUIRED/ *
* PREFERRED option *
* with DFSMSHsm hosts on z/OS V2R1 which *
* share SMS configurations, DFSMSHsm CDSs, *
* and applicable catalog information data *
* sets (CIDS) with hosts running at V1R13, *
* V1R12, V1R11, and/or V1R10. *

* PROBLEM DESCRIPTION: Changes in the fast replication *
* function in z/OS V2R1 require that *
* z/OS V1R10 through V1R13 DFSMSHsm *
* hosts to be updated. This APAR *
* provides toleration support for the *
* DFSMSHsm FlashCopy consistency group *
* function and CIDS compatibility. *

* RECOMMENDATION: *

This PTF must be installed and active on all DFSMSHsm systems using fast replication function in the SYSPLEX before V2R1 is installed into an existing SYSPLEX or before OA41298 can be applied to any system within the SYSPLEX, if DFSMSHsm fast replication with CAPTURE CATALOG INFORMATION REQUIRED/PREFERRED function is used.

.
An HSM restart is required to activate this fix on each system of the SYSPLEX, however, a rolling restart is sufficient to

accomplish the activation.

.
Additional keyword:
ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

DFSMSHsm has been modified such that

1. the new FlashCopy consistency group specification for z/OS V2R1 can be detected and handled appropriately.
2. When the Capture Catalog Information REQUIRED or PREFERRED option is specified for the copy pool, catalog information data sets (CIDS) created in either V2R1 or prior releases can be handled appropriately.

.
The following function has been updated: Fast Replication Backup (FRBACKUP). Copy pools that have the new FlashCopy Consistency Group field set to YES are required to be backed up by a V2R1 or later host.
FRBACKUP will fail backup of copy pools defined with FlashCopy Consistency Group = YES and issue existing MSGARC1806E with new RC89.
Backup versions with FRBF_FCCG_VER flag set (indicating it was created with with FlashCopy Consistency Group option at time of backup) will be deleted or rolled off normally.

.
LIST and ARCXTRCT functions will ignore the FRBF_FCCG_VER setting. Other fast replication command processing remains the same.

.
+HOLD(DOC)
The following publication updates will be made:
z/OS V2R1 MVS System Messages, Vol 2 (ARC-ASA), SA22-7632
SRL: SA22763218, SA22763219, SA22763220, SA22763221
Add new RC89 to the existing message ARC1806E.

ARC1806E FAST REPLICATION {BACKUP | RECOVERY | *****} HAS
FAILED FOR {COPY POOL cpname | VOLUME volser | *****},
RC=retcode

Explanation:

Retcode Meaning

89 FlashCopy consistency group setting of Yes was defined for the copy pool but the function is not supported in the host environment where the FRBACKUP command was being processed.

Application Programmer Response:

Retcode Meaning

89 The fast replication backup command for copy pool cpname cannot be processed in a z/OS

pre-V2R1 environment. Issue the FRBACKUP command for this copy pool on a system that supports the FlashCopy consistency group option.

.
++HOLD(A0)
SPECIAL CONDITIONS:
 AO: MSG=ARC1806E CHANGE=New RSN 89
 AO: ENDAO COMMENT='***End of the list of changes
 that could affect your automated
 operations. The DOC text has
 details of the changes.***'

.
++HOLD(MTS)
This PTF must be installed and active on all DFSMSHsm systems using fast replication function in the SYSPLEX before V2R1 is installed into an existing SYSPLEX or before OA41298 can be applied to any system within the SYSPLEX, if DFSMSHsm fast replication with CAPTURE CATALOG INFORMATION REQUIRED/PREFERRED function is used.
.
An HSM restart is required to activate this fix on each system of the SYSPLEX, however, a rolling restart is sufficient to accomplish the activation.

MODULES/MACROS: ARCFRBM ARCZCSI

SRLS: SA22763218 SA22763219 SA22763220 SA22763221

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA40259 EF V2

Example B-27 contains the cover letter for APAR OA40259.

Example B-27 OA40259

APAR Identifier	OA40259	Last Changed	13/06/03
NEW FUNCTION - TOLERATION			

Symptom	NF NEW FUNCTION	Status	CLOSED UR1
---------------	-----------------	--------------	-------------

Severity	3	Date Closed	13/04/17
Component	5695DF117	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	SYSTEM DATA MOV	Special Notice	
Current Target Date	..13/05/15	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release C10	:	UA68825 available 13/05/02 (F305)
Release D10	:	UA68826 available 13/05/03 (F305)

Parent APAR: OA39869

Child APAR list:

ERROR DESCRIPTION:

Coexistence for future new function.

LOCAL FIX:

PROBLEM SUMMARY:

```

*****
* USERS AFFECTED: Users of DFSMSdss copy and restore          *
*                   processing single striped multi-volume     *
*                   extended format sequential data sets.      *
*****
* PROBLEM DESCRIPTION: This APAR provides toleration support  *
*                   for single striped multi-volume           *
*                   extended format sequential Version 2      *
*                   data sets.                                 *
*****
* RECOMMENDATION:                                             *
*****

```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

DFSMSdss will be modified to process version 2
extended format single striped multi-volume sequential
data sets during copy, dump, and restore. Processing

version 2 extended format single striped data sets that were defined in z/OS V2.1 in a prior release may result in the version type to be converted. Whether the data set is converted or not the data set will be successfully copied, dumped, and restored if no errors are encountered.

During logical copy and restore if the target has to be allocated it will result in the target data set not defined as a versions 2 type. If the target is preallocated the version type of the target will be preserved. If the target has to be scratched and reallocated the version type will not be preserved.

During physical data set copy and restore, the target version type will be the version type of the source data set.

KEYWORDS: D/T2105 D/T2107 ZOS0201C/K

MODULES/MACROS: ANTS8000 ANTS8001

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA40587 RLS

Example B-28 contains the cover letter for APAR OA40587.

Example B-28 OA40587

APAR Identifier	OA40587	Last Changed	13/05/03
LARGE NUMBER OF ALTER REQUESTS IXC530I AND IXC534I FOR RLS CACHE STRUCTURES			
Symptom	IN INCORROUT	Status	CLOSED PER
Severity	2	Date Closed	13/03/01
Component	5695DF122	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	VSAM REC LEV SH	Special Notice	
Current Target Date	..13/04/30	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release C10 : UA68236 available 13/04/10 (F304)

Release D10 : UA68237 available 13/04/10 (F304)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

The data element to dir entry ratio for RLS caches is controlled by RLS and when necessary RLS will issue an ALTER request to adjust the ratio. These requests result in IXC530I and IXC534I indicating an XCF Alter was requested.

Some Alter activity is expected but in some cases can become excessive due to an improperly used variable in the RLS algorithm. This apar may reduce the number of ALTER's experienced due to RLS element/entry ratio adjustment requests

Additional Keywords:

RLSPSP/K

LOCAL FIX:

n/a

PROBLEM SUMMARY:

* USERS AFFECTED: All VSAM RLS users. *

* PROBLEM DESCRIPTION: Users will see a high number of XCF *
* messages IXC530I and IXC534I for *
* RLS cache structures. A high number *
* of these messages indicate that *
* SMSVSAM is issuing ALTER requests *
* to adjust the data element to directory *
* entry ratio too often. This could also *
* cause a CPU usage increase. *

* RECOMMENDATION: *

The data element to directory entry ratio for RLS caches structures is controlled by an RLS algorithm. When necessary RLS will issue an ALTER request to adjust the ratio. Some alter requests are expected but in some cases they can become excessive due to an improperly used variable in the RLS algorithm.

PROBLEM CONCLUSION:

The algorithm to adjust the element to directory ratio has been fixed to use the right variable. This reduces the number of cache structure ALTER requests initiated by SMSVSAM.

KEYWORDS: RLSPSP/K

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IGWSDADR

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA40844 PDSE V2

Example B-29 contains the cover letter for APAR OA40844.

Example B-29 OA40844

APAR Identifier OA40844 Last Changed 13/06/03
DF102 (ISITMGD) TOLERATION APAR FOR LI3039

Symptom NF NEWFUNCTION	Status CLOSED UR1
Severity 2	Date Closed 13/04/18
Component 5695DF102	Duplicate of
Reported Release C10	Fixed Release 999
Component Name BASE ACCESS MET	Special Notice
Current Target Date ..	Flags
SCP	
Platform	

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release C10 : UA68849 available 13/05/07 (F305)

Release D10 : UA68850 available 13/05/07 (F305)

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All PDSE users that have the ability to      *
*                  create PDSE V2 (z/OS V2R1) and want to be    *
*                  able to access in lower releases (R12 and    *
*                  R13).                                         *
*****
* PROBLEM DESCRIPTION: This APAR will allow to process PDSE V2 *
*                  datasets in Releases 12 and 13. PDSE V2     *
*                  datasets are created in V2R1 and above      *
*                  releases.                                     *
*****
* RECOMMENDATION:                                             *
```

Modify BAM code to allow the process of PDSE V2 datasets in R12 and R13

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

Allow the process of PDSE V2 datasets in lower releases (R12 and R13)
ZOS0201C/K - coexistence keyword for z/OS v2.1

MODULES/MACROS: IGGONOPL IGG0194S IGWCIMGD

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA41517 AMS

Example B-30 contains the cover letter for APAR OA41517.

Example B-30 OA41517

APAR Identifier	OA41517	Last Changed	13/05/03
NEW FUNCTION TOLERATION APAR			
Symptom	NF NEWFUNCTION	Status	CLOSED UR1
Severity	3	Date Closed	13/04/12
Component	5695DF103	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	ACCESS METHOD S	Special Notice	
Current Target Date	..13/05/31	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	:	UA68787	available	13/04/15	(		)
Release B10	:	UA68788	available	13/04/15	(		)
Release C10	:	UA68789	available	13/04/16	(F304	)	
Release D10	:	UA68790	available	13/04/16	(F304	)	

Parent APAR: OA36492

Child APAR list:

ERROR DESCRIPTION:

New function.

CATKEYS: CAT2012 CATRLS CATNEW

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: New Function *
*****
* PROBLEM DESCRIPTION: TOLERATION SUPPORT for RLS for Catalog *
*****
* RECOMMENDATION: *
```

Summary: TOLERATION SUPPORT for RLS for Catalog function.
FIXCAT KEYWORD: ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
None

MODULES/MACROS: IDCLC01

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA41790 PDSE V2

Example B-31 contains the cover letter for APAR OA41790. This APAR provides a MACRO that is required by OA39530 and OA40844.

Example B-31 OA41790

APAR Identifier	OA41790	Last Changed	13/06/03
COEXISTANCE APAR FOR PDSE			
Symptom		NF NEWFUNCTION	Status
Severity		2	CLOSED UR1
Component		5695DF102	Date Closed
Reported Release		C10	13/04/17
Component Name		BASE ACCESS MET	Duplicate of
Current Target Date		..13/05/31	Fixed Release
SCP			999
Platform			Special Notice
			Flags

Status Detail: APARCLOSURE - APAR is being closed.

PE PTF List:

PTF List:

Release C10 : UA68849 available 13/05/07 (F305)

Release D10 : UA68850 available 13/05/07 (F305)

Parent APAR: OA39530

Child APAR list:

ERROR DESCRIPTION:

Coexistence Apar for PDSE

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All PDSE users that have PDSE V2 installed *
*                  (Multiple releases with V2R1 or higher as *
*                  one of the releases) *
*****
* PROBLEM DESCRIPTION: This is a toleration APAR that will go *
*                  along with the zOSV2R1 changes made for *
*                  a new PDSE Version2 *
*****
* RECOMMENDATION: *
*****
Toleration APAR for PDSE Version 2 development item
```

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:

Modified IS it Managed macro to be used by PDSE Toleration
OA39530 and BAM Toleration APAR OA40844 for PDSE Version 2
development line item.

ZOS0201C/K - coexistence keyword for z/OS v2.1

MODULES/MACROS: IGWCISM

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA41864 PDSE

Example B-32 contains the cover letter for APAR OA41864.

Example B-32 OA41864

APAR Identifier	OA41864	Last Changed	13/09/30
PDSE ABENDOF4 with RSN14C6A424, RSN14C6A426, RSN13240323 or RSN150A001E			
Symptom	NF ABEND	Status	CLOSED UR1
Severity	2	Date Closed	13/04/16
Component	5695DF115	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	EXTENDED DATA S	Special Notice	HIPER
Current Target Date	..13/06/30	Flags	
SCP			
Platform			DATALOSS

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	: UA68807 available 13/04/29 ()
Release B10	: UA68808 available 13/04/29 ()
Release C10	: UA68809 available 13/04/30 (F304)
Release D10	: UA68810 available 13/04/30 (F304)

Parent APAR: OA39530

Child APAR list:

ERROR DESCRIPTION:

An error in the code that performs PDSE index record compression may result in PDSE data set index corruption

Possible symptoms include, but are not limited to:

ABENDOF4 RC24 RSN14C6A424 out of the module IGWIVRJ2
ABENDOF4 RC24 RSN14C6A426 out of the module IGWIVRJ2
ABENDOF4 RC24 RSN13240323 out of the module IGWLHV01
ABENDOF4 RC20 RSN150A001E out of in the module IGWBITS1
Messages from IEBPDSE or IGWPIT indicating doubly allocated index pages and including but not limited to:
ABENDOF4 RC24 RSN01188011 out of the module IGWDAV00

=====

This item is a coexistence Apar for PDSE 2.1, however; the underlying issue exists in all supported releases and previously releases of PDSE. Therefore all PDSE user need to apply this maintenance.

LOCAL FIX:
None

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: Users sharing PDSE's in fix release with      *
*                   release 2.1 and later.                        *
*****
* PROBLEM DESCRIPTION: Possible symptoms are abends at          *
*                   IGWLHV01+1418, IGWBITX1+10F8,                *
*                   IGWIVRJ2+13B6 and doubly allocated           *
*                   pages from IGWPIT.                           *
*                   TITLE=COMPID=DF115,CSECT=IGWIVRJ2+13B6      *
*                   ,DATE=03/04/13,MAINTID=0A39530              *
*                   ,ABND=0F4,RC=00000004,RSN=14C6A426          *
*                   TITLE=COMPID=DF115,CSECT=IGWLHV01+1418      *
*                   ,DATE=03/18/11,MAINTID= NONE                *
*                   ,ABND=0F4,RC=00000024,RSN=13240323          *
*                   TITLE=COMPID=DF115,CSECT=IGWBITX1+10F8      *
*                   ,DATE=03/18/11,MAINTID= NONE                *
*                   ,ABND=0F4,RC=00000020,RSN=150A001E          *
*                   *****                                     *
* RECOMMENDATION:                                               *
*****
The PDSE is broken because an index record insert caused an
incorrect recompression of subsequent records.
```

PROBLEM CONCLUSION:

ZOS0201C/K - coexistence keyword for z/OS V2.1

TEMPORARY FIX:

COMMENTS:

The recompression has been corrected to account for earlier recompression attributes.
ZOS0201C/K - coexistence keyword for z/OS V2.1

MODULES/MACROS: IGWICPIN

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42058 RMM

Example B-33 contains the cover letter for APAR OA42058.

Example B-33 OA42058

APAR Identifier OA42058 Last Changed 13/06/03
AE OA35808 FIX COMPLETION

Symptom AE	Status CLOSED PER
Severity 2	Date Closed 13/04/23
Component 5695DF186	Duplicate of
Reported Release A10	Fixed Release 999
Component Name DFSMSRMM	Special Notice
Current Target Date ..	Flags
SCP	
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10	: PTF not available yet
Release B10	: PTF not available yet
Release C10	: UA68803 available 13/05/03 (F305)
Release D10	: UA68804 available 13/05/03 (F305)

Parent APAR: OA35808

Child APAR list:

ERROR DESCRIPTION:

OA35808 AE fix completion.

LOCAL FIX:

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: NONE                                     *
*****
* PROBLEM DESCRIPTION: OA35808 fix completion              *
*****
* RECOMMENDATION:                                          *
*****
```

PROBLEM CONCLUSION:

The problem has been fixed

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: EDGMFIO EDGMLCS

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42043 SAM EF V2

Example B-34 contains the cover letter for APAR OA42043.

Example B-34 OA42043

APAR Identifier	OA42043	Last Changed	13/09/04
NEW FUNCTION - Coexistence for DFSMSshm SAM EF V2 data set support			
Symptom	NF INCORROUT	Status	CLOSED UR1
Severity	4	Date Closed	13/07/25
Component	5695DF170	Duplicate of	
Reported Release	C10	Fixed Release	999
Component Name	DFSMSHSM, ISMF	Special Notice	ATTENTION

Current Target Date ..13/08/15 Flags
SCP NEW FUNCTION
Platform

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:
Release C10 : UA70008 available 13/08/07 (F308)
Release D10 : UA70009 available 13/08/07 (F308)

Parent APAR:
Child APAR list:

ERROR DESCRIPTION:
NEW FUNCTION

LOCAL FIX:
NEW FUNCTION

PROBLEM SUMMARY:

* USERS AFFECTED: All users of DFSMSHsm in an HSMplex with *
* V2R1 and lower level DFSMSHsm releases. *

* PROBLEM DESCRIPTION: z/OS V1R12 and V1R13 levels of *
* DFSMSHsm need to fail ARECOVER of SAM *
* Extended Format (EF) Version 2 (V2) *
* data sets on the ALLOCATE list. *
* *

* RECOMMENDATION: *

z/OS V1R12 and V1R13 levels of DFSMSHsm need to fail ARECOVER
of SAM EF V2 data sets on the ALLOCATE list.
New release Keyword
z/OS V2R1 ZOS0201C/K

PROBLEM CONCLUSION:

TEMPORARY FIX:

COMMENTS:
On V2R1 SAM Extended Format (EF) Version 2 (V2) data sets
are fully supported. The z/OS V1R12 and V1R13 levels of

DFSMSHsm need to fail ARECOVER of this type of data set when it is on the ALLOCATION list with the existing message ARC6172E (MSGARC6172E). ARECOVER of this data set on the INCLUDE list will be successful.

SAM EF V2 data sets migrated and backed up on a V2.1 system will be able to be recalled and recovered on V1R12 and V1R13 levels of DFSMSHsm.

New release	Keyword
z/OS V2R1	ZOS0201C/K

++HOLD DOC
V1R12 and V1R13
MVS System Messages Vol 2 (ARC-ASA)
SRL: SA22763220, SA22763221

ARC6172E message - a new text will be added to the Application Programmer Response section:
For ARECOVER, ensure that all data sets specified in the INCLUDE, ALLOCATE, or ACCOMPANY lists are acceptable for processing:
- SAM data set which has the Extended Format of version 2 specified. Use DFSMSHsm V2R1 or higher to perform ARECOVER for this data set.

MODULES/MACROS:
ARCNVRFY

SRLS:
SA22763220
SA22763221

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42529 DFSMSHsm

Example B-35 contains the cover letter for APAR OA42529.

Example B-35 OA42529

APAR Identifier	OA42529	Last Changed	13/10/02
-----------------------	---------	--------------------	----------

ARC0187I I/O ERROR REPORTED FOR A 'D' RECORD BUT ERROR MESSAGE
SHOWS VSR RECORD WITH OA36576 APPLIED TO A 13/06/13 PTF PECHANGE

Symptom	MS MSGARC0187I	Status	CLOSED PER
Severity	2	Date Closed	13/08/14
Component	5695DF170	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	DFSMSHSM, ISMF	Special Notice	PE
Current Target Date	..13/09/12	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for
shipment.

PE PTF List: UA68717 UA68716

PTF List:

Release A10	: PTF not available yet
Release B10	: PTF not available yet
Release C10	: UA70225 available 13/08/23 (F308)
Release D10	: UA70229 available 13/08/23 (F308)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

With OA36576 applied to at least one HSM image in an HSMplex,
ARC0187I I/O error messages can occur during HSM Secondary
Space Management. The message will indicate that the error
occured on a 'D' type record, although the record key shows that
HSM was trying to read a VSR ('S'-type record):

ARC0187I I/O ERROR POSITIONING TO AN DFSMSHSM CONTROL DATA SET D
RECORD, KEY= VSRABC001 , RC=0012

SSM will continue to process. The VSR records that received the
ARC0187I message will not be checked for expiration processing
during that run of SSM.

LOCAL FIX:

Apply OA36576 to the HSM image that performs SSM.

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All z/OS V1R10,V1R11,V1R12 and V1R13          *
*                   DFSMSHsm users.                             *
*****
* PROBLEM DESCRIPTION: With OA36576 applied to at least          *
*                   one DFSMSHsm image in an HSMplex,           *
*                   ARC0187I I/O error messages can occur        *
```

```

*                during DFSMSHsm Secondary Space      *
*                Management (SSM).                      *
*                OA42529 was created to resolve this    *
*                issue.                                  *
*****

```

* RECOMMENDATION: *

```

*****
With OA36576 applied to at least one DFSMSHsm image in an
HSMplex, ARC0187I I/O error messages can occur during DFSMSHsm
Secondary Space Management (SSM).
OA42529 was created to resolve this issue.

```

PROBLEM CONCLUSION:

OA42529 was created to back off DSR/VSR records to pre-OA36576 length and prepare systems for the application of the coexistence enablement APAR OA42562. OA42529 must be applied to all systems in an HSMplex before OA42562 is applied.

++HOLD(MULTSYS) - The PTFs for OA42529 must be installed and active on all systems in the SYSPLEX before the PTFs for APAR OA42562 can be applied to any system within the sysplex.
KEYWORDS: ZOS0201C/K

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: ARCASTAI ARCAUDIT ARCAZWC ARCCPQST ARCCSTAI
ARCCTL ARCCVSR ARCDCOLL ARCDSTAI ARCESTAI ARCFCRE ARCFDEL
ARCFSTAI ARCGSTAI ARCISTAT ARCJSTAE ARCJSTAI ARCMLCLN ARCMSCLN
ARCMSTAI ARCMVCLN ARCPRLOG ARCRPT ARCUDATA ARCUTIL ARCXSTAI
ARCYSTAI ARSZMSG

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42541 Catalog

Example B-36 contains the cover letter for APAR OA42541.

Example B-36 OA42541

APAR Identifier OA42541 Last Changed 13/10/02
ABEND S130 IN IDA0200T AFTER INSTALLING PTFS FOR OA36916

Symptom	IN INCORROUT	Status	CLOSED PER
Severity	2	Date Closed	13/08/02
Component	5695DF106	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	DFSMS VSAM	Special Notice	PE
Current Target Date	..13/08/31	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List: UA68644 UA68643

PTF List:

Release A10	:	PTF not available yet
Release B10	:	PTF not available yet
Release C10	:	UA70130 available 13/08/28 (F308)
Release D10	:	UA70131 available 13/08/28 (F308)
Release 210	:	UA70146 available 13/08/28 (F308)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

An abend S130 may occur in IDA0200T at offset X'3600' after installing the PTFS for OA36916. The error is a dequeue for a resource that is not held.

The dequeue will show a request to dequeue SYSVSAM with a catalog name as the resource and ending in a "N".

LOCAL FIX:

Remove the PTFS for OA36916

PEX/ UA68641

PEX/ UA68642

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All z/OS 1.10 VSAM users and above who      *
*                   OPEN and CLOSE a catalog as a normal      *
*                   VSAM Data Set.                             *
*****
* PROBLEM DESCRIPTION: While closing a user catalog that has  *
*                   been opened with multiple ACBs, an        *
*                   ABEND S130 will result.                    *
*****
```

* RECOMMENDATION: Apply PTF. *

VSAM CLOSE processing was not correctly verifying
the last close for a user catalog.

PROBLEM CONCLUSION:
VSAM CLOSE processing was modified to correctly
determine the last close for a user catalog.
KEYWORDS: ZOS0201C/K

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IDA0200T

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42551 RLS DSS

Example B-37 contains the cover letter for APAR OA42551.

Example B-37 OA42551

APAR Identifier	OA42551	Last Changed	13/10/02
ADR952E ADR957I DURING LOGICAL DUMP OF USERCATALOG WITH DSS DUE TO A FAILURE IN RLS DURING QUIESCE PROCESS 13/06/14 PTF PECHANGE			
Symptom	MS MSGADR952E	Status	CLOSED PER
Severity	2	Date Closed	13/06/21
Component	5695DF122	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	VSAM REC LEV SH	Special Notice	PE
Current Target Date	..13/07/31	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for

shipment.

PE PTF List: UA68643 UA68644

PTF List:

Release A10 : PTF not available yet
Release B10 : PTF not available yet
Release C10 : UA69623 available 13/07/23 (F307)
Release D10 : UA69624 available 13/07/23 (F307)
Release 210 : UA69625 available 13/07/23 (F307)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

A logical dump of a usercat defined with a large number of extents (approximatly 30 and more) will fail with DSS messages:

ADR952E (001)-DTDSC(01), THE IDAQDMP MACRO FAILED DURING QUIESCE PROCESSING FOR FOR CLUSTER dsname WITH RETURN CODE 00000008 AND REASON CODE 61FF0001

This issue occurs after application of OA36422 and OA36403

Problem is due to workarea shortage during RLS locate request for catalog information.

ADDITIONAL KEYWORDS:

RLSPSP/K
SMSVSAM RLS VSAMRLS

LOCAL FIX:

You can use DSS patch to circumvent the RLS quiesce processing described in DFSMSdss Storage Administration
1.14.30 Bypassing RLS processing (OW32817)
PEX/ UA68641
PEX/ UA68642

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All Catalog users who issue DSS DUMP          *
*                   command to backup user catalog data sets.    *
*****
* PROBLEM DESCRIPTION: When performing the DSS DUMP function     *
*                   to backup user catalog data sets with        *
*                   about 30 extents, the following error        *
*                   is generated:                                 *
*                   ADR952E (001)-DTDSC(01), THE IDAQDMP         *
*                   MACRO FAILED DURING QUIESCE PROCESSING       *
```



```

*                                FOR CLUSTER dsname WITH RETURN CODE      *
*                                00000008 AND REASON CODE 61FF0001          *
*****
* RECOMMENDATION:                                                         *
*****
If a user catalog data set has approximately 30 or more extents,
any attempt to back it up using the DSS DUMP will failed with
ADR952E error message.

```

```

PROBLEM CONCLUSION:
The error is due to the work area being too small to handle a
user catalog with many extents. The fix will allocate a larger
work area and retry the operation when detecting the work area
too small error.
KEYWORDS: RLSPSP/K ZOS0201C/K

```

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IDAVQCAT

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42562 DFSMSHsm Storage Tiers

Example B-38 contains the cover letter for APAR OA42562.

Example B-38 OA42562

```

APAR Identifier ..... OA42562      Last Changed ..... 13/09/13
STORAGE TIERS COEXISTENCE ENABLEMENT

```

```

Symptom ..... IN INCORROUT          Status ..... CLOSED PER
Severity ..... 2                     Date Closed ..... 13/08/15
Component ..... 5695DF170           Duplicate of .....
Reported Release ..... A10           Fixed Release ..... 999

```

Component Name DFSMSHSM, ISMF Special Notice
Current Target Date ..13/09/12 Flags
SCP
Platform

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release A10 : UA70331 available 13/08/21 ()
Release B10 : UA70332 available 13/08/21 ()
Release C10 : UA70338 available 13/08/23 (F308)
Release D10 : UA70373 available 13/08/23 (F308)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

Storage Tiers coexistence enablement

LOCAL FIX:

N/A

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: All V1R10,V1R11,V1R12 and V1R13 DFSMSHsm      *
*                   users.                                         *
*****
* PROBLEM DESCRIPTION: Enlarge DSR/VSR record lengths for      *
*                   for Storage Tiers coexistence.              *
*****
* RECOMMENDATION:                                               *
*****
Enlarge DSR/VSR record lengths for Storage Tiers coexistence.
```

PROBLEM CONCLUSION:

DFSMSHsm will be changed to increase DSR/VSR record lengths for Storage Tiers coexistence.

++HOLD(MULTSYS) - The PTF(s) for OA42529 must be installed and active on all systems in the SYSPLEX before the PTF(s) for APAR OA42562 can be applied to any system within the sysplex. The PTFs for OA42529 are UA70197, UA70198, UA70225, UA70229

++HOLD(MULTSYS) - OA42562 PTF(s) will not be fully effective on the system it is being applied to until the PTF(s) for this APAR are applied to all systems in the SYSPLEX.

New release Keyword
z/OS V2R1 ZOS0201C/K

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS:

ARCASTAI ARCAUDIT ARCAZWC ARCCPQST ARCCSTAI ARCCTL ARCCVSR
ARDCOLL ARCDSTAI ARCESTAI ARCFCRE ARCFDEL ARCFSTAI ARCGSTAI
ARCISTAT ARCJSTAE ARCJSTAI ARCMLCLN ARCMSCLN ARCMSTAI ARCMVCLN
ARCPRLOG ARCRPT ARCUATA ARCUTIL ARCXSTAI ARCYSTAI ARCZMSG

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42667 XTLOT use

Example B-39 contains the cover letter for APAR OA42667. This APAR is applicable to DFSMS V1.13 as well.

Example B-39 OA42667

APAR Identifier	OA42267	Last Changed	13/10/01
IEC999I IFG0194D ABENDOC1 ABENDOC4 NON_VSAM_XTLOT DEVSUPXX			
TRKCALC RC04 IECOSCR1			
Symptom	AB ABENDOC4	Status	CLOSED PER
Severity	2	Date Closed	13/09/16
Component	5695DF107	Duplicate of	
Reported Release	D10	Fixed Release	999
Component Name	DATA MGMT SUPPO	Special Notice	
Current Target Date	..13/09/30	Flags	
SCP			
Platform			

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List:

PTF List:

Release D10 : UA70761 available 13/10/01 (1000)
Release 210 : UA70762 available 13/10/01 (1000)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

When keyword NON_VSAM_XTIOT is set to YES in PARMLIB member DEVSUPxx, IFG0194D +x'186A' UA68824 makes a TRKCALC call without LOC=ANY and the UCB is a 31-bit address pulled directly from the XTIOT. TRKCALC module IECOSCR1 takes an ABEND0C4 as a result. IFG0194D may also take an intentional ABEND0C1 to header IFG0194D TRKCALC_ERROR_SEE_R15 ERROR when register 15 is set to RC04. The joblog will show an IEC999I IFG0194D message.

LOCAL FIX:

Set NON_VSAM_XTIOT to NO in PARMLIB member DEVSUPxx

PROBLEM SUMMARY:

* USERS AFFECTED: All. *

* PROBLEM DESCRIPTION: Abend0c1 in IFG0194D. *

* RECOMMENDATION: *

IEC999I IFG0194D,POFFOR,POFFOR,TRKCALC ERROR SEE R15
due to an intentional abend0c1 after a TRKCALC failure.
TRKCALC is passed a 31bit UCB address without including
keyword: LOC=ANY in the macro invocation. OPEN is processing
with an XTIOT containing 31bit UCB addresses.

PROBLEM CONCLUSION:

IFG0194D now issues TRKCALC with the LOC=ANY keyword.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IFG0194D

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42846 Single Striped, Multi-volume SAM Tailored Compressed data set

Example B-40 contains the cover letter for APAR OA42846.

Example B-40 OA42846

APAR Identifier OA42846 Last Changed 13/09/12
APAR OA41459 CAN CAUSE LOOP IN IFG0232D

Symptom LP LOOP	Status CLOSED PER
Severity 2	Date Closed 13/08/05
Component 5695DF107	Duplicate of
Reported Release D10	Fixed Release 999
Component Name DATA MGMT SUPPO	Special Notice PE
Current Target Date ..13/09/15	Flags
SCP	
Platform	

Status Detail: SHIPMENT - Packaged solution is available for shipment.

PE PTF List: UA68273 UA68272 UA68824 UA68823

PTF List:

Release C10 : UA70205 available 13/09/12 (1000)
Release D10 : UA70204 available 13/09/12 (1000)
Release 210 : UA70203 available 13/09/12 (1000)

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

APAR OA41459 can cause a LOOP in IFG0232D when there are fewer than 5 used physical blocks on the last volume for tailored compressed data sets during CLOSE TYPE=T (TCLOSE), this LOOP will hold ENQ on SYSZTIOT.

LOCAL FIX:

remove PTF for OA41459

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: users of single striped, multi-volume sam      *
*                  tailored compressed data sets.                *
*****
* PROBLEM DESCRIPTION: users of single striped, multi-volume     *
*                  sam tailored compressed data sets may         *
*                  see loops when close type=t leave is          *
*                  issued when open for input or issued          *
*                  tiwce when open for output on the last        *
*                  volume and there is very little data          *
*                  (less than five physical blocks) on the       *
*                  current volume.                                *
*****
* RECOMMENDATION:                                                *
*****
users of single striped, multi-volume sam tailored compressed
data sets may see loops when close type=t leave is issued.
```

PROBLEM CONCLUSION:

Change code so that the count of physical blocks used by the tailored dictionary are ignored for the second and subsequent volumes of a single striped, multi-volume data set.

TEMPORARY FIX:

COMMENTS:

MODULES/MACROS: IFG0232D

SRLS: NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA42947 VSAM RLS

Example B-41 contains the cover letter for APAR OA42947. The DFSMS V2.1 equivalent APAR is OA43003.

Example B-41 OA42947

APAR Identifier OA42947 Last Changed 13/08/02
POSSIBLE INCORRECT RBA RETURNED AFTER ESDS PUT UPDATE OR DELETE

Symptom	IN INCORROUT	Status	OPEN
Severity	3	Date Closed	
Component	5695DF122	Duplicate of	
Reported Release	D10	Fixed Release	
Component Name	VSAM REC LEV SH	Special Notice	
Current Target Date	..14/03/30	Flags	
SCP			
Platform			

Status Detail: DESIGN/CODE - APAR solution is being designed
and coded.

PE PTF List:

PTF List:

Parent APAR:

Child APAR list: OA43003

ERROR DESCRIPTION:

For ESDS processing, it is possible for the RBA to be corrupted by application programs after it's initially passed in correctly. An application program specifies the correct RBA for a GET UPDATE request, VSAM RLS processes the GET UPDATE with the correct RBA successfully, then the application somehow corrupts the RBA, possibly by mistake, followed by a PUT UPDATE request. RLS also processes the PUT UPDATE successfully since RBA is not required. However, the incorrect RBA is returned at the end of the processing.

LOCAL FIX:

Avoid corrupting the RBA between requests.

APAR OA43537 ICF Catalog hang

Example B-42 contains the cover letter for APAR OA43537.

Example B-42 OA43537

APAR Identifier OA43537 Last Changed 13/10/21
IDC3009I 50-43 ERROR FOLLOWED BY SYSZVDS SYSIGGV2 HANG

Symptom	IN INCORROUT	Status	OPEN
---------------	--------------	--------------	------

Severity	2	Date Closed	
Component	5695DF105	Duplicate of	
Reported Release	D10	Fixed Release	
Component Name	ICF CATALOG & I	Special Notice	PE
Current Target Date	..14/01/31	Flags	
SCP			
Platform			

Status Detail: DESIGN/CODE - APAR solution is being designed and coded.

PE PTF List: UA68643 UA68644

PTF List:

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

When multiple DELETE / EXPORT DISCONNECT jobs are ran to remove user catalogs from the master catalog simultaneously, a 50-43 error can occur followed by a hang oinvolving SYSZVVDS and SYSIGGV2 for the master. This timing issue only occurs during maintenance related procedures that involve this type of workload and would not occur during typical production. the 50-43 was introduced in OW45703 to detect cases where a VVDS Get for Update was issued and the next request was not the corresponding Put or End Update. In this case, two tasks failed to serialize with each other on the VVDS parameter control block which resulted in put/end updates not being issued and enqueues not being released

CAS DYNAMIC DUMP-IGGOCLE0 RC50 RSN43

ISG343I 01.23.45 GRS STATUS 841

S=SYSTEMS SYSZVVDS volser

S=SYSTEMS SYSIGGV2 master.catalog

CATKEYS: CAT2013 CATHANG CATBCS CATVVDS

LOCAL FIX:

F CATALOG,LIST will help identify holders of resources. F CATALOG,ABEND(id) can be issued to abend a specific task. F CATALOG,RESTART can alternatively be used to free the hang.

APAR OA43583 PDSE (also applies to DFSMS V2.1)

Example B-43 contains the cover letter for APAR OA43583.

Example B-43 OA43583

APAR Identifier OA43583 Last Changed 13/10/21
ABEND0F4 RC14 RSN25400018 IGWDLCLS+OC32 @ UA68850

Symptom	AB ABEND0F4	Status	CLOSED	PER
Severity	2	Date Closed	13/10/21	
Component	5695DF115	Duplicate of		
Reported Release	D10	Fixed Release	999	
Component Name	EXTENDED DATA S	Special Notice		HIPER
Current Target Date	..14/01/15	Flags		
SCP				
Platform				DATALOSS

Status Detail: TESTPACKAGING - Packaged solution is being tested.

PE PTF List:

PTF List:

Release C10 : PTF not available yet
Release D10 : PTF not available yet
Release 210 : PTF not available yet

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

ABEND0F4 RC14 RSN25400018 out of the module IGWDLCLS+OC32 at
UA68850 maintenance level.

The reason code RSN25400018 translates to GETMAIN_FAILED.

VSM summary data (IP VERBX VSMDATA 'NOG SUMM') shows that
SMSPDSE1 extended private storage is almost exhausted:

LOCAL STORAGE MAP

Extended	80000000	<- Top of Ext. Private
LSQA/SWA/229/230	80000000	<- Max Ext. User Region
Address	32914000	<- ELSQA Bottom
(Free Extended Storage)	32908000	<- Ext. User Region Top
Extended User Region	32900000	<- Ext. User Region

An IP VERBX SMSXDATA 'F(POOLS) JOBNAME(SMSPDSE1)' will show an extremely large allocation for the "JCDM DSC POOL FOR LSTB DREFD" pool:

```
*****
      Below the Bar Storage
Exts  TotSize      TotCells  InUseCells SP  Key Description
*****
11315 1,205,002,240  1,131,500  1,131,500 124 50 JCDM DSC POOL
                                FOR LSTB
                                DREFD
```

The build up of the LSTB control structures is the result of PDSE index code that is attempting to join two index pages.

This failure can occur on PDSE data sets that are shared in a sysplex with mixed z/OS levels where at least one LPAR is at the HDZ2210 / z/OS 2.1 level.

LOCAL FIX:

Restart the SMSPDSE1 address space

PROBLEM SUMMARY:

```
*****
* USERS AFFECTED: PDSE index records can rarely be broken when *
*                  the data set is updated on a release HDZ2210 *
*                  which had been updated on a lower release.   *
*****
* PROBLEM DESCRIPTION: See Users Affected.                       *
*****
* RECOMMENDATION:                                              *
*****
PDSE index manager will be modified to now join index pages
which have created on HDZ2210 and a lower release.
```

PROBLEM CONCLUSION:

See Problem summary.

TEMPORARY FIX:

```
*****
* HIPER * AA45383
*****
```

COMMENTS:

MODULES/MACROS:

IGWIRCMT IGWIVRJ2

SRLS:
NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:

APAR OA43701 SAM EF V2 (also applies DFSMS V2.1)

Example B-44 contains the cover letter for APAR OA43701.

Example B-44 OA43701

APAR Identifier	OA43701	Last Changed	13/10/28
VOLUME LABEL MAY BE OVERWRITTEN BY SAM EXTENDED FORMAT VERSION 2			
DATA SET - Z/OS 2.1 ONLY			
Symptom	IN INCORROUT	Status	CLOSED PER
Severity	2	Date Closed	13/10/28
Component	5695DF102	Duplicate of	
Reported Release	210	Fixed Release	999
Component Name	BASE ACCESS MET	Special Notice	HIPER
Current Target Date	..13/12/15	Flags	
SCP			
Platform		PERVASIVE	DATALOSS

Status Detail: TESTPACKAGING - Packaged solution is being tested.

PE PTF List:

PTF List:

Release C10 : PTF not available yet
Release D10 : PTF not available yet
Release 210 : PTF not available yet

Parent APAR:

Child APAR list:

ERROR DESCRIPTION:

If a multi-volume single striped SAM Extended Format Version 2 data set is allocated using Guaranteed Space and then opened for EXTEND or opened for OUTPUT or OUTIN when allocated DISP=MOD, the volume label on the second to last volume may be overwritten by user data.

LOCAL FIX:

Do not use Version 2 if using Guaranteed Space allocation and either OPEN EXTEND or OPEN OUTPUT/OUTIN and DISP=MOD is used.

To bypass this problem you must do one of the following:

1) Specify DSNTYPE=(EXT,1) or DSNTYPE=(EXT) on the JCL

OR

2) Specify EXT_VERSION(1) in IGDSMSxx parmlib member AND do NOT specify DSNTYPE=(EXT,2) on the JCL.

The volume can be recovered by varying it offline to all systems and running the following job.

```
//*****  
/* THIS STEP REBUILDS CYL 0 HEAD 0 TO POINT TO VTOC */  
/* CHANGE THE VOLID TO THE VOLSER REQUIRED, UNIT TO DEVICE*/  
/* NUMBER AND THE START OF THE VTOC TO THE CYLINDER and */  
/* HEAD where it is  
//*****  
//RMT EXEC PGM=ICKDSF  
//SYSPRINT DD SYSOUT=*  
//SYSIN DD *  
 REFORMAT UNIT(dddd) NVFY VTOCPTR(Cyl,HD) PURGE VOLID(vvvvvv)  
/*
```

A successful vary online is a good indication all should be fine.

PROBLEM SUMMARY:

```
*****  
* USERS AFFECTED: All users of multi-volume single striped SAM *  
* extended format data sets allocated using *  
* guaranteed space. *  
*****  
* PROBLEM DESCRIPTION: Multi-volume single striped SAM *  
* extended format version 2 data sets *  
* allocated using guaranteed space may *  
* overlay the volume label on the second *  
* to last volume if the first OPEN is for *  
* EXTEND or for OUTPUT or OUTIN and *  
* DISP=MOD is specified. In some cases, *  
* such as ISPF browse, OPEN for INPUT or *  
* INOUT will position to the volume *  
* label. Though version 2 data sets may *  
* not be created on releases prior to *  
* HDZ2210, this problem could occur if *  
* the data set was allocated on HDZ2210 *  
* and first opened on an earlier release. *  
*****  
* RECOMMENDATION: *  
*****  
Multi-volume single striped SAM extended format version 2 data  
sets allocated using guaranteed space may overlay the volume  
label on the second to last volume.
```

PROBLEM CONCLUSION:

Changed code to correctly build internal control blocks to only allow access to the data set extents on the last volume.

TEMPORARY FIX:

* HIPER * AA43701

COMMENTS:

MODULES/MACROS:

IGG0193V

SRLS:

NONE

RTN CODES:

CIRCUMVENTION:

MESSAGE TO SUBMITTER:



C

Sample assembler code

This appendix contains source for sample programs developed during this project to exercise features and functions in z/OS V2.1 DFSMS.

IEBCOPY FAMS abend diagnosis

In Example C-1 together with continuation in subsequent examples we show sample code to analyze the IEBCOPY REGISTER 15 and REGISTER 0 data that could be returned if the FAMS routine has an abend.

This example is provided as an example of what could be done. If the code needs to be assembled and linked an example is provided in Example C-5 on page 401.

Example C-1 IEBCPYF program to display FAMS abend data 1 of 4

	MACRO		00010099
&NAME	SEGSTART		00020099
	AMODE 24		00030099
&NAME	STM 14,12,12(13)	SAVE HIS REGS IN HIS SAVE AREA	00040099
R0	EQU 0		00050099
R1	EQU 1		00060099
R2	EQU 2		00070099
R3	EQU 3		00080099
R4	EQU 4		00090099
R5	EQU 5		00100099
R6	EQU 6		00110099
R7	EQU 7		00120099
R8	EQU 8		00130099
R9	EQU 9		00140099
R10	EQU 10		00150099
R11	EQU 11		00160099
RB	EQU 12		00170099
R13	EQU 13		00180099
R14	EQU 14		00190099
R15	EQU 15		00200099
	BALR 12,0	SET UP ADDRESSABILITY	00210099
	USING *,12	USE REG 12 AS BASE REG	00220099
	ST 13,SAVEREGS+4	SAVE @ OF HIS SAVEAREA IN MINE	00230099
	LA 03,SAVEREGS	LOAD @ OF MY SAVE AREA IN REG 3	00240099
	ST 03,8(13)	SAVE @ OF MY SAVE AREA IN HIS	00250099
	LR 13,03	LOAD @ OF MY SAVE AREA IN REG 13	00260099
	MEND		00270099
	MACRO		00280099
&NAME	SEGEND		00290099
&NAME	L 13,SAVEREGS+4	LOAD REG13 WITH @ OF HIS SAVE	00300099
	LM 14,12,12(13)	RESTORE REGS FROM HIS SAVEAREA	00310099
	XR R15,R15		00320099
	BR 14	RETURN TO CALLING RTN VIA REG 14	00330099
SAVEREGS	DC 18F'0'	SET UP SAVE AREA	00340099
	MEND		00350099
	MACRO		00360099
&NAME	HEXTXT8 &KEY		00370099
	UNPK &KEY.(09),SAVR0(5)		00380099
	TR &KEY.(08),HEXTAB-240		00390099
	UNPK &KEY.+8(09),SAVR0+4(5)		00400099
	TR &KEY.+8(08),HEXTAB-240		00410099
	MVI &KEY.+16,X'40' BLANK THE EXTRA BYTE		00420099
	MEND		00430099

Example C-2 IEBCPYF program to display FAMS abend data 2 of 4

IEBCPYF	SEGSTART	00440099
*	THIS IS A SIMPLE PROGRAM TO LINK TO IEBCOPY THEN DISPLAY THE	00450099
*	CONTENTS OF REGISTER 15, AND OF REGISTER 0	00460099
*		00470099
*	R3 PARM	00480099
*	R4 IEBCREAS DSECT	00490099
*	R12 OVERALL BASE REGISTER	00500099
START	DS OH	00510099
	MVI TESTPARM,C'N'	00520099
	L R15,0(R1) GET ADDRESS OF PARM	00530099
	LH R3,0(R15) GET LENGTH OF PARM	00540099
	CH R3,=H'0' CHECK PARM LENGTH	00550099
	BE NOPARM	00560099
	SH R3,=H'1' REDUCE FOR EXECUTE	00570099
	EX R3,MVCPARM	00580099
	MVC SAVR0(8),SYSPARM+9	00590099
	MVI TESTPARM,C'Y'	00600099
NOPARM	DS OH	00610099
	LINK EP=IEBCOPY,PARAM=(OPTIONS),VL=1	00620099
*	SAVE R15 AND R0	00630099
	ST R15,SAVR15	00640099
	CLI TESTPARM,C'Y'	00650099
	BE SKIPRO	00660099
	STG R0,SAVR0	00670099
SKIPRO	DS OH	00680099
	OPEN (PRINTDCB,(OUTPUT))	00690099
	PUT PRINTDCB,PRINTHD1	00700099
*	PREPARE R15 FOR PRINTING AND PRINT	00710099
	L R15,SAVR15	00720099
	CVD R15,CVDWRK	00730099
	UNPK UNPKWRK+3(5),CVDWRK+4(4)	00740099
	OI UNPKWRK+7,X'F0'	00750099
	MVC MSG15V,UNPKWRK+4	00760099
	MVC PRTBUF,BLANK	00770099
	MVC PRTBUF(MSG15L),MSG15	00780099
*	PREPARE R0 FOR PRINTING AND PRINT	00790099
	LA R4,SAVR0	00800099
	USING IEBSNAREA,R4	00810099
	HEXTEXT8 MSGOV	00820099
	PUT PRINTDCB,PRTBUF	00830099
*	PRINT HEX FORM OF THE CONTENTS OF REGISTER 0	00840099
	MVC PRTBUF,BLANK	00850099
	MVC PRTBUF(MSGOL),MSGO	00860099
	PUT PRINTDCB,PRTBUF	00870099
	CLI TESTPARM,C'N'	00880099
	BE NOTTEST	00890099
*	IN TEST MODE	00900099
	MVC PRTBUF,BLANK	00910099
	PUT PRINTDCB,PRINTHD3	00920099

Example C-3 IEBCPYF program to display FAMS abend data 3 of 4

NOTTEST	DS	OH	00930099
* PREPARE ANALYSUS OF WHAT THE CONTENTS OF RO MEAN			00940099
TM	IEBRN_FLAGS,IEBRN_ABENDED		00950099
BNO	FINISH		00960099
PUT	PRINTDCB,PRINTHD2		00970099
* ABEND DID OCCUR IN FAMS			00980099
MVC	PRTBUF,BLANK		00990099
MVC	PRTBUF(MSG04L),MSG04		01000099
PUT	PRINTDCB,PRTBUF		01010099
MVC	PRTBUF,BLANK		01020099
* MVC MSG05V,IEBRN_ABNDRCODE			01030099
MVC	MSG05V,MSG0VAB		01040099
MVC	PRTBUF,BLANK		01050099
MVC	PRTBUF(MSG05L),MSG05		01060099
PUT	PRINTDCB,PRTBUF		01070099
* MVC MSG06V,IEBRN_ABNDRCN			01080099
MVC	MSG06V,MSG0VRC		01090099
MVC	PRTBUF(MSG06L),MSG06		01100099
PUT	PRINTDCB,PRTBUF		01110099
FINISH	DS	OH	01120099
	SEGEN		01130099
	ORG		01140099
WORKAREA	DC	C'***** WORKAREA *****'	01150099
	DS	OD	01160099
CVDWRK	DC	CL8'EMPTY'	01170099
UNPKWRK	DC	9XL1'FO'	01180099
SAVR0	DC	D'0'	01190099
SAVR15	DC	F'0'	01200099
OPTIONS	DC	H'0'	01210099
HEXTAB	DC	C'0123456789ABCDEF' TRANSLATE TABLE	01220099
BLANK	DC	CL133' '	01230099
PRTBUF	DS	CL133	01240099
MSG15	DC	C' ON EXIT FROM IEBCOPY R15='	01250099
MSG15V	DC	CL4'NNNN'	01260099
MSG15L	EQU	*-MSG15	01270099
MSG0	DC	C' ON EXIT FROM IEBCOPY R0='	01280099
MSG0V	DC	CL16'NNNNNNNNNNNNNNNN'	01290099
MSG0VAB	EQU	MSG0V+13	01300099
MSG0VRC	EQU	MSG0V+0	01310099
MSG0L	EQU	*-MSG0	01320099
MSG04	DC	C' IEBRN_ABENDED FLAG '	01330099
MSG04V	DC	C'IS ON INDICATING THAT AN ABEND OCCURRED IN FAMS'	01340099
MSG04L	EQU	*-MSG04	01350099
MSG05	DC	C' IEBRN_ABENDED FLAG '	01360099
	DC	C'IS ON AND ABEND CODE IS: '	01370099
MSG05V	DC	CL3'XXX'	01380099
MSG05L	EQU	*-MSG05	01390099

Example C-4 IEBCPYF program to display FAMSabend data 4 of 4

MSG06	DC	C' IEBSRN_ABENDED FLAG '	01400099
	DC	C'IS ON AND REASON CODE IS: '	01410099
MSG06V	DC	CL4'XXX'	01420099
MSG06L	EQU	*-MSG06	01430099
PRINTHD1	DC	CL133' REGISTER CONTENTS AFTER LINK TO IEBCOPY'	01440099
PRINTHD2	DC	CL133' REGISTER 0 CONTENTS ANALYSIS'	01450099
PRINTHD3	DC	CL133' *** WARNING - RUNNING WITH TEST PARM'	01460099
SYSPARM	DC	CL100'SYSPARM SYSPARM'	01470099
MVCPARM	MVC	SYSPARM(0),2(R15)	01480099
TESTPARM	DC	C'N'	01490099
PRINTDCB	DCB	DDNAME=PRINT,DSORG=PS,MACRF=(PM),LRECL=133	01500099
	ORG		01510099
	IEBCREAS		01520099
	END		01530099

In order to use the source code it must first be stored in a data set with member name IEBCPYF. The data set name used in this example is MHLRES2.IEBCOPY.DFSMS21.PDS. The data set can be PDSE or PDS. The only restriction is that the RECFM be *F* or *FB* and LRECL must be 80.

It is also necessary to create a data set to link the load module into. In this example the data set name used is MHLRES2.IEBCOPY.DFSMS21.LOAD. The LOAD data set can be PDS or PDSE and have RECFM U.

Once the source has been successfully assembled and linked, the program can be run using the example code in Example 7-27 on page 128.

In Example C-5 on page 401 we show sample JCL to assemble and link program IEBCPYF.

Example C-5 JCL to assemble and link program IEBCPYF

//MHLRES2E	JOB	(1234567,COMMENT),MHLRES2,TIME=10,	00010000
//	MSGCLASS=J,		00020000
//	MSGLEVEL=1,CLASS=A,		00030000
//	NOTIFY=&SYSUID		00040000
/*JOBPARM	S=*		00041000
//ASMHCL	PROC		00050000
//ASM	EXEC	PGM=ASMA90,REGION=0M,	00060000
//	PARM= 'OBJECT,NODECK'		00070000
//SYSLIN	DD	DSN=&&OBJ,DISP=(NEW,PASS),UNIT=SYSDA,	00080000
//	SPACE=(TRK,(10,2)),DCB=BLKSIZE=3120		00090000
//SYSLIB	DD	DISP=SHR,DSN=SYS1.MACLIB	00100000
//SYSPRINT	DD	SYSOUT=*	00110000
//SYSUT1	DD	DSN=&&SYSUT1,UNIT=SYSDA,SPACE=(CYL,(5,5))	00120000
//*			00130000
//LKED	EXEC	PGM=HEWL,REGION=2048K,COND=(8,LE,ASM),	00140000
//	PARM= 'XREF,LIST,LET'		00150000
//SYSLIN	DD	DSN=&&OBJ,DISP=(OLD,DELETE)	00160000
//	DD	DDNAME=SYSIN	00170000
//SYSPRINT	DD	SYSOUT=*	00180000
//SYSUT1	DD	DSN=&&SYSUT1,UNIT=SYSDA,SPACE=(CYL,(5,5))	00190000

// PEND	00200000
// EXEC ASMHCL	00210000
//ASM.SYSIN DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.PDS(IEBCPYFA)	00220007
/*	00230000
//LKED.SYSLMOD DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.LOAD	00240000
//LKED.SYSIN DD *	00250000
SETSSI 00002100	00260000
NAME IEBCPYF(R)	00270007

IEBCOPY User Exit Capability illustration sample code

In order to demonstrate the IECOPY user exit capability we generated two programs.

- ▶ The program (IEBCPYL) to invoke the Control Statement Exit
- ▶ The Control Statement Exit module (IEBCPYC)

In Example C-6 on page 402 we show the assembler source for the IEBCPYL program to load IEBCOPY and invoke the control statement exit IEBCPYC.

The use of the IEBCPARM DSECT=NO statement generates the work area passed to IEBCOPY. Once IEBCOPY has been invoked it will link to the IEBCPYC exit module until told to stop.

Example C-6 Assembler source for IEBCPYL program

```

                MACRO
&NAME          SEGSTART
                AMODE 24
&NAME          STM    14,12,12(13)          SAVE HIS REGS IN HIS SAVE AREA
R0              EQU    0
R1              EQU    1
R3              EQU    3
R4              EQU    4
RB              EQU    12
R13             EQU    13
R14             EQU    14
R15             EQU    15
                BALR   12,0                  SET UP ADDRESSABILITY
                USING  *,12                  USE REG 12 AS BASE REG
                ST      13,SAVEREGS+4        SAVE @ OF HIS SAVEAREA IN MINE
                LA      03,SAVEREGS          LOAD @ OF MY SAVE AREA IN REG 3
                ST      03,8(13)             SAVE @ OF MY SAVE AREA IN HIS
                LR      13,03                LOAD @ OF MY SAVE AREA IN REG 13
                MEND
                MACRO
&NAME          SEGEND
&NAME          L      13,SAVEREGS+4          LOAD REG13 WITH @ OF HIS SAVE
                LM      14,12,12(13)          RESTORE REGS FROM HIS SAVEAREA
                XR      R15,R15
                BR      14
SAVEREGS        DC    18F'0'                RETURN TO CALLING RTN VIA REG 14
                MEND
IEBCPYF         SEGSTART

```

```

* THIS IS A SIMPLE PROGRAM TO LINK TO IEBCOPY INSTRUCTING IT TO
* INVOKE THE CONTROL STATEMENT EXIT MODULE.
*
* R3    PARM
* R4    IEBCREAS DSECT
* R12   OVERALL BASE REGISTER
START   DS    0H
        LA    R1,CPARM_PARMLIST
        MVC   CPARM_OPTIONS_SIZE,=H'0'
        MVC   CPARM_DDNames_SIZE,=H'0'
        MVC   CPARM_PAGENUM_SIZE,=H'4'
        MVC   CPARM_PAGENUM_TEXT,=CL4'0004'
        LOAD  EPLOC=IEBCPYEC
        ST    R0,CPARM_EXIT_CONTROLSTMT_ADD
        OI    CPARM_PARMLIST_EXITS,CPARM_PARMLIST_LASTFLAG
        LA    R1,CPARM_PARMLIST
        LINK  EP=IEBCOPY
FINISH  DS    0H
        SEGEN
        ORG
        DS    0D
IEBCPYEC DC C'IEBCPYC ' EXIT FOR CONTROL STATEMENTS
        ORG
        IEBCPARM DSECT=NO
        IEBCREAS
        END

```

In Example C-7 on page 403 we show example JCL to generate program IEBCPYL. It processes the source from data set MHLRES2.IEBCOPY.DFSMS21.PDS(IEBCPYLA) and creates the program in MHLRES2.IEBCOPY.DFSMS21.LOAD which must be shown as a STEPLIB when executing program IEBCPYL.

Example C-7 JCL to assemble and link the IEBCPYL program

```

//MHLRES2E JOB (1234567,COMMENT),MHLRES2,TIME=10,
// MSGCLASS=J,
// MSGLEVEL=1,CLASS=A,
// NOTIFY=&SYSUID
/*JOBPARM S=*
//ASMHCL PROC
//ASM      EXEC PGM=ASMA90,REGION=0M,
//          PARM='OBJECT,NODECK'
//SYSLIN   DD DSN=&&OBJ,DISP=(NEW,PASS),UNIT=SYSDA,
//          SPACE=(TRK,(10,2)),DCB=BLKSIZE=3120
//SYSLIB   DD DISP=SHR,DSN=SYS1.MACLIB
//SYSPRINT DD SYSOUT=*
//SYSUT1   DD DSN=&&SYSUT1,UNIT=SYSDA,SPACE=(CYL,(5,5))
// *
//LKED     EXEC PGM=HEWL,REGION=2048K,COND=(8,LE,ASM),
//          PARM='XREF,LIST,LET'
//SYSLIN   DD DSN=&&OBJ,DISP=(OLD,DELETE)
//          DD DDNAME=SYSIN
//SYSPRINT DD SYSOUT=*
//SYSUT1   DD DSN=&&SYSUT1,UNIT=SYSDA,SPACE=(CYL,(5,5))
// PEND

```

```
// EXEC ASMHCL
//ASM.SYSIN DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.PDS(IEBCPYLA)
/*
//LKED.SYSLMOD DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.LOAD
//LKED.SYSIN DD *
SETSSI 00002100
NAME IEBCPYL(R)
```

Program IEBCPYL loads module IEBCPYC as the control statement exit.

In Example C-8 on page 404 we show the source for the IEBCPYC module.

The statement IEBCPLST is significant because it generates the map of the area that is passed by IEBCOPY as a result of being invoked with the PARAM that names the IEBCPYC exit.

Communication back to IEBCOPY is through updates to specific areas of the data area mapped by IEBCPLST, in particular note that R15 must not be used to send return codes to IEBCOPY.

This example code is intended to illustrate the possibilities for use of the control statement exit and as such contains diagnostic code.

The sections of code labeled CTRL1/2/3 and E manage the return of information to IEBCOPY then the last call to inform IEBCOPY not to return for further control statements.

Example C-8 Assembler source for IEBCPYC module

```
&NAME SEGSTART
MACRO
&NAME SEGSTART
AMODE 24
&NAME STM 14,12,12(13) SAVE HIS REGS IN HIS SAVE AREA
R1 EQU 1
R3 EQU 3
R4 EQU 4
R5 EQU 5
RB EQU 12
R13 EQU 13
R14 EQU 14
R15 EQU 15
BALR 12,0 SET UP ADDRESSABILITY
USING *,12 USE REG 12 AS BASE REG
ST 13,SAVEREGS+4 SAVE @ OF HIS SAVEAREA IN MINE
LA 03,SAVEREGS LOAD @ OF MY SAVE AREA IN REG 3
ST 03,8(13) SAVE @ OF MY SAVE AREA IN HIS
LR 13,03 LOAD @ OF MY SAVE AREA IN REG 13
MEND
MACRO
&NAME SEGEND
&NAME L 13,SAVEREGS+4 LOAD REG13 WITH @ OF HIS SAVE
LM 14,12,12(13) RESTORE REGS FROM HIS SAVEAREA
XR R15,R15
BR 14 RETURN TO CALLING RTN VIA REG 14
SAVEREGS DC 18F'0' SET UP SAVE AREA
```

```

MEND
IEBCPYC  SEGSTART
*  THIS IS A SIMPLE PROGRAM TO GENERATE IEBCOPY CONTROL
*  STATEMENTS TO PASS TO IEBCOPY.
*
*  R3    PARM
*  R4    WORK
*  R5    CPLST DATA AREA
*  R12   OVERALL BASE REGISTER
START    DS    0H
         LR    R3,R1
         WTO   'GOT TO IEBCPYEC'
         USING CPLST_PARMLIST,R3
         USING CPLST_CONTROL_AREA,R5
         TM    CPLST_PARMLIST_CONTROL_FLAGS,CPLST_CONTROL_INIT
         BO    INIT
         TM    CPLST_PARMLIST_CONTROL_FLAGS,CPLST_CONTROL_DATA
         BO    CONTROL
         DC    H'0'
INIT     DS    0H
         WTO   'CPLST_CONTROL_INIT'
         LA    R4,CPLST_RC_INIT
         B     SETRC
CONTROL  DS    0H
         CLI   CONTROLF,C'1'
         BE    CTRL1
         CLI   CONTROLF,C'2'
         BE    CTRL2
         CLI   CONTROLF,C'3'
         BE    CTRL3
         CLI   CONTROLF,C'E'
         BE    CTRL4
* LAST TIME IN
CTRL4    WTO   'CPLST_CONTROL_DATA - E'
         LA    R4,CPLST_RC_END
         LA    R4,CPLST_RC_FINISH
         B     SETRC
CTRL1    WTO   'CPLST_CONTROL_DATA - 1'
         MVI   CONTROLF,C'2'
         L     R5,CPLST_PARMLIST_ENTRY_ADD
         LA    R4,COPY1
         ST    R4,CPLST_CONTROL_DATA_ADD
         LA    R4,L'COPY1
         ST    R4,CPLST_CONTROL_DATA_LEN
         OI    CPLST_CONTROL_OFLAG1,CPLST_CONTROL_IN
         LA    R4,CPLST_RC_CONTROL
         B     SETRC
CTRL2    WTO   'CPLST_CONTROL_DATA - 2'
         MVI   CONTROLF,C'3'
         L     R5,CPLST_PARMLIST_ENTRY_ADD
         LA    R4,COPY1
         ST    R4,CPLST_CONTROL_DATA_ADD
         LA    R4,L'COPY1
         ST    R4,CPLST_CONTROL_DATA_LEN
         OI    CPLST_CONTROL_OFLAG1,CPLST_CONTROL_IN

```

```

                LA    R4,CPLST_RC_CONTROL
                B     SETRC
CTRL3          WTO   'CPLST_CONTROL_DATA - 3'
                MVI   CONTROLF,C'E'
                L     R5,CPLST_PARMLIST_ENTRY_ADD
                LA    R4,COPY2
                ST     R4,CPLST_CONTROL_DATA_ADD
                LA    R4,L'COPY2
                ST     R4,CPLST_CONTROL_DATA_LEN
                OI     CPLST_CONTROL_OFLAG1,CPLST_CONTROL_IN
                LA    R4,CPLST_RC_CONTROL
                B     SETRC
SETRC          DS     0H
                ST     R4,CPLST_PARMLIST_RC
FINISH         DS     0H
                SEGEND
COPY1          DC     CL72'  COPY INDD=SYSUTX,OUTDD=SYSUTY  '
COPY2          DC     CL72'  SELECT MEMBER=A                '
CONTROLF       DC     C'1'  NEXT ACTION, OR 'E' TO FINISH
                IEBCPLST
                ORG
                END

```

In Example C-9 on page 406 we show example JCL to generate module IEBCPYC. It processes the source from data set MHLRES2.IEBCOPY.DFSMS21.PDS(IEBCPYC) and creates the module in MHLRES2.IEBCOPY.DFSMS21.LOAD which must be shown as a STEPLIB when executing program IEBCPYL.

Example C-9 JCL to assemble and link the IEBCOPYC module

```

//MHLRES2E JOB (1234567,COMMENT),MHLRES2,TIME=10,
// MSGCLASS=J,
// MSGLEVEL=1,CLASS=A,
// NOTIFY=&SYSUID
/*JOBPARM S=*
//ASMHCL PROC
//ASM      EXEC PGM=ASMA90,REGION=0M,
//          PARM='OBJECT,NODECK'
//SYSLIN   DD DSN=&&OBJ,DISP=(NEW,PASS),UNIT=SYSDA,
//          SPACE=(TRK,(10,2)),DCB=BLKSIZE=3120
//SYSLIB   DD DISP=SHR,DSN=SYS1.MACLIB
//SYSPRINT DD SYSOUT=*
//SYSUT1   DD DSN=&&SYSUT1,UNIT=SYSDA,SPACE=(CYL,(5,5))
//*
//LKED     EXEC PGM=HEWL,REGION=2048K,COND=(8,LE,ASM),
//          PARM='XREF,LIST,LET'
//SYSLIN   DD DSN=&&OBJ,DISP=(OLD,DELETE)
//          DD DDNAME=SYSIN
//SYSPRINT DD SYSOUT=*
//SYSUT1   DD DSN=&&SYSUT1,UNIT=SYSDA,SPACE=(CYL,(5,5))
// PEND
// EXEC ASMHCL
//ASM.SYSIN DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.PDS(IEBCPYC)
/*
//LKED.SYSLMOD DD DISP=SHR,DSN=MHLRES2.IEBCOPY.DFSMS21.LOAD

```



```
//LKED.SYSIN DD *
SETSSI 00002100
NAME IEBCPYC(R)
```

Sample job to list VTOC to show DS1DSCHA flag

In Example C-10, and in the successive examples, we show the assembler source code for the program to display the Data set changed indicator (DS1DSCHA) for up to 20 data sets on a volume. This program also check various EAV indicators on the volume.

Example C-10 Source for CVSEQ8D part 1 of 19

CVSEQ8D TITLE 'CVAF CVAFSEQ TEST MODULE'	00010008
CVSEQ8D CSECT	00020008
CVSEQ8D AMODE 31	00030008
CVSEQ8D RMODE 24	00040008
*****	00050000
*	* 00060000
* CVSEQ8D - MODULE THAT ISSUES THE CVAFSEQ MACRO AND PROCESS A	* 00070008
* A VOLUME TO RETURN DSCBS, 20 AT A TIME, IN PHYSICAL	* 00080000
* SEQUENTIAL ORDER USING A STARTING CCHHR OF ZERO.	* 00090000
*	* 00100000
* THE CVAFSEQ MACRO CALL WILL BE ISSUED THREE TIMES	* 00110000
* USING THE FOLLOWING EADSCB KEYWORD SETTINGS:	* 00120000
* - EADSCB KEYWORD NOT CODED (DEFAULTS TO EADSCB=NOTOK)	* 00130000
* - EADSCB=NOTOK CODED	* 00140000
* - EADSCB=OK CODED	* 00150000
* CV4EADOK BIT SETTING WILL BE DETERMINED FOR EACH CALL.	* 00160000
*	* 00170000
* FOR A NON EAV VOLUME:	* 00180000
* WILL EXPECT ALL DSCBS ON THE VOLUME RETURNED IN THE	* 00190000
* WILL EXPECT ALL DSCBS ON THE VOLUME RETURNED IN THE	* 00190000
* ORDER THE DATASETS WERE CREATED.	* 00200000
*	* 00210000
* FOR A EAV VOLUME:	* 00220000
* WILL EXPECT RC04 / STAT082 RETURNED FROM CVAFSEQ CALL	* 00230000
* WHEN THE EADSCB KEYWORD IS NOT CODED OR EADSCB=NOTOK	* 00240000
* IS CODED.	* 00250000
* WILL EXPECT ALL DSCBS ON THE VOLUME RETURNED IN THE	* 00260000
* ORDER THE DATASETS WERE CREATED WHEN EADSCB=OK IS	* 00270000
* CODED.	* 00280000
*	* 00290000
* THIS PROGRAM WILL CREATE AN OUTPUT REPORT THAT	* 00300000
* SHOULD BE SIMILAR TO THE EXAMPLES LISTED BELOW.	* 00310000

Example C-11 Source for CVSEQ8D part 2 of 19

* -----	* 00320000
*	* 00330000
* EXAMPLE 1: NON EAV VOLUME WITH OS VTOC	* 00340000
* -----	* 00350000
*	* 00360000
* CVSEQ8D START OF OUTPUT MESSAGES	* 00370008
*	* 00380000
* CVAFSEQ CALL: EADSCB KEYWORD NOT CODED	* 00390000
* CV4EADOK BIT IS NOT SET / EADSCB=NOTOK	* 00400000
* RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL	* 00410000
* X"00" DEC"000" 00 - CVSTAT CODE VERIFIED	* 00420000
*	* 00430000
* CVAFSEQ CALL: EADSCB=NOTOK CODED	* 00440000
* CV4EADOK BIT IS NOT SET / EADSCB=NOTOK	* 00450000
* RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL	* 00460000
* X"00" DEC"000" 00 - CVSTAT CODE VERIFIED	* 00470000
*	* 00480000
* CVAFSEQ CALL: EADSCB=OK CODED	* 00490000
* CV4EADOK BIT SET / EADSCB=OK	* 00500000
* RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL	* 00510000
* X"00" DEC"000" 00 - CVSTAT CODE VERIFIED	* 00520000
*	* 00530000
* DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:	* 00540000
* DSN: CVS5MSC2.SEQ01	* 00550000
* DSN: CVS5MSC2.PDS01	* 00560000
* DSN: CVS5MSC2.VSAM01.DATA	* 00570000
* DSN: SYS1.VVDS.V1P9503	* 00580000
* DSN: CVS5MSC2.PDSE01	* 00590000
* DSN: CVS5MSC2.SEQ02	* 00600000
* DSN: CVS5MSC2.VSAM02.DATA	* 00610000
* DSN: CVS5MSC2.PDSE02	* 00620000
* END OF DATA REACHED - ALL DATASETS PROCESSED	* 00630000
*	* 00640000
* CVSEQ8D END OF OUTPUT MESSAGES	* 00650008
*	* 00660000

Example C-12 Source for CVSEQ8D part 3 of 19

* EXAMPLE 2: EAV VOLUME WITH OS VTOC	* 00670000
* -----	* 00680000
*	* 00690000
* CVSEQ8D START OF OUTPUT MESSAGES	* 00700008
*	* 00710000
* CVAFSEQ CALL: EADSCB KEYWORD NOT CODED	* 00720000
* CV4EADOK BIT IS NOT SET / EADSCB=NOTOK	* 00730000
* RC04 VERIFIED - LOGICAL ERROR STATUS IN CVSTAT	* 00740000
* X"52" DEC"082" 52 - CVSTAT CODE VERIFIED	* 00750000
*	* 00760000
* CVAFSEQ CALL: EADSCB=NOTOK CODED	* 00770000
* CV4EADOK BIT IS NOT SET / EADSCB=NOTOK	* 00780000
* RC04 VERIFIED - LOGICAL ERROR STATUS IN CVSTAT	* 00790000
* X"52" DEC"082" 52 - CVSTAT CODE VERIFIED	* 00800000
*	* 00810000
* CVAFSEQ CALL: EADSCB=OK CODED	* 00820000
* CV4EADOK BIT SET / EADSCB=OK	* 00830000
* RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL	* 00840000
* X"00" DEC"000" 00 - CVSTAT CODE VERIFIED	* 00850000
*	* 00860000
* DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:	* 00870000
* DSN: CVS5MSC2.SEQ01	* 00880000
* DSN: CVS5MSC2.PDS01	* 00890000
* DSN: CVS5MSC2.VSAM01.DATA	* 00900000
* DSN: SYS1.VVDS.V1P9503	* 00910000
* DSN: CVS5MSC2.PDSE01	* 00920000
* DSN: CVS5MSC2.SEQ02	* 00930000
* DSN: CVS5MSC2.VSAM02.DATA	* 00940000
* DSN: CVS5MSC2.PDSE02	* 00950000
* END OF DATA REACHED - ALL DATASETS PROCESSED	* 00960000
*	* 00970000
* CVSEQ8D END OF OUTPUT MESSAGES	* 00980008
*	* 00990000

Example C-13 Source for CVSEQ8D part 4 of 19

* EXAMPLE 3: NON EAV VOLUME WITH IX VTOC	* 01000000
* -----	* 01010000
*	* 01020000
* CVSEQ8D START OF OUTPUT MESSAGES	* 01030008
*	* 01040000
* CVAFSEQ CALL: EADSCB KEYWORD NOT CODED	* 01050000
* CV4EADOK BIT IS NOT SET / EADSCB=NOTOK	* 01060000
* RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL	* 01070000
* X"00" DEC"000" 00 - CVSTAT CODE VERIFIED	* 01080000
*	* 01090000
* CVAFSEQ CALL: EADSCB=NOTOK CODED	* 01100000
* CV4EADOK BIT IS NOT SET / EADSCB=NOTOK	* 01110000
* RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL	* 01120000
* X"00" DEC"000" 00 - CVSTAT CODE VERIFIED	* 01130000
*	* 01140000
* CVAFSEQ CALL: EADSCB=OK CODED	* 01150000
* CV4EADOK BIT SET / EADSCB=OK	* 01160000
* RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL	* 01170000
* X"00" DEC"000" 00 - CVSTAT CODE VERIFIED	* 01180000
*	* 01190000
* DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:	* 01200000
* DSN: SYS1.VTOCIX.VP9503	* 01210000
* DSN: CVS5MSC2.SEQ01	* 01220000
* DSN: SYS1.VVDS.V1P9502	* 01230000
* DSN: CVS5MSC2.PDS01	* 01240000
* DSN: CVS5MSC2.VSAM01.DATA	* 01250000
* DSN: CVS5MSC2.PDSE01	* 01260000
* DSN: CVS5MSC2.SEQ02	* 01270000
* DSN: CVS5MSC2.VSAM02.DATA	* 01280000
* DSN: CVS5MSC2.PDSE02	* 01290000
* END OF DATA REACHED - ALL DATASETS PROCESSED	* 01300000
*	* 01310000
* CVSEQ8D END OF OUTPUT MESSAGES	* 01320008
*	* 01330000

Example C-14 Source for CVSEQ8D part 5 of 19

* EXAMPLE 4: EAV VOLUME WITH IX VTOC	* 01340000
* -----	* 01350000
*	* 01360000
* CVSEQ8D START OF OUTPUT MESSAGES	* 01370008
*	* 01380000
* CVAFSEQ CALL: EADSCB KEYWORD NOT CODED	* 01390000
* CV4EADOK BIT IS NOT SET / EADSCB=NOTOK	* 01400000
* RC04 VERIFIED - LOGICAL ERROR STATUS IN CVSTAT	* 01410000
* X"52" DEC"082" 52 - CVSTAT CODE VERIFIED	* 01420000
*	* 01430000
* CVAFSEQ CALL: EADSCB=NOTOK CODED	* 01440000
* CV4EADOK BIT IS NOT SET / EADSCB=NOTOK	* 01450000
* RC04 VERIFIED - LOGICAL ERROR STATUS IN CVSTAT	* 01460000
* X"52" DEC"082" 52 - CVSTAT CODE VERIFIED	* 01470000
*	* 01480000
* CVAFSEQ CALL: EADSCB=OK CODED	* 01490000
* CV4EADOK BIT SET / EADSCB=OK	* 01500000
* RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL	* 01510000
* X"00" DEC"000" 00 - CVSTAT CODE VERIFIED	* 01520000
*	* 01530000
* DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENTIAL ORDER:	* 01540000
* DSN: SYS1.VTOCIX.VP9503	* 01550000
* DSN: CVS5MSC2.SEQ01	* 01560000
* DSN: SYS1.VVDS.V1P9502	* 01570000
* DSN: CVS5MSC2.PDS01	* 01580000
* DSN: CVS5MSC2.VSAM01.DATA	* 01590000
* DSN: CVS5MSC2.PDSE01	* 01600000
* DSN: CVS5MSC2.SEQ02	* 01610000
* DSN: CVS5MSC2.VSAM02.DATA	* 01620000
* DSN: CVS5MSC2.PDSE02	* 01630000
* END OF DATA REACHED - ALL DATASETS PROCESSED	* 01640000
*	* 01650000
* CVSEQ8D END OF OUTPUT MESSAGES	* 01660008
*	* 01670000
* NOTE: THIS MODULE REQUIRES THE USE OF THE CVSTCHK MODULE WHICH	* 01680000
* IS CALLED TO INTERPRET THE CVSTAT CODE RETURNED.	* 01690000
*	* 01700000
* -----	* 01710000

Example C-15 Source for CVSEQ8D part 6 of 19

```

***** 01720000
*
* CVSEQ8D - LOGIC NOTES * 01730000
*
* THIS MODULE WILL PERFORM THE FOLLOWING: * 01740008
*
* INITIALIZATION * 01750000
* - OBTAIN THE NECESSARY INFORMATION FROM THE DASD VOLUME * 01760000
* - OPEN THE OUTPUT FILE AND WRITE NECESSARY OUTPUT MESSAGES * 01770000
* - SET STARTING CCHHR TO ZERO * 01780000
* - SET END OF DATA SWITCH TO NO END OF DATA * 01790000
*
* MAINLINE * 01800000
* - LOAD THE TABLE WITH DSCB ADDRESSES TO USE FOR CVAFSEQ CALL * 01810000
* - INITIALIZE BUFFER LIST * 01820000
* - INVOKE CVAFSQ1 ROUTINE - EADSCB KEYWORD NOT CODED * 01830000
* - REPORT ON RETURN CODE AND CVSTAT CODE RETURNED FROM CALL * 01840000
* - INITIALIZE BUFFER LIST * 01850000
* - INVOKE CVAFSQ2 ROUTINE - EADSCB=NOTOK CODED * 01860000
* - REPORT ON RETURN CODE AND CVSTAT CODE RETURNED FROM CALL * 01870000
* - INITIALIZE BUFFER LIST * 01880000
* - INVOKE CVAFSQ3 ROUTINE - EADSCB=OK CODED * 01890000
* - INVOKE CVAFSQ3 ROUTINE - EADSCB=OK CODED * 01900000
* - REPORT ON RETURN CODE AND CVSTAT CODE RETURNED FROM CALL * 01910000
* - DO WHILE MORE TABLE ENTRIES TO PROCESS * 01920000
* - INVOKE PRTBRN ROUTINE TO PROCESS DSCBS RETURNED FROM CVAFSEQ * 01930000
* - PRINT OUT ALL DSNAMEs ON THE VOLUME IN SEQUENTIAL ORDER * 01940000
*
* FINALIZATION * 01950000
* - WRITE NECESSARY MESSAGES AND CLOSE THE OUTPUT FILE AND EXIT * 01960000
*
* CVSEQ8D - JOB INFORMATION * 01970000
*
* NORMAL END OF JOB: * 01980000
* - RC=00 AND OUTDD OUTPUT AS DETAILED ABOVE * 01990000
*
* ABNORMAL END OF JOB: * 02000000
* - ABEND 100 - ERROR OPENING VTOC ON THE DASD VOLUME THAT IS * 02010000
* ASSOCIATED WITH THE CVAFDD DD STATEMENT * 02020008
* - ABEND 101 - ERROR OPENING THE OUTDD DATASET * 02030000
* - ABEND 102 - ERROR CLOSING THE OUTDD DATASET * 02040000
*
* * 02050000
* * 02060000
* * 02070000
* * 02080000
* * 02090000
* * 02100000
* * 02110000

```

Example C-16 Source for CVSEQ8D part 7 of 19

```

***** 02120000
*
* HOUSEKEEPING
* - SAVE CALLER'S REGISTERS AND ESTABLISH A NEW REGISTER SAVE AREA
*
*          STM R14,R12,12(R13)  STANDARD LINKAGE CONVENTION
*          BALR R11,0           DCL R11 AS IMPLIED BASE REG
*          USING BASE,R11,R12   R12 IS ALSO BASE REG
BASE      L    R12,BASEADDR     SET UP ADDRESSING FOR R12
*          B    CV000000        BRANCH AROUND DECLARES
BASEADDR DC    A(BASE+4096)     ADDRESSING FOR R12
CV000000 DS    OH              CONTINUE...
*          ST   R13,SAVE+4      SAVE PTR TO CALLER'S SAVE AREA
*          LA   R14,SAVE        GET ADDRESS OF THE NEW SAVE AREA
*          ST   R14,8(,R13)     CHAIN CALLER'S AREA TO OURS
*          LR   R13,R14         ESTABLISH THE NEW SAVE AREA
*
***** 02280000
***** 02290000
*
* INITIALIZATION
*
INITIAL   DS    OH              INITIALIZATION SECTION
*          BAL  R14,IDVOLRTN     INVOKE RTN TO IDENTIFY THE VOLUME(S)
*          OPEN (OUTFILE,(OUTPUT)) OPEN THE OUTPUT MESSAGE FILE
*          TM    OUTFILE+48,X'10' IF OPEN OF OUTPUT FILE NOT OK
*          BO    OK1
*          ABEND 101             ISSUE USER ABEND 101
OK1       DS    OH              02380000
*          L    R1,UCBADD        02390000
*          MVC  VOLSER(6),28(R1)  02400000
*          PUT  OUTFILE,STRMSG    WRITE A RECORD TO THE OUTPUT FILE 02410000
*          PUT  OUTFILE,BLNKLINE  WRITE A RECORD TO THE OUTPUT FILE 02420000
*          MVC  CCHHRS,CCHHRO     INIT CCHHR START TO ZERO          02430000
*          MVI  SWEOD,NOEOD       SET SWITCH TO NO END OF DATA      02440000
*
***** 02450000
***** 02460000
***** 02470000
*
***** 02480000

```

Example C-17 Source for CVSEQ8D part 8 of 19

```

* MAINLINE                                     * 02490000
*                                               02500000
MAINLINE DS    OH                               MAINLINE SECTION      02510000
      BAL  R14,LDTABRTN      INVOKE LDTABRTN TO LOAD TABLE      02520000
      BAL  R14,INITBRTN      INVOKE INITBRTN TO INIT BUFF LIST  02530000
      BAL  R14,CVAFSQ1       INVOKE CVAFSQ1 TO ISSUE CVAFSEQ     02540000
      PUT  OUTFILE,BLNKLINE  WRITE THE REC TO OUTPUT FILE        02550000
      BAL  R14,INITBRTN      INVOKE INITBRTN TO INIT BUFF LIST  02560000
      BAL  R14,CVAFSQ2       INVOKE CVAFSQ2 TO ISSUE CVAFSEQ     02570000
      PUT  OUTFILE,BLNKLINE  WRITE THE REC TO OUTPUT FILE        02580000
      BAL  R14,INITBRTN      INVOKE INITBRTN TO INIT BUFF LIST  02590000
      BAL  R14,CVAFSQ3       INVOKE CVAFSQ3 TO ISSUE CVAFSEQ     02600000
      PUT  OUTFILE,BLNKLINE  WRITE THE REC TO OUTPUT FILE        02610000
      PUT  OUTFILE,MSGO      WRITE THE REC TO OUTPUT FILE        02620000
*                                               02630000
      CLI  SWEOD,EOD         DOWHILE TABLE DATA TO PROCESS    02640000
      BE   TABDONE           02650000
      BAL  R14,PRTBRTN       INVOKE PRTBRTN TO PROCESS TBL      02660000
*                                               02670000
TABDONE DS    OH                               02680000
      PUT  OUTFILE,EODMSG    WRITE THE REC TO OUTPUT FILE      02690000
*                                               02700000
***** 02710000
*                                               * 02720000
* FINALIZATION                               * 02730000
*                                               02740000
FINAL   DS    OH                               FINALIZATION SECTION 02750000
      PUT  OUTFILE,BLNKLINE  WRITE A RECORD TO THE OUTPUT FILE  02760000
      PUT  OUTFILE,ENDMSG    WRITE A RECORD TO THE OUTPUT FILE  02770000
      CLOSE (OUTFILE)        CLOSE OUTPUT FILE                  02780000
      C     R15,RCODE00      IF FILE CLOSE IS NOT OK             02790000
      BE   OK2               02800000
      ABEND 102              ISSUE USER ABEND 102              02810000
OK2     DS    OH                               02820000
      L     R13,4(R13)       RESTORE REGISTER                    02830000
      LM    R14,R12,12(R13)  RESTORE CALLERS REGISTERS          02840000
      LA    R15,0            SET RC TO 0                         02850000
      BR    R14              RETURN TO CALLER                    02860000
*                                               02870000
***** 02880000

```

Example C-18 Source for CVSEQ8D part 9 of 19

*	IDVOLRTN	* 02890000
*	- OBTAIN THE NECESSARY INFORMATION FROM THE DASD VOLUME	* 02900000
IDVOLRTN DS	OH IDENTIFY VOLUME ROUTINE	02910000
ST	R14, IDVLSAVE STORE C(R14) INTO SAVE AREA	02920000
RDJFCB	(VTOCDCB, (INPUT)) READ JFCB / OPEN VTOC	02930000
MVI	JFCB1, X'04' PUT IN ID FOR FORMAT 4	02940000
MVC	JFCB1+1(43), JFCB1 SETUP FOR VTOC OPEN	02950000
OPEN	(VTOCDCB, (INPUT)), TYPE=J OPEN VTOC (OPEN TYPE=J)	02960000
TM	VTOCDCB+48, X'10' IF OPEN OF VTOC NOT OK	02970000
BO	OK3	02980000
	ABEND 100 ISSUE USER ABEND 100	02990000
OK3 DS	OH	03000000
SLR	R3, R3 INIT R3 FOR DEB PTR	03010000
SLR	R4, R4 INIT R4 FOR UCB PTR	03020000
ICM	R3, B'0111', VTOCDCB+45 GET DEB ADDRESS	03030000
ST	R3, DEBADD SAVE DEB ADDRESS	03040000
ICM	R4, B'0111', 33(R3) GET UCB ADDRESS	03050000
ST	R4, UCBADD STORE UCB ADDRESS	03060000
IDVLEXIT DS	OH EXIT FROM IDVOLRTN	03070000
L	R14, IDVLSAVE LOAD C(IDVLSAVE) INTO R14	03080000
BR	R14 EXIT	03090000
*		03100000
*****		03110000
*	LDTABRTN	* 03120000
*	- LOAD 20 ENTRY TABLE WITH DSCB ADDRESSES TO USE	* 03130000
*		03140000
LDTABRTN DS	OH LOAD TABLE ROUTINE	03150000
ST	R14, LDTBSAVE STORE C(R14) INTO SAVE AREA	03160000
LA	R4, DSCB01 LOAD R4 WITH ADDRESS OF DSCB01	03170000
ST	R4, TDSCB01 STORE ADDRESS OF DSCB01 INTO TABLE	03180000
LA	R4, DSCB02 LOAD R4 WITH ADDRESS OF DSCB02	03190000
ST	R4, TDSCB02 STORE ADDRESS OF DSCB02 INTO TABLE	03200000
LA	R4, DSCB03 LOAD R4 WITH ADDRESS OF DSCB03	03210000
ST	R4, TDSCB03 STORE ADDRESS OF DSCB03 INTO TABLE	03220000
LA	R4, DSCB04 LOAD R4 WITH ADDRESS OF DSCB04	03230000
ST	R4, TDSCB04 STORE ADDRESS OF DSCB04 INTO TABLE	03240000
LA	R4, DSCB05 LOAD R4 WITH ADDRESS OF DSCB05	03250000
ST	R4, TDSCB05 STORE ADDRESS OF DSCB05 INTO TABLE	03260000
LA	R4, DSCB06 LOAD R4 WITH ADDRESS OF DSCB06	03270000
ST	R4, TDSCB06 STORE ADDRESS OF DSCB06 INTO TABLE	03280000
LA	R4, DSCB07 LOAD R4 WITH ADDRESS OF DSCB07	03290000
ST	R4, TDSCB07 STORE ADDRESS OF DSCB07 INTO TABLE	03300000
LA	R4, DSCB08 LOAD R4 WITH ADDRESS OF DSCB08	03310000
ST	R4, TDSCB08 STORE ADDRESS OF DSCB08 INTO TABLE	03320000
~~~~~ST	R4, TDSCB08 STORE ADDRESS OF DSCB08 INTO TABLE	03320000
LA	R4, DSCB09 LOAD R4 WITH ADDRESS OF DSCB09	03330000
ST	R4, TDSCB09 STORE ADDRESS OF DSCB09 INTO TABLE	03340000
LA	R4, DSCB10 LOAD R4 WITH ADDRESS OF DSCB10	03350000
ST	R4, TDSCB10 STORE ADDRESS OF DSCB10 INTO TABLE	03360000

Example C-19 Source for CVSEQ8D part 10 of 19

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```

~~~~~LA    R4,DSCB11      LOAD R4 WITH ADDRESS OF DSCB11      03370000
 ST R4,TDSCB11 STORE ADDRESS OF DSCB11 INTO TABLE 03380000
 LA R4,DSCB12 LOAD R4 WITH ADDRESS OF DSCB12 03390000
 ST R4,TDSCB12 STORE ADDRESS OF DSCB12 INTO TABLE 03400000
 LA R4,DSCB13 LOAD R4 WITH ADDRESS OF DSCB13 03410000
 ST R4,TDSCB13 STORE ADDRESS OF DSCB13 INTO TABLE 03420000
 LA R4,DSCB14 LOAD R4 WITH ADDRESS OF DSCB14 03430000
 ST R4,TDSCB14 STORE ADDRESS OF DSCB14 INTO TABLE 03440000
 LA R4,DSCB15 LOAD R4 WITH ADDRESS OF DSCB15 03450000
 ST R4,TDSCB15 STORE ADDRESS OF DSCB15 INTO TABLE 03460000
 LA R4,DSCB16 LOAD R4 WITH ADDRESS OF DSCB16 03470000
 ST R4,TDSCB16 STORE ADDRESS OF DSCB16 INTO TABLE 03480000
 LA R4,DSCB17 LOAD R4 WITH ADDRESS OF DSCB17 03490000
 ST R4,TDSCB17 STORE ADDRESS OF DSCB17 INTO TABLE 03500000
 LA R4,DSCB18 LOAD R4 WITH ADDRESS OF DSCB18 03510000
 ST R4,TDSCB18 STORE ADDRESS OF DSCB18 INTO TABLE 03520000
 LA R4,DSCB19 LOAD R4 WITH ADDRESS OF DSCB19 03530000
 ST R4,TDSCB19 STORE ADDRESS OF DSCB19 INTO TABLE 03540000
 LA R4,DSCB20 LOAD R4 WITH ADDRESS OF DSCB20 03550000
 ST R4,TDSCB20 STORE ADDRESS OF DSCB20 INTO TABLE 03560000
LDTBEXIT DS OH EXIT FROM LDTABRTN 03570000
 L R14,LDTBSAVE LOAD C(LDTBSAVE) INTO R14 03580000
 BR R14 EXIT 03590000
* 03600000
***** 03610000
* INITBRTN * 03620000
* - INITIALIZE THE BUFFER LIST * 03630000
* 03640000
INITBRTN DS OH INITIALIZE BUFFER LIST ROUTINE 03650000
 ST R14,INITSAVE STORE C(R14) INTO SAVE AREA 03660000
 LA R7,BUFLISTE LOAD R7 WITH ADDRESS OF BUFLIST ENTRY 03670000
 USING BFLE,R7 ESTABLISH ADDRESSABILITY TO BFLE 03680000
 LA R8,BUFLISTH LOAD R8 WITH ADDRESS OF BUFLIST HDR 03690000
 USING BFLHDR,R8 ESTABLISH ADDRESSABILITY TO BFLHDR 03700000
 LA R2,TABLE LOAD R2 WITH ADDRESS OF TABLE 03710000
 USING TBLMAP,R2 ESTABLISH ADDRESSABILITY USING TBLMAP 03720000
 XC BFLHDR(BFLHLN+TBLNBR*BFLELN),BFLHDR CLEAR BUFLIST 03730000
 OI BFLHFL,BFLHDSCB DSCBS TO BE READ WITH BUFFER LIST 03740000
 MVC BFLEARG,CCHHRS MOVE STARTING CCHHR TO ARG 03750000
 MVI BFLHNOE,TBLNBR MOVE NBR OF TBL ENTRIES TO BUFF NBR 03760000
 SLR R4,R4 INIT R4 WITH ZERO 03770000
 IC R4,BFLHNOE NBR OF BUFFER ENTRIES IN R4 03780000
 ST R4,COUNT NBR OF BUFFER ENTRIES IN COUNT 03790000
* 03800000

```

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Example C-20 Source for CVSEQ8D part 11 of 19

INIT0010	DS	0H	INIT BUFFER LIST WITH DSCB ADDR/LENG	03810000
	L	R5,DSCBA	LOAD R5 WITH DSCB ADDRESS FROM TABLE	03820000
	ST	R5,BFLEBUF-BFLE(R7)	PLACE IN BUFFER LIST	03830000
	MVI	BFLELTH-BFLE(R7),DSCBLTH	FULL DSCB READ	03840000
	OI	BFLEFL,BFLECHR	CCHHR TO BE RETURNED	03850000
	LA	R2,TBLLNG(R2)	POINT TO NEXT TABLE ENTRY	03860000
	LA	R7,BFLELN(R7)	POINT TO NEXT BUFFER LIST ENTRY	03870000
	BCT	R4,INIT0010	BRANCH TO INIT0010 IF C(R4) GT ZERO	03880000
	DROP	R2,R8	DROP R2,R7,R8	03890010
*	DROP	R2,R7,R8	DROP R2,R7,R8	03891010
INITEXIT	DS	0H	EXIT FROM INITBRTN	03900000
	L	R14,INITSAVE	LOAD C(INITSAVE) INTO R14	03910000
	BR	R14	EXIT	03920000
*				03930000
*****				03940000
*		PRTBRTN		* 03950000
*	-	PROCESS TABLE WHICH CONTAINS ADDRESS OF DSCB FOR EACH ENTRY		* 03960000
*		RETURNED FROM CVAFSEQ CALL.		* 03970000
*		TABLE IS CURRENTLY 20 ENTRIES.		* 03980000
*				03990000
PRTBRTN	DS	0H	PROCESS TABLE ENTRIES	04000000
	ST	R14,PRTBSAVE	STORE C(R14) INTO SAVE AREA	04010000
	L	R4,COUNT	LOAD COUNT IN R4	04020000
	LA	R2,TABLE	LOAD ADDRESS OF TABLE INTO R2	04030000
	USING	TBLMAP,R2	ESTABLISH ADDRESSABILITY TO TABLE	04040000
PRTB0000	DS	0H	PROCESS ENTRIES	04050000
	L	R3,DSCBA	ADDRESSABILITY TO DSCBA	04060000
	CLC	0(1,R3),FMT4	IS IT A FMT4?	04070000
	BNE	PRTB0010	NO, THEN CONTINUE TO PROCESS DSN	04080000
	B	PRTB0060	YES, BRANCH TO POINT TO NEXT ENTRY	04090000
PRTB0010	DS	0H	FMT5 CHECK	04100000
	CLC	0(1,R3),FMT5	IS IT A FMT5?	04110000
	BNE	PRTB0020	NO, THEN CONTINUE TO PROCESS DSN	04120000
	B	PRTB0060	YES, BRANCH TO POINT TO NEXT ENTRY	04130000
PRTB0020	DS	0H	FMT7 CHECK	04140000
	CLC	0(1,R3),FMT7	IS IT A FMT7?	04150000
	BNE	PRTB0024	NO, THEN CONTINUE TO PROCESS DSN	04160000
	B	PRTB0060	YES, BRANCH TO POINT TO NEXT ENTRY	04170000
PRTB0024	DS	0H	FMT9 CHECK	04180000
	CLC	0(1,R3),FMT9	IS IT A FMT9?	04190000
	BNE	PRTB0030	NO, THEN CONTINUE TO PROCESS DSN	04200000
	B	PRTB0060	YES, BRANCH TO POINT TO NEXT ENTRY	04210000
PRTB0030	DS	0H	DETERMINE IF END OF DATA WAS REACHED	04220000
	CLC	0(1,R3),NODSN	IS THERE '00' IN FIRST BYTE	04230000
	BNE	PRTB0040	NO, THEN CONTINUE TO PROCESS DSN	04240000
	B	PRTB0060	YES, BRANCH TO POINT TO NEXT ENTRY	04250000

Example C-21 Source for CVSEQ8D part 12 of 19

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PRTB0040	DS	OH	PROCESS DSN - FORMAT	04260000
	USING	IECDSL1,R3		04261011
	MVC	DSNMSG(44),DS1DSNAM	OVE DSN TO PRINT LINE	04270010
	MVI	DSNCHA,C'N'		04270113
	TM	DS1DSIND,DS1DSCHA	IS THE CHANGED BIT ON	04270213
*	TM	DS1DSCHA,DS1IND02	IS THE CHANGED BIT ON	04270313
	BNO	DSNCHANO		04270413
	MVI	DSNCHA,C'Y'		04270513
DSNCHANO	EQU	*		04270613
*	MVC	DSNMSG(44),0(R3)	MOVE DSN TO PRINT LINE	04271011
PRTB0050	DS	OH	PROCESS / FORMAT CCHHR	04280000
	PUT	OUTFILE,MSG1	WRITE A RECORD TO THE OUTPUT FILE	04290000
PRTB0060	DS	OH	PREPARE TO PROCESS NEXT TABLE ENTRY	04300000
	LA	R2,TBLLNG(R2)	POINT TO NEXT TABLE ENTRY	04310000
	BCT	R4,PRTB0000	BRANCH TO PRTB0000 IF C(R4) GT ZERO	04320000
	MVI	SWEOD,EOD	YES, SET SWITCH TO END OF DATA	04330000
PRTBEXIT	DS	OH	EXIT FROM PRTBRTN	04340000
	L	R14,PRTBSAVE	LOAD C(PRTBSAVE) INTO R14	04350000
	BR	R14	EXIT	04360000
*				04370000
*****				04380000
*		TSTRCRTN		* 04390000
*		- TEST RETURN CODE FROM CVAFSEQ		* 04400000
*		- FORMAT AND PRINT MESSAGES AS NEEDED		* 04410000
*		- INVOKE CVSTAT MODULE TO CHECK CVSTAT CODE		* 04420000
*				04430000
TSTRCRTN	DS	OH	CHECK RETURN CODE ROUTINE	04440000
	ST	R14,TSTRSAVE	STORE C(R14) INTO SAVE AREA	04450000
	L	R15,RETCODE	LOAD R15 WITH SAVED RETURN CODE	04460000
	C	R15,RCODE16		04470000
	BNL	PRCERMSG		04480000
	B	PROCESS(R15)		04490000
	SPACE			04500000
PROCESS	EQU	*		04510000
	B	PRC00MSG	0 RC MESSAGE	04520000
	B	PRC04MSG	4 RC MESSAGE	04530000
	B	PRC08MSG	8 RC MESSAGE	04540000
	B	PRC12MSG	12 RC MESSAGE	04550000
	B	PRC16MSG	16 RC MESSAGE	04560000
	SPACE			04570000
PRC00MSG	PUT	OUTFILE,RC00MSG	WRITE RC00 MESSAGE	04580000
	B	PUTMSG		04590000

---

Example C-22 Source for CVSEQ8D part 13 of 19

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PRC04MSG	PUT	OUTFILE,RC04MSG	WRITE RC04 MESSAGE	04600000
	B	PUTMSG		04610000
PRC08MSG	PUT	OUTFILE,RC08MSG	WRITE RC08 MESSAGE	04620000
	B	PUTMSG		04630000
PRC12MSG	PUT	OUTFILE,RC12MSG	WRITE RC12 MESSAGE	04640000
	B	PUTMSG		04650000
PRC16MSG	PUT	OUTFILE,RC16MSG	WRITE RC16 MESSAGE	04660000
	B	PUTMSG		04670000
PRCERMSG	PUT	OUTFILE,RCERMSG	WRITE RC ERROR MESSAGE	04680000
PUTMSG	DS	0H		04690000
	MVC	WCVSTCD(1),CVSTAT	MOVE CVSTAT CODE TO WCVSTCD	04700000
	BAL	R14,CALLCVST	INVOKE CALLCVST TO CALL CVSTCHK MOD	04710000
TSTREXIT	DS	0H	EXIT FROM TSTRCRTN	04720000
	L	R14,TSTRSAVE	LOAD C(TSTRSAVE) INTO R14	04730000
	BR	R14	EXIT	04740000
*				04750000
*****				04760000
*		CALLCVST		* 04770000
*	-	ROUTINE TO CALL CVSTCHK TO INTERPRET CVSTAT TABLE		* 04780000
*				04790000
CALLCVST	DS	0H	CALL TO CVSTCHK ROUTINE	04800000
	ST	R14,CALLSAVE	STORE C(R14) INTO SAVE AREA	04810000
	LA	R1,PARADDR1	LOAD R1 WITH PARMLIST FOR CALL	04820000
	L	R15,=V(CVSTCHK)	SET UP CALL TO CVSTCHK	04830000
	BALR	R14,R15	INVOKE CVSTCHK EXTERNAL SUBPROGRAM	04840000
	PUT	OUTFILE,WFMTRC1	WRITE A RECORD TO THE OUTPUT FILE	04850000
CALLEXIT	DS	0H	EXIT FROM CALLCVST	04860000
	L	R14,CALLSAVE	LOAD C(CALLSAVE) INTO R14	04870000
	BR	R14	EXIT	04880000
*				04890000
*****				04900000
*		CVAFSQ1		* 04910000
*	-	INVOKE THE CVAFSEQ MACRO AND READ THE DSCBS		* 04920000
*	EADSCB	KEYWORD NOT CODED - DEFAULTS TO EADSCB=NOTOK		* 04930000
*				04940000
CVAFSQ1	DS	0H	CVAFSEQ - ROUTINE1	04950000
	ST	R14,CVS1SAVE	STORE C(R14) INTO SAVE AREA	04960000
	LA	R2,CVPLIST	LOAD R2 WITH ADDR OF CVPLIST	04970000
	L	R3,UCBADD	LOAD R3 WITH UCB ADDRESS	04980000
	PUT	OUTFILE,CALLMR11	WRITE A REC TO OUTPUT FILE	04990000
*				05000000
	CVAFSEQ	ACCESS=GT,		X05010000
		UCB=(R3),		X05020000
		BRANCH=(YES,PGM),		X05030000
		BUFLIST=BUFLISTH,		X05040000
		MF=(E,(R2))		05050000
*				05060000
	ST	R15,RETCODE	STORE RC INTO RETCODE	05070000
*				05080000

---

Example C-23 Source for CVSEQ8D part 14 of 19

---

```

* REPORT ON CV4EADOK BIT SETTING 05090000
* 05100000
 TM CVFL4,CV4EADOK SELECT BASED ON FLAG4 05110000
 BNO NOTOK1 05120000
 PUT OUTFILE,OKMSG WRITE OK MSG RECORD 05130000
 B OK4 05140000
NOTOK1 PUT OUTFILE,NOTOKMSG WRITE NOTOK MSG RECORD 05150000
OK4 DS OH 05160000
* 05170000
 BAL R14,TSTRCRTN INVOKE ROUTINE TO CHECK RC/STAT 05180000
* 05190000
CVS1EXIT DS OH EXIT FROM CVAFSQ1 05200000
 L R14,CVS1SAVE LOAD C(SAVE AREA) INTO R14 05210000
 BR R14 EXIT 05220000
* 05230000
***** 05240000
* CVAFSQ2 * 05250000
* - INVOKE THE CVAFSEQ MACRO AND READ THE DSCBS * 05260000
* EADSCB=NOTOK IS CODED * 05270000
* 05280000
CVAFSQ2 DS OH CVAFSEQ - ROUTINE2 05290000
 ST R14,CVS2SAVE STORE C(R14) INTO SAVE AREA 05300000
 LA R2,CVPLIST LOAD R2 WITH ADDR OF CVPLIST 05310000
 L R3,DEBADD LOAD R3 WITH DEB ADDRESS 05320000
 PUT OUTFILE,CALLMR12 WRITE A REC TO OUTPUT FILE 05330000
* 05340000
 CVAFSEQ ACCESS=GT, X05350000
 DEB=(R3), X05360000
 BRANCH=(YES,PGM), X05370000
 BUFLIST=BUFLISTH, X05380000
 EADSCB=NOTOK, X05390000
 MF=(E,(R2)) 05400000
* 05410000
 ST R15,RETCODE STORE RC INTO RETCODE 05420000
* 05430000
* REPORT ON CV4EADOK BIT SETTING 05440000
* 05450000
 TM CVFL4,CV4EADOK SELECT BASED ON FLAG4 05460000
 BNO NOTOK2 05470000
 PUT OUTFILE,OKMSG WRITE OK MSG RECORD 05480000
 B OK5 05490000
NOTOK2 PUT OUTFILE,NOTOKMSG WRITE NOTOK MSG RECORD 05500000
OK5 DS OH 05510000
* 05520000
 BAL R14,TSTRCRTN INVOKE ROUTINE TO CHECK RC/STAT 05530000
* 05540000
CVS2EXIT DS OH EXIT FROM CVAFSQ2 05550000
 L R14,CVS2SAVE LOAD C(SAVE AREA) INTO R14 05560000
 BR R14 EXIT 05570000
* 05580000
***** 05590000

```

---

Example C-24 Source for CVSEQ8D part 15 of 19

---

```

* CVAFSQ3 * 05600000
* - INVOKE THE CVAFSEQ MACRO AND READ THE DSCBS * 05610000
* EADSCB=OK IS CODED * 05620000
* * 05630000
CVAFSQ3 DS OH CVAFSEQ - ROUTINE3 05640000
 ST R14,CVS3SAVE STORE C(R14) INTO SAVE AREA 05650000
 LA R2,CVPLIST LOAD R2 WITH ADDR OF CVPLIST 05660000
 L R3,DEBADD LOAD R3 WITH DEB ADDRESS 05670000
 PUT OUTFILE,CALLMR13 WRITE A REC TO OUTPUT FILE 05680000
* * 05690000
 CVAFSEQ ACCESS=GT, X05700000
 DEB=(R3), X05710000
 BRANCH=(YES,PGM), X05720000
 BUFLIST=BUFLISTH, X05730000
 EADSCB=OK, X05740000
 MF=(E,(R2)) 05750000
* * 05760000
 ST R15,RETCODE STORE RC INTO RETCODE 05770000
* * 05780000
* REPORT ON CV4EADOK BIT SETTING 05790000
* * 05800000
 TM CVFL4,CV4EADOK SELECT BASED ON FLAG4 05810000
 BNO NOTOK3 05820000
 PUT OUTFILE,OKMSG WRITE OK MSG RECORD 05830000
 B OK6 05840000
NOTOK3 PUT OUTFILE,NOTOKMSG WRITE NOTOK MSG RECORD 05850000
OK6 DS OH 05860000
* * 05870000
 BAL R14,TSTRCRTN INVOKE ROUTINE TO CHECK RC/STAT 05880000
* * 05890000
CVS3EXIT DS OH EXIT FROM CVAFSQ3 05900000
 L R14,CVS3SAVE LOAD C(SAVE AREA) INTO R14 05910000
 BR R14 EXIT 05920000
***** 05930000
* WORKING STORAGE * 05940000
 DS OD 05950000
 DC CL36'CVSEQ8D-WORKING STORAGE BEGINS HERE' 05960008
* * 05970000
***** 05980000

```

---

Example C-25 Source for CVSEQ8D part 16 of 19

---

```

* EQUATES * 05990000
R0 EQU 0 06000000
R1 EQU 1 06010000
R2 EQU 2 06020000
R3 EQU 3 06030000
R4 EQU 4 06040000
R5 EQU 5 06050000
R6 EQU 6 06060000
R7 EQU 7 06070000
R8 EQU 8 06080000
R9 EQU 9 06090000
R10 EQU 10 06100000
R11 EQU 11 06110000
R12 EQU 12 06120000
R13 EQU 13 06130000
R14 EQU 14 06140000
R15 EQU 15 06150000
***** 06160000
* SAVE AREAS * 06170000
SAVE DC 18F'0' MAIN PROGRAM SAVE AREA 06180000
IDVLSAVE DC F'0' IDENTIFY VOLUME ROUTINE SAVE AREA 06190000
CALLSAVE DC F'0' CALL CVSTAT ROUTINE SAVE AREA 06200000
LDTBSAVE DC F'0' LOAD TABLE ROUTINE SAVE AREA 06210000
INITSAVE DC F'0' INIT BUFFER ROUTINE SAVE AREA 06220000
PRTBSAVE DC F'0' PRINT TABLE ROUTINE SAVE AREA 06230000
TSTRSAVE DC F'0' TEST RETURN CODE ROUTINE SAVE AREA 06240000
CVS1SAVE DC F'0' CVAFSEQ READ 1 ROUTINE SAVE AREA 06250000
CVS2SAVE DC F'0' CVAFSEQ READ 2 ROUTINE SAVE AREA 06260000
CVS3SAVE DC F'0' CVAFSEQ READ 3 ROUTINE SAVE AREA 06270000
***** 06280000
* CONSTANTS * 06290000
RCODE00 DC F'0' RETURN CODE 0 06300000
RCODE04 DC F'4' RETURN CODE 4 06310000
RCODE08 DC F'8' RETURN CODE 8 06320000
RCODE12 DC F'12' RETURN CODE 12 06330000
RCODE16 DC F'16' RETURN CODE 16 06340000
CCHHRS DC XL5'0000000000' STARTING CCHHR 06350000
CCHHRO DC XL5'0000000000' INIT WITH ZERO 06360000
FMT4 DC XL1'04' FMT4? 06370000
FMT5 DC XL1'05' FMT5? 06380000
FMT7 DC XL1'07' FMT7? (ONLY CERTAIN DEVICE TYPES) 06390000
FMT9 DC XL1'09' FMT9? (ONLY CERTAIN DEVICE TYPES) 06400000
NODSN DC XL1'00' END OF DATA? 06410000
***** 06420000

```

---



Example C-26 Source for CVSEQ8D part 17 of 19

* PROGRAM MESSAGES			* 06430000
BLNKLINE	DC	CL133' '	06440000
STRMSG	DC	CL133' CVSEQ8D START OF OUTPUT MESSAGES	' 06450008
VOLTEXT	ORG	STRMSG+33	06460000
	DC	C'VOLUME: '	06470000
VOLSER	DC	CL6' '	06480000
	ORG		06490000
ENDMSG	DC	CL133' CVSEQ8D END OF OUTPUT MESSAGES	' 06500008
CALLMR11	DC	CL133' CVAFSEQ CALL: EADSCB KEYWORD NOT CODED	' 06510000
CALLMR12	DC	CL133' CVAFSEQ CALL: EADSCB=NOTOK CODED	' 06520000
CALLMR13	DC	CL133' CVAFSEQ CALL: EADSCB=OK CODED	' 06530000
RC00MSG	DC	CL133' RC00 VERIFIED - THE REQUEST WAS SUCCESSFUL	' 06540000
RC04MSG	DC	CL133' RC04 VERIFIED - LOGICAL ERROR STATUS IN CVSTAT	' 06550000
RC08MSG	DC	CL133' RC08 VERIFIED - LOGICAL ERROR STATUS IN CVSTAT	' 06560000
RC12MSG	DC	CL133' RC12 VERIFIED - LOGICAL ERROR STATUS IN CVSTAT	' 06570000
RC16MSG	DC	CL133' RC16 VERIFIED - LOGICAL ERROR STATUS IN CVSTAT	' 06580000
RCERMSG	DC	CL133' UNEXPECTED RETURN CODE RETURNED FROM CVAFSEQ	' 06590000
OKMSG	DC	CL133' CV4EADOK BIT SET / EADSCB=OK	' 06600000
NOTOKMSG	DC	CL133' CV4EADOK BIT IS NOT SET / EADSCB=NOTOK	' 06610000
EODMSG	DC	CL133' END OF DATA REACHED - ALL DATASETS PROCESSED	' 06620000
MSG0	DS	0CL133	06630000
	DC	CL50' DATASETS ON THE VOLUME LISTED IN PHYSICAL SEQUENT	' 06640000
	DC	CL83' IAL ORDER: '	06650000
MSG1	DS	0CL133	06660000
	DC	CL7' DSN: '	06670000
DSNMSG	DS	CL44' '	06680000
DSNCHA	DS	C' '	06681012
	DC	CL82' '	06690000
*****			06700000
* WORK AREAS			* 06710000
BUFLST	DS	0F	06720000
		BUFFER LIST WORK AREA	
BUFLISTH	DC	2F'0'	06730000
		BUFFER LIST HEADER (2 FULL WORD)	
BUFLISTE	DC	60F'0'	06740000
		20 BUFFER LIST ENTRIES (3 WORDS EACH)	
*			06750000
UCBADD	DC	F'0'	06760000
		UCB ADDRESS SAVE AREA	
DEBADD	DC	F'0'	06770000
		DEB ADDRESS SAVE AREA	
COUNT	DC	F'0'	06780000
		TABLE COUNTER	
	DS	D	06790000
DSCB01	DS	XL140	06800000
		DSCB01 BUFFER AREA	
DSCB02	DS	XL140	06810000
		DSCB02 BUFFER AREA	
DSCB03	DS	XL140	06820000
		DSCB03 BUFFER AREA	
DSCB04	DS	XL140	06830000
		DSCB04 BUFFER AREA	
DSCB05	DS	XL140	06840000
		DSCB05 BUFFER AREA	
DSCB06	DS	XL140	06850000
		DSCB06 BUFFER AREA	
DSCB07	DS	XL140	06860000
		DSCB07 BUFFER AREA	
DSCB08	DS	XL140	06870000
		DSCB08 BUFFER AREA	
DSCB09	DS	XL140	06880000
		DSCB09 BUFFER AREA	
DSCB10	DS	XL140	06890000
		DSCB10 BUFFER AREA	
DSCB11	DS	XL140	06900000
		DSCB11 BUFFER AREA	
DSCB12	DS	XL140	06910000
		DSCB12 BUFFER AREA	
DSCB13	DS	XL140	06920000
		DSCB13 BUFFER AREA	

Example C-27 Source for CVSEQ8D part 18 of 19

DSCB14	DS	XL140	DSCB14 BUFFER AREA	06930000
DSCB15	DS	XL140	DSCB15 BUFFER AREA	06940000
DSCB16	DS	XL140	DSCB16 BUFFER AREA	06950000
DSCB17	DS	XL140	DSCB17 BUFFER AREA	06960000
DSCB18	DS	XL140	DSCB18 BUFFER AREA	06970000
DSCB19	DS	XL140	DSCB19 BUFFER AREA	06980000
DSCB20	DS	XL140	DSCB20 BUFFER AREA	06990000
*				07000000
RETCODE	DC	F'999'	RETURN CODE SAVE AREA	07010000
WFMTRC1	DS	CL133' '	WORK FORMAT RECORD FOR OUTPUT	07020000
WCVSTCD	DC	XL1'FF'	CVSTAT CODE WORK AREA	07030000
*****				07040000
*		PARAMETER LIST FOR EXTERNAL SUBROUTINE		* 07050000
	DS	OD		07060000
PARADDR1	DC	A(WCVSTCD,WFMTRC1)		07070000
*****				07080000
*		DCB - OUTPUT FILE (OUTFILE)		* 07090000
OUTFILE	DCB	DDNAME=OUTDD,		X07100000
		DSORG=PS,RECFM=FB,LRECL=133,BLKSIZE=1330,MACRF=PM		07110000
*****				07120000
*		TABLES		* 07130000
TABLE	DC	OH	START OF TABLE DSCB ADDR	07140000
TDSCB01	DS	F'0'	DSCB01 ADDRESS (140 BYTE DSCB)	07150000
TBLLNG	EQU	*-TABLE	LENGTH OF TABLE ENTRY	07160000
TDSCB02	DS	F'0'	DSCB02 ADDRESS (140 BYTE DSCB)	07170000
TDSCB03	DS	F'0'	DSCB03 ADDRESS (140 BYTE DSCB)	07180000
TDSCB04	DS	F'0'	DSCB04 ADDRESS (140 BYTE DSCB)	07190000
TDSCB05	DS	F'0'	DSCB05 ADDRESS (140 BYTE DSCB)	07200000
TDSCB06	DS	F'0'	DSCB06 ADDRESS (140 BYTE DSCB)	07210000
TDSCB07	DS	F'0'	DSCB07 ADDRESS (140 BYTE DSCB)	07220000
TDSCB08	DS	F'0'	DSCB08 ADDRESS (140 BYTE DSCB)	07230000
TDSCB09	DS	F'0'	DSCB09 ADDRESS (140 BYTE DSCB)	07240000
TDSCB10	DS	F'0'	DSCB10 ADDRESS (140 BYTE DSCB)	07250000
TDSCB11	DS	F'0'	DSCB11 ADDRESS (140 BYTE DSCB)	07260000
TDSCB12	DS	F'0'	DSCB12 ADDRESS (140 BYTE DSCB)	07270000
TDSCB13	DS	F'0'	DSCB13 ADDRESS (140 BYTE DSCB)	07280000
TDSCB14	DS	F'0'	DSCB14 ADDRESS (140 BYTE DSCB)	07290000
TDSCB15	DS	F'0'	DSCB15 ADDRESS (140 BYTE DSCB)	07300000
TDSCB16	DS	F'0'	DSCB16 ADDRESS (140 BYTE DSCB)	07310000
TDSCB17	DS	F'0'	DSCB17 ADDRESS (140 BYTE DSCB)	07320000
TDSCB18	DS	F'0'	DSCB18 ADDRESS (140 BYTE DSCB)	07330000
TDSCB19	DS	F'0'	DSCB19 ADDRESS (140 BYTE DSCB)	07340000
TDSCB20	DS	F'0'	DSCB20 ADDRESS (140 BYTE DSCB)	07350000
TBLNBR	EQU	(*-TABLE)/TBLLNG	NBR OF TABLE ENTRIES	07360000
*****				07370000
*		SWITCHES		* 07380000
SWEOD	DC	XL1'00'	SWITCH - END OF DATA ?	07390000
EOD	EQU	X'FF'	END OF DATA DETECTED	07400000
NOEOD	EQU	X'00'	END OF DATA NOT DETECTED	07410000
*****				07420000

*Example C-28 Source for CVSEQ8D part 19 of 19*

* VTDC DCB AREA			* 07430000
VTDCDCB	DCB	DDNAME=CVAFDD,MACRF=E,EXLST=XLST1,DSORG=PS,DCBE=VTDCDCBE	07440000
XLST1	DC	X'87'	07450000
	DC	AL3(JFCB1)	07460000
JFCB1	DS	OCL176	07470000
TESTNAME	DS	CL44	07480000
	DS	CL8	07490000
	DS	BL1	07500000
	DS	CL123	07510000
VTDCDCBE	DCBE	EADSCB=OK	07520000
*****			07530000
* DSECTS			* 07540000
	ICVAFBFL	DSECT=YES	07550000
* IECSDSL1 (1)			07560000
	IECSDSL1	(1)	07570000
DSCBLTH	EQU	*-IECSDSL1 LENGTH OF DSCB	07580000
TBLMAP	DSECT	DUMMY CONTROL SECTION FOR TABLE MAP	07590000
DSCBA	DS	F DSCB ADDRESS ENTRY	07600000
*****			07610000
* CVAF PARAMETER LISTS			* 07620000
CVSEQ8D	CSECT		07630008
CVPLIST	CVAFSEQ	MF=L	07640000
	ORG	CVPLIST	07650000
CVPLMAP	ICVAFPL	DSECT=NO	07660000
	END	CVSEQ8D END OF CVSEQ8D	07670008

## Sample job to initialize data set for OCE Partial Release

In Example C-29 we show the assembler that was used for the OCE Partial release scenarios in Chapter 7, “DFSMSdfp enhancements” on page 105. It is not intended to be run and is presented as illustration of how the scenarios were set up.

*Example C-29 Assembler code to initialise a data set for demonstration (1 of 2)*

&NAME	SEGSTART		00001099
	MACRO		00010000
&NAME	SEGSTART		00020000
	AMODE	24	00030000
&NAME	STM	14,12,12(13) SAVE HIS REGS IN HIS SAVE AREA	00040000
R1	EQU	1	00060000
R3	EQU	3	00080000
R4	EQU	4	00090000
RB	EQU	12	00170000
R13	EQU	13	00180000
R14	EQU	14	00190000
R15	EQU	15	00200000
	BALR	12,0 SET UP ADDRESSABILITY	00210000
	USING	*,12 USE REG 12 AS BASE REG	00220000
	ST	13,SAVEREGS+4 SAVE @ OF HIS SAVEAREA IN MINE	00230000
	LA	03,SAVEREGS LOAD @ OF MY SAVE AREA IN REG 3	00240000

	ST	03,8(13)	SAVE @ OF MY SAVE AREA IN HIS	00250000
	LR	13,03	LOAD @ OF MY SAVE AREA IN REG 13	00260000
	MEND			00270000
	MACRO			00280000
&NAME	SEGEND			00290000
&NAME	L	13,SAVEREGS+4	LOAD REG13 WITH @ OF HIS SAVE	00300000
	LM	14,12,12(13)	RESTORE REGS FROM HIS SAVEAREA	00310000
	XR	R15,R15		00320000
	BR	14	RETURN TO CALLING RTN VIA REG 14	00330000
SAVEREGS	DC	18F'0'	SET UP SAVE AREA	00340000
	MEND			00350000
GENREC1	SEGSTART			00440099
*	THIS IS A SIMPLE PROGRAM TO GENERATE RECORDS TO FILL UP A DATA SET.			00450099
*	UNITL THE TASK GETS SB37 BY DEFAULT.			00460099
*	PARM=SHORT WILL CAUSE THE PROGRAM TO WRITE ONLY 1000 RECORDS.			00461099
*				00470000
*	R3	PARM		00480000
*	R4	RECORD COUNT		00490099
*	R12	OVERALL BASE REGISTER		00500000
START	DS	0H		00510000
	MVI	SHORTRUN,C'N'		00520099
	L	R15,0(R1)	GET ADDRESS OF PARM	00530000
	LH	R3,0(R15)	GET LENGTH OF PARM	00540000
	CH	R3,=H'0'	CHECK PARM LENGTH	00550000
	BE	EXPARM		00560099
	SH	R3,=H'1'	REDUCE FOR EXECUTE	00570000
	EX	R3,MVCPARM		00580099
	CLC	SYSPARM(5),=C'SHORT'		00590099
	BNE	EXPARM		00591099

---

*Example C-30 Assembler code to initialise a data set for demonstration (2 of 2)*

~~~~~	MVI	SHORTRUN,C'Y'	00600099
	LA	R4,1000	00601099
EXPARM	DS	0H	00610099
	OPEN	(GENDCB,(OUTPUT))	00690099
LOOPBK	DS	0H	00691099
	PUT	GENDCB,PRINTHD1	00700099
	CLI	SHORTRUN,C'N'	00701099
	BE	LOOPBK	00710099
	KEEP GOING UNTIL THE DATA SET IS FULL		
*	ONLY	DO 1000 WRITES	00710199
	BCT	R4,LOOPBK	00711099
FINISH	DS	0H	01120000
	SEGEND		01130000
	ORG		01140000
PRINTHD1	DC	CL80' TEST DATA RANDOM AS IT CAN BE WITHOUT THINKING'	01440099
GENDCB	DCB	DDNAME=GENDCB,DSORG=PS,MACRF=(PM)	01500099
SHORTRUN	DS	C	01501099
SYSPARM	DC	CL100'SYSPARM SYSPARM'	01502099
MVCPARM	MVC	SYSPARM(0),2(R15)	01503099
	ORG		01510000
	END		01530000

---

# Related publications

The publications listed in this section are considered particularly suitable for a more detailed discussion of the topics covered in this book.

## IBM Redbooks

The following IBM Redbooks publications provide additional information about the topic in this document. Note that some publications referenced in this list might be available in softcopy only.

- ▶ *z/OS DFSMSHsm Primer*, SG24-5272
- ▶ *z/OS V1.13 DFSMS Technical Update*, SG24-7961
- ▶ *DFSMSrmm Primer*, SG24-5983

You can search for, view, download or order these documents and other Redbooks, Redpapers, Web Docs, draft and additional materials, at the following website:

[ibm.com/redbooks](http://ibm.com/redbooks)

## Other publications

These publications are also relevant as further information sources:

- ▶ *z/OS DFSMS OAM Planning, Installation, and Storage Administration Guide For Tape Libraries*, SC23-6867
- ▶ *z/OS DFSMS Installation Exits*, SC23-6850
- ▶ *z/OS DFSMSHsm Storage Administration*, SC23-6871
- ▶ *z/OS DFSMSdss Storage Administration*, SC23-6868
- ▶ *z/OS DFSMSHsm Diagnosis*, GC52-1387
- ▶ *z/OS DFSMSrmm Reporting*, SC23-6875
- ▶ *z/OS DFSMSHsm Implementation and Customization Guide*, SC23-6869
- ▶ *z/OS MVS System Messages, Volume 2 (ARC-ASA)*, SA22-7632
- ▶ *z/OS DFSMS Using New Functions*, SC23-6857
- ▶ *z/OS DFSMSdfp Storage Administration*, SC23-6860
- ▶ *z/OS DFSMSdfp Diagnosis*, SC23-6863
- ▶ *MVS System Messages, Vol 2 (ARC-ASA)*, SA38-0669
- ▶ *z/OS DFSMS Managing Catalogs*, SC23-6853
- ▶ *z/OS DFSMS Access Method Services Commands*, SC23-6846
- ▶ *z/OS MVS Initialization and Tuning Guide*, SA23-1379
- ▶ *z/OS MVS Initialization and Tuning Reference*, SA23-1380

## Online resources

These websites are also relevant as further information sources:

- ▶ z/OS V2.1 Information Center  
<http://pic.dhe.ibm.com/infocenter/zos/v2r1/index.jsp>
- ▶ z/OS Hot Topics  
<http://www-03.ibm.com/systems/z/os/zos/library/hot-topics/hot-topics.html>
- ▶ TS7700 Customer Information Center  
<http://publib.boulder.ibm.com/infocenter/ts7700/cust/index.jsp>

## Help from IBM

IBM Support and downloads

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**Understand the features and functions of z/OS V2.1 DFSMS**

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